

A two-phase model of galaxy formation: IV. Seeding and growing supermassive black holes in dark matter halos

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Accepted XXX. Received YYY; in original form ZZZ

ABSTRACT

We present a theoretical framework for seeding and growing supermassive black holes (SMBHs) in dark matter halos along their assembly histories. Seeds are bred out of Pop-III stars formed during the first collapse of pristine gas in mini-halos that have reached the H₂-cooling limit, modulated by UV radiation from star formation and dynamical heating from fast halo assembly. Such breeding persists until the enrichment of the intergalactic medium (IGM) enables Pop-II stars to form. Post-seeding growth of black holes (BHs) is driven by distinct channels, starting with episodic super-Eddington accretion associated with nuclear bursts induced by global disturbances of galaxies, followed by sustained sub-Eddington accretion via capturing sub-clouds formed in self-gravitating gas clouds (SGCs) in halos of fast assembly, and ending with merger-dominated, quiescent growth. We implement the model in subhalo merger trees to build a coherent framework to follow SMBH-galaxy-halo co-evolution across the whole history and structural hierarchy. BH seeds are bred with a broad mass spectrum of $M_{\text{BH}} = 10\text{--}10^5 M_{\odot}$ at $z \approx 20\text{--}30$ in mini-halos with masses of $10^5\text{--}10^8 M_{\odot}$. Nuclear bursts provide the key condition for seeds to grow into SMBHs. The $M_{\text{BH}}\text{--}M_{\ast}$ relation is a multi-piece, redshift-dependent function shaped by the interplay among different growth channels. Our model predictions are broadly consistent with existing observations; especially, a population of BHs reminiscent of ‘little red dots’ (LRDs) discovered by JWST naturally results from the seeding and growing processes. Potential future tests of the model are discussed.

Key words: galaxies: high-redshift – galaxies: haloes – galaxies: formation – quasars: supermassive black holes

1 INTRODUCTION

The observable Universe exhibits a structure hierarchy composed of populations of objects across multiple scales (e.g. Mo et al. 2010). Galaxies, luminous stellar islands characterized by distinct luminosities and morphologies, constitute the building blocks of this hierarchy (Peacock 2003; Abazajian et al. 2003; Wright et al. 2016). On larger scales, galaxies are gravitationally clustered in groups and clusters of galaxies (Abell et al. 1989; Lumsden et al. 1992; Dalton et al. 1997; Yang et al. 2003; Van Den Bosch et al. 2003; Yang et al. 2007; Tinker 2021; Wang et al. 2020), which themselves are connected through complex structures such as filaments and sheets to form the cosmic web (Bond et al. 1996; Hahn et al. 2007; Wang et al. 2016; Zhang et al. 2025a). Within individual galaxies, the microscopic architecture, including individuals and clusters of stars and black holes (BHs), exhibits diverse properties and interacts with the host galaxy through environmental (Springel & Hernquist 2005; Dekel et al. 2009; Pimbblet 2011; Wang et al. 2018; Wise et al. 2019; Latif et al. 2022a; Wang et al. 2023; Meng et al. 2023; Puskás et al. 2025) and feedback processes (Croton et al. 2006; Bower et al. 2006; Somerville et al. 2008; Booth & Schaye 2009; Vogelsberger et al. 2013). According to

modern cosmology, these objects do not exist statically as they are observed. Instead, they are all seeded from primordial density perturbations in the early Universe, amplified and shaped by various physical processes over the long history of the Universe. With the advent of powerful telescopes, such as the James Webb Space Telescope (JWST) and Atacama Large Millimeter Array (ALMA), it is now possible to observe these objects from their infancy all the way to the present day (Finkelstein et al. 2023; Bezanson et al. 2022; Parlanti et al. 2023; Rowland et al. 2024; Fujimoto et al. 2025a; Suess et al. 2024; Naidu et al. 2024; Wang et al. 2025; Jia et al. 2025; Scholtz et al. 2025). This allows us to gather discrete evolutionary snapshots in observations to form a unified picture of structure formation. Theoretically, great efforts have been made in understanding the origin and evolution of these objects in the Λ cold dark matter (Λ CDM) paradigm (e.g. Peebles 1980; Davis et al. 1985; O’Shea & Norman 2008; Schaye et al. 2015; Stacy et al. 2016; Wang et al. 2015; Pillepich et al. 2018; Davé et al. 2019; Latif et al. 2021; Hopkins et al. 2023). This paradigm provides not only a unified cosmological background for interpreting observational data, but also initial conditions, matter/energy contents and physical principles to model the evolutionary processes and to predict the structure hierarchy over the entire history of the Universe. Substantial progress has been made in modeling structure formation within this paradigm using different approaches (e.g. Kauffmann et al. 1993; Peacock & Smith 2000; Springel & Hernquist 2003), but many

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problems remain unresolved. The main challenge comes from the many components that make up the structure hierarchy, the wide range of scales involved in the processes that drive the evolution, and the complex interplay among these processes.

A key step in tackling the challenge is, therefore, to build a coherent model that can cover the whole structure hierarchy and the full evolution history within Λ CDM, so that the connection and comparison between model and observation can be made across the whole structure hierarchy and over the entire history of the Universe. In a recent paper (Mo et al. 2024, hereafter *Paper-I*), we proposed a ‘two-phase’ scenario for galaxy formation in the context of current cosmology. The key hypothesis of this scenario is that the growth of each central galaxy can be divided into two phases according to the growth rate of its host halo: a fast phase, in which the halo accretion is fast and there is a rapid change in the gravitational potential, and a slow phase, in which the halo accretion is slow and the gravitational potential remains steady (Zhao et al. 2003; Lu et al. 2006; Zhao et al. 2009; Klypin et al. 2016). The distinct physical processes and the results expected from them in the two phases are discussed extensively and modeled in detail in the papers of this series. These include: (i) a reversed dynamical evolution in which a galaxy first builds its bulge in the fast phase and then grows its disk in the slow phase, and efficient growth of the central supermassive black hole (SMBH) via capturing dense sub-clouds formed in the turbulent, self-gravitating gas cloud (SGC) associated with bulge formation (*Paper-I*); (ii) a homology relation between bulge size and halo size (Chen et al. 2024, hereafter *Paper-II*); (iii) the two-channel formation of globular clusters (GCs; Chen et al. 2025a, hereafter *Paper-III*).

One important subject in understanding galaxy formation is to identify the co-evolution pattern of BHs and their host galaxies and halos (e.g. Kormendy & Ho 2013; Bower et al. 2017; Habouzit et al. 2021; Zhuang & Ho 2023; Graham & Sahu 2023; Li et al. 2025b). In observations, such evolutionary information is often not accessible for individual objects. Instead, properties of objects can be measured for representative samples within discrete intervals of redshift (z), and relations between the properties can be constructed to form a statistical description for galaxies at each snapshot of the Universe. Since the imprints of physical processes driving galaxy formation are encoded in these relations, albeit in a degenerate way, retrieving physical information from the relations has been a major endeavor in linking observations to formation processes. One of the most important relations is the $M_{\text{BH}}-M_*$ relation, which has been measured observationally in the local Universe (e.g. Kormendy & Ho 2013; Greene et al. 2020; Zhuang & Ho 2023; Graham & Sahu 2023) and has been recently extended to high z by observations with e.g. JWST and ALMA (Izumi et al. 2021; Übler et al. 2023; Harikane et al. 2023b; Maiolino et al. 2024b; Onoue et al. 2025; Chen et al. 2025b). Theoretical efforts have been made to recover the observed relation, to predict its extension to higher z , and to link the relation to underlying processes driving the co-evolution (e.g. Bower et al. 2017; Habouzit et al. 2021; Li et al. 2025b; Hong et al. 2023; Voit et al. 2024; Kido et al. 2025; Jiang et al. 2025; Dekel et al. 2025a; Bellows 2025).

A critical condition for the build-up of the $M_{\text{BH}}-M_*$ relation is effective regulation by the feedback from the accreting SMBH (e.g. Weinberger et al. 2018; Hong et al. 2023; Habouzit et al. 2021; *Paper-I*; Li et al. 2025b). However, this can happen only when the SMBH has grown sufficiently massive so that the radiative efficiency of the associated AGN, which is $\propto M_{\text{BH}}$, is sufficient to affect gas cooling and collapse. One way to achieve this in modeling is to grow BHs from massive seeds, usually with $M_{\text{BH}} \gtrsim 10^5 M_\odot$, which are on or above the observed $M_{\text{BH}}-M_*$ relation, and to rely on regulations by the stellar and AGN feedback to drive them down to and to move along

the relation. This approach is commonly adopted in cosmological hydrodynamic simulations (e.g. Schaye et al. 2015; Bower et al. 2017; Weinberger et al. 2018; Davé et al. 2019; Habouzit et al. 2021; Li et al. 2025b). However, it is unclear whether and how such seeds can be bred. Alternatively, one can start with seeds that are much less massive, and boost their growth via various mechanisms, such as mergers with star clusters brought in by dynamical friction, as suggested by Dekel et al. (2025b), or the capture of the low angular-momentum sub-clouds that are regenerated continuously by injecting disturbance into the turbulent gas, as suggested by *Paper-I* (see their §4.3.3 and fig. 7; see also Shi et al. 2023). Thus, a realistic model for the formation of BH seeds and the early growth in the post-seeding era is required to break this seeding-growing degeneracy.

In this paper, we extend the model presented previously in the papers of this series by developing prescriptions to seed and grow BHs in dark matter halos within the current cosmological context. We show how these prescriptions can be incorporated coherently into the two-phase scenario of galaxy formation, and implemented semi-analytically in halo merger trees to make model predictions. Our main goals are (i) to derive when, where and how BH seeds are bred in the early universe and what masses these seeds have when they are produced; (ii) to understand the early growth of the seeds that eventually makes them supermassive, as well as their late growth that shapes the scaling relations observed at lower redshift; (iii) to differentiate effects from seeding and growing on the observed BH populations at different z and to understand how different environmental factors affect the products; (iv) to predict observable quantities and relations for halos, galaxies, star clusters and (SM)BHs at different redshifts, both statistically for populations and individually for representative objects.

The paper is organized as follows. In §2, we start with a review of the cosmological context relevant to the modeling, and then introduce the two components of the model, the seeding procedure and the growing procedure. The main text focuses on the motivation, the key concepts and the outline of the model, and refers to the appendices for the detailed prescriptions. Analytical estimates are provided as much as possible in the appendices so that intuition and order-of-magnitude quantification can be obtained without diving into the details. In §3, we synthesize the physical prescriptions semi-analytically by implementing them into halo merger trees in a cosmological volume, present the results predicted by the model, and make comparisons with observations. In §4, we summarize our main results, and discuss their implications and potential follow-up investigations prompted by them. Throughout this paper, we adopt a flat Λ CDM cosmology given by Planck Collaboration et al. (2016): $h = 0.6774$, $\Omega_{\text{m},0} = 0.3089$, $\Omega_{\Lambda,0} = 0.6911$, $\Omega_{\text{b},0} = 0.0486$, and a Gaussian initial power spectrum specified by $n_s = 0.9667$ and $\sigma_8 = 0.8159$. We use BH to refer to the central BH of a host galaxy, unless specified otherwise.

2 MODELING BLACK-HOLE SEEDS AND THEIR GROWTH IN DARK MATTER HALOS

Fig. 1 outlines the key components of our model and the pipeline that links them together. Briefly, the model starts with the merger trees of dark matter halos (§2.1) and processes of galaxy formation expected in such trees (§2.2). A seeding procedure is then applied to the merger trees to determine when, where and how BH seeds are bred in mini-halos (§2.3). The breeding of the seed marks the beginning of the formation of a galaxy in the halo and initiates the growing procedure that follows the subsequent evolution of the galaxy and the BH (§2.4). The output of the model thus covers the full lifetimes of BHs and their host galaxies in a cosmological volume.

Outline of the model

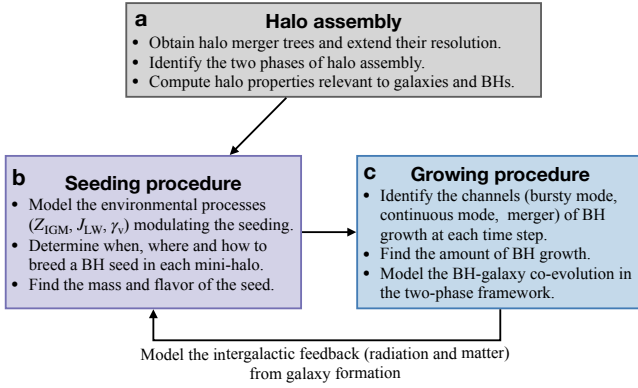


Figure 1. A brief outline of the model. The model takes halo merger trees in a cosmological volume as input, uses the ‘seeding procedure’ to set the initial condition for every mini-halo, and follows a ‘growing procedure’ to predict the post-seeding evolution over cosmic time. See §2 for an introduction of the model. A more detailed version of this outline is available in Fig. D1, and the details of numerical implementation are provided in Appendix D.

2.1 Assembly of dark matter halos

In the current Λ CDM paradigm of galaxy formation, galaxies form and evolve in dark matter halos. The early assembly of halos at high z is thus crucial for determining when, where and how the formation of the first generation of stars and the seeds of BHs occurs, as well as the subsequent growth of these seeds to SMBHs. To model the full lifetimes of BHs, the assembly of halos has to be tracked up to very high z , at which the halos are in their infancy (hereafter ‘mini-halos’). In Appendix A, we present a detailed review of the halo population and methods used to obtain their assembly histories. Here we summarize the results most relevant to our modeling and discussion.

Throughout this paper, a halo is defined by a spherical volume of virial radius R_v , within which the mean mass density is $\Delta_v = 200$ times the critical density of the Universe (Navarro et al. 1997; see e.g. Han 2026 for alternative definitions). The assembly history of a halo can then be described by the evolution of various virial properties, such as virial mass (M_v), virial density (ρ_v), virial velocity (V_v), virial temperature (T_v), dynamical timescale ($t_v \equiv R_v/V_v$), and accretion rate ($\dot{M}_v$). Halo density profiles are modeled by the NFW form (Navarro et al. 1997) specified by a scale radius R_s and a concentration parameter $c_v = R_v/R_s$ that is related to the assembly history (Zhao et al. 2009).

Fig. 2 shows the median assembly histories of halos of given masses at $z = 0$, obtained from large ensembles of halo merger trees sampled by a Monte Carlo algorithm (see Appendix D1). Typically, more massive halos assemble at higher rates, have higher M_v , R_v , V_v and T_v and smaller c_v across the cosmic history. However, halos with $M_{v,z=0} > 10^{12} M_\odot$ have similar progenitors at $z \gtrsim 15$, with mass $M_v \sim 10^8 M_\odot$, $R_v \sim 1$ kpc, $V_v \sim 20$ km s $^{-1}$ and $T_v \sim 10^{4.5}$ K. These ‘mini-halos’, with a typical density $\rho_v \sim 0.1 M_\odot \text{pc}^{-3}$, are places where BH seeds are bred, as we will see below.

A prominent property of CDM halos is their ‘two-phase’ assembly (Zhao et al. 2003; see also Lu et al. 2006; Zhao et al. 2009; Klypin et al. 2016): an earlier, fast-accretion phase (hereafter *fast phase*) where the gravitational potential of a halo rapidly deepens with a roughly constant concentration parameter $c_v = c_f \approx 4$, followed by a slow-accretion regime (hereafter *slow phase*) where the halo grows in the outskirts while the inner radius $R_f \approx c_f R_s$ established in the *fast*

phase changes little so that c_v increases with time. These two phases are evident in Fig. 2, where the growth of M_v slows over time and V_v even decreases over time in later epochs. The epoch separating the two phases is visible in panel b as the deviation of c_v from the constant value of c_f . In addition to the median assembly history, variations in individual halos can cause temporary excursions between the fast and slow phases. Our model includes such excursions by using simulated assembly histories, as described in the following.

To obtain a realistic sample of halos, we use halos from a cosmological N-body simulation, TNG100-1-Dark (Pillepich et al. 2018; Nelson et al. 2019). This simulation uses 1820^3 dark matter particles, each with a mass of $6.0 \times 10^6 h^{-1} M_\odot$, to sample the density field in a periodic box with a side length of $L_{\text{box}} = 75 h^{-1} \text{Mpc}$. Halos, subhalos and their merger trees were obtained from the simulation using the friends-of-friends (FoF) algorithm with a scaled linking length of 0.2 (Davis et al. 1985), the SUBFIND algorithm (Springel et al. 2001), and the SUBLINK algorithm (Springel 2005), respectively. The lower limit for the FoF halo mass is about $2 \times 10^8 h^{-1} M_\odot$, insufficient to resolve the mini-halos needed to breed the seeds. To overcome this, we apply a Monte Carlo algorithm to the simulated merger trees to extend them down to the mini-halo regime (see Appendix D1). The merger trees in our sample are those rooted in subhalos with host halo mass $M_v \geq 10^{9.5} M_\odot$ which is evaluated at $z = 0$ for a central subhalo and at the infall time for a satellite subhalo. Subhalos above this mass limit are expected to grow above the hydrogen atomic-cooling limit at some epochs, and the choice of this limit thus ensures that galaxies with substantial star formation after the cosmic reionization are included in our modeling (Lu et al. 2014). Halos that do not grow above this limit may also have some star formation when, for example, the intergalactic medium (IGM) has sufficiently high neutral fraction to shield UV radiation (e.g. Dijkstra et al. 2004; Benson et al. 2006; Ma et al. 2025c), which may produce extremely low-mass galaxies at low z that are missed by our model. Some of the halo mass limits relevant for the formation of early galaxies and BH seeds are plotted in Fig. 2 and will be discussed later.

2.2 The two-phase formation of galaxies

As a halo forms, gas within it is heated by virialization processes and must first cool down before collapsing further to fragment and form stars. The competition between heating and cooling processes thus determines galaxy formation in the halo. A critical consequence of the two-phase assembly of halos, as proposed in Paper-I and examined in Ma et al. (2025a), is the two-phase formation of galaxies. During the first phase, gas cools effectively before the halo reaches $M_v \approx M_{\text{cool}} \sim 10^{12} M_\odot$, collapses rapidly into the halo center, and forms a self-gravitating gas cloud (SGC) before rotation-support becomes important. Density perturbations are seeded and amplified by turbulence in the SGC, causing the SGC to fragment into sub-clouds that are dense enough to be free of drag force from ambient gas and cloud-cloud collision. The mixing of angular momentum among the sub-clouds is inefficient, so that the SGC sustains a dynamically-hot structure with a wide distribution of specific angular momentum (j). Stars formed in the sub-clouds inherit the hot dynamics, forming a dynamically hot, bulge-like stellar system with a homology relation between the sizes of the bulge and halo, $r_{\text{bulge}} \approx (1/100)R_f$, that is independent of halo mass and redshift (Chen et al. 2024). Stable cold gaseous/stellar disks form in the slow phase as the formation of sub-clouds stalls due to the stabilized inner halo potential and the reduced gas fraction by feedback. To highlight the distinct galactic dynamics during the two phases, we refer to the two phases interchangeably as the ‘*bulge phase*’ and the ‘*disk phase*’, respectively.

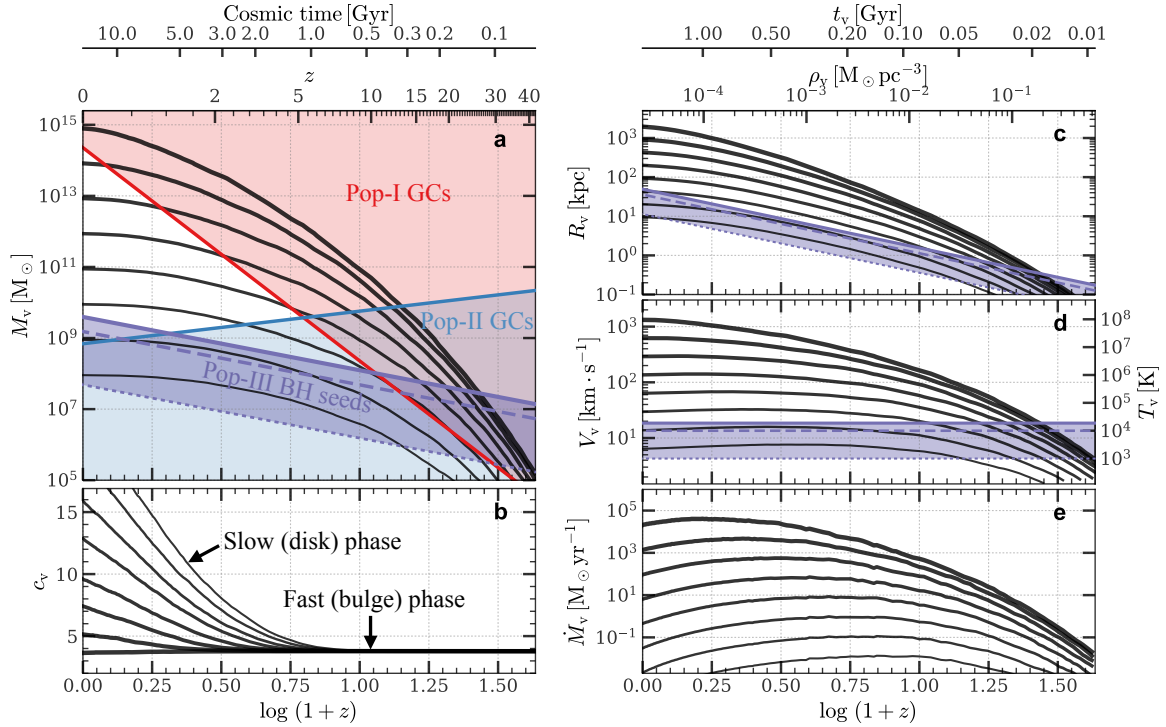


Figure 2. Assembly histories of dark matter halos. Here we show the evolution of different virial properties: **a**, mass (M_v); **b**, concentration (c_v); **c**, radius (R_v , in physical scale); **d**, velocity and temperature (V_v and T_v); **e**, accretion rate ($\dot{M}_v$). The assembly histories are computed along the main branches of halos at $z = 0$ sampled by a Monte Carlo algorithm, and a median is taken over the halos with each given $M_{v,z=0}$ from $10^8 M_\odot$ to $10^{15} M_\odot$ (**black curves** from thin to thick). **Purple bands** in **a**, **c** and **d** indicate the regime where the first collapse of an SGC is expected to be triggered so that Pop-III stars can form and a BH seed can be bred, with the associated purple **dotted**, **dashed** and **solid** lines representing M_{v,H_2} (the mass for H_2 -cooling to be effective; Eq. 1), $M_{v,H}$ (the mass for H-cooling to be effective; Eq. 3), and $f_b^{-1} M_{J,max}$ (the mass for the SGC with strong turbulence support to be Jeans unstable; Eq. 4), respectively. The exact location for a halo to be seeded within this band depends on its assembly history and radiation environment. For reference, **red** and **blue** regions in **a** show the regimes where Pop-I and Pop-II GCs can actively form, respectively, according to Paper-III, highlighting the role of IGM enrichment in modulating the stellar population and seed-formation pathway. This figure summarizes the properties of halos over wide ranges of mass and redshift (see §2.1), which will be used as input to our model to determine when, where and how the first generation of stars and BH seeds form (see §2.3), as well as their post-seeding growth (see §2.4).

The formation of sub-clouds in fast-accreting halos provides the condition for forming and growing different types of compact sub-structures. As proposed in Paper-III, the density of sub-clouds can be elevated by two distinct channels, the compression by supersonic turbulence and the slow cooling due to low metallicity, before they fragment to form stars. Some sub-clouds can reach a ‘supernova-free’ density, $n_{snf} \approx 10^{3.5} \text{ cm}^{-3}$, at which the free-fall timescale (t_{ff}) of the gas falls below $t_{snf} \approx 1 \text{ Myr}$, so that the gas is converted into stars before the onset of supernova feedback. The two channels of density elevation thus lead to the formation of two populations of globular clusters (GCs), Pop-I and Pop-II, that preferentially occupy halos in different evolutionary stages. Fig. 2a shows the ‘atlas’ of GC formation, where the red and blue regions indicate the regimes dominated by the two channels, respectively. In the extreme case where the halo is pristine, Pop-III stars are expected to form and seed a BH in the halo (see §2.3 for details). Some of the sub-clouds formed in the halo may sink to the halo center to feed the central BH, providing a channel for BH growth (see §2.4 for details).

2.3 Producing black-hole seeds in mini-halos

To seed a BH in a mini-halo, we identify key processes of cooling and heating so that conditions for the halo gas to collapse and fragment can be determined. Once these conditions are met, we consider the feedback from proto-stars to determine the properties of the first

generation stars that form in the halo. We then follow the evolution tracks of these stars to predict their fates and the properties of the BHs left as their remnants. The output of the seeding procedure is the flavor (representing the formation pathway) and mass of the BH seed. Fig. 3 sketches the key factors in the cosmological context to be considered in the seeding procedure. A detailed description of the seeding procedure is presented in Appendix B. The following is a brief summary of the key points.

At high z where the intergalactic environment is pristine, the first important coolant during the growth of a mini-halo is molecular hydrogen (H_2), which can keep the gas temperature at a level of $T_{H_2} = 10^3 \text{ K}$ (Latif & Khochfar 2019). The first collapse of halo gas is thus expected when the halo grows to $T_v \approx T_{H_2}$, corresponding to a halo mass of

$$M_{v,H_2} \approx \left[3.87 \times 10^5 M_\odot \right] (1+z)_{25}^{-3/2} T_{H_2,3}^{3/2}, \quad (1)$$

where $(1+z)_{25} \equiv (1+z)/25$ and $T_{H_2,3} \equiv T_{H_2}/(10^3 \text{ K})$. At $1+z = 25$, the collapse proceeds on a timescale of $t_v \approx 20.7 \text{ Myr}$, leading to the formation of a compact SGC with a mass $M_{SGC} \approx 6.08 \times 10^4 M_\odot$, a radius $R_{SGC} = 3.63 \text{ pc}$, a density $n_{SGC} = 1.03 \times 10^4 \text{ cm}^{-3}$, and a free-fall timescale of $t_{ff,SGC} = 0.463 \text{ Myr}$ (see Appendix B1.2). The thermal sound speed at T_{H_2} is $c_s \approx 3.4 \text{ km s}^{-1}$, implying a Jeans mass of

$$M_{J,H_2} = \left[2.30 \times 10^4 M_\odot \right] (1+z)_{25}^{-3/2} T_{H_2,3}^{3/2}. \quad (2)$$

Cosmological context of black-hole seeding

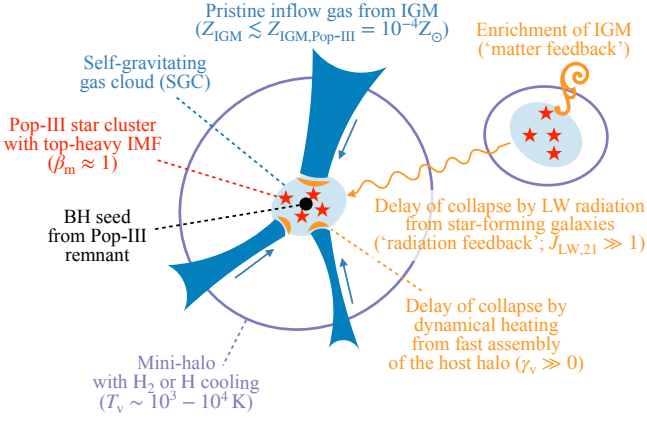


Figure 3. Schematic diagram summarizing the cosmological context of BH seeding. Pristine gas in a mini-halo that has reached the H_2 -cooling limit ($T_v \approx 10^3$ K; Eq. 1) can start to collapse to form a SGC at the halo center, contract further to form a Jeans core, and fragment to form a Pop-III star cluster. A light-weight BH seed can be bred when the most massive star (dominant star) in the cluster ends its life by CCSN. Three environmental effects can modulate the seeding process (orange texts): UV radiation and dynamical heating can delay the collapse and raise the mass of the dominant star so that PISN or DCBH can appear and leave a more massive seed; enrichment in IGM can promote the cooling, cause Pop-II stars to form and leave a less massive seed. A radiation-regulated scenario is proposed here to model the IMF of Pop-III stars, and the mass of BH seed is determined accordingly by following the stellar evolution of these stars. Our scenario suggests that the environmental effects – concepts that have been built for galaxy formation at low z , have already been ubiquitous in the early Universe at $z \gtrsim 20$. See §2.3 for details.

This Jeans mass is comparable to the SGC mass, indicating that further contraction of the SGC proceeds globally without fragmentation until other cooling channels are opened at a higher density.

With a primordial composition, the contraction leads to a core in the SGC with a density of $n = n_{\text{core}} \approx 10^6 \text{ cm}^{-3}$, at which three-body reactions rapidly cool the gas, increasing the production of H_2 and making the core unstable (see §3.3 of [Latif & Khochfar 2019](#)). The Jeans mass in the core is $M_{J,\text{core}} \approx 1.88 \times 10^3 M_\odot \ll M_{v,H_2}$, corresponding to a size of $R_{\text{core}} \approx 0.25 \text{ pc}$ for the density in question. Within such a core, conditions for star formation are satisfied, and fragmentation starts from the center and leads to the formation of a central (proto-)star followed by the formation of other stars. Subsequent rapid inflow and frequent mergers are expected to increase the mass of the central star, producing a top-heavy IMF for the resulting Pop-III star cluster whose mass is dominated by the central star (see e.g. [Stacy et al. 2016](#); [Latif et al. 2022b](#)). The central star dies in a Myr timescale in a core-collapse supernova (CCSN) explosion and leaves behind a ‘light-weight’ BH seed with a mass of a few $10 M_\odot$ (see e.g. §3.1 of [Costa et al. 2023](#)).

Various environmental effects can modulate the above process, promoting or delaying the seeding in a mini-halo, and modifying the mass of the resulting seed. We consider three such effects:

- (i) **UV radiation** from galaxies surrounding the mini-halo;
- (ii) **dynamical heating** associated with fast assembly of the mini-halo;
- (iii) **metal enrichment** in the inter-galactic environment.

The first two can delay the collapse of the halo gas, leading to a more massive SGC and thus to a more massive seed. The last one leads to earlier fragmentation of the SGC, prevents the formation of Pop-III stars, and produces a less massive seed. These effects are

summarized as orange texts in Fig. 3. In the following, we briefly describe our modeling of these effects and their consequences for seeding, referring details to Appendix B1.

UV radiation in the Lyman-Werner band 11.2–13.6 eV (hereafter LW radiation) has been proposed (e.g. §2.1 of [Haiman et al. 2000](#)) as a mechanism to dissociate H_2 , suppress gas cooling, and delay the first collapse of gas in mini-halos. Radiation in this band, mainly produced by young stars in nearby galaxies around a mini-halo, can escape the attenuation from neutral hydrogen in the IGM and propagate over long distances to provide external feedback to the mini-halo. In Appendix D5, we detail the modeling of the LW emission and propagation through the IGM. The intensity of LW radiation irradiating a mini-halo, J_{LW} , is defined as the energy in the LW band received at a location per unit time per unit area per unit frequency and per unit solid angle, and is conveniently expressed as $J_{\text{LW}} = J_{\text{LW},21} 10^{-21} \text{ erg s}^{-1} \text{ cm}^{-2} \text{ Hz}^{-1} \text{ sr}^{-1}$. The threshold mass for gas collapse in a H_2 -cooling mini-halo is expected to increase with increasing J_{LW} . Under a strong radiation of $J_{\text{LW},21} \gg 1$, the collapse of the halo gas can be delayed until atomic cooling starts to operate, i.e., when $T_v \approx T_H \approx 10^4$ K. This corresponds to a halo mass of

$$M_{v,H} = \left[1.22 \times 10^7 M_\odot \right] (1+z)^{-3/2} T_{H,4}^{3/2}, \quad (3)$$

where $T_{H,4} \equiv T_H/(10^4 \text{ K})$. In other cases with moderate J_{LW} , the collapse is expected to be delayed until M_v reaches a value between M_{v,H_2} and $M_{v,H}$, depending on the intensity of the radiation. We therefore parameterize the collapse threshold, $M_{v,LW}$, as a function of J_{LW} , so that it smoothly transits from M_{v,H_2} (at $J_{\text{LW}} = 0$) to $M_{v,H}$ (at $J_{\text{LW}} \gg 1$), and calibrate the parameterization with results from zoom-in hydrodynamic simulations (see Appendix B1.3 for details).

The SGC in a fast-accreting halo is expected to be a turbulence-supported system (see [Paper-I](#)). The situation is the same here because mini-halos hosting seeds are almost certainly in the fast-accreting regime, and because the lack of coolant in the primordial gas makes the turbulence hard to dissipate. We refer to the energy injection associated with fast accretion as ‘dynamical heating’, regardless of the details of how the energy subsequently cascades in the SGC. To model the effect of dynamical heating in delaying the collapse of the SGC, we first consider an extreme case where the heating is so efficient that radiative cooling is inefficient to dissipate it. Taking a turbulence velocity of $v_t = 20 \text{ km s}^{-1}$ reported by [Latif et al. \(2022a\)](#), the SGC has a Jeans mass of

$$M_{J,\text{max}} = \left[4.94 \times 10^6 M_\odot \right] (1+z)^{-3/2}, \quad (4)$$

above which it can collapse globally. The corresponding host halo mass, $M_{v,\gamma} = M_{J,\text{max}}/f_b = 3.14 \times 10^7 M_\odot$, is slightly larger than $M_{v,H}$ given by Eq. (3), indicating that dynamical heating can delay the formation of the first generation of compact objects by a timescale similar to that implied by a strong UV radiation field. In other cases of weaker dynamical heating, we parameterize the threshold mass, $M_{v,\gamma}$, for the SGC to collapse as a function of the halo specific assembly rate, $\gamma_v \equiv \dot{V}_v/[V_v H(z)]$ (Eq. 9 in [Paper-I](#)), so that it changes smoothly from $f_b^{-1} M_{J,H_2}$ (at $\gamma_v = 0$) to $f_b^{-1} M_{J,\text{max}}$ (at $\gamma_v \gg$ some critical value, γ_{crit}). See Appendix B1.4 for details.

In the presence of both UV radiation and dynamical heating, the collapse threshold is computed as the higher of the two:

$$M_{v,\text{th}} = \max [M_{v,LW}, M_{v,\gamma}], \quad (5)$$

With either of the two effects significantly strong, the central proto-star formed in the Jeans core is expected to be fueled by a higher gas accretion rate of $\lesssim 10.7 M_\odot \text{ yr}^{-1}$. The accretion can be sustained at an Eddington-level, causing the formation of a supermassive star

(SMS), until the feedback from the star itself becomes strong enough to quench the accretion (Appendix B1.5). At sufficiently high mass, the central star ends its life by a pair-instability supernova (PISN) explosion, leaving no remnant, or by a direct collapse to a BH, leaving a ‘heavy-weight’ seed. We thus use a *feedback-regulated model* to obtain the initial mass function (IMF) of Pop-III stars in the star cluster formed during the first collapse in a mini-halo and determine the mass of the BH seed by following the stellar evolution of these stars (see Appendix B2 for details).

Star formation in the early universe produces metals that enrich the IGM through galactic outflows and elevate the cooling of gas in nearby mini-halos that have not yet formed stars. To model the IGM enrichment, we evaluate the metal production by star formation in galaxies, and follow the expansion of SN-driven bubbles around individual galaxies to determine the zones of enrichment and the metallicity within them. We adopt a threshold value of $Z_{\text{IGM,Pop-III}} = 10^{-4} Z_{\odot}$ for the formation of Pop-III stars (Bromm & Loeb 2003; Valiante et al. 2011; Costa et al. 2023; Spinoso et al. 2023). We also assume that halos embedded in the IGM with metallicity $Z_{\text{IGM}} > Z_{\text{IGM,Pop-III}}$ form their first generation of stars in the form of a Pop-II star cluster, leaving a light-weight BH seed with $M_{\text{BH}} \sim 10 M_{\odot}$. (see Appendix B2 for details).

Fig. 1b outlines the steps of the seeding procedure. The procedure takes halo merger trees as input and models the above three environmental effects that modulate the seeding. A list of key conditions regarding the competition between cooling and heating is examined for every mini-halo to determine whether the gas can collapse to form the first generation of stars and the BH seed. The output of the procedure is the mass of the seed, as well as the ‘*flavor*’ that indicates the formation pathway of the seed: being modulated by strong UV radiation or fast assembly before the collapse; yielding ‘Pop-III’ or ‘Pop-II’ stars at formation; experiencing radiation regulation at sub-Eddington or Eddington level; and triggering CCSN, PISN or DCBH in the production of the seed. Once seeded, we assume that the galaxy is self-enriched and follow the growing procedure to be detailed below to track the subsequent evolution of the galaxy and the BH.

Fig. 2a shows an ‘*atlas*’ of BH seeds. Here, the purple band covers the region in the M_v - z plane to breed Pop-III seeds. The intersection of the band with the median assembly history of halos of a given $M_{v,z=0}$ (black curve) provides a reference for when (at which z), where (at which M_v) and how (modulated by which factor) BHs can be seeded. However, the exact location for a mini-halo to be seeded in the M_v - z plane within the band depends on the detailed assembly history and radiation environment, which varies from halo to halo. The regime of Pop-III seeds is almost fully (partly) covered by those of Pop-II (Pop-I) GC formation, highlighting the critical role of IGM enrichment in determining the first population of stars and BH seed that form in a mini-halo. In §3.1.3, we will come back to the seeding atlas with all these effects self-consistently modeled in a cosmological volume.

2.4 The growth from seeds to supermassive black holes

The formation of a BH seed is expected to be associated with gas depletion in the host mini-halo due to radiative and potentially SN feedback, causing a transient phase of quiescence in the star formation and AGN activity, until replenished gas reactivates the growth. As detailed in Paper-I, the critical condition for growing a BH is to break the barrier of angular momentum (denoted by J , or by j for the specific value per unit mass) so that gas with low j can reach the zone of influence of the BH within which the gravity is dominated by the BH.

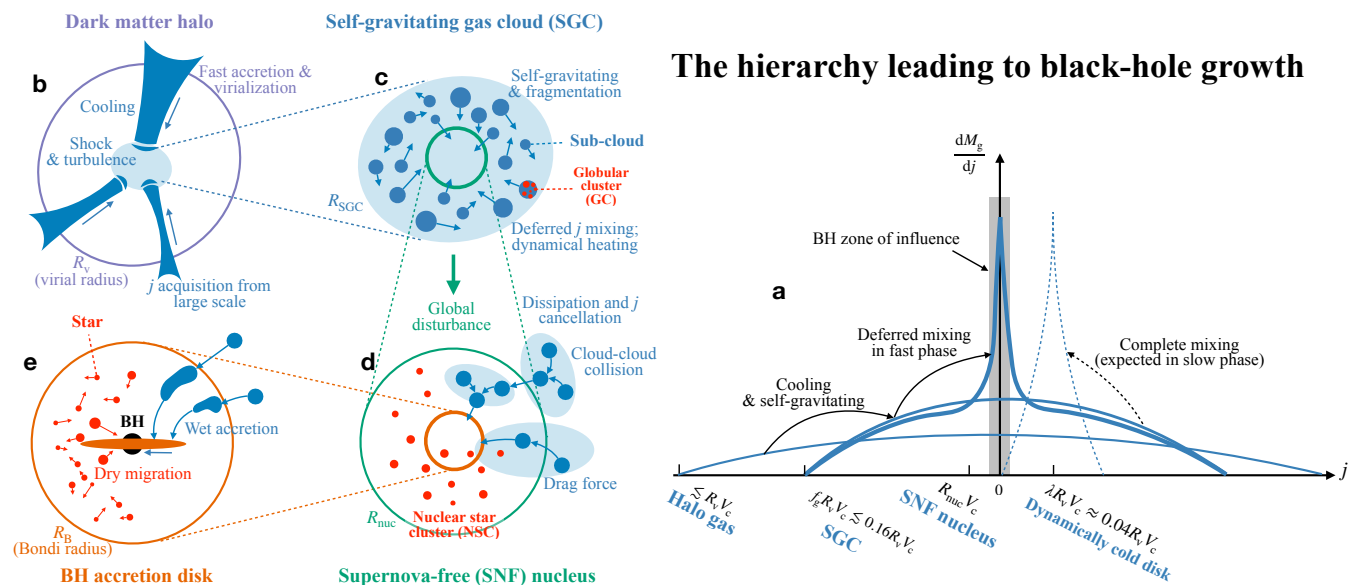
Such a condition is naturally achieved in an SGC owing to the early fragmentation that leads to the ballistic motion of sub-clouds (see also Hobbs et al. 2011; Gaspari et al. 2015), making the continuous capturing of sub-clouds the main channel of BH growth (referred to as the *continuous mode* thereafter), in contrast to *galaxy-galaxy mergers* that emerge as the main channel of BH growth at later epochs (e.g. Li et al. 2025b).

Toward the galactic nucleus, however, the situation for sub-clouds is expected to differ due to additional processes operating at small scales. One such process is BH feedback that regulates the amount of gas that can be sustained in the nucleus to power the growth of the BH itself. The second process is the regeneration of low- j sub-clouds that would otherwise be absent after the BH has captured them all or destroyed them via feedback (Sivasankaran et al. 2025, see their fig. 5). The third process is the mixing of j , as the gas density involved in the nucleus can be high enough to impart a significant drag force and frequent cloud-cloud collisions. These additional processes may cause the gas structure and dynamics to deviate from those expected from modeling the outer part of the SGC, thereby modifying BH accretion.

Here we extend the scenario of Paper-I by considering these nuclear processes. The numerical model for the nuclear processes is then built upon these considerations and implemented as a refinement engine working with much finer time steps on top of the two-phase scenario. Fig. 4 outlines the four levels of the hierarchy from the dark matter halo to the BH accretion disk for the BHs experiencing super-Eddington accretion in an MW-size galaxy and its progenitors experiencing fast assembly. The fast assembly of the halo and the formation of the SGC make up the first two levels of the hierarchy (panels b and c), and their effects on sustaining a broad distribution of j are indicated in panel a. These provide a boundary condition for the activation of nuclear processes that establish the last two levels of the hierarchy (panels d and e) and further redistribute j . A detailed description of the growing procedure and the related analytic estimation is presented in Appendix C, and the remainder of this section is a brief summary of the key points.

To avoid ambiguity, we define the nuclear region of an SGC as the inner part where the gas density is above a ‘supernova-free’ (SNF) threshold, $n_{\text{snf}} \equiv 10^{3.5} \text{ cm}^{-3}$ (Chen et al. 2025a). This corresponds to a free-fall timescale of $t_{\text{ff,g}} \lesssim 1 \text{ Myr}$ within the nucleus, implying that gas dynamics there proceeds faster than the lifetime of massive stars born therein and is thus largely free of SNe. In the following, we refer to this volume as the *SNF nucleus*, and denote its radius and gas mass by R_{nuc} and M_{nuc} , respectively. As estimated by Dekel et al. (2023), other feedback channels from stars, such as radiation and wind from both inside and outside the nucleus, are also ineffective in dispersing the gas at such a high density. Thus, the only ways remaining to quench an SNF nucleus are (i) gas consumption by star formation/BH; (ii) gas ejection by BH feedback. The consequence of the two is either intense star formation to grow an extremely compact nuclear star cluster (NSC) with a density $\rho_* \sim \rho_{\text{snf}} \equiv \mu m_p n_{\text{snf}}$, or rapid BH growth, or both. The growth of both NSC and BH is expected to be a runaway process due to the short $t_{\text{ff,g}}$. We therefore refer to such a bursty event as a *nuclear burst*, and the channel opened by such bursts to grow BH is referred to as the *bursty mode*.

As an approximation, we divide the processes in an SNF nucleus into two stages, a formation stage where gas inflow is boosted to replenish the nucleus, and a dispersion stage where the nuclear burst consumes and disperses the gas in the nucleus. The trigger of the formation of an SNF nucleus in a galaxy is modeled as the time when the host halo experiences a temporary excursion to a high accretion rate (see Appendix C2). Such an excursion is expected to be able to



impart a global disturbance on the galaxy, generating a certain level of dynamical hotness, removing the barrier of j and driving a strong gas inflow towards the nucleus. The boosted inflow then overpowers (at least over a short period of time) the gas consumption by star formation, the accretion of the BH and the ejection by the associated feedback, causing gas to pile up and establishing the SNF nucleus. The structure of the SNF nucleus is expected to be supported by the turbulence generated by the rapid inflow, and the gas profile within the nucleus ($r \leq R_{\text{nuc}}$) is therefore modeled as a smooth extrapolation from the outer SGC ($r > R_{\text{nuc}}$), with one parameter to characterize the turbulence structure.

The turbulent and high-density environment in the SNF nucleus provides ideal conditions to channel gas through the nucleus towards the BH, which we take into account in our model for the dispersion stage. Sub-clouds in the nucleus are expected to experience frequent collisions with other sub-clouds as well as strong drag forces from the ambient gas. Efficient cooling implied by the high density makes such interactions highly dissipative, causing further condensation and loss of j of the sub-clouds. A net inflow toward the BH is thus generated as the sub-clouds sink their orbits (Fig. 4d). The essential role of turbulence can thus be summarized as a ‘deferred mixing’: in the outer SGC, turbulence seeds the formation of sub-clouds and amplifies their density. Inefficient mixing among sub-clouds preserves the broad distribution of j , allowing sub-clouds with low j to approach the nucleus. As the low- j sub-clouds approach the inner SGC, efficient mixing drives a gas inflow to feed the BH. The effect of deferred mixing on the distribution of j is schematically illustrated in Fig. 4a, where a sharply peaked and centralized distribution forms and

powers efficient BH accretion. This is different from the case where a dynamically-cold disk forms, where the distribution of j is peaked but off-centered, leaving little fuel for the BH (dashed distribution in Fig. 4a).

The modeling of the dispersion stage for an SNF nucleus follows the continuity equation of the gas density within the nucleus, taking a spherical approximation for the relevant profiles. The evolution of gas in this equation is contributed by gas inflow from the outer part of the SGC into the nucleus, star formation in the nucleus, accretion of gas into the zone of influence of the BH, gas ejection by BH feedback, and a hard time window for gas depletion set by the delay time of SNe. Timescale analysis suggests significant competition among the sink terms: BH feedback is almost always the dominant contributor to gas depletion; star formation outpaces (is outpaced by) BH accretion if $M_{\text{BH}} \ll M_{\text{nuc}}$ ($M_{\text{BH}} \gg M_{\text{nuc}}$). A **turbulence-modified Bondi accretion model** (Bondi & Hoyle 1944) is used to account for the role of turbulence in driving efficient gas inflow, as discussed above, and to determine the exact accretion rate ($\dot{M}_{\text{B}}$, the Bondi rate) of gas into the zone of influence ($r \leq R_{\text{B}}$, the Bondi radius) of the BH. Given the high Bondi rate, a **turbulent and magnetized accretion disk model** (Hopkins et al. 2024c, thereafter H24, for the details) is used to solve the structure of the accretion disk, the star formation within R_{B} , and the resulting BH growth rate in such a ‘wet channel’; stars formed in the nucleus are also expected to experience frequent dynamical interactions and contribute a ‘dry channel’ to the BH growth (Fig. 4e). Numerical treatment is applied to evolve the BH mass and the gas and stellar densities within the nucleus, taking the structure established in the formation stage as the initial condition.

The nuclear region is assumed to enter a quiescent phase after a burst, waiting for the next excursion event of halo assembly to trigger the formation of the next SNF nucleus. Super-Eddington accretion is found to be achieved in most nuclear bursts owing to the inefficiency of all feedback channels (see Appendix C3 and Fig. C1). The growth history of a BH thus consists of flash episodes of super-Eddington accretion and intermittent quiescent stages where other channels of BH growth dominate. The co-growth of NSC and BH is also a natural consequence of the model.

To summarize, the following channels of BH growth are considered in our model after seeding:

- (i) **Bursty mode**, which appears as episodic super-Eddington accretion of the BH and intense starburst in the SNF nucleus formed due to global disturbances caused by excursions of halo assembly. Two distinct channels, the wet channel and the dry channel, are expected to contribute to the BH growth in this mode.
- (ii) **Continuous mode**, in which the BH continuously captures subclouds formed in the SGC during the fast phase of a halo. The associated star formation in the SGC is expected to build a dynamically hot stellar component (bulge) of the host galaxy. In contrast, during the slow phase of a halo, the BH stops growing, and star formation builds up the dynamically cold stellar component (disk).
- (iii) **Merger**, in which two galaxies and their central BHs merge to form a new galaxy and a new central BH.

As we will show, these channels dominate BH growth at different epochs in the history of a halo, thereby shaping a redshift-dependent, multi-piece $M_{\text{BH}}-M_*$ relation.

An interesting conclusion can be reached by considering the evolution of structures at different levels of the cosmic hierarchy. The structure of SNF nuclei formed at different epochs in the history of a halo is related to the assembly history of the halo. For example, the derivative of the gas surface density of the SNF nucleus, Σ_{nuc} , can be derived analytically as

$$\frac{d \ln \Sigma_{\text{nuc}}}{d \ln a} \approx \frac{2\Gamma_v - 3}{6}, \quad (6)$$

where $a = 1/(1+z)$, and $\Gamma_v \equiv \dot{M}_v/(M_v H)$ is the specific growth rate of the halo mass. The derivative changes its sign at a critical value of $\Gamma_v = 1.5$, coinciding with the epoch found by Boco et al. (2023) that separates the fast- and slow-accretion phases of a halo. The same argument can also be applied to R_{nuc} and M_{nuc} , yielding the same critical value for Γ_v . Moving deeper into the SNF nucleus, the Bondi radius, the Bondi-accretion rate and the structure of BH accretion disk are all determined by Σ_{nuc} . Hence, the cosmological information can propagate through galaxy ecosystems to the innermost regions of galaxies, and the transitions of halo assembly also lead to transitions in BH accretion. This is a highly non-trivial result of our model, as it links the cosmological assembly of structures over many orders of magnitude in spatial scale through physical processes.

The details of analytical estimations and the evolution of the structure hierarchy are presented in Appendix F. The numerical implementation of the model in a cosmological N-body simulation is described in Appendix D. In the following, we show the numerical results predicted by the *Default* model and discuss their physical and observational implications. Several variants of the model are defined and studied in Appendix G to show the degrees of freedom allowed by the physical prescriptions and to pinpoint the model space to be explored in the future.

3 MODEL PREDICTIONS AND COMPARISON WITH OBSERVATIONAL DATA

3.1 The expected seed populations

3.1.1 Seed mass functions

We define the seed mass function as the number of BH seeds per unit logarithmic mass interval per unit comoving volume formed before a given redshift z :

$$\Phi(M_{\text{BH,seed}}, z) = \frac{1}{V_u} \frac{dN_{\text{seed}}(>z)}{d \log M_{\text{BH,seed}}}, \quad (7)$$

where V_u is the comoving volume of the simulation, and $dN_{\text{seed}}(>z)$ is the number of BH seeds that have been bred by z within the mass interval $dM_{\text{BH,seed}}$ in the volume.

Fig. 5a–c show the total seed mass function (black) and the contributions of different flavors (colored) at $z = 20$. The peak of Φ is at $M_{\text{BH,seed}} \approx 500 M_\odot$, recovering the typical mass assumed in theoretical and observational studies of Pop-III stars (Schaerer 2002; Wang et al. 2024b). The shape of Φ is quite broad, demonstrating the diversity of the environment that modulates the seeding process (see below). PISN is preferentially generated at high mass, which destroys the central star, leaves a less massive BH seed (the remnant of another star surviving PISN), and produces a prominent bimodality with a valley at $M_{\text{BH,seed}} \approx 100 M_\odot$, in Φ . DCBH is preferentially generated at even higher mass and produces the seeds that dominate the high-mass end of Φ . In particular, a non-negligible fraction of seeds, with an expected number density of $\Phi \lesssim 10^{-3} \text{ Mpc}^{-3} \text{ dex}^{-1}$, falls in the mass range of the SMS ($M_{\text{BH,seed}} \gtrsim 10^4 M_\odot$). SMSs are believed to be promising candidates for the progenitors of bright quasars observed at $z \sim 6$ (e.g. Wise et al. 2019; Latif et al. 2022a; see also our discussion of archaeology in §3.2.4), and our model implies that SMSs are indeed sufficiently abundant for such a link, provided that some of the seeds can grow further via different channels (see Figs. 11 and 12).

A galaxy is expected to experience a short period of quiescent phase immediately after a seed is bred, because the associated AGN feedback strongly suppresses star formation and BH growth (see §3.2 for details). Since BHs with higher mass are more efficient in feedback (in terms of the energy released per unit of accreted mass), seeds with the DCBH flavor may quench the galaxy for a longer period of time, while seeds with the CCSN and PISN flavors may allow the galaxy to rejuvenate earlier. The bimodal distribution of the seed mass is thus expected to produce a bimodal distribution in the age of stars after the seeding. Observations targeting the first generation of quiescent galaxies may thus be able to reveal the properties of the infant galaxies that have just completed the seeding.

At $z = 20$, the peak value of Φ is above $0.1 \text{ Mpc}^{-3} \text{ dex}^{-1}$, higher than the number density of dwarf galaxies with $M_* = 10^8 M_\odot$ at $z = 0$ (see e.g. fig. 15 of Chen et al. 2019), suggesting that most such galaxies may have their lifetime starting at $z \gtrsim 20$ (see also Fig. 6). These stars can either remain in the inner galaxy, be dynamically heated to higher orbits, be ejected as halo stars, or merge into other galaxies. The search for fossils of these stars has been carried out in and around the Milky Way (MW) and has yielded interesting cases with Pop-III-like metallicity (see e.g. Bonifacio et al. 2025, for a review), or with a peculiar PISN-compatible chemical pattern (e.g. Aguado et al. 2023; Xing et al. 2023). Our model suggests that these stars may be used to reveal the formation of the MW back to $z \approx 20$, the epoch when the Universe is younger than 200 Myr.

Both dynamical heating and UV radiation delay the collapse of an SGC, elevate the gas mass available for accretion, bring the accretion

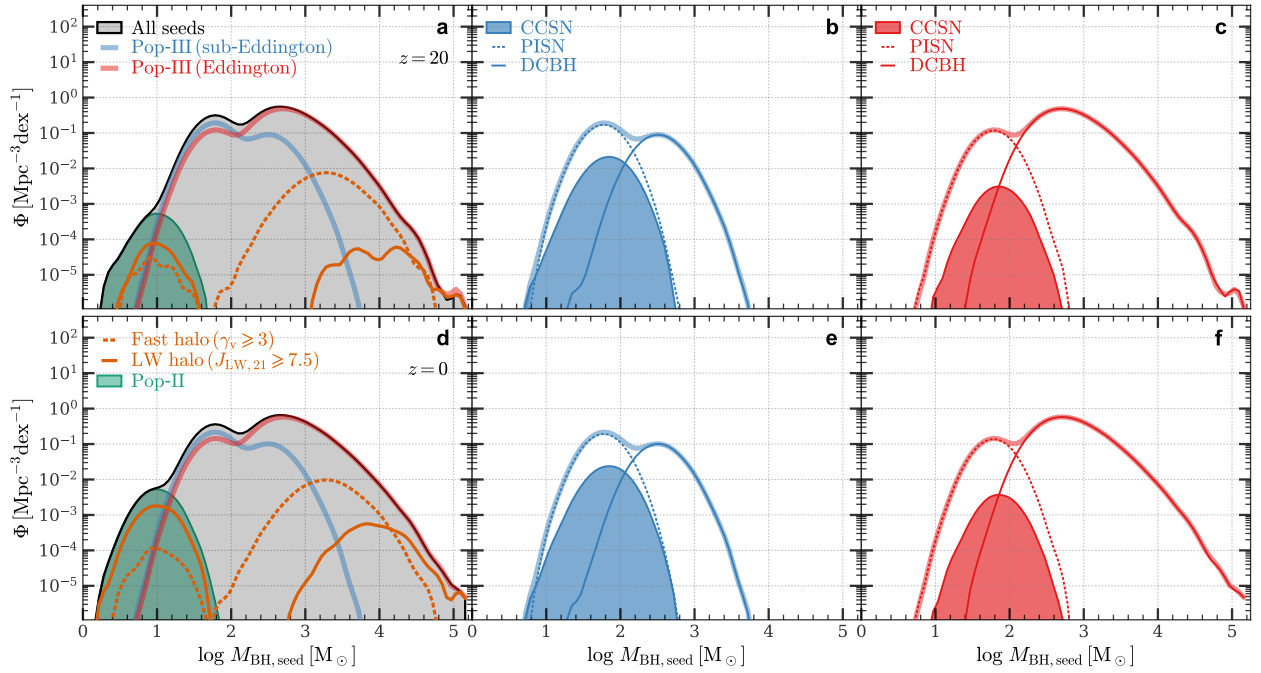


Figure 5. Mass function of black-hole seeds. **a**, $\Phi(M_{\text{BH,seed}}, z = 20)$, mass functions of all BH seeds that have been bred until $z = 20$ in the simulation volume (black curve enclosing gray area) and of seeds with different ‘flavors’ (formation pathways): descendants of Pop-III stars that experienced sub-Eddington (blue) or Eddington (red) accretion during their formation, those born from dynamically heated SGC in ‘fast halos’ (i.e. fast-accreting halos with $\gamma_v \geq 3$; orange dashed) or from ‘LW halos’ (i.e. LW-irradiated halos with $J_{\text{LW},21} \geq 7.5$; orange solid), and descendants of Pop-II stars fed by enriched IGM (green). **b**, **c**, mass functions of sub-categories of Pop-III seeds. Here each category of Pop-III seeds (sub-Eddington or Eddington) is further divided into three sub-categories according to the stellar evolution (CCSN, PISN or DCBH) that ends the life of the dominant star in the host Pop-III star cluster. **d**–**f**, similar but including all seeds bred until $z = 0$. The mass spectrum of BH seeds is quite broad and exhibits two main modes, suggesting that galaxy bimodality has already appeared at $z \approx 20$. Environmental processes have a divergent impact on the masses of BH seeds, producing both high-mass and low-mass bumps in the mass function, with DCBH and Pop-II flavors, respectively. See §3.1.1 for details.

of the central star to the Eddington level, and grow it to higher mass. However, the relative importance of these two environmental factors differs between halos. Dynamical heating is stronger in halos with faster assembly and is thus more important at higher M_v and higher z (e.g. Fig. 2). LW radiation is stronger if galaxies with higher SFRs are present in the neighborhood, and is thus more important at lower z when a larger number of massive, star-forming galaxies have formed in over-density regions. To illustrate the difference between these two factors, we select seeds bred in (i) ‘*fast halos*’, defined as those with $\gamma_v \geq 3$ at the seeding epoch; (ii) ‘*LW halos*’, defined as those with $J_{\text{LW},21} \geq 7.5$. The thresholds are chosen so that the expected collapse threshold is raised to $M_v \approx 10^7 M_\odot$ and massive seeds can be bred thereafter (see Fig. B1). As shown in Fig. 5, seeds in both types of halos tend to produce accretion at the Eddington level, highlighting the importance of the large-scale environment in modulating the seeding process. The masses of seeds in LW halos are higher than those in halos of fast assembly, consistent with the expectation that the LW radiation is more efficient at lower z when halos that have not been seeded can reach higher masses.

For comparison, we show $\Phi(M_{\text{BH,seed}}, z = 0)$, the seed mass function at $z = 0$ in Fig. 5d–f. This is similar to that at $z = 20$ in terms of normalization, bimodal shape and contributions of different flavors. A noticeable difference is that Φ at $z = 0$ is broadened towards both the low- and high-mass ends. Both tails are dominated by LW halos and fast halos, implying that both environmental factors are important in delaying the seeding. The environmental factors now have a *divergent effect* on the masses of BH seeds: on the one hand, it delays the collapse of an SGC, so that more massive DCBHs are

bred owing to the higher gas mass available for the formation of Pop-III stars (red thin curve in panel f); on the other hand, it gives more time for SNe from nearby galaxies to enrich the mini-halo, so that Pop-II seeds are more likely to form via efficient cooling provided by the metals (green shade in panel d). In fact, most Pop-II seeds are bred at $z \lesssim 20$, in contrast to the early formation of Pop-III seeds (mostly bred at $z \approx 20$ –30; see also Fig. 6a). The interplay between dynamical heating and the two forms of feedback (matter and radiation) from galaxy formation to the large-scale environment may thus leave interesting imprints on the seed populations.

3.1.2 The cosmic seeding history

To quantify the redshift distribution of seed formation, we define the cosmic seeding rate density as the number of BH seeds bred per unit logarithmic cosmic scale factor per unit comoving volume:

$$\frac{dn_{\text{seed}}}{d \log(1+z)} = \frac{1}{V_u} \frac{dN_{\text{seed}}(>z)}{d \log(1+z)}. \quad (8)$$

The cumulative value of the cosmic seeding rate density,

$$n_{\text{seed}}(>z) = N_{\text{seed}}(>z)/V_u, \quad (9)$$

is referred to as the cosmic cumulative seeding density. This mimics the definition of cosmic star formation history (Madau & Dickinson 2014), but is specific to BH seeding.

Fig. 6a–c show the evolution of seeding rate density in total and for seeds with different flavors, and Fig. 6d–f show the corresponding cumulative distributions. The total seeding rate density increases

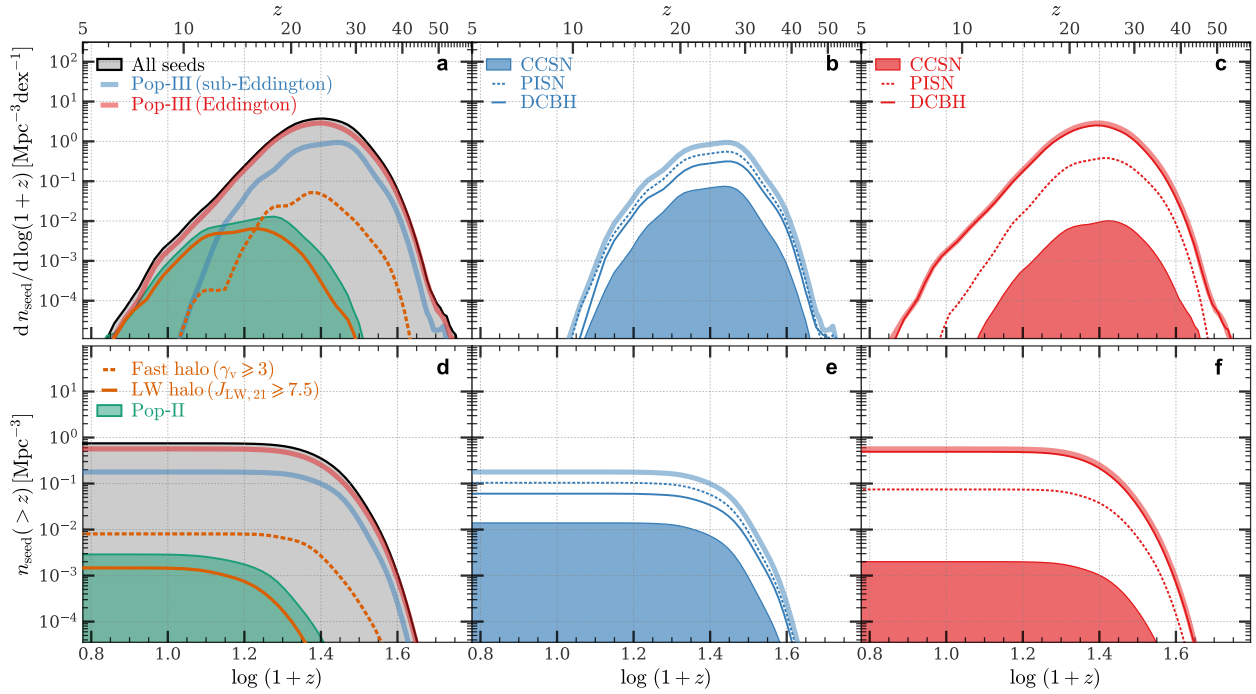


Figure 6. Cosmic black-hole seeding history. **a–c**, cosmic seeding rate density (i.e. number of BH seeds bred per unit logarithmic scale factor per unit comoving volume) as a function of z . **d–f**, cosmic cumulative seeding density (i.e. number of BH seeds bred before z) as a function of z . The results are shown for all seeds (black curve) and for seeds with different ‘flavors’ defined in the same way as in Fig. 5. At $z \approx 20$, most seeds have already been bred. Pop-II seeds form at systematically lower redshift than Pop-III seeds, and account for a significant fraction of the seed population at $z \lesssim 10$. See §3.1.2 for details.

steeply with time, reaches a peak at $z \gtrsim 25$, and then decreases rapidly. Most seeds have already been bred by $z \approx 20$, consistent with the expectation of the seed mass functions (see Fig. 5). The fraction of Pop-II seeds is less than 1/100 at $z = 20$, but increases significantly with time, again reflecting the divergent effect of heating sources on seed mass (see §3.1.1).

For Pop-III seeds, those formed with Eddington accretion of the progenitor stars dominate the seeding rate at all z , and become increasingly so at lower z ; those formed with Eddington (sub-Eddington) accretion are more likely to have DCBH (PISN) flavor. The seeding rates of fast halos and LW halos cross over at $z \approx 15$ – 20 , with the former (latter) dominating at higher (lower) z , reinforcing the conclusion drawn in §3.1.1 that the two environmental factors work in different regimes. Most seeds bred at $z \lesssim 15$ are delayed by the LW radiation, implying that the optimal strategy for searching for ongoing formation of Pop-III stars and BH seeds, as well as their immediate descendants, is to target the surroundings of over-dense regions. Recent JWST projects (e.g. Matthee et al. 2023; Naidu et al. 2024; Fujimoto et al. 2025b; Hsiao et al. 2025) have already been able to cover such high redshifts, and our model provides a theoretical framework to optimize observational strategies and interpret their findings.

3.1.3 The seeding atlas: when, where and how seeds are bred

The purple band in the M_v - z plane (Fig. 2a) provides an atlas to identify when and where seeds are expected to be bred. Implementing the model into the cosmological N-body simulation, we are now able to track the diverse environments of individual mini-halos and use them to dissect the atlas into regimes where seeds with distinct flavors preferentially occupy. Fig. 7 shows such a refined atlas, where individual seeds are color-coded by the mass ($M_{\text{BH,seed}}$) or by one of the three environmental properties (Z_{IGM} , J_{LW} and γ_v).

The first generation of stars and BH seeds emerges at $z \gtrsim 40$ when the progenitors of extremely massive halos ($M_{v,z=0} \gtrsim 10^{14} M_\odot$) reach M_{v,H_2} (Eq. 1). In this early era of galaxy formation, IGM has not been enriched (panel b), LW field has not been built (panel c), and dynamical heating is the main environmental factor that modulates seeding (panel d). A mini-halo with a temporary excursion to slow accretion gets a chance to escape the dynamical heating and form a low-mass, Pop-III seed, while a mini-halo sustaining fast accretion breeds a more massive seed later on. A direct prediction of our model is that cosmic star formation and seeding activity start to prevail at the locations of extremely massive halos that reach M_{v,H_2} in pristine regions and have temporary excursions to low γ_v at high z .

Given the extremely compact sizes of SGCs at such high z (a few pc; see the texts below Eq. 3) and the high density threshold for the pristine gas to fragment, these earliest Pop-III star clusters are expected to be more compact than the Pop-I/II GCs formed later. The low masses of their host mini-halos also imply low masses of the BH seeds (mostly $\lesssim$ a few hundred $M_\odot$), and thus the dominance of CCSNe and PISNe in their flavors. There are also cases where mini-halos form very early and remain at high γ_v at $z \gtrsim 40$, and our model predicts that DCBHs with masses $\gtrsim$ a few hundred $M_\odot$ can be bred in these mini-halos (panel a).

If a massive halo can sustain a high γ_v over a long period, seeding can be further delayed and the seed mass can be elevated (yellow dots in panel d). For most halos with $M_{v,z=0} \gtrsim 10^{13} M_\odot$, the seeding epoch is $z \gtrsim 20$ and the seed mass spans from a few tens to about $10^4 M_\odot$ (panel a). This wide range of seed mass also implies a diverse flavor (formation pathway) of seeds. The LW field has not been established and is thus unimportant for the seeding in these halos (panel c). Individual halos may violate the general trend due to the diverse assembly histories and environments of massive halos (see e.g. §4.2 of Chen et al. 2023b).

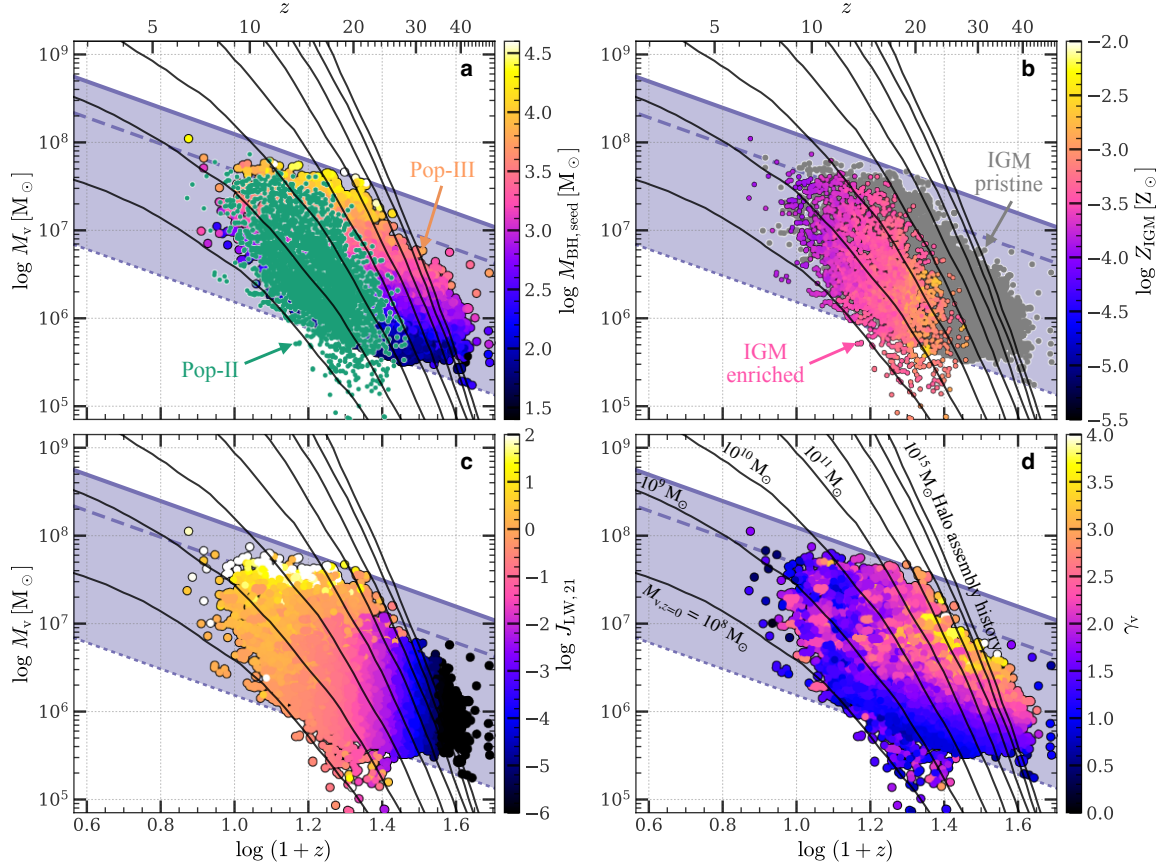


Figure 7. The atlas of black-hole seeds. Here we show the distribution of BH seeds in the M_v - z plane (host halo mass versus redshift at the seeding epoch), color-coded by the mass of the BH seed ($M_{\text{BH,seed}}$, **a**), the metallicity of the IGM surrounding the subhalo (Z_{IGM} , **b**), the intensity of LW radiation irradiating the subhalo ($J_{\text{LW},21}$, **c**), or the specific halo growth rate (γ_v , **d**), all evaluated at the seeding epoch. In **a**, green dots represent Pop-II seeds, while the others represent Pop-III seeds. In **b**, grey dots represent those embedded in pristine IGM ($Z_{\text{IGM}} = 0$), while the others are color-coded by Z_{IGM} . All seeds bred in the simulation volume are included in this plot. Kernel smoothing is applied to the colored properties to clarify the trends. Halo assembly histories (**black curves**) and the regime of Pop-III seeds (**purple band**) are overlaid for reference (the same as those in Fig. 2). This figure illustrates when, where and how BH seeds are bred, highlighting the role of environmental processes (quantified by Z_{IGM} , J_{LW} and γ_v) in modulating the seeding process. See §3.1.3 for details.

Less massive halos with $M_{v,z=0} \lesssim 10^{13} M_\odot$ follow a similar trend driven by dynamical heating, albeit with some upper-left shift in the M_v - z plane. In the atlas, dynamical heating thus defines a fusiform-shaped regime filled with tilted, stripe-like equal- $M_{\text{BH,seed}}$ or equal- γ_v contours (panels a and d). LW radiation starts to modulate seeding in these halos due to the higher cosmic SFR density (see Fig. 10 and Madau & Dickinson 2014) and less effective IGM attenuation (see Appendix D5) at lower z . These LW-illuminated halos have the most significant delay of seeding, reach the highest seed masses, and populate a tip-like regime in the atlas (panel c).

The divergent effect of dynamical heating and LW radiation on seed mass (see §3.1.1) is also reflected in the atlas. Many low-mass halos ($M_{v,z=0} \lesssim 10^{12} M_\odot$, particularly those with $M_v \lesssim 10^{10} M_\odot$) have enriched seeding environments, and Pop-II seeds with low masses are therefore bred (green cloud in panel a). However, metal diffusion is much slower than radiation propagation. The enriched IGM thus only appears for halos already affected by the LW radiation. Hence, the distribution of Pop-II seeds in the atlas appears as an elongated stripe pulled off by the two sources of heating.

In summary, the seeding atlas can be dissected into three regimes according to the environmental processes that modulate the seeding:

- (i) **The dynamical-heating regime**, which spans a fusiform area in the M_v - z plane.

- (ii) **The LW-radiation regime**, a tip-shaped structure that extends the low- z , high- M_v end of the dynamical-heating regime.
- (iii) **The regime of enriched IGM**, which is a two-component stripe on top of the low- z parts covered by the above two regimes.

Seeds modulated by different environmental effects or bred with different masses differ in spatial distribution, since processes establishing the environments operate on different spatial scales and have different temporal persistence. The clustering pattern of seeds may thus encode information about how they were bred, and this pattern may persist for a long time, allowing the seed properties to be retrieved from low- z observations targeting the descendants (see Appendix E for details).

3.2 The co-evolution of galaxies and black holes

Our model is built on the seeding and the multiple channels of BH growth and star formation (see §2.4). Here we show the contribution of these channels to the growth of individual galaxies and to the cosmic history of the galaxy population. As we will show, the evolution pathways of galaxies also provide a tool for galaxy archaeology and futurology that link observed galaxies across cosmic time.

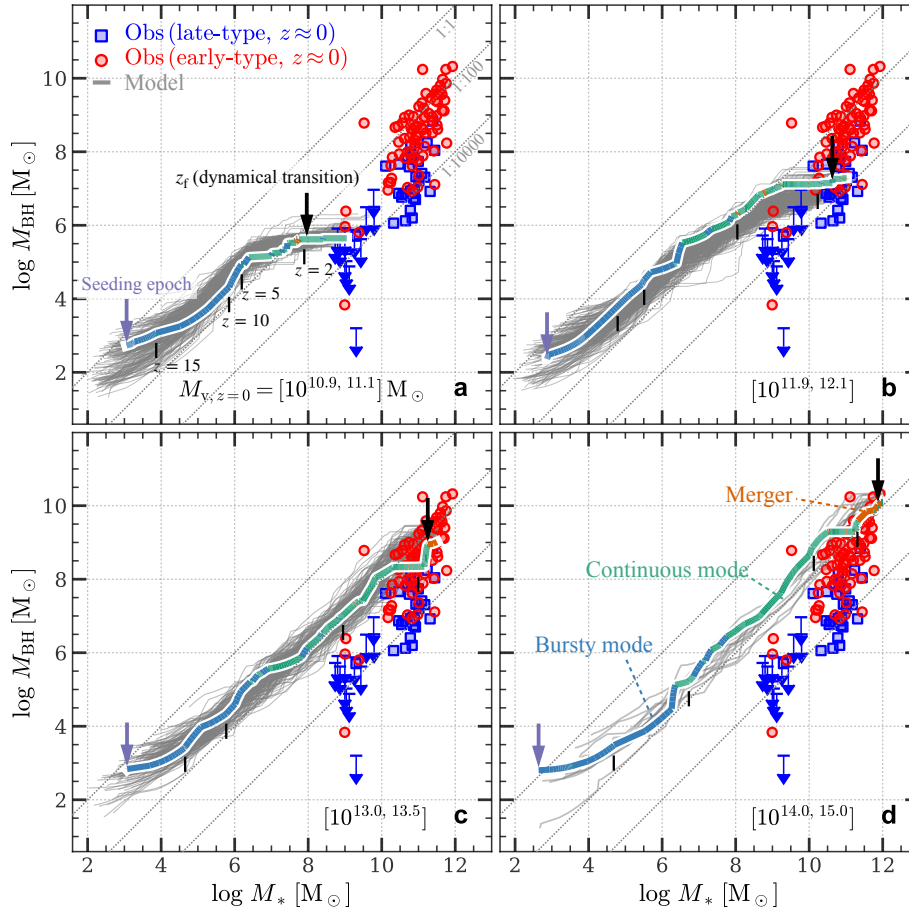


Figure 8. Predicted growth paths of individual galaxies in the $M_{\text{BH}}-M_*$ plane. a–d, prediction for central galaxies with different halo masses at $z = 0$, as labeled. In each panel, **grey** curves show individual modeled galaxies, and a **colored** curve shows one example galaxy with the channel dominating the growth of the BH at each time step indicated by the color of the path segment (**blue**, bursty mode, wet channel; **green**, continuous mode; **brown**, merger). Note that the dry channel in the bursty mode is always subdominant compared with the wet channel, and is thus totally absent. Critical epochs for the example galaxy are marked (**purple arrow**: seeding epoch; **black arrow**: z_t ; **black ticks**, $z = 15, 10, 5$ and 2). For reference, **blue** and **red** scatters represent individual late-type and early-type galaxies, respectively, at $z \approx 0$, compiled by [Greene et al. \(2020\)](#). Tilted grey lines indicate the loci of $M_{\text{BH}}:M_* = 1:1, 1:100$ and $1:10000$, respectively. In **a, b**, a random subsample of 256 galaxies is shown for clarity. The channels that dominate the growth of a BH emerge sequentially in time, and are expected to produce different observational signatures as the galaxy ecosystem evolves from early to late stages. See §3.2.1 for details.

3.2.1 Growth channels

Fig. 8 shows the evolution of individual central galaxies in the $M_{\text{BH}}-M_*$ plane. In each range of $M_{v,z=0}$, a representative galaxy is highlighted, with each colored segment indicating the channel that contributes the most to the growth of M_{BH} during a time step. In more detail, Fig. 9 shows the growth of each mass component, M_X ($X = \text{halo, BH, stars or gas}$), for three of the highlighted galaxies in Fig. 8. The specific growth rate, $\dot{M}_X/M_X/H(z)$, for BH, stars and gas, and γ_v for halo is shown for reference. The contribution of growth from each channel is shown separately and in total.

Most dwarf-size galaxies ($M_{v,z=0} \approx 10^{11} M_\odot$) are seeded between $z = 20$ and 30 , roughly the duration when the cosmic seeding activity peaks (Fig. 6a). The highlighted galaxy (Fig. 8a) has a seed mass of $M_{\text{BH,seed}} \leq 10^3 M_\odot$ and a DCBH flavor. Dwarfs are born in the $M_{\text{BH}}-M_*$ plane close to the line of $M_{\text{BH}}:M_* = 1:1$, due to the top-heavy IMF assumed for the Pop-III stars. After seeding, gas accretion into the SGC is efficient due to the fast halo assembly (Fig. 9a7,a8). However, the over-massive BH produces efficient AGN feedback that, together with the efficient stellar feedback in a low-mass halo, prevents the accumulation of star-forming gas and quenches star formation and

BH growth at $z \gtrsim 20$ (Fig. 9a3–a6). The continuous mode contributes negligibly to the growth of M_{BH} and M_* at $z \gtrsim 10$ (with a specific growth rate of $\lesssim 0.1$). Frequent excursions of the halo assembly (Fig. 9a2) globally disturb the galaxy, cause episodic formation of SNF nuclei, trigger nuclear bursts, and drive rapid rises of M_{BH} and M_* at $z \approx 10$ – 20 . The wet channel in the bursty mode dominates the total growth at $z \gtrsim 5$ and keeps $M_{\text{BH}}/M_* \approx 1/10$. At $z \approx 5$, the galaxy has reached $M_* \approx 10^6 M_\odot$ and $M_{\text{BH}} \approx 10^5 M_\odot$. Thus, the evolution during the bursty stage has two distinct features:

- (i) the back-and-forth switch between star-forming and quiescent states;
- (ii) the formation of a compact, globular-like object expected from the dominance of the nuclear star formation within the SGC.

Our model thus predicts that the formation of *the first generation of quiescent galaxies* is as early as $z \approx 20$ – 30 soon after the seeds are bred, that the population of quiescent galaxies is ubiquitous at $z \gtrsim 10$ as they fall into gaps between nuclear bursts, and that galaxies with *compact morphology and prominent NSCs* are common in that era.

At $z \lesssim 5$, the bursty mode diminishes in the dwarf due to the less frequent halo excursion and the large ratio of $M_{\text{BH}}/M_{\text{nuc}}$. A few nuclear bursts, each growing M_{BH} (M_*) by about $10^5 M_\odot$ ($10^6 M_\odot$), are seen at $z = 2$ – 3 when halo excursions are triggered by halo-halo

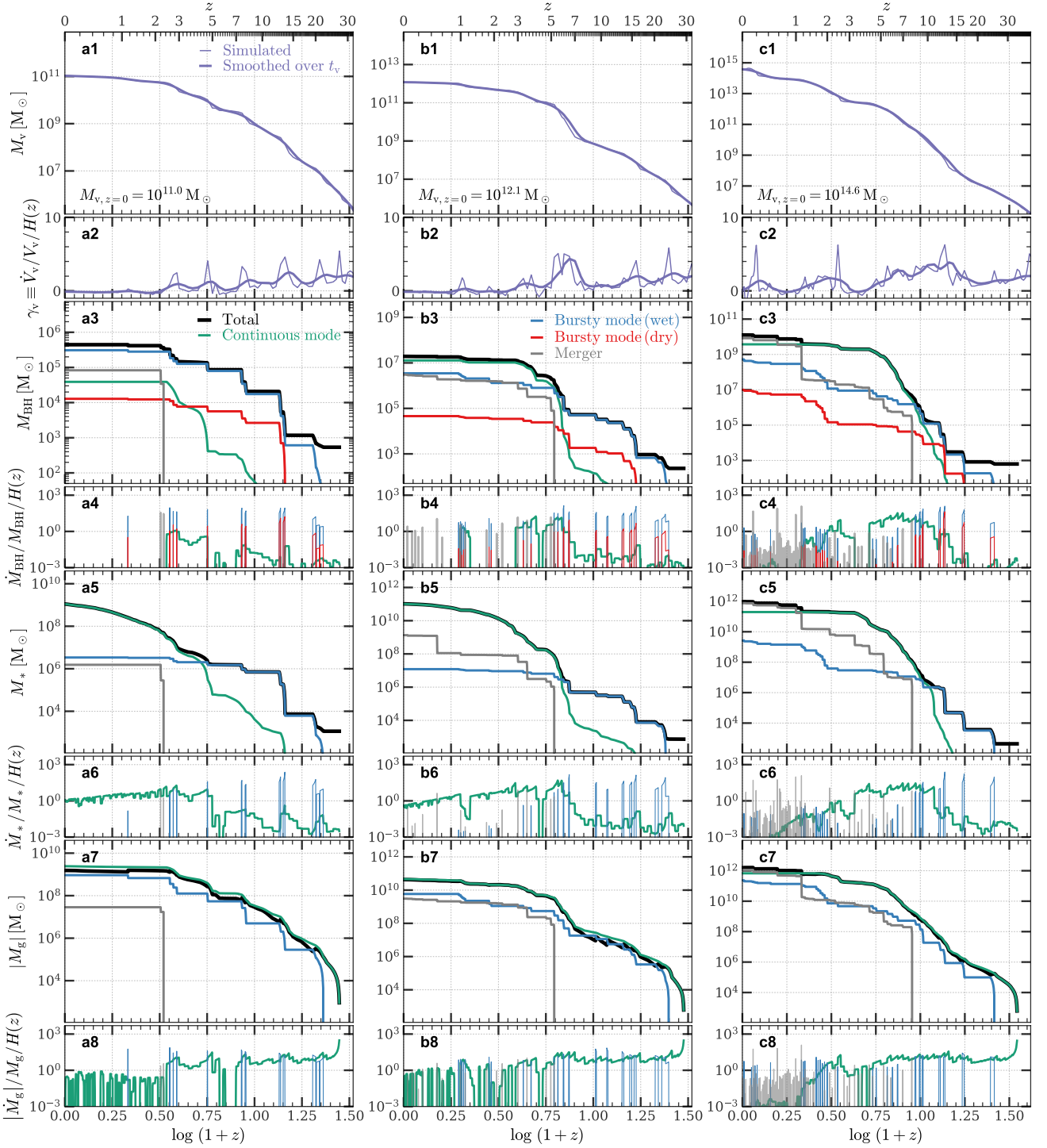


Figure 9. Evolution of different mass components and contributions from different channels for three example galaxies. Columns a–c, three example central galaxies (the same galaxies as the colored ones in Fig. 8a,b,d), respectively, predicted by our model, with their $M_{v,z=0}$ indicated in the top row. For each component of mass (halo mass, M_v ; BH mass, M_{BH} ; stellar mass, M_* ; gas mass, M_g), we show the cumulative growth and specific growth rate in two rows, respectively. For M_v , the simulated growth history and that smoothed over t_v are both shown. Curves with different colors represent the growth contributed by different channels (continuous mode, wet and dry channels in bursty mode, and merger). The total growth, obtained by summing the contributions from all the channels (including the seeds), is shown in **black**. The contribution of nuclear burst to M_g is negative due to the consumption of gas, so we show the absolute value. The growth rates by nuclear burst (both wet and dry) and merger are evaluated by averaging over 10 Myr, given their bursty nature. See §3.2.1 for the details.

mergers. These mergers also bring satellites that later merge into the central, contributing small amounts of M_{BH} and M_* through the merger channel. The continuous mode dominates the growth of M_* at $z \lesssim 5$ and keeps the galaxy in the star-forming main sequence (with the specific SFR $\gtrsim 1$). The specific growth rate of M_{BH} by the continuous mode never exceeds 1 due to the regulation of the BH and stellar feedback, and M_{BH}/M_* is reduced to $\sim 1/100$ at $z \approx 2$, a value comparable to local massive ETGs (Fig. 8a). At $z \lesssim 2$, the halo assembly transits to a slow phase, causing the galaxy to transit to a dynamically cold disk. BH growth is quenched due to inefficient viscosity to drive inward gas migration, and a horizontal path of the galaxy is seen in the $M_{\text{BH}}-M_*$ plane (Fig. 8a). The paths of dwarfs end at $M_{\text{BH}}/M_* \lesssim 1/1000$ at $z = 0$, intersecting with the upper bounds of the observational data. Since nuclear bursts dominate the overall growth of M_{BH} in dwarfs, *precise measurements of the masses of intermediate-mass black holes (IMBHs) in dwarfs may put stringent constraints on the physical processes involved in the bursts.*

MW-sized galaxies ($M_{\text{v},z=0} \approx 10^{12} M_{\odot}$) follow a growth pattern similar to that of dwarfs, but with systematic shifts in the values of physical properties. The seeding in an MW-sized galaxy is earlier due to the earlier formation of the host halo, and the chance to breed a Pop-II seed is lower due to the reduced intersection with the Pop-II regime in the seeding atlas (Fig. 7a). The post-seeding growth is still dominated by the wet channel in the bursty mode (Fig. 8b; Fig. 9b3,b5), but the continuous mode starts to dominate the growth of M_* at an earlier epoch ($z \approx 5-10$). The late-time growth of M_{BH} at $z \lesssim 5$ is mainly contributed by the continuous mode (Fig. 9b3), indicating a stronger resistance to feedback by the deeper potential well of the host halo – a conclusion also reached by hydro simulations (Habouzit et al. 2021; Li et al. 2025b; Bower et al. 2017) and analytical estimations (Bower et al. 2017; Hong et al. 2023; Paper-I). The transition of the host halo to slow assembly is earlier (at $z \approx 3$; Fig. 9b2), and the high star formation efficiency (SFE) in MW-mass disks (e.g. Yang et al. 2003, 2012; Behroozi et al. 2013), together with the cessation of BH growth, drives a horizontal migration in the $M_{\text{BH}}-M_*$ plane (Fig. 8b). Such a migration is significant among all galaxies, results in a minimal ratio of $M_{\text{BH}}/M_* \approx 1/10000$, and causes the $M_{\text{BH}}-M_*$ relation of LTGs to systematically shift to the right compared to that for ETGs, as seen also in observations (Fig. 8b).

The brightest cluster galaxies (BCGs; $M_{\text{v},z=0} \gtrsim 10^{13} M_{\odot}$) follow the general growth pattern found above, but again with systematic shifts in the values of physical properties. However, a distinct, merger-driven evolution in M_{BH} and M_* appears in their late stages (Fig. 8d). The host halo of the example galaxy remains in the fast phase throughout the history (Fig. 9c3,c5), drives effective growth of M_{BH} in the continuous mode, causes early quenching by BH feedback at $z \approx 1.5$, and leaves mergers as the only channel to grow M_{BH} and M_* at later times. A similar conclusion for the dominance of mergers at low z was reached by Li et al. (2025b, see their figs. 2 and 10) using a cosmological hydro simulation (see also Weinberger et al. 2018; Habouzit et al. 2021; Ma et al. 2025b).

One interesting conclusion is that BCGs are the only type of (central) galaxies where all the channels of BH growth are relevant sequentially in the history and become dominant in some stages (Fig. 8d). **Dynamical hotness** in each of the stages plays a fundamental role in maintaining the growth of the BH: the fast assembly of halo delays seeding and elevates seed mass; temporary excursions of halo assembly in the post-seeding era induce global disturbances that trigger episodic nuclear bursts and super-Eddington accretion; accretion in the fast phase sustains a dynamically hot SGC and a continuous mode of BH growth; once the galaxy is quenched, the large number of satellites that have been brought in by the fast assembly

of the halo sustain the growth of the BH through dry mergers. The sequential dominance of these channels results in a high but constant ratio of $M_{\text{BH}}/M_* \approx 1/100$ throughout the post-seeding history of a BCG, and results in a tight $M_{\text{BH}}-M_*$ relation with a slope of ≈ 1 (Fig. 8c and d).

As a brief summary, Figs. 8 and 9 show that the dominant channels sequentially emerging in the growth history of a central galaxy are the nuclear burst (wet and dry channels), the continuous growth within the self-gravitating gas cloud/disk (with a bursty-to-continuous transition mostly at $z \approx 5-15$), and the ex-situ growth via merger (most significant in BCGs after quenching). The sequence is universal for all galaxies, while some of the later channels may be absent in the growth of M_{BH} or M_* due to, e.g. dynamical transition to a cold disk and infrequent merger in a low-mass halo.

3.2.2 The cosmic histories of black-hole accretion and star formation

The universal sequence of the dominant channels in the histories of individual galaxies implies that the cosmic history of galaxy formation (BH accretion or star formation) is also sequentially dominated by these channels. This conclusion can be reached by considering the cosmic history as a convolution of the halo mass function at $z = 0$ and the histories of individuals with different $M_{\text{v},z=0}$, plus some contributions from satellite galaxies. The transition epoch of the cosmic history from one channel to the next is thus a weighted average of the transition epochs of individuals. This leads to the expectation that the cosmic history can be divided into the following eras based on the driving channel:

- (i) **The seeding era** ($z \gtrsim 20$, during which most seeds have been bred; see §3.1.2 and Fig. 6), in which the cosmic history is dominated by the seeding process in mini-halos (formation of Pop-III stars and BH seeds). Over-massive BHs with $M_{\text{BH}}/M_* \approx 1$ are expected. The latest seeding events in our sample are at $z \approx 5$, in the progenitors of dwarfs (see Fig. 7). The products of galaxy formation in this era constitute only a small fraction of the total M_{BH} and M_* to be observed at $z \approx 0$, given the low masses of Pop-III star clusters. If star formation stops in a mini-halo after seeding, the remnant would be an ultra-faint dwarf. The search for such a population in the low- z universe is thus challenging.
- (ii) **The bursty era** ($5 \lesssim z \lesssim 20$, before the transition of dwarf-size galaxies to the continuous mode; see §3.2.1 and Figs. 8 and 9), in which the BH accretion and star formation take place mostly as episodic nuclear bursts. Compact and concentrated morphology, dominance of NSCs, and over-massive BHs with $M_{\text{BH}}/M_* \approx 1/10$ are expected for galaxies in this era. Quiescent galaxies are expected to be ubiquitous between the gaps of nuclear bursts.
- (iii) **The continuous era** ($z \lesssim 5$), in which both the BH accretion and the star formation are dominated by the continuous mode. Galaxies in this era first appear as bulges and may later transit to form extended disk components. Super-Eddington accretion is much less frequent, and the ratio of M_{BH}/M_* is reduced to about $1/100-1/10000$, depending on the dynamical hotness of the galaxy.

Fig. 10 shows the evolution of the cosmic BH accretion rate (BHAR) density, $\dot{\rho}_{\text{BH}}$, and the SFR density, $\dot{\rho}_*$, defined respectively as

$$\dot{\rho}_{\text{BH}} = \frac{1}{V_{\text{u}}} \sum_i \dot{M}_{\text{BH},i}, \quad (10)$$

$$\dot{\rho}_* = \frac{1}{V_{\text{u}}} \sum_i \dot{M}_{*,i}/(1-R), \quad (11)$$

where $R = 0.4$ is the returned mass fraction given by the Chabrier IMF (Bruzual & Charlot 2003) assumed in our model for Pop-I/II stars,

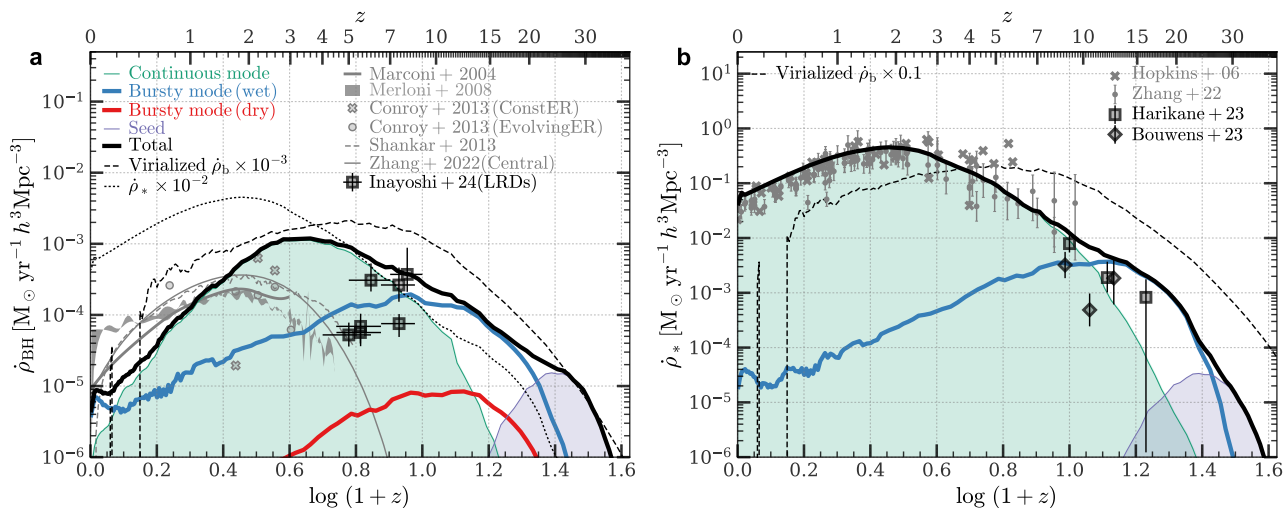


Figure 10. Evolution of cosmic black hole accretion rate (BHAR) density and star formation rate (SFR) density. **a**, cosmic BHAR density (ρ_{BH}). **b**, cosmic SFR density (ρ_*). Both densities are defined as the mass formed per unit comoving volume per unit cosmic time. In each panel, **black thick** curve represents the total rate predicted by our model, while colored curves show the contributions from different channels (**green**, continuous mode; **blue**, wet channel in bursty mode; **red**, dry channel in bursty mode). The contribution from the seeding procedure is also included and shown by **purple** curve. For comparison, we show a fixed fraction of the growth rate of the virialized baryon mass density (virialized ρ_b ; see Eq. 12 for definition). We also scale ρ_* by a factor of 10^{-2} and show it in **a**. For ρ_{BH} , we show the empirical results calibrated by observations obtained by a number of studies, including Marconi et al. (2004), Merloni & Heinz (2008), Conroy & White (2013, with either constant or redshift-dependent Eddington ratio distribution assumed), Shankar et al. (2013) and Zhang et al. (2022, including only central BHs), and the results compiled by Inayoshi & Ichikawa (2024) based on recent JWST observations for the LRDs (Akens et al. 2024; Kokorev et al. 2024; Matthee et al. 2024; Greene et al. 2024). For ρ_* , we show the observational data points compiled by Hopkins & Beacom (2006) and Zhang et al. (2022), and recent observational results obtained by Harikane et al. (2023a) and Bouwens et al. (2023) based on JWST. This figure shows that the cosmic histories of BH accretion and star formation can be divided into three sequential eras: the seeding era ($z \gtrsim 20$), the bursty era ($5 \lesssim z \lesssim 20$) and the continuous era ($z \lesssim 5$), with each era dominated by a distinct channel of growth. See §3.2.2 for details.

and the summations are over all galaxies predicted by our model in the simulated volume, V_u , in each given snapshot. Contributions from different channels are shown in total and separately. For completeness, the contribution from seeding is included in the total densities. For reference, we also show the growth rate of the ‘virialized’ baryon mass density, defined as

$$\text{Virialized } \rho_b \equiv \frac{\dot{f}_b}{V_u} \frac{d}{dt} \sum_i M_{v,i}, \quad (12)$$

scaled down by factors of 10^{-3} and 10^{-1} in the two panels of Fig. 10, respectively, and the summation is over all halos in our sample at each given snapshot. The cosmic history and the transition between the eras follow the expectation above, with the following details worth noting.

At $z \gtrsim 10$, the contribution of the bursty mode to ρ_{BH} rises rapidly with time, due to the fact that halos accrete rapidly (with $\gamma_v \gtrsim 1$ for $M_{v,z=0} \gtrsim 10^{9.5} M_\odot$; see Fig. B1) and undergo frequent excursions, and densities of entire SGCs are close to the supernova-free threshold (Eq. B2). About 1/10, 000–1/1000 of the total virialized baryon mass is accreted into BHs (Fig. 10a). All the above factors promoting the bursty growth diminish with time. The contribution of the bursty mode to ρ_{BH} reaches a plateau of $\approx 10^{-4} M_\odot \text{yr}^{-1} h^3 \text{Mpc}^{-3}$ at $z \approx 10$ –5, is surpassed by that of the continuous mode at $z \approx 8$, and constitutes $\lesssim 1/10$ of the total ρ_{BH} at $z \lesssim 5$.

In the continuous era, ρ_{BH} reaches a peak of $\approx 10^{-3} M_\odot \text{yr}^{-1} h^3 \text{Mpc}^{-3}$ at $z \approx 3$ –4, and rapidly declines at $z \lesssim 2$ as most halos have now transitioned to the slow phase (see Fig. 2b) in which the BH growth is quenched. ρ_{BH} is less than 1% of ρ_* in this era, as expected from the low M_{BH}/M_* of the LTGs (Fig. 8). At $z \approx 0$, the bursty mode appears outpacing the continuous mode again, but the rate of nuclear bursts and the growth of BHs during each burst are

small, as seen from the large fluctuations in, and the small values of ρ_{BH} .

The evolution of ρ_* (Fig. 10b) has an evolution pattern similar to that of ρ_{BH} , albeit with different transition epochs between the eras and different relative contributions from different channels. The bursty mode dominates ρ_* at $z \gtrsim 10$, converting $\lesssim 1\%$ of the virialized baryon mass into stars. Such an efficiency is comparable to those of dwarfs, but is much lower than that in a MW-sized galaxy at $z \approx 0$, indicating a strong depletion of gas by AGN feedback in nuclear bursts (see Appendix C5.1). The dominance of the continuous mode in ρ_* starts at $z \approx 10$, slightly earlier than that in ρ_{BH} , due to the competition between BH accretion and star formation in nuclear bursts and the difference of feedback in regulating BH growth and star formation in the continuous mode. The subsequent evolution of ρ_* follows the observations, with a single peak at $z \approx 2$, the ‘cosmic noon’. The virialized ρ_b shows a decline with time due to the slowdown of the halo assembly. The values of ρ_b and ρ_* are comparable at $z < 2$, indicating that some of the baryons ejected by previous feedback processes can be recycled back to galaxies to fuel star formation.

Recent JWST observations have extended the constraints on ρ_{BH} and ρ_* to redshifts higher than previously possible. Remarkably, a population of ‘little red dots’ (LRDs), featured by their peculiar V-shaped spectral energy distributions (SEDs), extremely compact morphologies and large population density (about 100× that of UV-selected quasars), has been observed at $z \gtrsim 4$ (Akens et al. 2024; Barro et al. 2024; Greene et al. 2024; Kokorev et al. 2024; Matthee et al. 2024; Pacucci & Narayan 2024; Yue et al. 2024b; Chen et al. 2025b; Labbe et al. 2025; Li et al. 2025d; Perger et al. 2025). The compact morphologies and the presence of broad emission lines suggest that they may harbor over-massive BHs with $M_{\text{BH}}/M_* \gtrsim$ a few percent (Chen et al. 2025b). This, if confirmed, will increase the cosmic ρ_{BH} to

be significantly higher than extrapolations from low- z results, as seen by comparing the black scatters and grey curves/scatters in Fig. 10a. Interestingly, the predicted $\dot{\rho}_{\text{BH}}$ of the bursty mode is consistent with the observational results based on the LRDs, thus offering a potential explanation of LRDs as the mixture of NSCs and over-massive BHs formed in nuclear bursts.

The JWST observations of the cosmic $\dot{\rho}_*$ also suggest a moderate enhancement of the SFE at high z (Naidu et al. 2022; Chen et al. 2023b; Yung et al. 2023; Harikane et al. 2023a; Xiao et al. 2024; Sabti et al. 2024; Wang et al. 2025). The predictions of our model at $z \gtrsim 10$ match the JWST results (Fig. 10b), even without a calibration using these observations. Given the dominance of the bursty mode at $z \gtrsim 10$, *the precise measurements of the cosmic $\dot{\rho}_{\text{BH}}$ and $\dot{\rho}_*$ in the bursty era may thus offer clean constraints on processes that affect the properties of nuclear bursts.*

3.2.3 Effects of seeding methods

The small contributions of BH seeds and Pop-III stars to $\dot{\rho}_{\text{BH}}$ and $\dot{\rho}_*$ (Fig. 10) imply that these earliest products of galaxy formation are soon mixed into, and outshone by the M_{BH} and M_* formed after seeding. Given the difficulty in targeting the seeds directly, it is interesting to see if seeds bred by different methods can preserve some observable properties and whether certain predictions of the model are sensitive to the seeding method.

Here we perform an *idealized and controlled experiment* for each target halo using two extreme variants of our model, *MinSeed* and *MaxSeed*, to bracket the seed masses expected from variants covering the entire parameter space. Specifically, *MinSeed* closes all the ‘heating’ channels that may delay seeding, breeds seeds once halos reach the H_2 -cooling threshold (Eq. 1), and yields minimal seed masses. *MaxSeed* forces the ‘heating’ to be maximally effective, thus maximizing the delay of seeding and the seed masses (see Appendix G for details). All other ingredients are controlled to be the same as the *Default* variant, so that the effects of seeding method can be isolated. To have a large halo sample covering a large range of M_v , here we use halo merger trees generated by the Monte-Carlo method described in Appendix D1. For any target $M_{v,z=0}$, we randomly generate a merger tree that is rooted in the halo. Since we focus on galaxies at high z when the difference between the variants is significant (see below), mergers do not play a significant role in the growth of M_{BH} and M_* (see §3.2.1 and Fig. 8). Thus, we only retain the main branch of each tree to save computational cost. This removes the ‘dry’ contribution to M_{BH} and M_* from side branches, i.e. masses contained in side branches and added to the main branch in mergers, but retains the ‘wet’ contribution, i.e. masses added by the continuous and bursty modes due to mergers. The two extreme variants are then applied independently to the main branch of the same target halo.

Fig. 11 shows the growth of M_{BH} and M_* in halos of several $M_{v,z=0}$. The predictions of *MaxSeed* and *MinSeed* for an example halo in each sample of $M_{v,z=0}$ are shown by the thick and thin lines of the same color, respectively. To show the variance due to the diversity of halo assembly, we apply the *MaxSeed* variant to a sample of 500 halos with the same $M_{v,z=0}$ and plot the $1-\sigma$ range as the shaded region. In the *MinSeed* variant, seeds are bred with $M_{\text{BH}} \approx 10^2 M_\odot$, mostly at $z \gtrsim 20$. In the *MaxSeed* variant, seeding is delayed and seeds (mainly DCBHs) are bred with $M_{\text{BH}} \gtrsim 10^4 M_\odot$. For the example of $M_{v,z=0} = 10^{10} M_\odot$, the delay is the most significant, to $z \approx 8-9$. This again highlights the critical role of environments in modulating seeding, as seen in the seeding atlas (Fig. 7).

For halos with $M_{v,z=0} > 10^{10} M_\odot$, post-seeding growth histories of M_{BH} predicted by the two variants are very similar. In *MinSeed*,

the early growth of each example BH, regardless of $M_{v,z=0}$, shows repeated jumps with large amplitudes, with each jump powered by super-Eddington accretion during a short period of $\lesssim 1$ Myr in a nuclear burst. In contrast, the jumps in the early growth predicted by *MaxSeed* have much smaller amplitudes so that the overall growth is much slower. Consequently, the original differences in M_{BH} between the two variants are almost completely erased shortly after seeding. However, the convergence in the predicted M_{BH} occurs at a redshift that depends on halo mass, because a halo with higher $M_{v,z=0}$ is seeded earlier and has a faster post-seeding growth. In fact, the growth rates predicted by *MinSeed* are around or above the Eddington limit for halos with $M_{v,z=0} \gtrsim 10^{13} M_\odot$, but below the Eddington limit for halos of lower masses. The growth of M_* in the controlled experiments also follows the same convergent pattern as that of M_{BH} , and the epoch of convergence is again earlier for higher $M_{v,z=0}$.

The convergence of the predicted post-seeding growth between the two model variants originates from the self-regulation of BH growth by the associated feedback. In a nuclear burst, a larger M_{BH} implies a higher Eddington luminosity (L_{Edd}), a faster depletion of the nuclear gas and a less efficient growth of M_{BH} and M_* (see Appendix C5 and Fig. C1). In the continuous mode, a larger M_{BH} also causes a stronger reduction in the accumulation rate of cold gas, thus limiting the growth of both M_{BH} and M_* (see Eq. D7). Thus, post-seeding growth in both channels is expected to be more suppressed by a higher $M_{\text{BH,seed}}$, which causes the difference in seeded mass to be erased by the growth. Such self-regulated growth of BH has also been found in some hydro simulations (Li et al. 2025b; Su et al. 2025; Sivasankaran et al. 2025), albeit with a focus on growth in later epochs.

The fact that the predicted post-seeding evolution of M_{BH} and M_* is insensitive to the adopted seeding method has important implications for using the model to interpret observations. The small difference in M_{BH} and M_* between the two extreme variants after convergence implies that the predictions at low z are robust against the seeding method. On the other hand, this also implies that it is difficult to use low- z observations (e.g. D’Antona et al. 2023; Pacucci et al. 2023; Onoue et al. 2025; Jeon et al. 2025) to constrain the seeding method due to the degeneracy between seeding and growing. The only exception is the case of $M_{v,z=0} \approx 10^{10} M_\odot$, in which the convergence of M_{BH} is never reached. This suggests that a potential way to distinguish different seeding scenarios is to observe IMBHs hosted by (the progenitors of) dwarf galaxies. The main challenge here is that IMBHs may be too faint to be captured by current observations, as expected from their low masses ($M_{\text{BH}} \lesssim 10^5 M_\odot$) and low accretion rates (implied by the flat growth paths in Fig. 11a1 at $z \lesssim 8$). Additional data, such as abundance patterns of relic stars, may provide useful information (e.g. Xing et al. 2023; Skúladóttir et al. 2024).

3.2.4 Archaeology and futurology: linking black holes and galaxies across cosmic time

The controlled experiments can provide some prior constraints for model inferences from observations. Here we demonstrate the idea by recovering the growth histories of observed galaxies through galactic archaeology and model forecasting (referred to as futurology). In Fig. 11, we overplot JWST-based measurements of z , M_{BH} and M_* for individual galaxies. By comparing the locations of the observed galaxies with the growth histories predicted by our model, useful conclusions for the general population of observed galaxies can be drawn. All observational data fall into the converged regions predicted by the *MinSeed* and *MaxSeed* variants, suggesting that these observations cannot be used to reconstruct the early growth history before the epoch of convergence and to distinguish seeding methods,

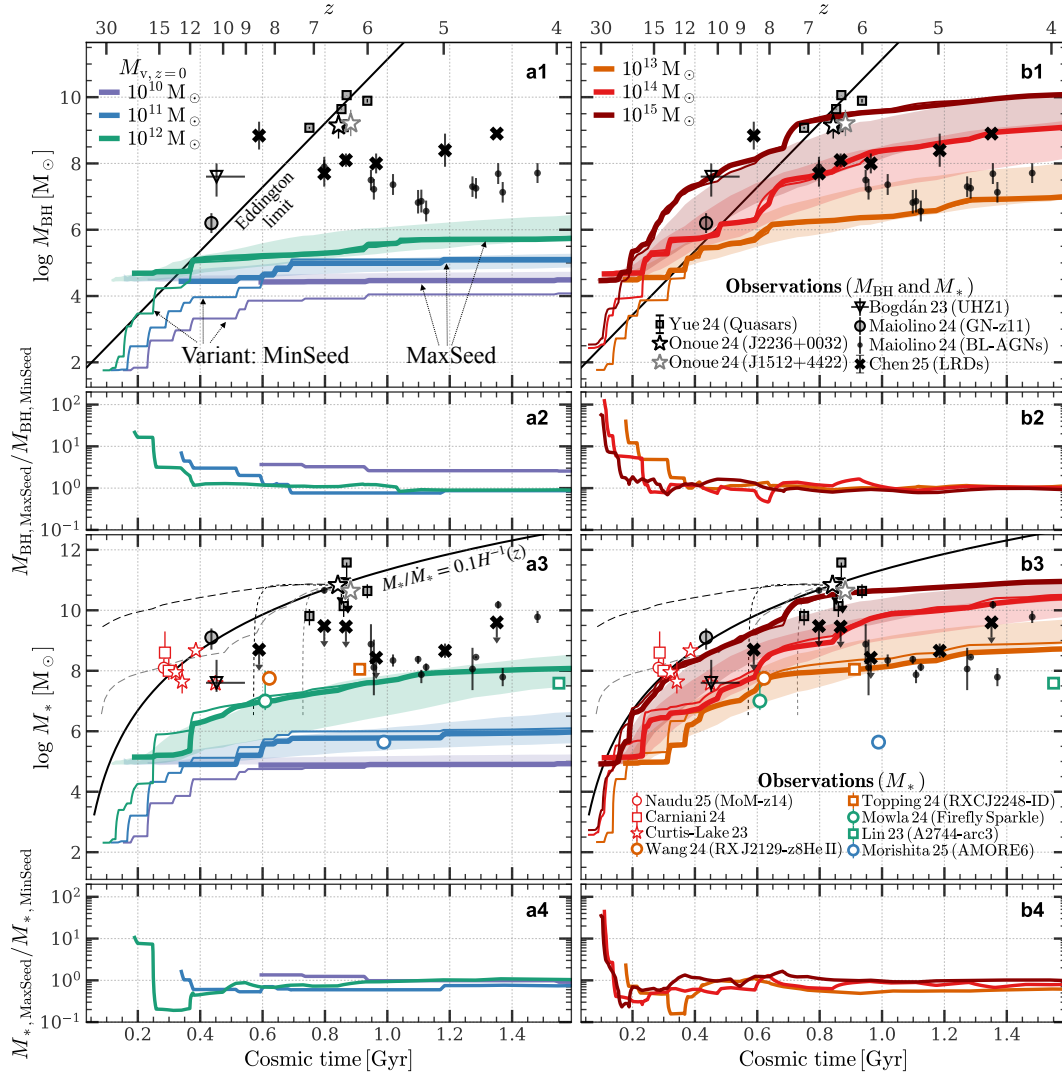


Figure 11. Idealized and controlled experiments showing effects of seeding methods and demonstrating the use of model predictions for archaeology and futurology. Each colored curve shows the growth history predicted by our model for the central galaxy in a halo with a given mass at $z = 0$. The results are separated into two columns for clarity: **a**, $M_{v,z=0} = 10^{10}, 10^{11}$ and $10^{12} M_{\odot}$; **b**, $M_{v,z=0} = 10^{13}, 10^{14}$ and $10^{15} M_{\odot}$. For each halo, the *MinSeed* and *MaxSeed* variants of our model are applied, and the results are shown by **thin** and **thick** curves, respectively. These two variants differ only in the seeding method, and produce the minimal and maximal masses of BH seeds, respectively, among all the variants in the entire parameter space. All the other ingredients of the model are strictly the same as those in the *Default* variant, thus isolating the effects of seeding methods on the growth of galaxies. Each row shows a property as a function of cosmic time: **1**, BH mass (M_{BH}); **2**, the ratio of M_{BH} between *MaxSeed* and *MinSeed*; **3**, stellar mass (M_*); **4**, the ratio of M_* between *MaxSeed* and *MinSeed*. **Shaded** region covers the 1- σ (16th–84th) percentiles, at each z , of the masses predicted by *MaxSeed* for a sample of 500 halos with $M_{v,z=0}$ the same as the example halo. In **a1** and **b1**, a tilted **black line** shows the path of M_{BH} expected for a BH growing at the Eddington rate (with a radiative efficiency of $\epsilon_r = 0.1$, i.e. $\dot{M}_{\text{BH}} = 10 M_{\text{BH}}/t_{\text{Sal}}$ in our notation; see Appendix C3.2). In **a3** and **b3**, a **black solid curve** shows the path of M_* expected for a galaxy with a specific star formation rate ($\text{sSFR} \equiv \dot{M}_*/M_*$) equal to $t_v^{-1} = 10H(z)$. For comparison, we show the observations of M_{BH} and M_* for high- z objects (Yue et al. 2024a; Onoue et al. 2025; Bogdán et al. 2023; Maiolino et al. 2024a,b; Chen et al. 2025b). In **a3** and **b3**, the long- and short-dashed curves connecting each post-starburst (PSB) galaxy represent the star formation histories, assuming non-parametric and delayed- τ forms, respectively, obtained by their spectrophotometric fitting method. Systems with only M_* measurements are also shown (Naidu et al. 2025; Carniani et al. 2024; Curtis-Lake et al. 2023; Wang et al. 2024b; Topping et al. 2024; Mowla et al. 2024; Lin et al. 2023; Morishita et al. 2025). Monte Carlo trees are used as input of the model for this figure to ensure a large sample size. This figure shows that the BHs produced by different seeding methods converge quickly to a similar M_{BH} owing to the self-regulated post-seeding growth, thus ensuring the robustness of the predictions (see §3.2.3). Matching the observed galaxies with predicted evolutionary tracks can be performed to reconstruct the formation histories of the galaxies, and to forecast their future growth (see §3.2.4).

as already mentioned in §3.2.3. The forward forecast of the growth histories, although not sensitive to seeding methods, can take into account the diversity and fluctuations in the growth histories of dark matter halos. For halos with $M_{v,z=0} = 10^{15} M_{\odot}$, the 1- σ range of M_{BH} (M_*) is about 2 dex (1 dex) at $z \gtrsim 4$. The growth history of an individual galaxy can also fluctuate significantly, often exhibiting

rapid jumps followed by flat periods due to the back-and-forth switch between star-forming and quiescent states (see §3.2.1). Such large intrinsic diversity and fluctuation imply a high probability of catching extreme objects in observations, such as those with M_{BH}/M_* much larger than the expectation from the mean $M_{\text{BH}}-M_*$ relation (Chen

et al. 2023b; Boylan-Kolchin 2023; Maiolino et al. 2024a; Pacucci & Loeb 2024).

Fig. 11 demonstrates that archaeological and forecasting information of individual galaxies can be obtained, and objects observed at different redshifts may be linked together to form a coherent picture of evolution. The two massive, post-starburst (PSB) quasar-host galaxies observed at $z \approx 6-7$ (Onoue et al. 2025) appear to overlap with the predicted evolution of M_{BH} and M_* of the example BCG with $M_{\text{v},z=0} = 10^{15} M_{\odot}$. The observed M_{BH} and M_* are slightly above the upper $1-\sigma$ (84th) percentile predicted for the BCG sample (shaded region), indicating that the two galaxies are massive outliers even among the BCG progenitors of Coma-size halos ($M_{\text{v}} = 10^{14.96} M_{\odot}$; Okabe et al. 2014). The quasar-host galaxies observed at $z \gtrsim 6$ (Yue et al. 2024a) occupy the ranges of M_{BH} and M_* similar to those of the two PSB galaxies, suggesting that they have similar origins and formation pathways. Among the sample of LRDs (Chen et al. 2025b), three (at $z = 4.46, 6.34$ and 6.76) are within the $1-\sigma$ ranges of M_{BH} and M_* predicted for BCG progenitors with $M_{\text{v},z=0} \approx 10^{14} M_{\odot}$; two (at $z = 4.96$ and 5.84) have M_* (or upper bound) close to the example BCG progenitor with $M_{\text{v},z=0} \approx 10^{13} M_{\odot}$, but their M_{BH} are over-massive, close to that for $M_{\text{v},z=0} \approx 10^{14} M_{\odot}$; one (at $z = 8.50$) has an obviously over-massive M_{BH} , even compared with the $1-\sigma$ range of the BCG progenitors with $M_{\text{v},z=0} \approx 10^{15} M_{\odot}$. The broad-line AGNs (Maiolino et al. 2024b) are consistent with BCG progenitors with $M_{\text{v},z=0} \approx 10^{13} M_{\odot}$, but some have under-massive M_* that fall into the $1-\sigma$ range of the modeled galaxies with $M_{\text{v},z=0} \approx 10^{12} M_{\odot}$.

Towards earlier epochs, the galaxy GN-z11 at $z = 10.60$ (Maiolino et al. 2024a; see also Tacchella et al. 2023; Ji et al. 2025) has an M_* close to the example BCG with $M_{\text{v},z=0} \approx 10^{15} M_{\odot}$ (Fig. 11, column b), making it a viable progenitor for galaxies like the two PSBs and other quasar hosts. Its M_{BH} is significantly under-massive, but still consistent with the $1-\sigma$ range of BCG progenitors with $M_{\text{v},z=0} \approx 10^{15} M_{\odot}$. A galaxy with such high M_{v} is predicted to have experienced repeated nuclear bursts by this epoch, and thus to have built a compact and massive NSC (see §3.2.1 and Fig. 9). This is consistent with the observational inference by e.g. D’Antona et al. (2023) that GN-z11 was formed out of pristine gas mixed with ejecta from asymptotic giant branch (AGB) stars in a massive GC or a NSC. A similar conclusion can be reached for the galaxy UHZ1 at $z \approx 10.3$, which hosts a heavily obscured but X-ray luminous quasar (Bogdán et al. 2023), and for the samples of Lyman-break galaxies at $z \gtrsim 10-14$ (Naidu et al. 2025; Carniani et al. 2024; Curtis-Lake et al. 2023) whose M_* align with the evolution predicted for BCG progenitors with $M_{\text{v},z=0} \approx 10^{15} M_{\odot}$. Thus, their fates should be bright quasars at $z \gtrsim 6$ and BCGs of Coma-like halos at $z \approx 0$.

Observations for low-mass galaxies in the early Universe are more challenging, but have been achieved by JWST with strong lensing. In the third row of Fig. 11, we show three examples ‘Firely Sparkle’, ‘A2744-arc3’ and ‘AMORE6’. The former two are occupied by compact star clusters, and the latter is a galaxy with a compact morphology. All three have a peculiar enrichment pattern: Firely Sparkle has ISM and stellar metallicities of about 1% of solar values; A2744-arc3 has ISM metallicity that appears to be lower than that of the CGM; AMORE6 has an extremely low ISM metallicity of $\lesssim 0.1\%$ of the solar value. The peculiarity in both morphology and metallicity of these galaxies suggests that they are in the early stages of evolution, where an extended, smooth stellar body and well-enriched, well-mixed gas environment have not yet been established. The observed stellar masses of these galaxies are consistent with the evolutionary paths of modeled galaxies with $M_{\text{v},z=0} \approx 10^{11}-10^{12} M_{\odot}$ (Fig. 11a3). The prediction is that a large fraction of their stellar mass is formed by nuclear bursts or in the form of globular clusters (Chen et al. 2025a,

see their figs. 17 and 18), which provides a plausible explanation for their peculiar properties.

The mismatch of individual galaxies between model predictions and observations is not unexpected, given that our model has not been calibrated to match the observations and that the observations themselves are subject to systematics, such as uncertainties in sample selection and SED fitting. Examples of such uncertainties are demonstrated in row 3 of Fig. 11 by the star formation histories of the two PSB galaxies shown in dashed curves. Their spectrophotometric fitting with two different prior forms for the star formation histories (non-parametric and delayed- τ) yields unconstrained results at high z , highlighting the challenges in observational data modeling. Observational data on galaxy environments may provide additional information about the formation history, as demonstrated by using 2PCCFs for galaxies with different seeds in our model (§3.1.3 and Fig. E1) and for galaxies with different structure parameters reported in a recent observational study (Zhang et al. 2025a). Similarly, our model can be implemented into a constrained N-body simulation (e.g. Wang et al. 2016; Stiskalek et al. 2026) in which large-scale dark-matter environments around individual halos are constrained, leaving the modeling of baryonic (Z_{IGM}) and radiative ($J_{\text{LW},21}$) environments for the seeding less uncertain.

3.3 The build-up of the black hole mass-stellar mass relation

The $M_{\text{BH}}-M_*$ relation is observationally accessible and encodes the co-evolution of BHs and their host galaxies. It is thus interesting to see how multi-channel growth in our model shapes the $M_{\text{BH}}-M_*$ relation over cosmic time. Fig. 12 shows the $M_{\text{BH}}-M_*$ relations at different z , as well as M_{BH} and M_* of individual galaxies predicted by the *Default* variant. In the last panel, we summarize the build-up of the $M_{\text{BH}}-M_*$ relation predicted by our model by showing (i) the distribution of seeds at their seeding epoch; (ii) a collection of modeled relations at different z ; (iii) the distribution of individual galaxies at $z = 0$, color-coded according to the channel that contributes the most to M_{BH} . In Appendix G, we explore the $M_{\text{BH}}-M_*$ relations in other model variants.

At $z = 15$, the $M_{\text{BH}}-M_*$ relation shows two branches, with $M_{\text{BH}} : M_* \approx 1 : 1$ and $M_{\text{BH}} : M_* \approx 1 : 100$, respectively, connected by a smooth transition at $M_* \approx 10^4 M_{\odot}$. All galaxies fall into the mass ranges of star clusters and dwarfs ($< 10^9 M_{\odot}$), with a few extremely massive outliers reaching $M_* \approx 10^9$ and $M_{\text{BH}} \gtrsim 10^7 M_{\odot}$. At $z \gtrsim 10$, the two branches persist, but more galaxies have evolved to the upper limit of dwarf mass ($M_* = 10^9 M_{\odot}$). The galaxy GN-z11 ($z = 10.60$; Tacchella et al. 2023; Maiolino et al. 2024a; Ji et al. 2025) falls into the most massive tip of the galaxy distribution, with an under-massive BH touching the lower boundary of M_{BH} of the modeled galaxies with similar M_* . The galaxy UHZ1 ($z \approx 10.32$; Bogdán et al. 2023) falls onto the other side of the same tip, with an over-massive BH touching the upper boundary of M_{BH} of the modeled galaxies with similar M_* . At $z = 5$ and 4 , the branch with $M_{\text{BH}} : M_* \approx 1 : 1$ fades away, leaving only a tail that joins the low-mass end of the other branch. Observed galaxies at these redshifts exhibit significant diversity in the $M_{\text{BH}}-M_*$ plane: at a given M_* , the observed M_{BH} can vary by about 3 dex. The most massive galaxies in our sample have $M_* \gtrsim 10^{10} M_{\odot}$ and $M_{\text{BH}} \gtrsim 10^8 M_{\odot}$, overlapping with the range covered by bright quasars (Izumi et al. 2021; Stone et al. 2023, 2024; Yue et al. 2024a) and massive PSB galaxies (Carnall et al. 2023; Onoue et al. 2025) found in observations. Less massive galaxies with $10^8 \lesssim M_*/M_{\odot} \lesssim 10^{10}$ partially overlap with the reported M_{BH} and M_* for faint AGNs (Pacucci et al. 2023; Harikane et al. 2023b; Übler et al. 2023; Maiolino et al. 2024b), LRDs (Chen et al. 2025b) and

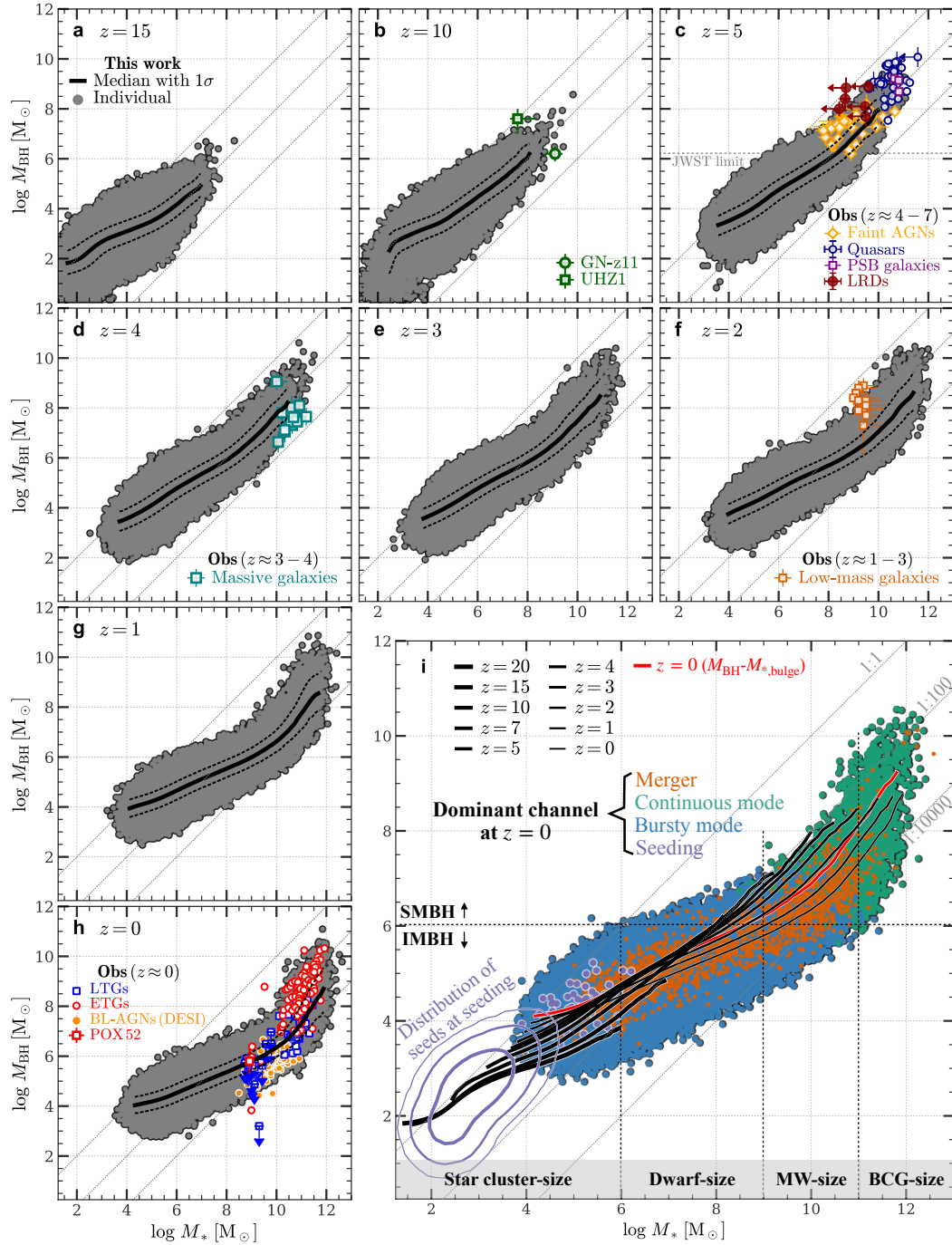


Figure 12. $M_{\text{BH}}-M_*$ relations. **a–h**, each showing the relation at a given redshift, as labeled. **Grey dots** represent individual galaxies predicted by the model, and **black curves** are the running median and $1-\sigma$ ($16^{\text{th}}-84^{\text{th}}$) percentiles of M_{BH} at given M_* . **i**, a summary of the predicted median $M_{\text{BH}}-M_*$ relations at different redshifts (**black curves**). For comparison, we also show the predicted median black hole mass-stellar bulge mass ($M_{\text{BH}}-M_{*,\text{bulge}}$) relation at $z = 0$ (**red curve**), and the distribution of BH seeds with their M_{BH} and M_* evaluated at the seeding epochs (**purple contours**; from inner to outer, enclosing 68%, 95% and 99.7% of the seeds, respectively). Dots represent galaxies at $z = 0$, with color of each indicating the channel that contributes the most to the M_{BH} (**purple**, seeding; **blue**, bursty mode; **green**, continuous mode; **orange**, merger). **Colored markers** in **a–h** show observations obtained/compiled by different studies (Maiolino et al. 2024a; Bogdán et al. 2023; Pacucci et al. 2023; Harikane et al. 2023b; Übler et al. 2023; Maiolino et al. 2024b; Chen et al. 2025b; Onoue et al. 2025; Carnall et al. 2023; Izumi et al. 2021; Stone et al. 2023, 2024; Yue et al. 2024a; Li et al. 2025a; Mezcua et al. 2024; Greene et al. 2020; Pucha et al. 2025; Kawamuro et al. 2024; Sun et al. 2025). A horizontal dashed line in **c** indicates the detection limit of JWST at $z \approx 5$. In all panels, the tilted grey lines indicate constant ratios of M_{BH}/M_* . A log-normal noise with a standard deviation of 0.3 dex is added to the M_{BH} and M_* of each modeled galaxy to mimic the observational uncertainties (e.g. Behroozi et al. 2013; Kormendy & Ho 2013). This figure demonstrates that the $M_{\text{BH}}-M_*$ relation is a redshift-dependent, multi-piece relation, with each piece driven by a distinct channel of galaxy formation. See §3.3 for the details.

massive galaxies hosting normal-mass BHs (Li et al. 2025a), indicating that each sample only represents a biased sub-population of galaxies. At $z \lesssim 3$, the surviving branch starts to deviate from the line of $M_{\text{BH}} : M_* \approx 1 : 100$. This deviation is mass-dependent, with galaxies around the MW-mass ($M_* \approx 10^{10} M_\odot$) shifted most significantly to the right. The observed low-mass galaxies with $M_* \lesssim 10^{10} M_\odot$ at $z \approx 1-3$ (Mezcua et al. 2024) appear to overlap with the upper boundary of the $M_{\text{BH}}-M_*$ scatters. At $z \lesssim 1$, the $M_{\text{BH}}-M_*$ relation stabilizes as a down-bending, bow-shaped curve. The observed ETGs and LTGs compiled by Greene et al. (2020) occupy the high- M_* end and the knee of the bow, with M_{BH} above and below the median relation at the corresponding M_* , respectively (see also Graham & Sahu 2023). The broad-line AGN candidates in low-mass galaxies (Pucha et al. 2025) from the Dark Energy Spectroscopic Instrument (DESI) survey, and POX 52, a local early-type dwarf showing evidence for IMBH (Kawamuro et al. 2024; Sun et al. 2025), also occupy the knee of the relation.

The broad consistency between model predictions and observations indicates the plausibility of the physical processes implemented. The predicted $M_{\text{BH}}-M_*$ relation (Fig. 12i) at any z is not a single power-law but consists of separable pieces. Thus, it is interesting to dissect the relation and reveal the underlying physical mechanisms that shape individual pieces. As shown in Figs. 8 and 9, the halo mass ($M_{\text{V},z=0}$) appears to be an important factor that determines the growth stage reached by a galaxy at a given z , and the dominant growth channels (seeding, bursty or continuous mode, or merger) emerge sequentially as a galaxy evolves from one stage to another, establishing a distinct $M_{\text{BH}}-M_*$ relation in each of the stages (see §3.2.1). Thus, the overall $M_{\text{BH}}-M_*$ relation at a given z is a convolution between the halo mass function at $z = 0$ and the relations of the sub-populations in different stages at the redshift z in question.

During the seeding era ($z \approx 20-30$; §3.2.2), BH seeds are bred with $M_{\text{BH}} \sim M_*$ due to the top-heavy IMF of Pop-III stars adopted by our model. This sets the starting loci of the $M_{\text{BH}}-M_*$ relation and produces the highest ratio, $M_{\text{BH}}/M_* \approx 1$, that can ever be achieved in the growth history of galaxies. Most seeds are IMBHs, waiting for the post-seeding channels to boost them to SMBHs. The post-seeding growth of each galaxy starts with the bursty mode, consisting of episodic nuclear bursts triggered by global disturbances on the galaxy. The M_{BH}/M_* ratio is $\sim 1/10$ during this era (Fig. 8), and a few nuclear bursts are sufficient to boost a BH seed to $\gtrsim 10^6 M_\odot$ (Fig. 9, rows 3 and 4). *The bursty mode turns out to be the key condition for raising the masses of all BHs, especially low-mass seeds, to the regime of SMBHs*, so that the subsequent growth of BHs in the continuous mode is initialized with sufficiently high M_{BH} and galaxies are well-regulated by the AGN feedback to sustain an $M_{\text{BH}}-M_*$ relation similar to that observed at low z . The bursty mode thus provides a physical explanation for the origin of the ‘enhancement factor’, inferred phenomenologically in Paper-I (see their fig. 7 and §4.3.3), needed to boost the BHs in low-mass halos to the regime of SMBHs.

Following the bursty era, the growth of BHs via nuclear bursts is self-limited by the associated feedback and the continuous-mode growth gradually takes over and keeps M_{BH}/M_* around $1/100$ (Fig. 8). Towards lower z , the efficiency of AGN feedback at a given M_{V} increases due to the shallower gravitational potential (quantified by V_{max} of the halo) and the stronger coupling between feedback energy and the SGC (see Appendix D4). The growth of BHs in dynamically hot SGCs at lower z is thus expected to have stronger self-regulation, leading to a $M_{\text{BH}}-M_*$ relation with a lower amplitude, $M_{\text{BH}}/M_* \approx 1/1000$, as shown by the red curve in Fig. 12i at $z = 0$. A logarithmic slope ≥ 1 is seen at the high-mass end, as expected from the energy argument of AGN feedback (Mo et al. 2024, §4.3 therein).

The net effect is that the $M_{\text{BH}}-M_*$ relation around $M_* \sim 10^{10} M_\odot$ bends downward at $z \lesssim 5$.

At $z \approx 2$, halos hosting progenitors of MW-size galaxies (with $M_{\text{V},z=0} \approx 10^{12} M_\odot$ and $M_{*,z=0} \approx 10^{10.5} M_\odot$) start to transit from the fast phase to the slow phase (see Fig. 2b). Galaxies then start to grow their dynamically cold (disk) components effectively in the continuous mode, but the BH accretion is terminated almost completely. These MW progenitors thus migrate horizontally in the $M_{\text{BH}}-M_*$ plane (Fig. 8), and eventually reach the knee of the bow-shaped $M_{\text{BH}}-M_*$ relation at $z = 0$. Halos hosting less massive galaxies (e.g. dwarfs) transit to the slow phase earlier (Fig. 2b), but the post-transition star formation is less efficient than that in progenitors of MW-size galaxies due to their shallower gravitational potentials to resist the feedback, so that the horizontal shift in the $M_{\text{BH}}-M_*$ plane is smaller. Halos hosting more massive galaxies (e.g. BCGs) transit to the slow phase later, and both the SFE and the duration for star formation in the dynamically cold phase in these halos are smaller than those in MW-size galaxies, which reduces the amount of horizontal migration in the $M_{\text{BH}}-M_*$ plane. The mass-dependence of halo transition time and the post-transition SFE thus gradually ‘draw the bow’ at $z \lesssim 3$, and eventually build up the bow-shaped $M_{\text{BH}}-M_*$ relation at $z = 0$ (Fig. 12i). Dry mergers of galaxies are not important in most parts of the $M_{\text{BH}}-M_*$ relation, since the mass added by each merger only takes a temporary dominance in a short period after the merger and is soon surpassed by the growth via other channels (see the brown segments in Fig. 8). The exception is in the high-mass end ($M_* \gtrsim 10^{11} M_\odot$) of the relation at low z ($z \lesssim 2$), where repeated mergers drive galaxies towards a relation of $M_{\text{BH}} \propto M_*$ and produce the scatter in the high-mass tip in Fig. 12i (see also Fig. 8d).

As a summary, our model predicts a **redshift-dependent, multi-piece $M_{\text{BH}}-M_*$ relation**, and the growth of galaxies within each piece is driven by a distinct channel. The snapshot views provided by observations, although subject to various uncertainties, appear to support the model predictions. Observations of dwarfs ($M_* \leq 10^9 M_\odot$), which are still missing, are expected to be the key to testing the model. In particular, galaxies of star-cluster size ($M_{*,z=0} \lesssim 10^6 M_\odot$) are predicted to be close to the seeding locations in the $M_{\text{BH}}-M_*$ plane, and a small fraction of them have M_{BH} dominated by the seeded mass (Fig. 12i, purple dots). The post-seeding growth of these galaxies is the least important among all galaxies, and the search for IMBHs in this population may provide crucial constraints on the seeding model.

4 SUMMARY AND DISCUSSION

In this paper, we have developed a model for the formation and co-evolution of (SM)BHs with their host galaxies and halos (see Fig. 1 for an outline). The model is built upon the two-phase framework of galaxy formation established in Paper-I (see §2.2 for a summary), with extensions specific to the seeding and the early growth of BHs. The model covers wide ranges of spatial, temporal and mass scales ($r \approx 10^{-3}-10^6$ pc, $z \approx 40-0$, $M_{\text{BH}} \approx 10-10^{10} M_\odot$), so that the entire lives of SMBHs and their host galaxies can be self-consistently initialized and followed within a cosmological context. We have detailed physical processes relevant to the modeling, derived equations describing each process and presented analytical approximations to obtain order-of-magnitude estimates. The main features of the model are summarized as follows.

- (i) The model takes halo merger trees (§2.1; Fig. 2) in a cosmological volume as input to determine when, where and how BH seeds are bred: the seeding epoch of a mini-halo is determined by the operation

of key coolants in pristine gas (H and H₂) and environmental effects known to modulate the cooling (LW radiation, dynamical heating and IGM enrichment); a seed is bred as the remnant in the first star cluster formed from the cooled and collapsed gas in the mini-halo; a feedback-regulated model is adopted to derive the mass of the dominant star and, together with the assumed top-heavy IMF, to give the mass spectrum of Pop-III stars; the evolution of these stars is tracked, and the most massive BH remnant is selected as the seed. This seeding procedure determines not only the mass of each seed ($M_{\text{BH,seed}}$), but also the flavor that represents the formation pathway of the seed (§2.3; Fig. 3).

- (ii) The post-seeding growth of BHs is modeled as a multi-channel process. Each channel emerges naturally within the cosmological context when a certain level of dynamical hotness is generated to remove the angular-momentum (j) barrier for gas to accrete onto a BH. A new channel proposed in this paper is the bursty mode, consisting of episodic nuclear bursts, each triggered by a global disturbance on the host galaxy to form a dense, gas-rich, turbulent and supernova-free nucleus in the galactic center. During a nuclear burst, super-Eddington accretion powers rapid growth of the BH, and intense nuclear star formation grows the NSC. Other channels of BH growth are also included in the model: a continuous mode, in which a BH grows via capturing low- j sub-clouds formed in the turbulent SGC during the fast phase of the host halo, and a merger channel, in which a BH grows via merging with another BH brought in by a galaxy-galaxy merger (§2.4; Fig. 4).
- (iii) To resolve the wide range of scales required to model the post-seeding growth of BHs, we use a ladder-like profile to characterize the density around each BH. The ladder consists of five steps, each describing one of the five levels in the cosmic hierarchy: the Universe, halos, galaxies, galactic nuclei, and the accretion disks of BHs. Equations governing each level and relating one level to the next are derived, so that one can start from the outer boundary and ‘climb up’ the ladder to obtain the gas structure deep into the vicinity of a BH (see Appendix C1 and Fig. A1). Such a hierarchy propagates cosmological information on halo scales all the way down to regions close to the central BHs, thus establishing a connection between halo assembly and BH growth (§2.4).

We have implemented the model into the merger trees constructed from a N-body simulation (see Fig. 1 for an outline), presented our model predictions, and demonstrated the potential applications to be explored in the future. Our main results and conclusions are summarized as follows.

- (i) The BH seed mass function is broad (covering $\sim 10\text{--}10^5 M_\odot$) and bimodal (with a valley at $M_{\text{BH}} \approx 10^2 M_\odot$ created by PISNe). DCBHs from massive Pop-III progenitors dominate the high-mass peak (around $500 M_\odot$), and seeds resulting from CCSNe/PISNe dominate the low-mass peak (around $50 M_\odot$). LW radiation and dynamical heating have a divergent effect, producing both high-mass and low-mass bumps with DCBH and Pop-II flavors, respectively, in the mass function (§3.1.1; Fig. 5).
- (ii) The cosmic seeding history shows that most BH seeds are bred at $z = 20\text{--}30$ and most Pop-II seeds are bred at $z = 10\text{--}20$. The seeding rate becomes lower at $z \lesssim 10$ and negligibly small at $z \lesssim 5$, making observational search for ongoing seeding activity at $z \lesssim 5$ challenging. LW radiation and dynamical heating act mainly on mini-halos at $z \lesssim 20$ and $z \gtrsim 10$, respectively. Most seeds bred at $z \lesssim 10$ are delayed by LW radiation, highlighting the importance of radiation feedback in producing late-formed Pop-III seeds that are targets of some recent JWST programs and future surveys (§3.1.2; Fig. 6).
- (iii) An environment-modulated ‘seeding atlas’ is constructed to show how seeds modulated by different environmental effects are distributed

in the $M_V\text{--}z$ plane. The atlas can be dissected into three regimes, each dominated by one environmental effect: a fusiform area shaped by dynamical heating, a tip-like structure shaped by LW radiation, and a two-component stripe shaped by IGM enrichment (§3.1.3; Fig. 7).

- (iv) These channels driving the post-seeding growth emerge sequentially and become dominant successively in the growth history of any given galaxy, provided that the galaxy can reach the mass scale relevant to a channel at some epoch and its dynamically hot state is sustained to that epoch. The bursty mode dominates the growth of BHs soon after their seeding ($z \approx 20\text{--}30$ in most cases), and the continuous mode takes over at $z \approx 5\text{--}15$ when M_{BH} reaches about $10^6 M_\odot$. The bursty-to-continuous transition occurs earlier in more massive halos. In the continuous stage, star formation builds the dynamically hot (bulge) stellar component first followed by the dynamically cold (disk) component. Mergers become important only in BCG-size galaxies at $z \lesssim 2$ (§3.2.1; Figs. 8 and Fig. 9).
- (v) Galaxy evolution during the bursty stage has two distinct features, one is the back-and-forth switching between star-forming and quiescent states, and the other is the formation of a compact, globular-like NSC. Our model therefore predicts that the formation of the first generation of quiescent galaxies can be as early as $z \approx 20\text{--}30$, soon after the seeds are bred, and that quiescent galaxies and galaxies with extremely compact morphology are ubiquitous at $z \gtrsim 10$ (§3.2.1).
- (vi) The cosmic histories of BH accretion and star formation are compositions of those of individual galaxies. As a result, the cosmic histories are also driven, sequentially, by the above channels, and can be divided into three eras: the seeding era ($z \approx 20\text{--}30$), the bursty era ($z \approx 5\text{--}20$), and the continuous era ($z \lesssim 5$). The predicted evolution of the cosmic SFR density and BHAR density is broadly consistent with JWST observations (§3.2.2; Fig. 10).
- (vii) A set of idealized and controlled experiments is performed with two variants of our model, *MinSeed* and *MaxSeed*, both adapted from the *Default* model with extreme assumptions that minimize and maximize the masses of seeds, respectively. The self-regulated nature of BH accretion tends to erase the memory of the seeding method, making the differences in M_* and M_{BH} at low z between the two variants indistinguishable. Halos with $M_{V,z=0} \lesssim 10^{11} M_\odot$ are the only population whose difference in M_{BH} caused by the seeding method can persist to $z \lesssim 8$ (§3.2.3; Fig. 11).
- (viii) The ensemble of BHs and galaxies, and their growth histories predicted by our model allow the archaeology and futurology to link observed BHs and galaxies across the cosmic history. We demonstrate the idea by comparing the predicted evolution of M_{BH} and M_* to the BHs and galaxies observed by JWST. The bright galaxies (massive BHs) observed at $z \gtrsim 10$ and the bright quasars at $z \gtrsim 6$ seem to align with the predicted evolution of galaxies hosted by massive halos with $M_{V,z=0} \approx 10^{15} M_\odot$, and are expected to evolve to be the massive BCGs in Coma-like clusters in the local Universe. The faint AGNs (including LRDs) observed at $z \gtrsim 4$ appear to be hosted by halos with $M_{V,z=0} \approx 10^{13} M_\odot$. The origins and fates of a small fraction of low-mass galaxies with extraordinary morphology and enrichment pattern at $z \gtrsim 4$ are also inferred using our model (§3.2.4; Fig. 11).
- (ix) Our model predicts an $M_{\text{BH}}\text{--}M_*$ relation that covers the full lifetimes of galaxies (from the seeding era, $z \gtrsim 20$, to present day) and the full spectrum of mass (from star cluster-size galaxies with $M_* < 10^6 M_\odot$ to BCG-size galaxies with $M_* \gtrsim 10^{11} M_\odot$). The predicted relation is redshift-dependent and multi-piece, with each piece shaped by one of the growth channels. BH seeds populate the low-mass end, $M_* \lesssim 10^5 M_\odot$, of the relation, with $M_{\text{BH}}/M_* \approx 1$. The bursty mode provides the key condition to raise the mass of all BH seeds to the regime of SMBHs, and can sustain a relation of $M_{\text{BH}}/M_* \approx 1/10$ at $M_{\text{BH}} \lesssim 10^6 M_\odot$. The continuous mode takes over at higher mass, and

can sustain a relation of $M_{\text{BH}}/M_* \approx 1/100$ during the dynamically hot phase of galaxies. The transition of galaxies to the dynamically cold phase prevents BHs from growing further and drive a horizontal migration in the $M_{\text{BH}}-M_*$ plane. Such a migration is mass-dependent, with a maximum at the MW mass that reaches $M_{\text{BH}}/M_* \approx 1/10^4$, thus drawing the relation to a down-bending, bow-shaped one at $z \lesssim 3$. A comparison with observations shows that the prediction of the model recovers the observed $M_{\text{BH}}-M_*$ relation from $z \approx 10$ to $z \approx 0$ (§3.3; Fig. 12).

The model developed in this paper provides a framework to examine the co-evolution pattern of BHs and their host galaxies and halos across cosmic time, to distinguish the physical processes driving the evolution in different regimes (redshift, mass and assembly of halo, and environment), and to interpret the origin and predict the fate of observed BHs and galaxies. Clearly, the model is not definitive; it leaves many interesting questions to be addressed in the future using both theoretical and observational approaches. Here we discuss a few of them.

The environment provided by dark matter halos sets the fundamental context of galaxy formation in our model, and in most of the other empirical and semi-analytical models. However, the dynamical nature of halos and the effects of their dynamics have not been fully explored. The upper limit of the amount of baryon available for galaxy formation in a halo is assumed to be f_b (the cosmic baryon fraction) times the mass of the halo, and is used to determine the properties of the SGC hosted by the halo (Eqs. B1–B5). This assumption is certainly not valid in every case, especially for halos that interact strongly with other halos or with dense environments. The bullet cluster (e.g. Clowe et al. 2004; Markevitch et al. 2004) and dark matter-deficient galaxies (Guo et al. 2020; Lee et al. 2021; van Dokkum et al. 2022; Ogiya et al. 2022; Moreno et al. 2022; Contreras-Santos et al. 2024) observed in the local Universe provide two sets of examples for high- and low-mass halos, respectively, in which dark matter and baryon gas are detached due to interactions between halos and their environments so that the baryon fraction in the structure is actually higher than f_b . The situation is expected to be more common at high z when mergers between halos are frequent (Chen et al. 2023b) and the stream velocity between dark matter and baryons is significant compared to the small virial velocities of mini-halos (Stacy et al. 2011; Maio et al. 2011; Greif et al. 2011). It is unclear what fraction of halos with different M_v and at different z have elevated baryon contents and how such elevations change the seeding and early growth of BHs and galaxies. Detailed analyses using hydro simulations may provide some clues.

The collapse of gas within mini-halos is determined by environmental conditions on different scales. Our model includes the radiation and matter feedback from star formation to the large-scale environment, and the dynamical heating from the fast assembly of host halos (§2.3). The seeding atlas constructed from the model predictions shows how seeds modulated by these environmental effects are distributed in the M_v - z plane (§3.1.3). An empirical strategy of seeding to be applied to, e.g. other empirical/semi-analytical models and hydro simulations, may be constructed from the atlas, to partly account for the statistical expectation and variation of the seed population at given M_v and z , if the environmental information is fully or partly unavailable. A caveat to be noted is that our seeding procedure is achieved with simplified treatments that ignore many details. For example, the LW emission is assumed to be isotropic around the source galaxy, and the propagation of the radiation is assumed to depend only on redshift and the distance from the source to the target. In reality, the ISM, CGM and IGM have complex structures, which may cause the escape and propagation of the LW radiation to be highly anisotropic and

time-dependent (e.g. Pallottini et al. 2022). Radiation other than that in the LW band, such as that with energy > 0.76 eV, may detach the H^- and thus also affect the formation of H_2 (see the end of Appendix B1.3 and Latif & Khochfar 2019). Similar arguments also apply to metal ejection and diffusion. The effect of dynamical heating introduces additional complexity, as it is generated not only from halo assembly but also from the aforementioned stream velocity between dark matter and baryon gas. All these need to be revisited in detail using hydro simulations with the relevant processes implemented accurately.

Many questions regarding the formation, growth and evolution of Pop-III stars remain unsolved. The structure of Pop-III stars, especially supermassive stars (SMSs), can exhibit large fluctuations, making their evolution tracks and radiative feedback hard to model (Hosokawa et al. 2013). The accretion of gas onto Pop-III stars has been considered critical in modifying their mass spectrum (Omukai et al. 2008; Schleicher et al. 2013). However, such accretion flow is usually anisotropic, exhibiting disk-like structure with gas/stellar clumps (Chon & Omukai 2020; Latif et al. 2022b) and magnetic fields amplified by turbulence (Sharda et al. 2020). Resolving these issues remains to be a gap in our knowledge of stellar evolution and requires simulations with dynamical ranges beyond current computational power.

A key assumption of our model to grow BHs is that the formation of an SNF nucleus can be triggered by some global disturbance on the SGC, and such a disturbance is associated with the temporary excursion of the halo assembly to high γ_v . In reality, the sources of disturbance may be diverse, such as a close flyby of another galaxy (Li et al. 2025c), galaxy-galaxy merger (Luo et al. 2014; Puskás et al. 2025), counter-rotation of cold stream (Danovich et al. 2015), violent disk instability (Agertz et al. 2009; Meng et al. 2019; Renaud et al. 2021), or even the tidal force induced by the misalignment between the galaxy and the interior of the host halo (Chisari et al. 2017). Detailed examinations of these processes require simulations with both large volumes to provide statistical samples and controlled experiments to disentangle the effects of different processes. Such simulations with sufficiently high resolution and with implementations of all relevant processes can verify or falsify that nuclei with densities higher than n_{SNF} can indeed form as a result of the disturbances, as well as resolving the density profiles of the nuclei.

Super-Eddington accretion of BHs appears to be a natural outcome if SNF nuclei can indeed form around the BHs. Whether or not such a high accretion rate can be sustained depends on the structure (e.g. anisotropy and clumpiness) of the nucleus and the properties (geometry, magnetic field, radiation and jet feedback) of the accretion disk. The dry channel (i.e. BH-BH merger, and capture of stars) may also contribute to the growth of BHs, as explored by some simulations (Tremmel et al. 2015; Gaete et al. 2024; Fujii et al. 2024) and (semi-)analytical models (Partmann et al. 2024; Liu et al. 2024; Dekel et al. 2025b), and produce observable signatures such as tidal disruption events (TDEs) and gravitational waves (GWs). These processes are still poorly understood and require further refinement of the model to resolve (e.g. Anglés-Alcázar et al. 2021; Latif et al. 2022b; Hopkins et al. 2024a). Observational constraints can also help break the degeneracy in the model and tighten the posterior space to be explored.

The formation and growth of NSCs are by-products of nuclear bursts. Intuitively, the competition between star formation and BH accretion naturally leads to an anti-correlation between the NSC-BH mass ratio ($M_{\text{NSC}}/M_{\text{BH}}$) and the mass of BH (M_{BH}), as suggested by some observations (e.g. Georgiev et al. 2016; Nguyen et al. 2018; Neumayer et al. 2020). However, the dry evolution of any NSC, such

as that contributed by mergers with other star clusters (Tremaine et al. 1975; Capuzzo-Dolcetta & Miocchi 2008; Neumayer et al. 2011; Arca-Sedda & Capuzzo-Dolcetta 2014; Guillard et al. 2016; Tsatsi et al. 2017; Fahrion et al. 2022) and by the interaction between the NSC and the central BH (Antonini et al. 2015) can introduce significant contamination and needs to be modeled carefully.

The growth of BHs in the bursty mode provides a key condition for BH seeds to grow to SMBHs. However, as demonstrated in §3.2.3, the required boost in the BH mass degenerates with the seeding method, and current observations are unable to break the degeneracy. The key targets that can be used to isolate the effects of the bursty mode are galaxies at $z \gtrsim 10$, when the growth of BHs is dominated by nuclear bursts (see §3.2.2 and Fig. 10), and BHs in low-mass galaxies (e.g. dwarfs with $M_* < 10^9 M_\odot$ and especially star cluster-size galaxies with $M_* < 10^6 M_\odot$) where a large fraction of the total M_{BH} is contributed by the bursty mode (see §3.2.1, Figs. 8, 9 and 12). In addition, low-mass halos, such as those with $M_V \lesssim 10^{9.5} M_\odot$ at their infall times, are around or below the atomic-cooling limit (Fig. 2) and thus expected to host galaxies with little or no star formation after the cosmic reionization. Detections of galaxies hosted by this population of halos may thus provide fossil information of galaxy formation at $z \gtrsim 6$. A calibration for the seeding method and the growth in the bursty mode using these targets may be achievable, and should be explored further.

ACKNOWLEDGEMENTS

YC is funded by the China Postdoctoral Science Foundation (Grant No. 2022TQ0329). HYW is supported by the National Natural Science Foundation of China (Nos. 12192224) and CAS Project for Young Scientists in Basic Research (Grant No. YSBR-062). YC thanks Tao Wang, Kai Wang, Fangzhou Jiang, Ziwen Zhang, Yunjing Wu, Enci Wang, Long Wang, Qi Guo, Baitian Tang, Zhaozhou Li, Houzun Chen and Xi Kang for their valuable insights and discussions, and thanks Yi Mao and Chen Chen for their technical support. The authors would like to express their gratitude to the Tsinghua Astrophysics High-Performance Computing platform at Tsinghua University and the Supercomputer Center of the University of Science and Technology of China for providing the necessary computational and data storage resources that have significantly contributed to the research results presented in this paper.

The computations and presentations in this paper are supported by various software tools, including the HPC toolkits HIPP (Chen & Wang 2023)¹ and PyHIPP², interactive computation environment IPYTHON (Perez & Granger 2007), numerical libraries NUMPY (Harris et al. 2020), ASTROPY (Robitaille et al. 2013; Astropy Collaboration et al. 2018, 2022) and SCIPY (Virtanen et al. 2020), graphical library MATPLOTLIB (Hunter 2007), and correlation function calculator (Sinha & Garrison 2019, 2020)³. The implementation of the Monte Carlo trees is revised from the public code SATGEN⁴ (Jiang et al. 2021). This research has made extensive use of the arXiv and NASA's Astrophysics Data System. Data compilations used in this paper have been made much more accurate and efficient by the software WEBPLOTDIGITIZER.

DATA AVAILABILITY

The code repository TwoPhaseGalaxyModel⁵ implements the model described in this series of papers. All data used in this paper, including data points displayed in figures and observational results used for comparison, will be distributed along with the repository.

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⁵ <https://github.com/ChenYangyao/two-phase-galaxy-model>

¹ <https://github.com/ChenYangyao/hipp>

² <https://github.com/ChenYangyao/pyhipp>

³ <https://github.com/manodeep/Corrfunc>

⁴ <https://github.com/shergreen/SatGen>

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APPENDIX A: DARK MATTER HALOS AND THEIR ASSEMBLY

A key component in the current Λ CDM paradigm of galaxy formation is the assembly of dark matter halos – dynamically stable clumps

produced by gravitational instability of the cosmic density field, within which galaxies form and evolve. As to be detailed later, the early assembly of dark matter halos at high z is crucial input for the seeding (Appendix B) and growing procedures (Appendix C). Here we briefly summarize the properties and assembly histories of dark matter halos that are relevant to the description of our model.

In the adopted flat Λ CDM cosmology (see the end of §1), the cosmological quantities are defined as follows. The critical density at $z = 0$ is written as $\rho_{c,0} = 3H_0^2/(8\pi G)$, where $H_0 = 100h \text{ km s}^{-1} \text{ Mpc}^{-1}$ is the Hubble constant at the present day. The critical density at $z > 0$ is given by $\rho_c(z) = \rho_{c,0}E^2(z)$, where $E(z) = [\Omega_{m,0}(1+z)^3 + \Omega_{\Lambda,0}]^{1/2}$. The cosmic baryon fraction is $f_b \equiv \Omega_{b,0}/\Omega_{m,0} \approx 0.157$, while the cosmic baryon density at $z \geq 0$ is

$$\rho_b(z) = \rho_{c,0}\Omega_{b,0}(1+z)^3 = \left[4.19 \times 10^{-28} \text{ g cm}^{-3}\right] (1+z)_{10}^3, \quad (\text{A1})$$

with $(1+z)_{10} \equiv (1+z)/10$.

Fig. A1a shows the gas density as a function of the halo-centric distance at three epochs in the progenitor of an MW-size halo. The density profile exhibits a ladder-like structure, consisting of five steps that correspond to the five levels of the cosmic hierarchy. Grey segments here show the cosmic mean baryon density at three different z . It varies from about 10^{-4} cm^{-3} at $z = 9$ to 10^{-7} cm^{-3} at $z = 0$, reflecting the factor $(1+z)^3$ that encodes the expansion of the Universe. The Universe itself, given by the Λ CDM cosmology with the fiducial set of parameters, is the outermost step of the density ladder, and the large-scale structures, including dark matter halos, form through the gravitational instability in the density field of the Universe.

In our definition of a virialized halo (§2.1), the virial density, virial radius, virial velocity, and virial temperature are given respectively by

$$\begin{aligned} \rho_v &\equiv \Delta_v \rho_{c,0} E^2(z) \\ &\approx \left[7.91 \times 10^{-3} \text{ M}_{\odot} \text{ pc}^{-3}\right] (1+z)_{10}^3; \end{aligned} \quad (\text{A2})$$

$$\begin{aligned} R_v &\equiv \left(\frac{2GM_v}{\Delta_v}\right)^{1/3} H^{-2/3}(z) \\ &\approx [4.57 \text{ kpc}] M_{v,9.5}^{1/3} (1+z)_{10}^{-1}; \end{aligned} \quad (\text{A3})$$

$$\begin{aligned} V_v &\equiv (\Delta_v/2)^{1/6} [GM_v H(z)]^{1/3} \\ &\approx [54.5 \text{ km s}^{-1}] M_{v,9.5}^{1/3} (1+z)_{10}^{1/2}; \end{aligned} \quad (\text{A4})$$

$$\begin{aligned} T_v &= 0.75 \times \frac{\mu m_p}{2k_B} V_v^2 \\ &\approx \left[1.62 \times 10^5 \text{ K}\right] M_{v,9.5}^{2/3} (1+z)_{10}, \end{aligned} \quad (\text{A5})$$

where the first line of each expression is the definition of the quantity; the approximation in the second line is valid at $z \gg 1$; $M_{v,9.5} \equiv M_v/(10^{9.5} \text{ M}_{\odot})$. The expression of virial temperature is given by e.g. Bryan & Norman (1998) and Wise & Abel (2007), with an additional factor 0.75 calibrated by Fernandez et al. (2014) using their hydro simulations, m_p is the proton mass, and we assume the mean molecular weight $\mu = 1.2$. We model the density profiles of halos using the NFW form:

$$\rho(r) \propto \frac{1}{x(1+x)^2}, \quad (\text{A6})$$

where $x \equiv r/R_s$, with R_s being a scale radius related to R_v by the concentration parameter, $c_v = R_v/R_s$ (Navarro et al. 1997). As shown

Density ladder in the structure hierarchy

e SNF nucleus → BH accretion disk

$$R_B = \frac{GM_{BH}}{\sigma_{eff}^2(R_B)}; \dot{M}_B = 4\pi R_B^2 \rho_g(R_B) \sigma_{eff}(R_B);$$

$$v_t = \psi_t V_K r^{1/2-\zeta_t}; v_A = \psi_A V_K r^{1/2-\zeta_A};$$

$$\Sigma_g = \frac{\dot{M}_g}{2\pi\sqrt{GM_{BH}R_B}} r^{2\zeta_t-3/2};$$

$$H_g \sim \psi_A R_B r^{3/2-\zeta_A}; \rho_g \sim \frac{\Sigma_g}{2H_g}$$

$$Q = 2\sqrt{2}\psi_t \sqrt{1 + M_A^{-2} \frac{M_{BH}\Omega_B}{\dot{M}_g}} r^{-3\zeta_t}.$$

d SGC → SNF nucleus

$$\rho_g = \frac{2\rho_{snf}}{c_g x_r (1 + c_g x_r)}; (p = 0, q = 1/2)$$

$$c_g = \frac{R_{SGC}}{R_{nuc}}; \frac{2m_g(c_g)}{2m_g(c_g)} = \frac{\rho_{snf}}{\rho_{SGC}};$$

$$M_g = M_{SGC} \frac{m_g(c_g)}{m_g(1)}; M_{nuc} = M_{SGC} \frac{m_g(1)}{m_g(c_g)};$$

$$V_c = V_{nuc} \sqrt{\frac{m_g(c_g x_r)}{m_g(1) c_g x_r}}.$$

c Halo → SGC

$$R_{SGC} = f_r R_v; M_{SGC} = f_g M_v; \rho_{SGC} = f_g f_r^{-3} \rho_v; V_{SGC} = \sqrt{GM_{SGC}/R_{SGC}};$$

$$\rho_g \sim \rho_{SGC} (r/R_{SGC})^{-2}. (T \approx 10^4 \text{ K})$$

b Universe → Halo

$$V_v = (\Delta_v/2)^{1/6} [GM_v H(z)]^{1/3}; R_v = (2GM_v/\Delta_v)^{1/3} H^{-1/3}(z); \rho_v = \Delta_v \rho_{c,0} E^2(z);$$

$$\rho_g \sim f_B \rho_v (r/R_v)^{-2}. (\Delta_v = 200; T \approx T_v = 0.75 \times \frac{\mu m_p}{2k_B} V_v^2)$$

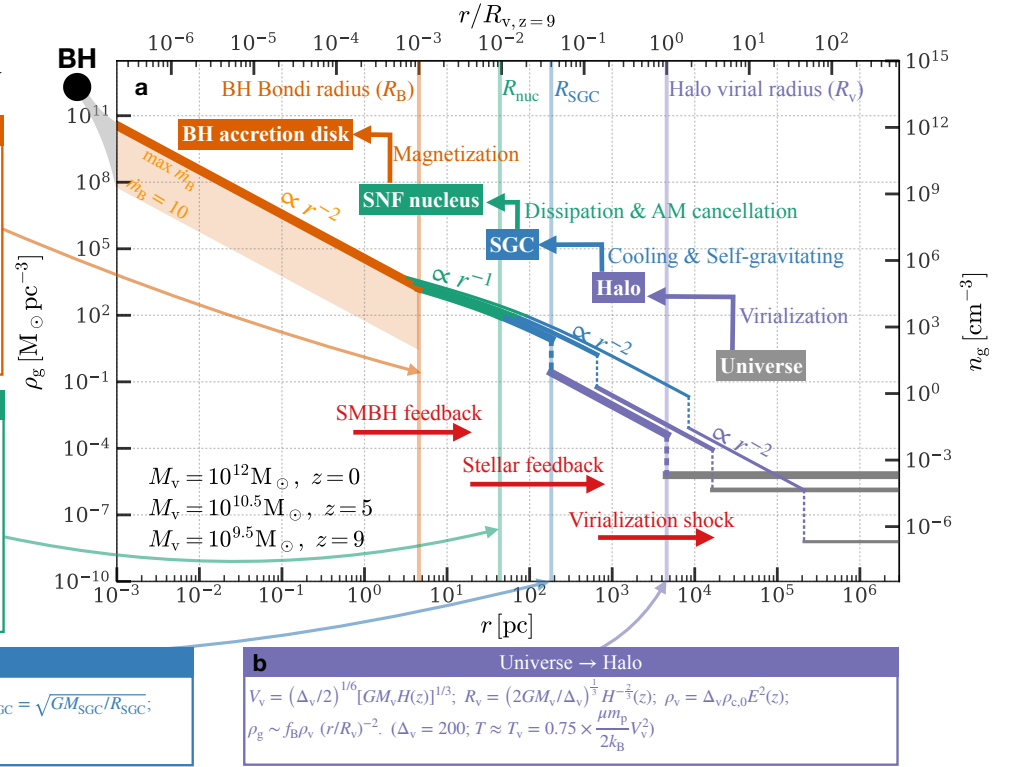


Figure A1. Density ladder of gas in the cosmic structure hierarchy. **a**, gas density profiles predicted by our model at three redshifts, $z = 0, 5$ and 9 (from thin to thick), in and around an MW-size galaxy and its progenitors under global disturbance. At each redshift, an SMBH with $M_{BH} = 10^6 M_\odot$ is put at the center of the galaxy. The profile exhibits a ladder-like structure, separated into **five segments**, each corresponding to a level in the structure hierarchy discussed in this paper. The innermost segment (**orange**) describes the middle-plane density of the accretion disk around the central BH, with a solid line indicating the profile expected in the beginning of the nuclear burst when the turbulence-modified Bondi accretion rate, $\dot{m}_{BH}$ (in the unit of Eddington rate), is the maximum (Eq. C30), and a shaded area covering the range of the profile during the burst until $\dot{m}_{BH}$ is reduced to 10. Each of the four **vertical lines** indicates the transition scale of the interface bridging two adjacent levels at $z = 9$. **b–d**, cards listing the sets of equations bridging the structure properties of any outer level to those of the adjacent inner level. Processes driving the formation of the structure hierarchy, and feedback resulting from the structure formation, are indicated by the texts. See Appendix C1 for details.

by Zhao et al. (2009), the value of c_v of a halo is closely related to the mass assembly history of the halo. Finally, we define the dynamical timescale of a halo as

$$t_v \equiv \frac{R_v}{V_v} = \frac{1}{10H(z)} \approx 81.9 \text{ Myr} (1+z)_{10}^{-3/2}. \quad (\text{A7})$$

As one can see, at a given redshift, ρ_v and t_v depend only on cosmology, while R_v , V_v and T_v depend on both cosmology and M_v . Thus, if we know the assembly history of a halo, i.e. its M_v as a function of z , we can follow its properties over its history. As shown by calibrations using numerical simulations (e.g. Dekel et al. 2013), the average accretion rate of a virialized halo of mass M_v at $z \gg 1$ can be approximated as

$$\dot{M}_v \approx \left[10.3 M_\odot \text{ yr}^{-1} \right] M_{v,9.5}^{1.14} (1+z)_{10}^{2.5}. \quad (\text{A8})$$

This can be integrated to obtain the average assembly histories and provides guidance for the growth of individual halos. For reference, Fig. 2 shows the median assembly histories of halos of given masses at $z = 0$, in terms of M_v (panel a), c_v (panel b), R_v (panel c), V_v and T_v (panel d), and $\dot{M}_v$ (panel e), as functions of z (or ρ_v). These curves are obtained by generating a large sample of halo merger trees with any given root mass at $z = 0$ by a Monte Carlo algorithm based on the extended Press-Schechter (EPS) formalism (see Appendix D1 for details), extracting the main branches from the trees, and computing the median of each property at each redshift. These curves show

how the properties of a halo evolve along its assembly history to the present day.

In addition to the assembly of dark matter halos, environments of halos can also affect the formation of BH seeds through radiation exposure (see Appendix B1.3) and metal contamination (see Appendix B2). Environmental variations are not well captured by the assembly of individual halos, as halo assembly and structure properties correlate only weakly with the environment (see fig. 9 of Chen et al. 2020; Gao & White 2007; Wang et al. 2011). To model environmental variations more accurately, we thus use halos and their merger trees from a cosmological N-body simulation, TNG100-1-Dark (see §2.1). The main progenitor of a subhalo is defined as the one with the most massive history among all progenitors (De Lucia & Blaizot 2007; Rodriguez-Gomez et al. 2015). The central subhalo of an FoF halo is defined as the one with the most massive history among all subhalos within the halo, and the main branch of an FoF halo corresponds to the main branch of its central subhalo. The specific run, TNG100-1-Dark, used here ensures a balance between statistical robustness and resolution in halo assembly histories.

In Fig. A1a, each purple segment shows the gas density at virialization of the halo, assuming an isothermal profile with temperature equal to T_v (Eq. A5; see also Croton et al. 2006; Guo et al. 2011; Lacey et al. 2016). The card in Fig. A1b summarizes the above equations governing halo properties. The formation of virialized, dark-matter-dominated halos causes a jump in the density to a level

about $\Delta_v \sim 200$ times the cosmic mean density. The gas density associated with the dark matter halo is also lifted to be about Δ_v times the cosmic mean baryon density, $\rho_b(z)$ (Eq. A1). At such density, the cooling of the gas is significantly enhanced due to the n^2 dependence of the two-body cooling processes, bringing the time needed for the condensation of the gas well below the Hubble timescale, $H^{-1}(z)$ (see e.g. the cooling diagram in fig. 2 of Paper-I). The formation of halos thus forms a necessary step, and sets the boundary condition for the subsequent evolution of gas, star formation and BH growth in galaxies, as we will discuss in the following.

APPENDIX B: BREEDING BLACK-HOLE SEEDS

B1 Producing black-hole seeds in mini-halos

B1.1 Formation of self-gravitating clouds

If the angular-momentum support is negligible, as is the case when the fraction of the cooled gas in a halo is not much lower than the universal baryon fraction, the collapse of gas towards the halo center is expected to lead to the formation of an SGC supported by supersonic turbulence, as shown in Paper-I. For a halo with a gas fraction $f_g \equiv M_{\text{SGC}}/M_v$, the gas becomes self-gravitating after contracting by a factor $\sim 1/f_g$, and the SGC eventually becomes virialized at a radius of $f_r R_v$, where f_r is a coefficient to be determined. According to Paper-III (see their §2.2), the radius, density, surface density and circular velocity of such an SGC can be written as

$$R_{\text{SGC}} = f_r R_v \approx [183 \text{ pc}] f_{r,0.04} M_{v,9.5}^{1/3} (1+z)_{10}^{-1}; \quad (\text{B1})$$

$$n_{\text{SGC}} = \frac{f_g}{\mu m_p f_r^3} \rho_v \approx [656 \text{ cm}^{-3}] f_{g,0.157} f_{r,0.04}^{-3} (1+z)_{10}^3; \quad (\text{B2})$$

$$\Sigma_{\text{SGC}} \sim \frac{M_{\text{SGC}}}{R_{\text{SGC}}^2} \approx [1.49 \times 10^4 \text{ M}_\odot \text{ pc}^{-2}] f_{g,0.157} f_{r,0.04}^{-2} M_{v,9.5}^{1/3} (1+z)_{10}^2; \quad (\text{B3})$$

$$V_{\text{SGC}} = \sqrt{\frac{GM_{\text{SGC}}}{R_{\text{SGC}}}} \approx [108 \text{ km s}^{-1}] f_{g,0.157}^{1/2} f_{r,0.04}^{-1/2} M_{v,9.5}^{1/3} (1+z)_{10}^{1/2}, \quad (\text{B4})$$

respectively, and the free-fall timescale ($t_{\text{ff,SGC}}$) and dynamical timescale ($t_{\text{dyn,SGC}}$) are

$$t_{\text{ff,SGC}} = \frac{\pi}{\sqrt{8}} t_{\text{dyn,SGC}} = \sqrt{\frac{3\pi}{32G\rho_{\text{SGC}}}} = [1.48 \text{ Myr}] n_{\text{SGC},3}^{-1/2} = [1.84 \text{ Myr}] f_{g,0.157}^{-1/2} f_{r,0.04}^{3/2} (1+z)_{10}^{-3/2}, \quad (\text{B5})$$

where the SGC density $\rho_{\text{SGC}} = \mu m_p n_{\text{SGC}}$, and we write $n_{\text{SGC}} = 10^3 \text{ cm}^{-3} n_{\text{SGC},3}$, $f_g = 0.157 \times f_{g,0.157}$ and $f_r = 0.04 \times f_{r,0.04}$.

The formation of the SGC produces the jump in density from the halo density produced by halo formation. Blue segments in Fig. A1a show the resulting density profiles at this step, where we have assumed an isothermal density profile (Henriques et al. 2020; Hopkins et al. 2024a) with a temperature, $T \approx 10^4 \text{ K}$, at which a cooling barrier is present (see e.g. the cooling functions in fig. 4 of Maio et al. 2007 and fig. 1 of Smith et al. 2008). The blue card in Fig. A1c summarizes the above equations relevant to this step. Depending on its internal properties and environment, an SGC can fragment to form sub-clouds which, in turn, can collapse further to form compact

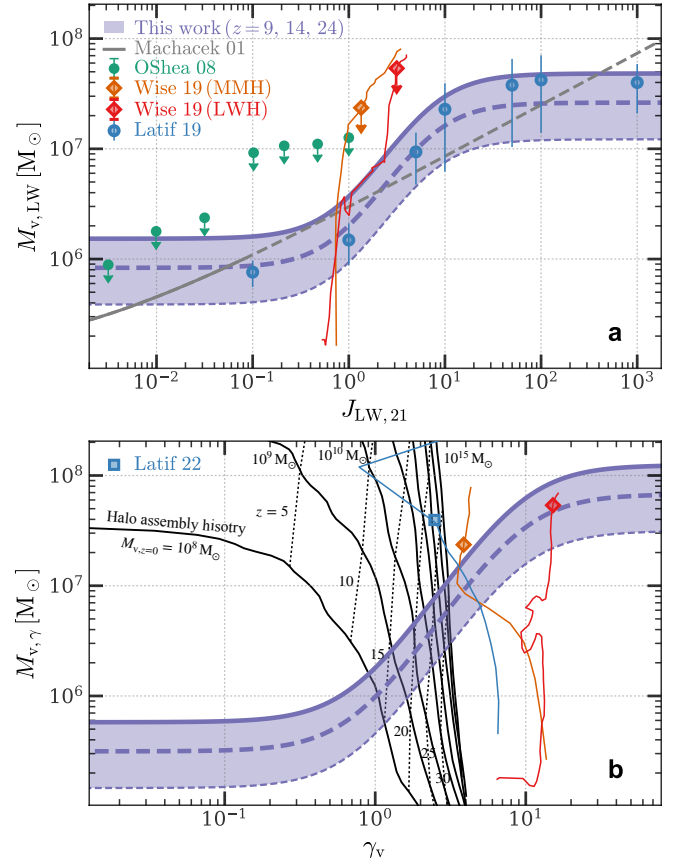


Figure B1. Delay of gas collapse in mini-halos. Here we show the threshold mass of halo expected for the gas to collapse, under **a**, different strength of UV background ($J_{\text{LW},21}$, measured in the LW band) and **b**, different specific rate of halo accretion (γ_v). The former measures the reduction of gas cooling due to the reduction of H_2 production, while the latter measures the dynamical heating due to fast accretion. In both panels, solid, dashed and dotted **purple curves** show our parameterization of the threshold mass at $z = 9, 14$ and 24 , respectively. **Black curves** in **b** show the assembly histories of halos in the M_v - γ_v plane, solid for halos with given masses at $z = 0$ (from 10^9 to $10^{15} \text{ M}_\odot$, each separated by 1 dex), and dashed for contours at given z (from 5 to 30), obtained by averaging a large set of Monte Carlo merger trees (see Appendix D1). The SGC in a pristine halo with dynamical heating thus collapses and forms its first generation of stars once the trajectory of the halo passes the purple band. **Colored markers** show simulation results obtained by O'Shea & Norman (2008), Wise et al. (2019), Latif & Khochfar (2019) and Latif et al. (2022a). The halos used by O'Shea & Norman (2008) and Wise et al. (2019) are extremely massive ones, dominated by the effect of dynamical heating, and thus we show their results as upper limits in **a**. **Colored curves** associated with the markers for Wise et al. (2019) and Latif et al. (2022a) are the histories of LW background (**a**) and halo assembly (**b**). Grey curve in **a** shows the fitting result, $M_{v,\text{cool}}/\text{M}_\odot = 1.25 \times 10^5 + 2.86 \times 10^6 J_{\text{LW},21}^{0.47}$, obtained by Machacek et al. (2001) using their simulations with $J_{\text{LW},21} \lesssim 0.1$, with dashed segment indicating the extrapolation. See Appendix B1 for details.

objects of different masses, such as dense gas cores and stars. Of particular interest are the formation of massive dense cores and the first massive stars at high z , and the possibility for them to produce BH seeds. In the rest of this subsection, we describe in more detail the formation of SGCs in low-mass halos at high z and discuss potential channels of BH seed production in SGCs.

B1.2 Gas collapse in mini-halos at high z

As discussed in §2.3, the collapse of pristine halo gas can occur when a low-mass mini-halo reaches $T_v \approx T_{H_2} = 10^3$ K. A more quantitative analysis by Yoshida et al. (2003) and Glover (2013), based on the production of H_2 and the required fraction of H_2 for halo gas to cool within a dynamical (free-fall) timescale ($t_{\text{cool}} \sim t_{\text{dyn}} \sim t_{\text{ff}}$) gives a similar value for the critical temperature. Using T_v given by Eq. (A5), we can get the critical halo mass for the gas to contract (Eq. 1). According to Eqs. (B1)–(B5), the mass, radius, density and free-fall timescale are derived (see the text below Eq. 1). The circular velocity of the SGC at $1+z=25$ is $V_{\text{SGC}} = 8.49 \text{ km s}^{-1}$, a small value that implies gas depletion after seeding. Purple dotted lines in Fig. 2 show M_{v,H_2} as a function of redshift (panel a) assuming $T_{H_2,3} = 1$, and the corresponding R_v (panel c), V_v and T_v (panel d) obtained from M_{v,H_2} using Eqs. (A3), (A4) and (A5). The temperature, $T = T_{H_2}$, corresponds to a thermal sound speed of $c_s \approx 3.4 \text{ km s}^{-1}$. The implied Jeans mass of the SGC,

$$M_{J,H_2} = \frac{\pi^{5/2}}{6} \frac{c_s^3}{\sqrt{G^3 \rho_{\text{SGC}}}} = \left[2.30 \times 10^4 M_{\odot} \right] f_{g,0.157}^{-1/2} f_{r,0.04}^{3/2} (1+z)^{-3/2} T_{H_2,3}^{3/2}, \quad (\text{B6})$$

sets the lower mass limit for the gas to collapse and fragment. The fact that this Jeans mass is comparable to the mass of the SGC and that the cooling by H_2 is not rapid ($t_{\text{cool}} \sim t_{\text{dyn}}$) implies that the contraction of the SGC proceeds globally without leading to fragmentation until other cooling channels are open. For gas with a primordial composition, where cooling from metals and dust grains is insignificant, the whole SGC cools and contracts gradually to a high density before fragmentation and star formation can occur.

The above discussion only takes into account radiative cooling, ignoring potential heating sources that may be present. In the presence of heating, the contraction of the halo gas to form an SGC, and the fragmentation and collapse of the SGC to form stars, are delayed so that the mass of the Jeans core is increased. The central part of the Jeans core may then collapse directly to form a BH with mass greater than a few $100 M_{\odot}$, producing a ‘heavy-weight’ seed to grow into an SMBH. Two sources, one from UV radiation in the environment and the other from the energy of the turbulence associated with the fast-accretion regime of the host halo, are particularly relevant to our problem and their effects are described below.

B1.3 Delay of collapse by UV radiation

For a given halo, UV radiation produced by young stars in nearby galaxies can provide external feedback that suppresses gas cooling and collapse in the halo. At high z when the Universe is still largely neutral, neutral hydrogen in the IGM can effectively absorb UV photons above 13.6 eV, the Lyman limit of H, but the IGM remains optically thin for photons below that limit. This leads to significant attenuation only in resonance frequencies, producing a saw-toothed spectrum (see e.g. §2.1 of Haiman et al. 2000). Such photons can dissociate H_2 via the Solomon process, $H_2 + \gamma \rightarrow H_2^* \rightarrow H + H$, where H_2^* is any of the resonance states above 11.2 eV (see e.g. §2 of Machacek et al. 2001). As H_2 is needed to cool the gas, UV photons within the window of 11.2–13.6 eV, usually referred to as the Lyman-Werner (LW) radiation, can suppress gas cooling and the formation of stars and BHs in the halo.

The threshold mass for a H_2 -cooling mini-halo to collapse is expected to increase with increasing J_{LW} . If the radiation is strong enough ($J_{\text{LW},21} \gg 1$), the collapse of the halo gas to form an SGC

can be delayed until atomic cooling kicks in, namely when $T_v \approx T_H \approx 10^4$ K. This gives another critical halo mass, $M_{v,H}$, for the gas under strong LW radiation to collapse (Eq. 3), which is about two orders of magnitude higher than the H_2 -cooling mass. According to Eqs. (B1)–(B5), the SGC produced has mass $M_{\text{SGC}} = f_b M_{v,H} = 1.92 \times 10^6 M_{\odot}$, radius $R_{\text{SGC}} = 11.5 \text{ pc}$, circular velocity $V_{\text{SGC}} = 26.9 \text{ km s}^{-1}$, and density and free-fall timescale the same as those of the H_2 -cooling SGC at a given redshift (see the text below Eq. 1). Purple dashed lines in Fig. 2a, c and d show $M_{v,H}$ and the corresponding R_v and V_v (T_v), respectively, as a function of redshift. They are parallel to but higher than the corresponding H_2 -cooling lines. More generally, we can parameterize the halo mass for effective cooling to drive the formation of SGC as a function of J_{LW} as

$$M_{v,\text{LW}} = \frac{M_{v,H_2} + f_J M_{v,H}}{1 + f_J}, \quad (\text{B7})$$

where f_J depends on the intensity of the LW radiation irradiated on the halo,

$$f_J = \left(\frac{J_{\text{LW},21}}{J_{\text{crit},21}} \right)^{\beta_J}. \quad (\text{B8})$$

We set $\beta_J = 1.5$ and $J_{\text{crit},21} = 7.5$ by default to match the results from zoom-in hydrodynamic simulations (see below). So modeled, $M_{v,\text{LW}}$ smoothly transits from the H_2 -cooling limit to the H-cooling limit at any given redshift. Fig. B1a shows our default results of $M_{v,\text{LW}}$ as a function of J_{LW} at different redshifts, in comparison to results obtained by other studies. Machacek et al. (2001) performed zoom-in hydrodynamic simulations for halos embedded in different levels of UV background with $J_{\text{LW},21} \lesssim 0.1$. They found that the threshold halo mass ($M_{v,\text{LW}}$) required for the gas to collapse increases with J_{LW} , and fitted the $M_{v,\text{LW}}-J_{\text{LW}}$ relation with a power-law function (grey curve in Fig. B1a, with dashed segment indicating the extrapolation). Latif & Khochfar (2019) performed zoom-in simulations for 6 halos, selected at $z \approx 23$ –26 with M_v a few times 10^5 – $10^6 M_{\odot}$, imposing different levels of UV radiation with $J_{\text{LW},21} = 0.1$ –1000. The collapse of the gas is significantly delayed at high $J_{\text{LW},21}$. The identified collapsing threshold $M_{v,\text{LW}}$ as a function of $J_{\text{LW},21}$ is shown by blue markers in Fig. B1a, with error bars indicating the $1\text{-}\sigma$ dispersion among halos. The threshold rises sharply at $J_{\text{LW},21} \gtrsim 1$, consistent with the results of Machacek et al. (2001), and saturates to a constant value at $J_{\text{LW},21} \gtrsim 10$. Given the wide range of J_{LW} covered by this set of simulations, we calibrate our parameterization in Eq. (B8) with it, yielding the default parameters of β_J and $J_{\text{crit},21}$. For comparison, green markers show the results obtained by O’Shea & Norman (2008) using a set of zoom-in simulations for one halo with different levels of J_{LW} ; orange and red markers show the results obtained by Wise et al. (2019) using two zoom-in simulations for two halos, respectively, with two curves passing the markers showing the histories of LW radiation produced mainly by nearby star-forming galaxies impacting the halos. Their halos are selected to be extremely massive ones in cosmological volumes, thus experiencing fast accretion and expected to have strong dynamical heating (see Appendix B1.4). Thus, we show their results for the collapsing threshold as upper limits in Fig. B1a.

Our modeling of BH-seed breeding takes into account effects of UV radiation in the LW band, as described above. Other effects of the UV radiation, such as the detachment of H^- by photons of $> 0.76 \text{ eV}$, can also affect the formation of H_2 . Unfortunately, a reliable model of such effects requires detailed modeling of the stellar spectral energy distribution (SED), which depends on the star formation and chemical enrichment history of the emitting galaxy, and radiative transfer, which depends on the properties of the interstellar medium (ISM), circumgalactic medium (CGM) and IGM, both beyond the scope of

this paper. A less sophisticated model of the UV radiation, as adopted in the hydrodynamic simulations of [Latif & Khochfar \(2019\)](#), is to assume that the emitted radiation has a black-body SED with some characteristic temperature in the range of 10^4 – 10^5 K. Our calibration using the results of [Latif & Khochfar \(2019\)](#) thus partially captures the H^- photodetachment. [Agarwal et al. \(2019\)](#) presented a more detailed model of the UV radiation by post-processing hydrodynamic simulations, synthesizing stellar SEDs, and adopting a critical curve for the DCBH in the 2-D plane of H_2 photodissociation and H^- photodetachment. Their results suggest that the net effect of the UV radiation on the delay of the SGC collapse is tightly correlated with the star formation rate (SFR) of emitting galaxies. This motivates us to use the SFR (or young stars) as a proxy of J_{LW} in the threshold variable f_j (Eq. B8) as an effective way to account for the net effect of the UV radiation. Since our model can predict star formation histories of individual galaxies in a cosmological volume, the information needed for the modeling of UV radiation is self-consistently included in the model. In Appendix D5, we will present the detailed steps for such a model. The UV radiation produced by young stars, and its effects on the delay of SGC formation, are the earliest ‘radiation feedback’ of galaxy formation to its environment.

B1.4 Delay of collapse by turbulence associated with fast accretion

An SGC is expected to be a turbulence supported system ([Paper-I](#)). Depending on the details of the gas collapse and the turbulence cascading, the partition between the kinetic and thermal energy can be different for different halos. [Wise et al. \(2019\)](#) reported results of two simulated halos at $z \approx 15$ as potential sites to form supermassive stars (SMSs). In each of the two cases, the external LW radiation is moderate, while the fast halo accretion heats the SGC and leads to the formation of a hot, thermally supported core with mass $\gtrsim 10^3 M_\odot$ (see their Extended Data fig. 3). [Latif et al. \(2022a\)](#) reported results of one primordial halo at $z \approx 25$ as potential site of SMS formation. The halo is not exposed to any external LW radiation, but its fast accretion leads to supersonic turbulence that supports the entire SGC. The energy of the turbulence in these examples all comes from gravity. We thus refer to this channel of energy injection by the fast accretion of halos collectively as ‘dynamical heating’, regardless of whether the injected energy is stored in kinetic or thermal form.

As briefly mentioned in §2.3, we first consider an extreme case where energy injection is so efficient that radiative cooling is inefficient to dissipate it. If the cooling timescale t_{cool} becomes comparable or longer than $V_v/\dot{V}_v$, the timescale for the potential energy of a halo to be totally flushed by infalling material, the gravitational energy associated with the mass accretion of the halo can generate a significant impact on the SGC over the cooling timescale. Thus, gas in a halo can collapse rapidly only when $t_{\text{cool}} \lesssim V_v/\dot{V}_v$. Equating the two timescales, we obtain a critical condition for the turbulence to affect the state of the SGC: $V_v/\dot{V}_v \approx t_{\text{cool}}$. Noticing that at the collapse of SGC, $t_{\text{cool}} \sim t_v$, and using Eq. (A7) for t_v , we have

$$\gamma_v \equiv \frac{\dot{V}_v}{V_v H(z)} = -\frac{d \ln V_v}{d \ln(1+z)} \approx 10. \quad (\text{B9})$$

To incorporate the additional support to SGC by turbulence, we replace the sound speed in the Jeans mass (Eq. B6) with an effective velocity dispersion, $\sigma_{\text{eff}} = (c_s^2 + v_t^2)^{1/2}$. The exact value of the turbulent velocity dispersion v_t is unknown. As a conservative estimate, we take the value, $v_t \approx 20 \text{ km s}^{-1}$, obtained by [Latif et al. \(2022a\)](#) from their simulation of a fast-accreting halo with H_2 cooling unable to dissipate the turbulence. This value of v_t , together with c_s given by $T = T_{H_2}$ and the SGC density given by Eq. (B2), leads to the

following Jeans mass:

$$M_{J,\text{max}} = \left[4.94 \times 10^6 M_\odot \right] f_{g,0.157}^{-1/2} f_{r,0.04}^{3/2} (1+z)^{-3/2}. \quad (\text{B10})$$

The SGC thus starts to collapse rapidly only when it reaches this mass scale. The corresponding radius and circular velocity are $R_{\text{SGC}} = 15.7 \text{ pc}$ and $V_{\text{SGC}} = 36.8 \text{ km s}^{-1}$, respectively. Purple solid lines in Fig. 2a, c and d show $M_v = M_{J,\text{max}}/f_b$ and the corresponding R_v and $V_v(T_v)$, respectively, as a function of redshift. They are parallel to but higher than the corresponding H_2 -cooling lines, and close to the corresponding H -cooling lines.

More generally, a halo can be somewhere between the two extremes, the one without turbulence ($v_t \ll c_s$) and the other with strong turbulence ($v_t \gg c_s$). The Jeans mass is thus between those given by Eq. (B6) and Eq. (B10), with a transition at $\gamma_v \approx 10$. We parameterize the halo mass at which the thermal and turbulent motion is no longer able to support the SGC against gravitational collapse as a function of γ_v :

$$M_{v,\gamma} = f_b^{-1} \frac{M_{J,H_2} + f_\gamma M_{J,\text{max}}}{1 + f_\gamma}, \quad (\text{B11})$$

where

$$f_\gamma = \left(\frac{\gamma_v}{\gamma_{\text{crit}}} \right)^{\beta_\gamma}, \quad (\text{B12})$$

and we adopt $\gamma_{\text{crit}} = 10$. The exact value for β_γ is not known, and we adopt $\beta_\gamma = 2$ as a demonstration of the effect of dynamical heating. Future calibrations using detailed hydrodynamic simulations are needed to provide a more reliable estimate of γ_{crit} and β_γ , and to account for the synergy of dynamical heating with other heating processes (e.g. the UV radiation discussed in Appendix B1.3).

Purple curves in Fig. B1b show our model of the collapsing threshold, $M_{v,\gamma}$, at three different redshifts, as a function of halo specific growth rate, γ_v . The threshold from $z = 24$ to 9 covers a narrow band of ≈ 0.3 dex due to the dependence of the Jeans mass on SGC density (thus redshift). The delay of SGC collapse by dynamical heating can lead up to ≈ 2 dex increase in M_v when $\gamma_v \gtrsim 10$. Since halos with higher M_v accrete faster than those with lower M_v at a given z , formation of BH seeds in the progenitors of massive halos at present day are most likely affected significantly by dynamical heating. To see this, we take the EPS-based halo assembly histories used to produce Fig. 2, compute the specific growth rate γ_v , and average over halos with given $M_{v,z=0}$. Solid black curves in Fig. B1b show the M_v - γ_v trajectories of halos with different $M_{v,z=0}$, while each dashed curve links a snapshot of these trajectories at a given redshift. Halos with $M_{v,z=0} \gtrsim 10^{12} M_\odot$ have very similar M_v and γ_v at $z \gtrsim 15$, consistent with the assembly histories shown in Fig. 2. γ_v exhibits slow reduction with redshift for all halos at high z . The significant reduction of γ_v occurs earlier for halos with lower mass, consistent with the findings in [Paper-I](#) (see their fig. B1) that z_f shows negative correlation with $M_{v,z=0}$.

The expected epoch of collapse, as well as host-halo properties at this epoch, of an SGC being dynamically heated by fast accretion (without taking into account other heating sources) can be identified by the intersection point of the M_v - γ_v trajectory with the purple band in Fig. B1b. Since the slopes of equal-time contours (dashed black) are higher than that of the purple band in the M_v - γ_v plane, SGCs in massive halos are expected to collapse earlier than those in less massive halos. Note that this is only a conclusion averaged over ensembles of halos, and halo-to-halo variations can be significant. Halos with $M_{v,z=0} = 10^{11} M_\odot$ (hosting dwarf galaxies with $M_* \approx 10^9 M_\odot$ at $z = 0$) reach the collapsing threshold at $z \approx 25$ when M_v is a few times $10^6 M_\odot$ while those with $M_{v,z=0} \gtrsim 10^{13} M_\odot$ (hosting BCGs at

$z = 0$) are expected to collapse at $z \approx 30$ with a comparable M_v . The values of M_v at collapse for these halos are thus ≈ 1 dex higher than that of H_2 -cooling halos (M_{v,H_2} , Eq. 1), implying that much more baryon gas is available for the formation of stars and BHs in the SGCs. Meanwhile, such a delay of collapse also gives the SGCs more chance to be irradiated by the UV photons from nearby star-forming galaxies, causing further delay, as we will discuss and model in Appendix B1.5.

For comparison, we show by colored curves in Fig. B1b the assembly histories of three pristine halos whose baryonic processes were resolved by hydrodynamic simulations. Two of them are the same halos as those in panel a, studied by Wise et al. (2019), and one by Latif et al. (2022a). All of them are progenitors of extremely massive halos, and were predicted to seed DCBHs and then evolve to quasars at $z \approx 6$. γ_v of these halos are extremely large, reflecting an accretion faster than that expected from progenitors of halos with $M_{v,z=0} \approx 10^{15} M_\odot$. Dynamical heating is thus expected to be very strong in these halos, causing significant delay of the SGC collapse. These are indeed seen in these simulations, as shown by the colored markers in Fig. B1b that denote the collapses they identified, close to the prediction of our model.

B1.5 Consequences of the delayed collapse

As shown in Fig. B1, both UV radiation and fast accretion can delay the collapse of SGCs, as well as the formation of the first generation of stars and the breeding of BH seeds. The effects are described by the two halo mass thresholds, $M_{v,LW}$ and $M_{v,\gamma}$, respectively, and the higher one is used as the final threshold for the collapse of each SGC (Eq. 5). If either J_{LW} or γ_v is sufficiently large, the collapse is delayed to the regime of atomic cooling (Eq. 3). Consequently, the total available baryon mass, $M_{g,th} \approx f_b M_{v,th}$, and the halo virial velocity, V_v , are much larger than those for a H_2 -cooling halo.

The purple band in Fig. 2a summarizes the regime in the z - M_v plane where the first collapse of an SGC in a halo can take place. The dotted, dashed and solid curves represent M_{v,H_2} (the mass for the H_2 -cooling to be effective; Eq. 1), $M_{v,H}$ (the mass for the H -cooling to be effective; Eq. 3), and $f_b^{-1} M_{J,max}$ (the halo mass for the SGC to be Jeans-unstable at $T = 10^3$ K and $v_t = 20$ km s $^{-1}$; Eq. B10), respectively. These masses are plotted on top of the median assembly histories of halos of different masses at $z = 0$ (black curves). Pop-III stars are expected to form in a primordial halo when the halo mass, $M_v(z)$, crosses M_{v,H_2} (dotted line) and enters the Pop-III band limited by J_{LW} (dashed line) or γ_v (solid line). Note that the region for Pop-III stars in the z - M_v plane is almost fully (partly) covered by that marked Pop-II (Pop-I) GC, the region in which Pop-II (Pop-I) globular clusters are expected to form (see Paper-III). Thus, only halos that have not been self-enriched by previous generations of stars or externally enriched by star formation in nearby galaxies can host Pop-III stars, while the others are likely to host globular clusters. In the progenitors of halos with $M_{v,z=0} \approx 10^9$ – $10^{10} M_\odot$, Pop-III stars can form at $z \lesssim 10$ through the first collapse of the SGC provided that the IGM around these halos has not been enriched and the UV radiation and/or dynamical heating are sufficiently strong to delay the collapse to that low redshift. These Pop-III stars of late-formation may thus be observable in JWST surveys, such as those conducted recently by Wang et al. (2024b), Hsiao et al. (2025), Suess et al. (2024), Naidu et al. (2024) and Fujimoto et al. (2025b). The enrichment history of both the ISM and IGM turns out to be critical in disentangling the two populations of stars in the z - M_v plane, as to be discussed in Appendix D6. For reference, purple bands in Fig. 2c and d show the first-collapse regime in the z - R_v and z - V_v (z - T_v) planes, respectively.

The free-fall timescale of an SGC at its first collapse is about 1 Myr (see the texts below Eq. 3), independent of halo mass (Eq. B5). This timescale is shorter than the ‘supernova-free’ timescale, $t_{snf} \approx 1$ Myr (Chen et al. 2025a; Dekel et al. 2023). Thus, supernova (SN) feedback is expected to be unimportant for the first collapse of the SGC. This is particularly true for the Jeans core that has $n \gtrsim 10^6$ cm $^{-3}$, corresponding to an even shorter t_{ff} of $\lesssim 0.05$ Myr. Without other types of feedback, the upper limit of the gas accretion rate onto the central star can then be estimated as

$$\dot{M}_g \leq \frac{f_b M_{v,th}}{t_{ff,SGC}} \leq \frac{M_{J,max}}{t_{ff,SGC}} \approx 10.7 M_\odot \text{yr}^{-1}, \quad (\text{B13})$$

independent of redshift due to the cancellation of the redshift dependence in $M_{J,max}$ (Eq. B10) and $t_{ff,SGC}$ (Eq. B5). As suggested by numerical simulations of the formation of supermassive stars (e.g. Hosokawa et al. 2012, 2013), a high accretion rate at $\gtrsim 10^{-2} M_\odot \text{yr}^{-1}$ can lead to very different evolution in the stellar structure than a lower rate. After the triggering of the Kelvin-Helmholtz (KH) contraction in the core, the radiation due to the release of gravitational energy causes the outer envelope to expand, leading to the formation of a stellar object with inhomogeneous structure and non-monotonic density profile. Hydrogen burning can be ignited in the core, but provides only minor contribution to the total energy flux. At a stellar mass $M_* \sim 100 M_\odot$, the outer radius is related to M_* as

$$R_* \approx 2.6 \times 10^4 R_\odot \left(\frac{M_*}{100 M_\odot} \right)^{1/2} = 12 \text{ au} \left(\frac{M_*}{100 M_\odot} \right)^{1/2}, \quad (\text{B14})$$

independent of the accretion rate. The stellar object, with its effective temperature locked at a low value of $T_{eff} \approx 5000$ K due to strong temperature dependence of the H^- absorption, does not evolve to the main sequence. The UV luminosity at such a surface temperature is low, and the radiative feedback ineffective. The stellar object can then continue to accrete and grow significantly in mass, up to $\sim 10^3$ – $10^5 M_\odot$, until the density of the stellar envelope drops to a level that significantly reduces the opacity of H^- . The upper limit of $\dot{M}_g$ given by Eq. (B13) for the central stellar object in an atomic-cooling halo is thus sufficient for it to grow into a supermassive star within a timescale less than 1 Myr, before an SN can detonate. Idealized simulations (Choi et al. 2013) and cosmological zoom-in simulations (Latif et al. 2013; Wise et al. 2019; Latif et al. 2021, 2022a) have shown that such high accretion rates are indeed achievable in a real physical setting and in the cosmological context. Such an SMS is expected to collapse when the gas pressure is no longer able to counterbalance its weight, leading to the formation of an intermediate-mass, direct-collapse black hole (DCBH), with a BH mass comparable to the SMS.

The above estimate ignores potential suppression of the accretion rate by feedback, angular momentum support and fragmentation of the gas to form low-mass stars. It thus only represents an upper limit of the mass that a DCBH can reach in a mini-halo. In the following, we consider a more realistic model that takes into account the effects of radiative feedback from the central star, and use it to derive the mass spectra of Pop-III stars and BH seeds expected to form during the first collapse of an SGC in a mini-halo.

B2 Radiation-regulated mass function of Pop-III stars and the mass spectrum of black-hole seeds

The description presented above provides the avenue to model the mass of individual Pop-III stars expected to form in halos where the gas state is regulated by different cooling and heating processes. In this subsection, we first use the results to construct a model for the initial mass function (IMF) of Pop-III stars. We then combine

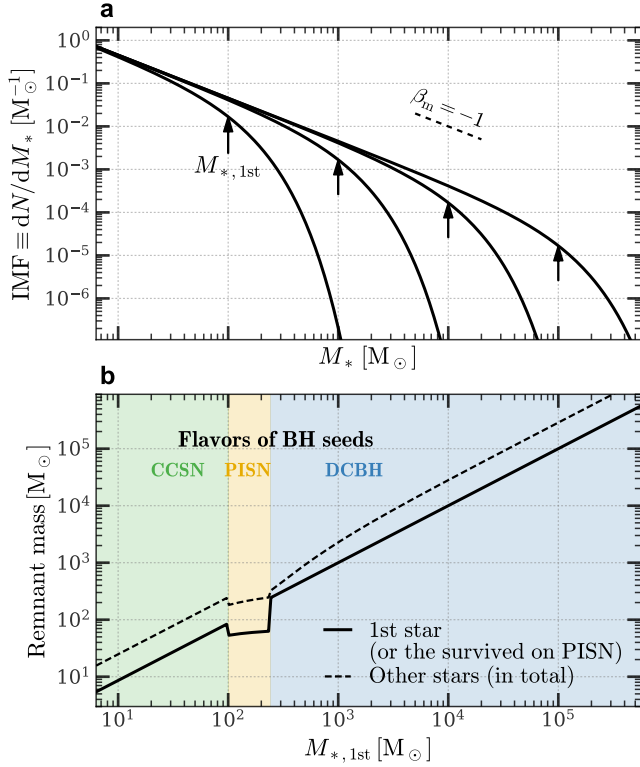


Figure B2. Masses of Pop-III stars and BH remnants. **a**, initial mass function (IMF) of stars in a Pop-III star cluster, each with a given mass of the dominant (1st) star ($M_{*,1st}$, indicated by the arrow). **b**, remnant mass of stars in a Pop-III star cluster as a function of $M_{*,1st}$ of the cluster, **solid** for the dominant star (i.e. the mass of BH seed) and **dashed** for the other stars of the cluster in total. The evolution (CCSN, PISN or DCBH) of the dominant star to end its life is determined by $M_{*,1st}$. If the dominant star triggers PISN, we take the remnant mass of the most massive star surviving PISN as a replacement. See Appendix B2 for the details.

the results with assumptions of stellar evolution to obtain the mass spectrum of BH seeds.

As seen above, the slow cooling of primordial gas has important implications to the star formation process. For the radiative cooling to effectively dissipate the energy generated by the gravitational collapse, the SGC has to contract to achieve a sufficiently high density. Gas fragmentation and star formation, which are possible only when the SGC can cool effectively, are thus delayed until the gas density is sufficiently high (see e.g. fig. 9 of Omukai et al. 2008). The situation is similar to that in the formation of Pop-II GCs in metal-poor ($Z \sim 10^{-2} Z_\odot$) sub-clouds with $n \gtrsim n_{\text{snf}} = 10^{3.5} \text{ cm}^{-3}$ (Chen et al. 2025a), except the more extreme condition produced by the fact that the temperature does not drop significantly even when the density reaches $n \gtrsim n_{\text{core}} = 10^6 \text{ cm}^{-3}$ in a Pop-III gas (see e.g. fig. 1 of Latif et al. 2022b). The dense environment favors the continuous accretion by the dominating central SMS, which suppresses the growth of off-center stars (Bonnell et al. 2001; Chon & Omukai 2020). Merger into the SMS is frequent due to its central position and large size (e.g. Stacy et al. 2016; Chon & Omukai 2020), which can also increase its mass. The IMF of the stars formed in the SGC is thus expected to be significantly affected by these processes and to evolve to be top-heavy. If additional ‘heating’ sources, such as the LW radiation and the turbulence generated by the fast accretion, are present, the formation of small-scale gas clumps can be further suppressed, making the IMF even more top-heavy.

As discussed in Appendix B1.5, although a gas cloud hosting Pop-III stars is expected to be free of SN feedback, radiative feedback from young Pop-III stars can still be present. Recent hydrodynamic simulations (e.g. Latif et al. 2021, 2022b) incorporating this source of feedback have found that it can significantly regulate the accretion rate of gas into the core of the SGC, and thus limit the total stellar mass of the star cluster formed there. The gas within the SGC was found to be highly dynamical, where gas inflow and feedback-driven outflow co-exist with comparable rates (see e.g. fig. 6 of Latif et al. 2021). Such strong feedback can ‘quench’ the star formation continuously and lead to an upper limit on the total amounts of stars that can form (see e.g. §3 of Latif et al. 2022b). Following the same strategy as that adopted by Paper-I in modeling the AGN-feedback effects on SMBH formation, and by Paper-III in modeling the stellar-feedback effects on globular-cluster formation, we relate the stellar mass of a Pop-III star cluster to the properties of its host SGC by a balance between the feedback energy/momentum and the SGC binding energy/force, with model parameters appropriate for the Pop-III regime. The top-heavy IMF, together with the fact that $L \propto M_*^2$ for massive stars concerned here, implies that the stellar feedback is dominated by the most massive star, referred to as the first (1st) star or dominant star in the following.

As one extreme case, we first consider a halo at the H_2 -cooling threshold, $M_{\text{v,H}_2}$ (Eq. 3). The 1st star in such a halo cannot grow too massive, and can approach the main sequence before the cloud is destroyed (see e.g. fig. 6 of Stacy et al. 2016). The zero-age main sequence (ZAMS) luminosity is

$$L_* = L_{\text{ZAMS}} \approx 1.4 \times 10^4 L_\odot \left(\frac{M_*}{10 M_\odot} \right)^2 \quad (\text{B15})$$

(e.g. Schaller et al. 1992; Hosokawa et al. 2010). The upper limit of the stellar mass, above which the gas accretion is stopped, can be estimated as

$$\frac{\eta_r L_*}{c} = \frac{G M_{\text{SGC}}^2}{R_{\text{SGC}}^2}, \quad (\text{B16})$$

where the free-parameter η_r stands for the coupling efficiency of the stellar radiation to the surrounding gas (and can be > 1 due to multiple scattering and the optically-thick nature of dense SGC). Taking $M_{\text{SGC}} = f_b M_{\text{v,H}_2}$, and using Eq. (B1) for R_{SGC} , we can estimate the star formation efficiency as

$$\begin{aligned} \epsilon_{\text{ZAMS}} &\equiv \frac{M_*}{M_{\text{SGC}}} = \left[\frac{G (10 M_\odot)^2 c}{(1.4 \times 10^4 L_\odot) \eta_r} \right]^{1/2} R_{\text{SGC}}^{-1} \\ &= 2.17 \times 10^{-3} \eta_{r,25}^{-1/2} \left(\frac{R_{\text{SGC}}}{3.63 \text{ pc}} \right)^{-1} \\ &= 2.17 \times 10^{-3} \eta_{r,25}^{-1/2} (1+z)_{25}^{3/2}, \end{aligned} \quad (\text{B17})$$

where our default value, $\eta_r = 25$, is chosen to be consistent with those obtained by the simulations of Latif et al. (2021), Latif et al. (2022b) and Latif et al. (2022a) for halos of 10^5 – $10^7 M_\odot$ at $z \approx 15$ – 30 . The corresponding stellar mass is

$$M_* = \epsilon_{\text{ZAMS}} M_{\text{SGC}} = 132 M_\odot \eta_{r,25}^{-1/2}. \quad (\text{B18})$$

Interestingly, this value does not depend on redshift, implying that the SGC collapse in a halo with mass at the H_2 -cooling threshold forms a fixed amount of stars. This is a consequence of the fact that the redshift dependencies of ϵ_{ZAMS} and $M_{\text{v,H}_2}$ cancel each other.

As the other extreme, we consider an SGC that is supported by strong turbulence in a halo with mass $f_b^{-1} M_{J,\text{max}}$ (Eq. B10). In this

case, the 1st star has sufficient gas supply to grow massive, with its luminosity limited by the Eddington luminosity,

$$L_* = L_{\text{Edd}} \approx 3.284 \times 10^6 L_{\odot} \left(\frac{M_*}{100 M_{\odot}} \right) \quad (\text{B19})$$

(Hosokawa et al. 2012). Substituting this luminosity into the momentum balance (Eq. B16), taking $M_{\text{SGC}} = M_{\text{J,max}}$, and using Eq. (B3) for Σ_{SGC} , we can estimate the star formation efficiency:

$$\begin{aligned} \epsilon_{\text{Edd}} &= \frac{G(100 M_{\odot})c}{(3.284 \times 10^6 L_{\odot})\eta_r} \Sigma_{\text{SGC}} \\ &= 5.29 \times 10^{-3} \eta_{r,25}^{-1} \left(\frac{\Sigma_{\text{SGC}}}{2.00 \times 10^4 M_{\odot} \text{pc}^{-2}} \right) \\ &= 5.29 \times 10^{-3} \eta_{r,25}^{-1} (1+z)_{25}^{3/2}. \end{aligned} \quad (\text{B20})$$

This efficiency is only slightly higher than that for a H_2 -cooling halo, and is a result of the balance between the higher binding force of the more massive SGC and the stronger radiative feedback of massive stars. The redshift dependence remains the same as that of ϵ_{ZAMS} , and the stellar mass of the 1st star,

$$M_* = \epsilon_{\text{Edd}} M_{\text{SGC}} = 2.62 \times 10^4 M_{\odot} \eta_{r,25}^{-1/2}, \quad (\text{B21})$$

is again independent of redshift.

Thus, the 1st star in a Pop-III cluster accounts for a few times 10^{-3} of the mass of an SGC at $z \approx 25$, and scales as

$$M_* = \epsilon M_{\text{SGC}} \propto M_{\text{SGC}}^{\beta} \quad (\text{B22})$$

at a given SGC density, where the star formation efficiency ϵ varies from $\epsilon_{\text{ZAMS}} \propto R_{\text{SGC}}^{-1}$ (Eq. B17) in pure H_2 -cooling halos to $\epsilon_{\text{Edd}} \propto \Sigma_{\text{SGC}}$ (Eq. B20) in extremely turbulent H -cooling halos. The index, β , thus varies from 2/3 to 4/3 between the two extremes. In both cases, the redshift-dependent factors in ϵ and M_{SGC} cancel out, leading to a time-independent M_* - M_{SGC} relation. The delay of the SGC collapse by either the UV radiation or the fast accretion thus affects the mass of the 1st star by elevating M_{SGC} ($\approx f_b M_v$), with the star formation efficiency constrained in a narrow range. Note that we assumed η_r to be the same for the two cases, which is only a crude approximation. In reality, an SMS may be unstable against pulsation, and its radius and temperature can experience episodic oscillations due to a strongly varying inflow rate, making the coupling efficiency, η_r , mass dependent (e.g. Latif et al. 2022a, see their Extended Data fig. 3). The asymmetry in the internal structure of stars and in the distribution of gas around the stars can also affect the coupling efficiency (e.g. Susa 2013; Latif et al. 2021) and cause halo-to-halo variations in the mass spectrum of Pop-III stars. Such details need to be modeled with numerical simulations.

More general cases are expected to fall between the two extremes described above, and we model them as follows. Once the halo mass for an SGC about to collapse is determined (Eq. 5), we obtain the properties of the SGC using Eqs. (B1)–(B4), the star formation efficiency $\epsilon = \epsilon_{\text{ZAMS}}$ using Eq. (B17), and M_* and $L_* = L_{\text{ZAMS}}$ of the 1st star using Eq. (B19). If L_* so obtained does not exceed L_{Edd} given by M_* , we take M_* as the mass of the 1st star; otherwise, we compute $\epsilon = \epsilon_{\text{Edd}}$ from Eq. (B20) and use it to find the mass of the 1st star.

Once the mass of the 1st star, $M_{*,1\text{st}}$, is determined through the radiation-regulated processes described above, we obtain the full IMF using the parametrization:

$$\frac{dN}{dM_*} = \phi_m \left(\frac{M_*}{M_{*,1\text{st}}} \right)^{-\beta_m} \exp \left(-\frac{M_*}{M_{*,1\text{st}}} \right). \quad (\text{B23})$$

For cases without UV heating and with a typical halo accretion rate, cosmological zoom-in simulations of mini-halos performed by Stacy

et al. (2016) and Latif et al. (2022b) taking into account radiative feedback from stars give $\beta_m \approx 1.2$. The simulation of Chon & Omukai (2020) assuming an intense LW background gives a similar value, $\beta_m \approx 1$, although the value of $M_{*,1\text{st}}$ is much higher. We thus use $\beta_m = 1$ as our default choice, which is also consistent with the choice of Liu et al. (2024) in their modeling of binary black holes. However, we note that the IMF of Pop-III stars is still poorly understood, and other values of β_m are possible (see e.g. Costa et al. 2023, for a number of alternatives). The normalization factor ϕ_m is determined by the constraint that the expected number of stars above $M_{*,1\text{st}}$ is unity:

$$1 = \int_{M_{*,1\text{st}}}^{\infty} \frac{dN}{dM_*} dM_*, \quad (\text{B24})$$

which gives $\phi_m = 4.56/M_{*,1\text{st}}$ for $\beta_m = 1$. Thus, at a given β_m , the only parameter needed to fully specify the IMF is $M_{*,1\text{st}}$, which we have already determined above from SGC properties and the feedback-regulated scenario. Fig. B2a shows the IMF of stars in a Pop-III star cluster with the default choice of β_m for a number of different $M_{*,1\text{st}}$ values in the range of 10^2 – $10^5 M_{\odot}$. At $M_* \ll M_{*,1\text{st}}$, all IMFs converge to the same logarithmic slope of -1 , specified by β_m , and the same normalization. A higher value of $M_{*,1\text{st}}$ gives a higher number of massive stars other than the 1st star. As we will see below, massive stars are expected to evolve along pathways different from the less massive ones, thus leaving BH seeds with different flavors.

The total stellar mass other than the 1st star is determined by

$$M_{*,\text{other}} = \int_{M_{*,\text{min}}}^{M_{*,1\text{st}}} M_* \frac{dN}{dM_*} dM_*, \quad (\text{B25})$$

where $M_{*,\text{min}}$ is the lower limit of the IMF. As shown in Fig. B1, a LW background of $J_{\text{LW},21} \gtrsim 10$ is able to delay the collapse of the SGC to the atomic-cooling regime. An even higher value of $J_{\text{LW},21}$ does not cause further delay, as the atomic cooling is effective to compensate the heating. However, in metal-free gas, a higher $J_{\text{LW},21}$ can prevent the formation of H_2 and can make the collapse closer to an isothermal contraction dominated by Lyman- α cooling. This can generate a much higher inflow rate, promoting the formation of the SMS and suppressing the formation of low-mass stars (Wise et al. 2019; Latif et al. 2021). A similar argument applies to cases where the rapid halo assembly can sustain significant amounts of turbulence to erase small clumps (Latif et al. 2022a). These factors can affect $M_{*,\text{min}}$, but the exact value appropriate for it is unknown because of the limited resolution of these simulations. Indeed, as discussed by, e.g. Stacy et al. (2016, see their §6.4), their simulation with a higher resolution can produce stars down to a lower $M_{*,\text{min}}$ and predicts an IMF that does not show a peak at $M_* \lesssim 1 M_{\odot}$. In contrast, Chon & Omukai (2020) found the presence of a peak, with the peak mass increasing with the Jeans mass. Despite the uncertainty, the total mass, $M_{*,\text{other}}$, in stars other than the 1st is very insensitive to $M_{*,\text{min}}$. Using $M_{*,\text{min}} = 0$, the integration of Eq. (B25) gives

$$M_{*,\text{other}} = 2.88 M_{*,1\text{st}}, \quad (\text{B26})$$

and the variation is less than $0.01 M_{*,1\text{st}}$ for any $M_{*,\text{min}} < 0.01 M_{*,1\text{st}}$.

The 1st star in a Pop-III star cluster is expected to die on a timescale of a few Myr and collapse into a BH (see e.g. §2.2 of Costa et al. 2023). As dynamical interactions among stars tend to sink the more massive object into the center, the BH originated from the 1st star is expected to become the central BH and to grow further by accreting gas and stars from its neighborhood, as to be detailed in Appendix C. The initial mass of the central BH is thus determined by the remaining mass of the 1st star after it dies. Here we adopt the result of Costa et al.

(2023, see their §3.1) who used the PARSEC stellar evolution code (Iorio et al. 2023) to follow the evolution tracks of Pop-III stars in different regimes defined by the initial stellar mass, M_* . For a massive star with $M_* \geq M_{*,\text{DCBH}} \equiv 242 M_\odot$, the remnant is a DCBH with mass $M_{\text{BH}} = M_*$. The equality holds as stellar wind is ineffective due to low line-driven opacity originated from the low metallicity (Vink et al. 2001; Baraffe et al. 2001), and the negligible mass loss due to pulsational instability compared with the large accretion rate (see §3.2 Hosokawa et al. 2013). With $100 M_\odot \equiv M_{*,\text{PISN}} \leq M_* < M_{*,\text{DCBH}}$, the pair-instability supernova (PISN) is the trigger to end the life of the star, which totally disrupts the star and leaves no remnant behind. For $M_* < M_{*,\text{PISN}}$, core-collapse supernova (CCSN) is the trigger to end the stellar evolution. We crudely approximate the remnant mass as $M_{\text{BH}} = f_{\text{rem}} M_*$, with $f_{\text{rem}} = 0.86$ to match the numerical result of Costa et al. (2023). A remnant below $3 M_\odot$ is classified as a neutron star, but is not expected owing to the large mass of the 1st star.

In the case where the PISN is triggered in the 1st star, we seed a central BH with a mass $M_{\text{BH}} = f_{\text{rem}} M_{*,\text{surv}}$, where $M_{*,\text{surv}}$ is the initial mass of the most massive star that can survive the PISN, obtained by solving

$$1 = \int_{M_{*,\text{surv}}}^{M_{*,\text{PISN}}} \frac{dN}{dM_*} dM_* . \quad (\text{B27})$$

The total mass of other stars able to survive is obtained using equations similar to Eqs. (B25) and (B26), but with the upper limit of integration replaced by $M_{*,\text{surv}}$. In our numerical implementation (Appendix D3), the mass loss due to stellar evolution for stars other than the 1st is also included when deriving the mass of their remnants. Note that our treatment of the stellar evolution is a simplification. For example, pulsational pair instability (PPI) can take place at a mass close to $M_{*,\text{PISN}}$ and cause fluctuation in the remnant mass (e.g. Farmer et al. 2019; Mapelli et al. 2020; Costa et al. 2023). Star-to-star variation is also expected, due to the dynamical nature of the stellar structure and the dense environment of the host star cluster that leads to frequent interactions among stars.

Fig. B2b shows the remnant mass, separately for the most massive star (1st star at the birth of the star cluster, or the most massive star that can survive the PISN) and other stars in total, as a function of $M_{*,1\text{st}}$ of the Pop-III star cluster. The remnant mass of the most massive star, i.e. the mass of the BH seed, follows closely $M_{*,1\text{st}}$, owing to the insignificant mass loss (zero for DCBH, and $1 - f_{\text{rem}} \approx 0.14$ for CCSN). The exception is for the case of PISN, where the 1st star at birth of the cluster is destroyed and the BH seed is bred by the remnant of the most massive star surviving the PISN. Total remnant mass of other stars also closely follows $M_{*,1\text{st}}$, except for the case of PISN, due to the simple relation between $M_{*,\text{other}}$ and $M_{*,1\text{st}}$ (Eq. B26).

The above discussion for the formation of SGCs, Pop-III star clusters and BH seeds assumes a pristine gas environment in which the SGCs reside. Such a condition is expected only if the IGM is not significantly enriched by previous generations of stars. Since our model can predict star formation of individual galaxies in a cosmological volume, the metallicity of the IGM can also be self-consistently predicted and used to determine when and where the formation of Pop-III stars is possible or prohibited. We present the details for the modeling of IGM enrichment in Appendix D6. Briefly, we find that at $z = 10$, most of the cosmic volume has already been filled by SN bubbles that carry metals produced by the star formation, and the median IGM metallicity goes above $Z_{\text{IGM}} \approx 10^{-4} Z_\odot$. This means that halos at $z \gtrsim 10$ form their first generation of stars mainly in the form of Pop-III, and are expected to seed their central BHs according to the model described above. The rapid star formation in galaxies during their fast phases then quickly enriches the IGM,

brings the cosmic star formation into the Pop-II regime, and ends the formation of Pop-III stars at $z \approx 10$. In this paper, we adopt a threshold value of $Z_{\text{IGM,Pop-III}} = 10^{-4} Z_\odot$ for the formation of Pop-III stars (Bromm & Loeb 2003; Valiante et al. 2011; Costa et al. 2023; Spinoso et al. 2023), and assume that halos embedded in the IGM with metallicity above this value form their first generation of stars in the form of Pop-II. The mass of BH seeds in these halos is expected to be low, assuming that Pop-II stars do not have a top-heavy IMF. We manually set the mass of BH seeds in these halos to be $M_{\text{BH}} = 10 M_\odot$, a typical value of the remnant mass of massive stars in normal star clusters.

The enrichment of the IGM by the outflow of metal produced by star formation, and its effect on the formation of stars and BH seeds in nearby halos, are the earliest ‘matter feedback’ of galaxy formation to its environment. This, together with the ‘radiation feedback’ described in Appendix B1.3, is represented by the orange markers in Fig. 3.

The branch in Fig. D1b summarizes the seeding procedure in the model pipeline. This procedure includes a list of key conditions regarding the competition between cooling and heating to be checked for the SGC in each halo. These include (i) whether a seed has already been bred in the halo, and thus whether the SGC is self-enriched to cool effectively; (ii) whether the IGM metallicity is high enough so that the SGC is externally enriched to cool effectively; (iii) whether the SGC is exposed to a strong, external UV background that stalls the cooling and delays the SGC collapse; (iv) whether dynamical heating is strong enough to prevent the collapse and fragmentation of the SGC. The fate of the SGC is then determined by the outcome of these conditions, and the BH seed is bred accordingly. Phenomenologically, condition (i) represents the internal effect of galaxy evolution on itself, while conditions (ii)–(iv) are external effects, from either halo assembly or large-scale environment, on galaxy evolution. Such conceptual separation of ‘nature’ and ‘nurture’ has been built for galaxy formation at low z (Peng et al. 2010; Niemi 2011; Casado et al. 2015; Paulino-Afonso et al. 2019; McGibbon & Khochfar 2022; Mucesh et al. 2024; Shi et al. 2024b), while our scenario suggests that all these effects have already been present ubiquitously in the early Universe at $z \gtrsim 20$. Intriguingly, the recent observations of JWST have pushed the discoveries of galaxies and the measurements of environment towards this high- z regime (e.g. Puskás et al. 2025; Shuntov et al. 2025; Lin et al. 2025; Arita et al. 2025), providing opportunities to test our predictions.

APPENDIX C: THE GROWTH FROM SEEDS TO SUPERMASSIVE BLACK HOLES

During the formation of a BH seed, gas can be temporarily expelled from the host SGC by the radiation and potentially also the SN explosion from Pop-III stars. This loss of gas fuel may lead to a quiescent phase in the star formation and AGN activity. Such a phase is expected to be transient, since rapid replenishment of gas is expected as a combined consequence of rapid halo assembly ($\gamma_v \gtrsim 1$, see Fig. B1) at the typical redshift of seed breeding ($z \gtrsim 10$) and efficient atomic cooling in mini-halos.

As summarized in §2.4, the critical condition for the growth of a BH is to break the barrier of angular momentum, and the contiguous capturing of sub-clouds proposed in Paper-I provides a natural mechanism to achieve this. The fraction of the gas captured by the BH thus depends on the mass and size of the SGC, which determine the width of the j -distribution, and on the black hole mass (M_{BH}), which determines j_{cap} . This, together with the regulation of

the total gas mass of the SGC by feedback from the active galactic nucleus (AGN), shapes the $M_{\text{BH}}-M_*$ relation for dynamically hot galaxies observed at $z \approx 0$. In contrast, the BH growth and its feedback effects become much weaker after the galaxy transits into the slow phase where the formation of a rotation-supported disk prevents gas from sinking towards the center. This produces an offset in the $M_{\text{BH}}-M_*$ relation for dynamically-cold galaxies relative to that for dynamically hot galaxies (see [Paper-I](#)). Galaxy mergers play an increasingly important role in the growth of both M_* and M_{BH} for galaxies residing in more massive halos, because of the reduced cold-gas fraction for in-situ growth and because of the increase of merging rate with halo mass.

This paper extends the above continuous mode of BH growth by taking into account the small-scale processes in the galactic nucleus. The key components include

- (i) the formation of a gaseous nucleus that is largely free of supernovae owing to the high density and short dynamical timescale;
- (ii) the turbulence-modified Bondi accretion that is highly efficient in driving gas inflow in the nucleus towards the BH;
- (iii) the formation of a turbulent and magnetized accretion disk around the BH due to the rapid inflow;
- (iv) the two modes of accretion, one ‘wet’ and the other ‘dry’, that can reach a super-Eddington rate and cause the BH to grow fast in short bursts.

The numerical model for the nuclear processes is then built upon these considerations and implemented as a refinement engine of the whole growing procedure so that BH growth can be resolved self-consistently within the context of the cosmic hierarchy. Fig. 4 shows the levels of hierarchy and processes relevant to BH growth at each level. Fig. A1 shows the gas density profile at each level of the hierarchy at three redshifts, in the history of a MW-size galaxy whose center BH is predicted to experience super-Eddington accretion at these epochs. The profile is built upon the outer levels established by the Universe, the halo (see Appendix A), and the SGC (see Appendix B1.1), with the inner levels established by the nuclear processes. In the following, we first review the entire hierarchy leading to BH growth, and then present the details of the key components listed above.

C1 The hierarchy leading to black-hole growth

A key property of CDM halos, as mentioned in §2.1, is their two-phase assembly. The early evolution of a dark matter halo is thus expected to consist of processes distinct from those later on. These include:

- (i) fast accretion, with $\gamma_v \gtrsim \gamma_f$, where the threshold value $\gamma_f \approx 0-3/8$ ([Zhao et al. 2003](#); [More et al. 2015](#); [Boco et al. 2023](#); see also §2 of [Paper-I](#) for a summary);
- (ii) efficient cooling, with $t_{\text{cool}} \lesssim t_v$, if halo mass has not grown above a redshift-independent threshold of $\approx 10^{12} M_\odot$ (see e.g. fig. 2 of [Paper-I](#));
- (iii) cold streams in halo gas, depending on the configuration of cosmic filaments connected to the halo ([Keres et al. 2005](#); [Dekel et al. 2009](#); [Ceverino et al. 2010](#); [Danovich et al. 2015](#)).

Processes (i) and (ii) in combination are sufficient to maintain rapid inflow and keep the halo center gas-rich before feedback can eject much of the gas. The high gas fraction then implies that the gas can become self-gravitating when the inflow reaches a radius of $f_g R_v \lesssim f_b R_v \approx 0.16 R_v$ before angular momentum becomes important to resist the collapse (at a scale of $\lambda_g R_v \approx 0.04 R_v$, assuming that the spin of the gas, λ_g , follows that of the dark matter). This leads to the formation of a self-gravitating gas cloud in the halo center, as shown schematically in Fig. 4b. The motion of gas elements in the SGC at the time of self-gravitation is thus dominated by random motion

instead of rotation, and the dispersion of j at an arbitrary direction is about $f_g R_v V_c$, where V_c is the circular velocity at the SGC scale (see Fig. 4a).

Process (ii) alone implies that the gas within the halo can fall freely towards the halo center, with a velocity of $\approx V_v$, without being significantly affected by thermal pressure. Shock is thus expected to be produced when the high-flux, high-speed inflow collides within the SGC. This, together with the frequent dynamical disturbance imparted by the rapid change of the gravitational potential during the fast-accretion regime, can generate supersonic turbulence, with the Mach number $M_s \equiv v_t/c_s \gg 1$, over the entire SGC. Fragmentation in the turbulent medium due to Jeans instability and shock compression produces sub-clouds with densities as high as $n_{\text{sc}} \approx M_s^2 n_{\text{SGC}}$, making both the cloud-cloud collision and the drag force from the surrounding gas negligible. The presence of process (iii) can further enhance the turbulence by increasing the fraction of kinetic energy that can penetrate the SGC, which can accelerate the formation of sub-clouds by promoting early fragmentation of gas in the filamentary inflow. As soon as the collision and drag become unimportant, the mixing of angular momentum among sub-clouds ceases, leaving a set of ballistically moving sub-clouds that support the structure of the SGC. Sub-clouds with highest density ($n_{\text{sc}} \gtrsim 10^{3.5} \text{ cm}^{-3}$) are predicted to be the sites to form globular clusters, as modeled in [Paper-III](#). Fig. 4c is a schematic plot showing the structure of an SGC, formed in a halo (panel b), together with its sub-clouds. As a consequence of the formation of sub-clouds, the broad distribution of j on the SGC scale is preserved, in contrast to the peaked, off-center distribution that would be expected for a dynamically-cold disk formed in a halo in the slow phase, as indicated in Fig. 4a. A fraction of the sub-clouds with low j in the broad distribution is thus expected to reach the influence zone of, and to be captured by, the central BH, as indicated by the vertical grey band in Fig. 4a.

The additional processes needed to be considered towards galactic nucleus have been summarized in §2.4. Among them, the regeneration of low- j sub-clouds is speculated as a consequence of ‘positive’ SN feedback and other sources of turbulence in [Paper-I](#), and is supported by the findings of hydro simulations that nuclear star formation is not passive (e.g. [Bower et al. 2017](#); [Habouzit et al. 2021](#)). Stellar feedback in the nucleus with high density may not be able to unbind the gas (e.g. [Dekel et al. 2023](#); [Hopkins et al. 2024a](#)), but can drive turbulence (e.g. [Ma et al. 2020](#)) and re-create sub-clouds to feed the BH. Another process in galactic nucleus, the mixing of angular momentum, can be clearly seen in the simulation of [Gaspari et al. \(2015, see their fig. 15\)](#), in which even weak turbulence promotes the redistribution of angular momentum in galactic nucleus and the BH feeding is expected to be replenished by the low angular-momentum sub-clouds generated by the redistribution. The results of all these processes, however, depend on the competition between gas heating and cooling. For example, SN feedback, cloud-cloud collision, drag force and AGN feedback can inject energy into the gas, which may destroy sub-clouds if cooling is ineffective, instead of promoting sub-cloud formation. Thus, these additional processes may cause the gas structure and dynamics to deviate from those expected from modeling the outer part of the SGC, thereby modifying the density ladder, angular momentum distribution and BH accretion rate. We therefore model the extension of the hierarchy to small scales in the following.

C2 The formation of supernova-free nuclei

The aforementioned processes within a galactic nucleus require an additional refinement on top of the SGC to capture their consequences

for gas dynamics, star formation and BH accretion, and motivate the definition of a new level, the SNF nucleus (§2.4), in the structure hierarchy. The threshold number density, n_{snf} , implies a mass density of

$$\rho_{\text{snf}} = \mu m_p n_{\text{snf}} \approx 6.35 \times 10^{-21} \text{ g cm}^{-3}. \quad (\text{C1})$$

Using Eq. (B5), the free-fall timescale of the gas at a density $\rho_g = \mu m_p n_g$ is

$$t_{\text{ff},g} = \sqrt{\frac{3\pi}{32G\rho_g}} = [0.836 \text{ Myr}] n_{g,3.5}^{-1/2}, \quad (\text{C2})$$

where $n_g \equiv 10^{3.5} \text{ cm}^{-3} n_{g,3.5}$. Thus, a density of $n_g \geq n_{\text{snf}}$ corresponds to a free-fall timescale of $t_{\text{ff},g} \leq t_{\text{snf}} \equiv 0.836 \text{ Myr}$, implying that gas dynamics within the nucleus proceeds faster than the lifetime of massive stars born in it. Such a nucleus is thus largely free of supernovae, and is thus referred to as the SNF nucleus in this paper. The argument in §2.4 suggests that a nuclear burst appears inevitable once such a nucleus is formed, and the consequence of the burst is a short period of rapid growth of the compact NSC, or BH, or both, together with gas ejection by BH feedback.

Fig. 4d shows the sketch of the SNF nucleus as a zoom-in view of the central part of the host SGC. The relations of the nucleus to the expected nuclear star cluster (NSC) and central BH are also highlighted. In the following, we describe the formation and structure of the SNF nucleus in detail.

C2.1 Criteria for the formation of supernova-free nuclei

Questions remain as to how the SNF nucleus can be formed in the first place. Several processes can present barriers for the formation, mainly,

- (i) the formation of dynamically-cold disk at the SGC scale, which can build an angular-momentum barrier for the gas flow towards the nucleus (e.g. Power et al. 2011; Rosas-Guevara et al. 2015; Curtis & Sijacki 2016);
- (ii) star formation and stellar feedback, which may consume/eject much of the gas before it can reach the nucleus (Meng & Gnedin 2020; Renaud et al. 2021; van Donkelaar et al. 2022, 2023; Garcia et al. 2023; Renaud et al. 2024; Mayer et al. 2024);
- (iii) feedback from the accreting BH, which may affect gas accumulation in the nucleus (Su et al. 2023; Dutta et al. 2024; Sivasankaran et al. 2025; Su et al. 2025) or may even eject/heat the gas in the SGC and halo (Zinger et al. 2020; Shi et al. 2022; Ayromlou et al. 2023; Weinberger et al. 2023; Wang et al. 2024a).

The formation of SNF nucleus is thus expected only when a set of conditions are met to break these barriers.

In our modeling relevant to the early fast accretion regime of the halo assembly, the creation of a dynamically hot, globally disturbed SGC is essential to remove those barriers. The ‘hotness’ is needed to prevent the formation of dynamically cold disks, so as to break the angular-momentum barrier (point i above). Global disturbance can make an SGC out of equilibrium, which can produce a strong gas inflow to overpower (at least over a short period of time) the gas consumption by star formation/BH accretion and ejection by the associated feedback (points ii and iii above). Once these conditions are met, an SNF nucleus can be formed as a consequence of the boosted inflow, and the formation of a compact stellar system and a period of rapid growth of the central BH seem inevitable.

The conditions for the hotness and global disturbance of an SGC arise naturally owing to the fluctuations in the halo assembly expected in the fast accretion regime (see examples in fig. 1 of Paper-I). The

temporary speed-up of the halo mass assembly implies an excessive amount of matter being added to the halo, which can perturb the SGC significantly through changing the gravitational potential and by the ram pressure of the infalling gas. Part of such disturbance is produced by mergers with other halos, which can perturb the SGC during a close encounter of the merging galaxies, a process known to be critical in transforming galactic structure (e.g. Springel & Hernquist 2005; Keselman & Nusser 2012; Puech et al. 2012; Yoon et al. 2022; Jackson et al. 2022; Wright et al. 2025; Sampaio et al. 2025; Proctor et al. 2025). A halo temporarily taking an excursion to a sufficiently high mass assembly rate is thus expected to be able to impart global disturbance and create the hotness required for the formation of an SNF nucleus in the SGC. These considerations motivate us to adopt the following criterion to trigger the formation of an SNF nucleus:

$$\gamma_v - \mathcal{S}(\gamma_v; t_v) \geq \delta_\gamma \gamma_f, \quad (\text{C3})$$

where $\mathcal{S}(\gamma_v; t_v)$ denotes the value of γ_v smoothed over a timescale of t_v along the main branch of the halo assembly history (see Appendix D4 for details), $\gamma_f \approx 0-3/8$ is the threshold that separates the fast and slow phases of halo assembly (Paper-I), and $\delta_\gamma \sim 1$ is a parameter that defines the strength of the excursion needed for the formation of an SNF nucleus. Here, we adopt $\gamma_f = 3/16$, the average of the two extremes, and we choose $\delta_\gamma = 1$ to identify excursions that have relaxation timescales longer than the typical dynamical time of the halo.

The frequency of the formation of SNF nuclei is expected to be high in the early Universe. This can be understood as a result of the specific merger rate of CDM halos (Dong et al. 2022) in combination with their faster assembly rate at higher z (see e.g. Fig. 2). As we will model in Appendix C5, the growth of the BH in an SNF nucleus is expected to be super-Eddington. This, together with the high frequency of formation of SNF nuclei, provides an efficient way for BH seeds to grow into SMBHs (see §3.2.1).

C2.2 Structure of the supernova-free nuclei

The violent gas inflow associated with the formation of SNF nuclei creates a situation similar to that for the formation of SGC, albeit with smaller volume, higher density, and shorter timescale. The large mass flux ($\sim M_{\text{SGC}}/t_{\text{dyn,SGC}}$) and high velocity ($\sim V_{\text{SGC}}$) are similar to the conditions of the SGC formation provided by a halo of fast assembly and efficient cooling, and are also expected to drive supersonic turbulence in the nucleus. Such a gaseous nucleus is thus dynamically hot (with turbulence velocity $v_t \gg c_s$), with structure supported by the turbulent motion. This gives some clue about the structure expected from an SNF nucleus.

However, the gas dynamics within an SNF nucleus is not entirely the same as that in an SGC. One key difference is the lack of regulation by supernova and AGN feedback in the nucleus, since the formation of the nucleus, as detailed in Appendix C2.1, requires temporarily boosted inflow so that star formation, BH accretion and their feedbacks are incapable of balancing the inflow rate. Instead, the main processes driving energy transfer during the formation of an SNF nucleus are the release of gravitational potential energy carried by the inflow and the dissipation of the turbulent energy by interacting gas elements. The situation here thus resembles that in hydrodynamic simulations of isothermal turbulence with a single driving source, such as that mimicked by a stochastic force term or a shaking box in such simulations (Eswaran & Pope 1988; Schmidt et al. 2009; Federrath et al. 2021). The real gas profiles concerned here may still be different from those in the simulations, as the SNF nuclei may not be able to fully develop an equilibrium, in which energy

injection balances dissipation, before nuclear star formation and BH accretion are elevated to disperse the gas. The lack of regulation and the runaway nature of SNF nuclei during formation may thus cause uncertainties in the modeling based on the assumption of equilibrium.

As an approximation, we divide the processes in an SNF nucleus into two stages. The first, referred to as the formation stage, is the period when the gas inflow is boosted to replenish the gas content in the nucleus and the regulation from the feedback has yet to be established. The second, referred to as the dispersion stage, is the period of a nuclear burst, in which star formation and BH accretion are elevated so that the gas consumption and dispersion by them overpower the inflow before the gas density within the nucleus is reduced to the original level. For the formation stage, we will model the structure equations of the gas around and within an SNF nucleus by a parametric approach in the following, with a number of free parameters to capture the unknown degrees of freedom. For the dispersion stage, we model the amount of the star formation and BH accretion in the nuclear burst together with the reduction of the gas in Appendix C5.

We parameterize the gas density (ρ_g) as a function of the galactocentric distance (r) within the entire SGC, including that within the SNF nucleus during the formation stage, by the following profile:

$$\rho_g(r) = \frac{2^{p+2q} \rho_{\text{snf}}}{(c_g x_r)^{2-2q} (1 + c_g x_r)^{p+2q}}, \quad (\text{C4})$$

where the scaled radius $x_r = r/R_{\text{SGC}}$; the concentration parameter $c_g \equiv R_{\text{SGC}}/R_{\text{nuc}}$; R_{nuc} is the outer radius of the SNF nucleus; ρ_{snf} is the supernova-free threshold (Eq. C1); $q \leq 1$ is the structure parameter of the turbulence (see below); $p \geq 0$ characterizes the density profile at large radius. It can then be shown that $\rho_g(R_{\text{nuc}}) = \rho_{\text{snf}}$. Thus, $\rho_g \geq \rho_{\text{snf}}$ for any $r \leq R_{\text{nuc}}$, consistent with the requirement that the nucleus is supernova-free.

The double power-law form of the profile divides the gas structure of the SGC into two regimes: one at $r \gg R_{\text{nuc}}$, where the density $\rho_g \propto r^{-(2+p)}$, and the other at $r \ll R_{\text{nuc}}$, where the density $\rho_g \propto r^{-(2-2q)}$. This separation is plausibly produced by distinct processes. Specifically, the parameter q can be understood in terms of the velocity structure of a turbulent medium, $\sigma_v \propto r^q$, namely the velocity dispersion σ_v of the gas as a function of the scale r . Assuming virialization, the mass enclosed within r is $M_g(r) \propto r \sigma_v^2 \propto r^{2q+1}$. The density profile is thus $\rho_g \equiv dM_g / (4\pi r^2 dr) \propto r^{-(2-2q)}$, which is the same as our parameterization (Eq. C4) at small r . The mean density within r can then be written as $\bar{\rho}_g(r) = 3\rho_g(r)/(2q+1)$. We choose $q = 1/2$ as the default value. This choice is commonly used for well-relaxed, isothermal and supersonic turbulence (Federrath et al. 2021), and we adopt it because of the supersonic nature of the SNF nuclei expected from the rapid inflow driven by the global disturbance. The choice of q thus implies that Eq. (C4) applies to a well-relaxed state of the SNF nucleus, where dissipation by radiative cooling is balanced by the energy injection into the gas. This balance is expected to be transitional, however, as the bursty star formation and BH accretion will soon end the formation stage and take the nucleus into the dispersion stage.

We adopt $p = 0$ as the default value, which corresponds to an isothermal profile, $\rho_g \propto r^{-2}$, for the volume of SGC well outside the nucleus. Such a gas profile is seen in hydro simulations (e.g. Hopkins et al. 2024a), and is adopted by semi-analytical models of galaxy formation (e.g. Henriques et al. 2020). With $p = 0$ and $q = 1/2$ given above, Eq. (C4) becomes

$$\rho_g(r) = \frac{2\rho_{\text{snf}}}{c_g x_r (1 + c_g x_r)}, \quad (\text{C5})$$

and the corresponding asymptotic behaviors are $\rho_g \propto r^{-2}$, $\bar{\rho}_g(r) = 3\rho_g(r)$, $M_g \propto r$, and $V_c = V_{\text{SGC}} = \text{const.}$ for $r \gg R_{\text{nuc}}$; $\rho_g \propto r^{-1}$, $\bar{\rho}_g(r) = 3\rho_g(r)/2$, $M_g \propto r^2$, and $V_c \propto r^{1/2}$ for $r \ll R_{\text{nuc}}$.

To determine the structure of an SNF nucleus from the global properties of the host SGC (Eqs. B1–B5), we can substitute the constraint $M(R_{\text{SGC}}) = M_{\text{SGC}}$ into the profile Eq. (C4), and obtain the concentration parameter c_g by solving the following implicit equation:

$$\frac{2}{c_g} \frac{m_g(c_g)}{c_g^2} = \frac{\rho_{\text{SGC}}}{\rho_{\text{snf}}}, \quad (\text{C6})$$

where $m_g(x) \equiv 3[x - \ln(1+x)]$. The radius of the SNF nucleus is then given by the definition, $R_{\text{nuc}} = R_{\text{SGC}}/c_g$. The mass enclosed within r is obtained by integrating $4\pi r'^2 \rho_g(r')$ over $r' \in [0, r]$:

$$M_g(r) = M_{\text{SGC}} \frac{m_g(c_g x_r)}{m_g(c_g)}. \quad (\text{C7})$$

The mass of the SNF nucleus, $M_{\text{nuc}} \equiv M_g(R_{\text{nuc}})$, is then

$$M_{\text{nuc}} = M_{\text{SGC}} \frac{m_g(1)}{m_g(c_g)}, \quad (\text{C8})$$

where $m_g(1) \approx 0.921$, and the circular velocity profile is

$$V_c(r) \equiv \sqrt{\frac{GM_g(r)}{r}} = V_{\text{nuc}} \sqrt{\frac{m_g(c_g x_r)}{m_g(1)c_g x_r}}, \quad (\text{C9})$$

with $V_{\text{nuc}} \equiv (GM_{\text{nuc}}/R_{\text{nuc}})^{1/2}$. For $r \ll R_{\text{nuc}}$ (i.e. $c_g x_r \ll 1$), V_c can be approximated as

$$V_c \approx \sqrt{1.5 c_g x_r / m_g(1)} V_{\text{nuc}} \approx 1.28 (c_g x_r)^{1/2} V_{\text{nuc}}. \quad (\text{C10})$$

To obtain numerical estimates of some quantities in the above equations, we consider $c_g \gg 1$, which holds for SGC at $z \lesssim 10$ (see Fig. F1b). In this case, $m_g(c_g) \approx 3c_g$, and the above equations for the structure of the SNF nucleus reduce to

$$\frac{3}{m_g(1)} \frac{M_{\text{nuc}}}{M_{\text{SGC}}} \approx \frac{R_{\text{nuc}}}{R_{\text{SGC}}} = \frac{1}{c_g} \approx \left(\frac{\rho_{\text{SGC}}}{6\rho_{\text{snf}}} \right)^{1/2}. \quad (\text{C11})$$

Taking the estimates of R_{SGC} and n_{SGC} in Eqs. (B1) and (B2), we obtain

$$c_g^{-1} \approx 0.239 f_{g,0.157}^{1/2} f_{r,0.04}^{-3/2} n_{\text{snf},3.5}^{-1/2} (1+z)_{10}^{3/2}; \quad (\text{C12})$$

$$R_{\text{nuc}} \approx [43.6 \text{ pc}] f_{g,0.157}^{1/2} f_{r,0.04}^{-1/2} n_{\text{snf},3.5}^{-1/2} M_{v,9.5}^{1/3} (1+z)_{10}^{1/2}; \quad (\text{C13})$$

$$M_{\text{nuc}} \approx \left[6.00 \times 10^7 M_{\odot} \right] f_{g,0.157}^{3/2} f_{r,0.04}^{-3/2} \times n_{\text{snf},3.5}^{-1/2} M_{v,9.5} (1+z)_{10}^{3/2}, \quad (\text{C14})$$

where $M_{v,9.5}$ is the halo mass scaled by $10^{9.5} M_{\odot}$, roughly the value of the progenitor at $z = 9$ of an MW-size halo at $z = 0$ ($M_{v,z=0} \approx 10^{12} M_{\odot}$; see Fig. 2). The free-fall timescale and angular frequency of the nucleus are

$$t_{\text{ff,nuc}} = \sqrt{\frac{3\pi}{32G\bar{\rho}_g(R_{\text{nuc}})}} \approx 0.616 \text{ Myr } n_{\text{snf},3.5}^{-1/2} \quad (\text{C15})$$

$$\Omega_{\text{nuc}} \equiv V_{\text{nuc}}/R_{\text{nuc}} \approx 1.80 \text{ Myr}^{-1} n_{\text{snf},3.5}^{1/2}, \quad (\text{C16})$$

respectively. Using the structure equations of the nucleus (Eq. C11)

and the numerical estimate of Σ_{SGC} (Eq. B3), the surface density of the SNF nucleus is

$$\Sigma_{\text{nuc}} \sim \frac{M_{\text{nuc}}}{R_{\text{nuc}}^2} \approx \frac{m_g(1)}{3} c_g \Sigma_{\text{SGC}} \approx 3.15 \times 10^4 \text{ M}_{\odot} \text{pc}^{-2} [\chi_g(1+z)_{10}]^{1/2} M_{v,9.5}^{1/3}, \quad (\text{C17})$$

where $\chi_g = f_{g,0.157n_{\text{snf},3.5}f_{r,0.04}^{-1}}$. The circular velocity of the nucleus, V_{nuc} , is related to that of the SGC, V_{SGC} , by $V_{\text{nuc}} = \sqrt{m_g(1)/3} V_{\text{SGC}}$.

Green segments in Fig. A1a extend the density profiles to the nuclear volumes of SGCs in MW-like progenitors by assuming that SNF nuclei have formed. For reference, in Appendix F, we show the properties of SNF nuclei in halos with a wide range of mass over an extended range of redshift. The two pieces of each SGC profile at $r \ll R_{\text{nuc}}$ and $r \gg R_{\text{nuc}}$ are joined smoothly by the function given by Eq. (C5). The inner and outer slopes (in logarithmic scale) are -1 and -2, respectively, as an outcome of our assumption on the structure parameters, p and q . The nucleus size, $R_{\text{nuc}} \lesssim 100 \text{ pc}$, shows only slow evolution from $z = 9$ to $z = 0$, much slower than the evolution of R_{SGC} of the host SGC (and R_v of the host halo). This is due to the requirement of a constantly high density, $\rho_g(R_{\text{nuc}}) = \rho_{\text{snf}}$, and the cancellation of two effects (see Eq. C11): the growth of R_{SGC} (and R_v) with galaxy (and halo) assembly, and the decline of ρ_{SGC} (and ρ_v) with the cosmic expansion. The nucleus size is roughly 1/4 of the SGC size at $z = 9$ (1/80 at $z = 0$), corresponding to 1/100 (1/2000) of the halo virial radius, respectively. This indicates that the SNF nucleus constitutes a significant part of the host SGC at high z , but becomes only a tiny part at low z . This also implies that SNF nuclei play the dominating role in supplying BH growth at high z , and a negligible role at low z (see §3.2.1 and Figs. 8 and 9). The card in Fig. A1d summarizes equations that link the properties of the SNF nucleus with those of the host SGC.

As discussed at the end of §2.4, the equations derived above provide a way to link the assembly of structures at different levels of the hierarchy. For example, we consider the time evolution of structural properties of the SNF nucleus in an SGC. This can be quantified by taking the time derivative of, e.g. Σ_{nuc} (Eq. C17), along the main branch of a halo:

$$\frac{d \ln \Sigma_{\text{nuc}}}{d \ln a} \approx -\frac{1}{2} + \frac{1}{3} \frac{d \ln M_v}{d \ln a} = \frac{2\Gamma_v - 3}{6}, \quad (\text{C18})$$

Note that this equation does not describe the evolution of a single nucleus; instead it describes the difference between nuclei formed at different epochs in the main branch of a halo. The derivative of Σ_{nuc} changes its sign at a critical value of $\Gamma_v = 1.5$, coinciding with the epoch found by Boco et al. (2023) that separates the fast- and slow-accretion phases of a halo (equivalent to $\gamma_v = 3/8$, our ‘Low- z_f ’ variant defined in §2 of Paper-I). The same arguments applied to R_{nuc} and M_{nuc} using Eqs. (C13) and (C14), respectively, yield the same critical value of Γ_v . Note that the presence of a critical value of Γ_v , and its exact value of 1.5, are owing to the assumption of the isothermal profile in the outer SGC (and thus the index of 1/2 in Eq. C11) and independent of the assumption made for the profile at $r \lesssim R_{\text{nuc}}$. The drag term, $-1/2$, in Eq. (C18) comes from the decrease of the density of a virialized halo with the cosmic time, and thus encodes the effect of cosmological expansion on the growth of structures deeply embedded in centers of dark matter halos.

As we will show later in Appendix C3, the turbulent motion of gas within the nucleus can induce strong inflow toward the central BH, and both the Bondi radius (R_B ; Eq. C29) and the Bondi accretion rate ($\dot{m}_B$; Eq. C30) of the BH are determined by Σ_{nuc} . The structure of the accretion disk of the BH, as to be quantified in Appendix C4 by

the profiles of the surface density (Σ_g ; Eq. C42), the volume density (ρ_g ; Eq. C43), and the Toomre Q parameter (Eq. C44), all depend on Σ_{nuc} through the Bondi properties. This leads to the conclusion made at the end of §2.4 that the transition of the halo assembly (at a scale of $r \gtrsim 100 \text{ kpc}$ for $M_v \gtrsim 10^{11} \text{ M}_{\odot}$ at $z = 0$) also leads to a transition in the BH-accretion properties (at a scale of $r \lesssim$ a few pc for $M_{\text{BH}} \lesssim 10^6 \text{ M}_{\odot}$).

The mass scale of an SNF nucleus, M_{nuc} , is typically a few times $10^7 \text{ M}_{\odot}$ (Eq. C14), about one order of magnitude larger than that of a sub-cloud at a comparable density (see Eq. 29 of Paper-III). The mass scale is also much larger than the typical mass of BH seeds, including massive DCBHs (see Eq. B21 and Fig. B2), and is comparable to massive star clusters observed in the local Universe (e.g. Baumgardt & Hilker 2018; Krumholz et al. 2019; Neumayer et al. 2020; Brown & Gnedin 2021). The formation of SNF nuclei thus implies that a large amount of dense gas is available for star formation and BH growth, as to be described below.

C3 The turbulence-modified Bondi accretion of black holes

C3.1 The role of turbulence

As summarized in §2.4, the turbulent environment in the SGC is critical for the gas to efficiently flow through the SNF nucleus, cross the radius of influence (i.e. Bondi radius) of the central BH, and eventually feed the BH.

In the outer volume of an SGC, the formation and condensation of sub-clouds make collision among them infrequent and drag force from surrounding gas negligible, as shown in §3.3 of Paper-I. This leads to the ballistic motion of sub-clouds, so that angular-momentum mixing between them is weak. Towards the inner volume of an SGC, however, the collision frequency for a sub-cloud, ν_{coll} , increases. The number of collisions expected for a sub-cloud per dynamical time is

$$\mathcal{N}_{\text{coll}} = \nu_{\text{coll}} t_{\text{dyn}} \propto V_c t_{\text{dyn}} R_{\text{sc}}^2 \rho_g \propto \rho_g^{1/2} R_{\text{sc}}^2 \propto r^{-1} R_{\text{sc}}^2, \quad (\text{C19})$$

where R_{sc} is the typical size of a sub-cloud, and an isothermal profile is assumed for ρ_g of the SGC. Thus, $\mathcal{N}_{\text{coll}}$ increases rapidly with decreasing r , unless R_{sc} shrinks. Of particular significance is the case when a sub-cloud is close to or within the SNF nucleus, $r \lesssim R_{\text{nuc}}$, where the environmental density is $\rho_g \approx \rho_{\text{snf}}$. Such a density is comparable to that of sub-clouds themselves, including even the densest ones that host globular clusters (see e.g. fig. 6 of Paper-III). Thus, sub-clouds in the nuclear volume are expected to interact frequently. A similar estimate can be made for the drag force by the ambient gas on a sub-cloud. The force is stronger at smaller r and becomes significant at $r \lesssim R_{\text{nuc}}$.

The dissipation has important consequences for gas dynamics in inner SGC. First, the collision between sub-clouds can compress the gas. Owing to the more effective cooling at higher gas density, the compressed gas does not push back, but becomes denser and more massive than the original sub-clouds. This leads to further fragmentation due to the reduced Jeans mass. Second, the loss of orbital energy and angular momentum due to the collision causes the gas to sink to a smaller orbit. Statistically, for a large number of sub-clouds that experience repeated collision, the orbit sinking generates a net inflow of gas from the outer part of the SGC to the nucleus, and from nucleus to the central BH. This can be understood by considering an extreme case where two sub-clouds have the same mass, move on circular orbits with the same speed but opposite direction. The head-on collision between them totally cancels the angular momentum, and enables the gas to fall freely towards the central BH. Thus, the net effect of collision of sub-clouds on the gas

dynamics is to cause (i) mergers of sub-clouds to form denser ones; (ii) the conversion of kinetic energy to radiation; (iii) the loss of angular momentum; and (iv) mass inflow through the SNF nucleus towards the central BH. These conclusions hold even if the SGC is metal-poor, since the shock compression can raise the gas temperature above 10^4 K, the regime of efficient cooling by hydrogen and helium atoms. A similar conclusion can be reached by considering the interaction between sub-clouds and the ambient gas. Sub-clouds can survive the disruption by the surrounding gas due to cooling in the turbulence mixing layer, if the sub-cloud density is significantly higher than that of the ambient gas (e.g. [Gronke & Oh 2020](#)). The interaction with the surrounding gas thus provides ram-pressure that compresses sub-clouds, and a friction force that drags sub-clouds inwards.

The essential role of turbulence can thus be summarized as ‘deferred mixing’, as discussed in §2.4. In the absence of a strong disturbance, the inflow is expected to be steady, producing a quasi-equilibrium state in which the gas inflow balances gas consumption by star formation, BH growth and associated feedback. This is the continuous mode of BH growth defined in this paper. However, in the presence of a strong disturbance, such as that associated with a strong merger, the collision of sub-clouds and the mixing of angular momentum can occur over a large volume of the SGC. This produces a temporary surge in the gas inflow that can overwhelm the outflow generated by the feedback from the existing BH and stars, leading to the formation of a massive SNF nucleus that can power a ‘burst’ of BH growth. In this case, the feedback may become ‘positive’, since the outflow generated by it provides an additional source to compress and mix the gas (see [Paper-I](#)). This mode of BH growth is referred to as the bursty mode in the paper. The effect of the deferred mixing of sub-clouds on the distribution of j has been schematically illustrated in Fig. 4a: a sharply peaked, centralized distribution forms via efficient mixing in the inner region of an SGC, on top of a broad distribution expected from the lack of mixing in the outer SGC. Note that the net angular momentum of the SGC is not necessarily different from that of a dynamically-cold disk. Instead, it is the turbulence that continuously redistributes j and generates low- j sub-clouds to feed the central region.

C3.2 The turbulence-modified Bondi accretion rate

The SNF nuclei formed through the rapid collapse of sub-clouds into halo centers are expected to be dynamically hot and turbulent. This motivates the use of modified Bondi accretion ([Hoyle & Lyttleton 1939](#); [Bondi & Hoyle 1944](#); [Bondi 1952](#)) to model the amount of gas to be accreted into the zone of influence of the central BH. The isotropic condition of the Bondi accretion may be met here due to the dynamical hotness, but the velocity dispersion of gas elements is now dominated by the turbulent motion rather than the thermal motion. Modified by the turbulence, the Bondi accretion rate becomes:

$$\dot{M}_B = 4\pi R_B^2 \rho_g(R_B) \sigma_{\text{eff}}(R_B), \quad (\text{C20})$$

where σ_{eff} is the effective velocity dispersion of the gas defined by $\sigma_{\text{eff}}^2 = c_s^2 + v_t^2$, accounting for both thermal motion (quantified by the sound speed, c_s) and turbulent motion (quantified by turbulent velocity dispersion v_t). The Bondi radius, R_B , is the radius of the zone of influence of the BH, within which the gravity of the BH dominates the total gravity:

$$R_B = \frac{GM_{\text{BH}}}{\sigma_{\text{eff}}^2} = [0.430 \text{ pc}] M_{\text{BH},4} \sigma_{\text{eff},10}^{-2}, \quad (\text{C21})$$

where $\sigma_{\text{eff}} = 10 \text{ km s}^{-1} \sigma_{\text{eff},10}$.

In the supersonic, dispersion-dominated regime that is expected for an SNF nucleus,

$$V_c \approx \sigma_{\text{eff}} \approx v_t \gg c_s. \quad (\text{C22})$$

Using the general density profile of the SGC (Eq. C4), we can express the circular velocity profile at $r \ll R_{\text{nuc}}$ as

$$V_c = \psi_{\text{nuc}} V_{\text{nuc}} \left(\frac{r}{R_{\text{nuc}}} \right)^q, \quad (\text{C23})$$

where $\psi_{\text{nuc}} \sim O(1)$ is a normalization factor. In the default case of $q = 1/2$, $\psi_{\text{nuc}} \approx 1.28$ (Eq. C10). The density profile is given by

$$\rho_g(r) = \frac{(2q+1)\psi_{\text{nuc}}^2 \Omega_{\text{nuc}}^2}{4\pi G} \left(\frac{r}{R_{\text{nuc}}} \right)^{2q-2}. \quad (\text{C24})$$

The Bondi radius can then be obtained from Eq. (C21):

$$R_B = R_{\text{nuc}} \left(\frac{M_{\text{BH}}}{\psi_{\text{nuc}}^2 M_{\text{nuc}}} \right)^{1/(2q+1)}. \quad (\text{C25})$$

We define the velocity and angular frequency at R_B as the Bondi velocity and the Bondi angular frequency, respectively:

$$V_B \equiv V_c(R_B) = \psi_{\text{nuc}} V_{\text{nuc}} \left(\frac{R_B}{R_{\text{nuc}}} \right)^q; \quad (\text{C26})$$

$$\begin{aligned} \Omega_B &\equiv \frac{V_B}{R_B} = \psi_{\text{nuc}} \Omega_{\text{nuc}} \left(\frac{R_B}{R_{\text{nuc}}} \right)^{q-1} \\ &= \psi_{\text{nuc}} \Omega_{\text{nuc}} \left(\frac{M_{\text{BH}}}{\psi_{\text{nuc}}^2 M_{\text{nuc}}} \right)^{\gamma_{\text{nuc}}}, \end{aligned} \quad (\text{C27})$$

where $\gamma_{\text{nuc}} = (q-1)/(2q+1)$ ranges from -1 ($q=0$) to $1/2$ ($q \rightarrow +\infty$). Using Eq. (C20) for $\dot{M}_B$, we obtain the Eddington ratio of the Bondi accretion rate:

$$\begin{aligned} \dot{m}_B &\equiv \frac{\dot{M}_B}{\dot{M}_{\text{Edd}}} = (2q+1) t_{\text{Sal}} \Omega_B \\ &= (2q+1) \psi_{\text{nuc}} t_{\text{Sal}} \Omega_{\text{nuc}} \left(\frac{M_{\text{BH}}}{\psi_{\text{nuc}}^2 M_{\text{nuc}}} \right)^{\gamma_{\text{nuc}}}, \end{aligned} \quad (\text{C28})$$

where $\dot{M}_{\text{Edd}} \equiv M_{\text{BH}}/t_{\text{Sal}}$ is the Eddington accretion rate⁶, and $t_{\text{Sal}} \equiv \sigma_{\text{TC}}/(4\pi G m_p) \approx 450 \text{ Myr}$ is the Salpeter timescale. Since $\Omega_{\text{nuc}} \sim 1 \text{ Myr}^{-1}$, the factor $(2q+1)\psi_{\text{nuc}} t_{\text{Sal}} \Omega_{\text{nuc}}$ in $\dot{m}_B$ is of the order of 10^3 . For the default choice ($q=1/2$), $\dot{m}_B \gtrsim 10^3$ for $M_{\text{BH}} \lesssim M_{\text{nuc}}$ (or $R_B \lesssim R_{\text{nuc}}$; see Eq. C25). This implies a super-Eddington accretion of the BH in the beginning of nuclear burst, provided the gas flowing across the Bondi radius can reach the BH (see Appendix C4).

In the default case ($q=1/2$, $\gamma_{\text{nuc}}=-1/4$), R_B and $\dot{m}_B$ (Eqs. C25 and C28) can be expressed neatly as functions of Σ_{nuc} and M_{BH} . Using Eq. (C17), the dependence on Σ_{nuc} can then be transformed into a dependence on Σ_{SGC} or M_v . This gives the following two chains of

⁶ Since we do not impose a constant value on the radiative efficiency (see Appendix C5), the definition of the Eddington rate of BH accretion in this paper assumes a radiative efficiency of unity for simplicity. We will clarify alternative definitions when discussing studies using them.

equations:

$$\begin{aligned}
 R_B &= \psi_{\text{nuc}}^{-1} \left(\frac{M_{\text{BH}}}{\Sigma_{\text{nuc}}} \right)^{1/2} \\
 &= \psi_{\text{nuc}}^{-1} \left(\frac{3}{m_g(1)} \frac{M_{\text{BH}}}{c_g \Sigma_{\text{SGC}}} \right)^{1/2} \\
 &\approx 4.56 \text{ pc} \frac{M_{\text{BH},6}^{1/2}}{[\chi_g(1+z)_{10}]^{1/4} M_{v,9.5}^{1/6}};
 \end{aligned} \tag{C29}$$

$$\begin{aligned}
 \dot{m}_B &= 2\psi_{\text{nuc}}^{3/2} t_{\text{Sal}} G^{1/2} \frac{\Sigma_{\text{nuc}}^{3/4}}{M_{\text{BH}}^{1/4}} \\
 &= 2\psi_{\text{nuc}}^{3/2} t_{\text{Sal}} G^{1/2} \left[\frac{m_g(1)}{3} \right]^{3/4} \frac{(c_g \Sigma_{\text{SGC}})^{3/4}}{M_{\text{BH}}^{1/4}} \\
 &\approx 6010 [\chi_g(1+z)_{10}]^{3/8} \frac{M_{v,9.5}^{1/4}}{M_{\text{BH},6}^{1/4}},
 \end{aligned} \tag{C30}$$

Each of the chains is composed of three equations that link quantities at four different radii: the Bondi radius (R_B), the radius of the SNF nucleus (R_{nuc}), the radius of the SGC (R_{SGC}), and the virial radius of the halo (R_V). Note that R_B depends weakly on M_V and z due to the weak dependence of Σ_{nuc} on M_V and z (see the texts below Eq. C17). This implies that the Bondi radius in an SNF nucleus is mainly controlled by the BH mass. The power-law indices on Σ_{nuc} in both chains are less than 1. Combined with the nearly flat evolution of Σ_{nuc} during the assembly history of any halo (see Fig. F1c), this implies that both R_B and $\dot{m}_B$ evolve very slowly for given M_{BH} and $M_{v,z=0}$.

The inner ends of the green segments in Fig. A1a show the predicted Bondi radii of BHs with $M_{\text{BH}} = 10^6 M_\odot$ embedded in the SNF nuclei that are expected to form in an MW-size halo and its progenitors. The Bondi radii are a few pc, and as expected, evolve very little from $z = 9$ to $z = 0$. For reference, we show the Bondi properties for wide ranges of halo masses and redshifts in Appendix F and Fig. F2.

C4 Gas structure inside the Bondi radius

Within the Bondi radius, the central BH dominates the total gravity. Gas clouds moving across this radius are tidally torn apart, forming a new gaseous structure that fuels the BH. As suggested by hydro simulations (Anglés-Alcázar et al. 2021; Hopkins et al. 2024a,b; Shi et al. 2024c), the gas dynamics in this region is highly asymmetric, eventually leading to the build-up of an accretion disk surrounding the BH. In the regime of the high accretion rate expected in an SNF nucleus, the accretion disk does not resemble an α -disk (Shakura & Sunyaev 1973) in the following aspects (see Hopkins et al. 2024c, thereafter H24, for the details):

- (i) The disk is turbulent, with a sonic Mach number $\mathcal{M}_s = v_t/c_s \gg 1$, a consequence of the rapid (super-Eddington) inflow. In contrast, turbulence in an α -disk is subsonic.
- (ii) Magnetic field can be strong, with an Alfvén velocity $v_A \equiv B/\sqrt{4\pi\rho_g} \gg c_s$. The strong magnetic field appears to be inevitable. Even if the seed field is initially small, it can be amplified by the turbulent dynamo and reach to equal partition with the gas kinetic energy ($v_A \sim v_t$, with the Alfvén Mach number $\mathcal{M}_A \equiv v_t/v_A \sim 1$), over a number of dynamical timescales, as suggested by MHD simulations (Martin-Alvarez et al. 2018; Hopkins et al. 2024a) and analytical studies (Pariev et al. 2003; Begelman & Pringle 2007; Begelman & Silk 2023).

- (iii) The disk is thickened by the large v_t and v_A , in contrast to an α -disk which is thin ($H_g/r \ll 1$) and supported by thermal and radiation pressure.
- (iv) The gas inflow rate in the disk towards the BH, $\dot{M}_g$, is large, which is a consequence of the large kinematic viscosity contributed by the large v_t^2 (Reynolds stress) and v_A^2 (Maxwell stress).
- (v) The disk is stable against fragmentation and star formation, as the Toomre Q parameter is expected to be large (see Eq. C44).

The transition of the gas from less structured turbulence at $r \gtrsim R_B$ to an ordered accretion disk at $r \lesssim R_B$ implies an additional level in the cosmic hierarchy. In particular, the properties of the accretion disk determine the possibility of star formation of the gas around the BH and eventually the growth rate of the central BH. We thus explicitly include the accretion disk in our modeling, with the outer boundary conditions set by the Bondi properties at R_B obtained in Appendix C3 and the inner boundary conditions set by the BH itself. Briefly, the boundary conditions include M_{BH} of the BH, $\dot{M}_g$, v_t/V_K and v_A/V_K , all evaluated at R_B , and some normalization factors. Here, $V_K \approx V_c$ is the Keplerian velocity, which approximates the circular velocity at $r \leq R_B$ as the gravity is dominated by the BH in this region. The output of this model is v_t/V_K , v_A/V_K and Σ_g , and other variables deducible from them, such as the surface gas density of the disk (Σ_g), the disk height (H_g), and the Toomre Q parameter, all as functions of the BH-centric distance r within the domain of $r_s \ll r \leq R_B$. Here r_s is the Schwarzschild radius, defined as

$$\begin{aligned}
 R_s &= \frac{2GM_{\text{BH}}}{c^2} \\
 &= 10^{-4} \text{ pc } M_{\text{BH},9} = 20 \text{ au } M_{\text{BH},9} = 0.02 \text{ au } M_{\text{BH},6},
 \end{aligned} \tag{C31}$$

at which relativistic effects become significant. The model we adopt is developed by H24, which was shown to be capable of reproducing results of their hyper-refinement zoom-in simulations carried out in a realistic cosmological context. We refer the readers to their paper for details.

The basic equation can be derived by considering the energy conservation that dissipation of the turbulence/magnetic energy within the disk is driven by the release of the gravitational energy of the inflow gas. Equating the dissipation rate per unit area, $dE_{\text{t+A}}/(dA dt)$, and the gravitational power per unit area, $dE_{\text{grav}}/(dA dt)$, we obtain

$$\frac{3}{2\psi_t^2} \Sigma_g \Omega_K v_t^2 = \frac{3}{4\pi} \Omega_K^2 \dot{M}_g. \tag{C32}$$

Here, $\Omega_K = V_K/r$ is the angular frequency of a circular orbit, and $\dot{M}_g$ is the net inflow (inflow minus outflow) rate of the gas. As we will see below (Eq. C44), the accretion disk does not have significant star formation, and thus the Bondi accretion rate is preserved across the accretion disk, i.e. $\dot{M}_g = \dot{M}_B = \dot{M}_{\text{BH}}$. The condition $\psi_t^2 \sim \mathcal{O}(1)$ fuses the normalization factors of both sides in the equation. The total specific energy to dissipate is about $(1/2)(v_t^2 + v_A^2) \sim v_t^2$, where the approximation holds for a magnetic field close to the equal-partition level. The dissipation timescale is $2\psi_t^2/(3\Omega_K)$. More formally, this equation has to be derived by the conservation of angular momentum (see Pariev et al. 2003, for details). An equivalent formalism is $\dot{M}_g = 3\pi\Sigma v_{\text{visc}}$, where the viscosity is $v_{\text{visc}} = 2v_t^2/(3\psi_t^2\Omega_K)$. The inflow velocity, $|v_r|$, can be deduced by its definition,

$$|v_r| \equiv \frac{\dot{M}_g}{2\pi r \Sigma_g} = \left(\frac{v_t}{\psi_t V_K} \right)^2 V_K. \tag{C33}$$

To form a closed set for Eq. (C32), we need an equation to describe the turbulence structure, $v_t(r)$. Here we seek for similarity solutions and assume that the radial dependence of the turbulent velocity can be parameterized as $v_t \propto r^{-\zeta_t}$. We further assume that the gas falls

freely at R_B , i.e. $|v_r(R_B)| = V_K$, which is used to fix the normalization. With these assumptions, the equation for the turbulence structure is

$$\frac{v_t}{V_K} = \psi_t y_r^{1/2 - \zeta_t}, \quad (\text{C34})$$

where $y_r \equiv r/R_B$. Substituting Eq. (C34) into Eq. (C32), we obtain

$$\Sigma_g = \frac{\dot{M}_g}{2\pi\sqrt{GM_{BH}R_B}} y_r^{2\zeta_t - 3/2}. \quad (\text{C35})$$

Interestingly, Σ_g does not depend on ψ_t .

The vertical structure of the disk, as suggested by H24, is contributed by both turbulence and magnetic field, with the latter being even more dominant at smaller radius. The vertical structure of the disk is thus best parameterized through the magnetic field, as $H_g/r \sim v_A/V_K$. Based on their simulations, Hopkins et al. (2024a,b) found that the magnetic field within R_B can be described by the flux-frozen/advection scenario extrapolated from the outer boundary conditions set at R_B . They thus proposed an effective equation of state (eEOS) for the magnetic field, as $B^2 \propto \rho_g^{\gamma_B}$, or $v_A \propto \rho_g^{(\gamma_B - 1)/2}$, where $\rho_g \sim \Sigma_g/(2H_g) \sim \Sigma_g\Omega_K/(2v_A)$ is the density at the mid-plane of the accretion disk. Reorganizing v_A to one side of the equation, and using Eqs. (C32) and (C34) for Σ_g , we obtain

$$\frac{v_A}{V_K} = \psi_A y_r^{1/2 - \zeta_A}, \quad (\text{C36})$$

where $\zeta_A = (3 - 2\zeta_t)(\gamma_B - 1)/(\gamma_B + 1)$, and ψ_A is a normalization factor. With this equation, the height of the disk is $H_g \sim \psi_A R_B y_r^{3/2 - \zeta_A}$.

Eqs. (C32), (C34) and (C36) form a closed set describing the density, turbulence and magnetic field structure of the accretion disk. The set is fixed once the structure index ζ_t and ζ_A (or γ_B) are given/calibrated, and the outer boundary R_B and the boundary conditions on M_{BH} , $\dot{M}_B$, ψ_t and ψ_A are obtained from the modeling described earlier. Other by-products can be derived using these equations, such as the Toomre Q parameter of the disk:

$$Q \equiv \frac{\sigma_{\text{eff}} \kappa}{\pi G \Sigma_g} \approx 2\sqrt{2}\psi_t \sqrt{1 + M_A^{-2}} \frac{M_{BH}\Omega_B}{\dot{M}_g} y_r^{-3\zeta_t}, \quad (\text{C37})$$

where we have used $\kappa = \sqrt{2}\Omega_K$, $\Omega_B = \Omega_K(R_B)$, and $\sigma_{\text{eff}}^2 = v_t^2 + v_A^2$, with the sound speed neglected.

A special case of this model (the default model of H24), which we adopt in this paper, is given by a flux-frozen magnetic field ($\gamma_B = 4/3$) and a trans-Alfvénic, supersonic turbulence ($M_A \sim 1$, $M_s \gg 1$). These imply that $\zeta \equiv \zeta_t = \zeta_A = 1/3$, $\psi \equiv \psi_t \sim \psi_A$, and thus

$$\frac{H_g}{r} \sim \frac{v_A}{V_K} \sim \frac{v_t}{V_K} = \psi y_r^{1/6}; \quad (\text{C38})$$

$$\Sigma_g = \frac{\dot{M}_g}{2\pi\sqrt{GM_{BH}R_B} y_r^{5/6}}; \quad (\text{C39})$$

$$\rho_g \sim \frac{\dot{M}_g}{4\pi\psi\Omega_B R_B^3 y_r^2}; \quad (\text{C40})$$

$$Q \sim \frac{4\psi M_{BH}\Omega_B}{\dot{M}_g y_r}. \quad (\text{C41})$$

Taking R_B from Eq. (C29), we obtain

$$\begin{aligned} \Sigma_g &= 24.7 \text{ M}_\odot \text{pc}^{-2} \frac{M_{BH,6}^{1/2} \dot{m}_{B,10}}{R_{B,4.56}^{1/2} y_r^{5/6}} \\ &= 24.7 \text{ M}_\odot \text{pc}^{-2} [\chi_g(1+z)_{10}]^{1/8} \frac{M_{v,9.5}^{1/12} M_{BH,6}^{1/4} \dot{m}_{B,10}}{y_r^{5/6}}; \end{aligned} \quad (\text{C42})$$

$$\begin{aligned} \rho_g &\sim 2.70 \text{ M}_\odot \text{pc}^{-3} \psi^{-1} \frac{M_{BH,6}^{1/2} \dot{m}_{BH,10}}{R_{B,4.56}^{3/2} y_r^2} \\ &= 2.70 \text{ M}_\odot \text{pc}^{-3} \psi^{-1} [\chi_g(1+z)_{10}]^{3/8} \frac{M_{v,9.5}^{1/4} \dot{m}_{B,10}}{M_{BH,6}^{1/4} y_r^2}; \end{aligned} \quad (\text{C43})$$

$$\begin{aligned} Q &\sim 1240\psi \frac{M_{BH,6}^{1/2}}{R_{B,4.56}^{3/2} \dot{m}_{B,10} y_r} \\ &= 1240\psi [\chi_g(1+z)_{10}]^{3/8} \frac{M_{v,9.5}^{1/4}}{M_{BH,6}^{1/4} \dot{m}_{B,10} y_r}, \end{aligned} \quad (\text{C44})$$

where $\dot{m}_{B,10} \equiv \dot{m}/10$ is the Eddington ratio of the Bondi accretion rate assuming a radiative efficiency of $\chi_{\text{AGN}} = 0.1$ (Crain et al. 2015; Rosas-Guevara et al. 2015). The properties of the accretion disk depend weakly on M_v and M_{BH} , but strongly on $\dot{m}_B$ and y_r . $Q \gg 1$ is implied by Eq. (C44) at any $y_r \leq 1$, consistent with the assumption that the disk is stable against fragmentation (see the texts below Eq. C32). This is true even when we consider the extremely high accretion rate of $\dot{m}_B \sim 6000$ at the beginning of the nuclear burst (Eq. C30).

Fig. 4e schematically shows the structures within the Bondi radius – the last level in the cosmic hierarchy. The accretion disk is turbulent, magnetized and thickened. Star formation is suppressed in the inner volume of this regime, considering the r^{-1} dependence of the Toomre Q parameter (Eq. C44). Stars might be present in the outer volume, which is not properly captured by the approximation of the accretion-disk model. These stars might also redistribute their energy and angular momentum via dynamical processes, and contribute to the BH growth. Such contribution is referred to as the ‘dry’ channel, in complement to the ‘wet’ channel of gas accretion. Each orange segment in Fig. A1a shows the mid-plane density of the accretion disk formed during the nuclear burst in an MW-size halo or its progenitors, assuming a central BH with $M_{BH} = 10^6 \text{ M}_\odot$ and a maximal accretion rate given by Eq. (C30) at the beginning of the burst. The orange shade covers the range of ρ_g as $\dot{m}_{BH}$ varies from the Eddington level ($\dot{m}_{BH} = 10$) to the maximum. The profile of ρ_g varies very little from $z = 0$ to $z = 10$, due to the weak evolution of the Bondi radius and Bondi accretion rate (see the texts below Eqs. C29 and C30). For reference, the card in Fig. A1e summarizes the equations for the accretion disk.

The role of turbulence in BH growth is again highlighted here. It magnifies the magnetic field, suppresses star formation, and preserves the rapid inflow at R_B through the accretion disk to feed the BH. As outlined in the beginning of Appendix C2, such a rapid process is expected to be transient, associated with bursty star formation, BH accretion and feedback. The evolution of the SNF nucleus in the dispersion stage is thus a process expected to be largely asynchronous with the evolution of the host galaxy/halo. This suggests that a refinement engine working at much finer spatial scales and temporal steps than those for modeling the host galaxy/halo is needed to follow the evolution of the hierarchy below the SNF nuclei, as to be described in the following.

C5 The bursty nature of black-hole growth and star formation in the nuclei

Here we build a model for the evolution of the SNF nucleus in the dispersion stage. Nuclear burst, including the bursty star formation and BH accretion, and their feedback are modeled together. A number of depletion timescales in the nucleus are analytically derived and compared with each other to gain insights into the mechanisms driving the evolution. The equations governing the evolution can be solved numerically, and will be incorporated as a refinement engine to complete the whole modeling procedure (see Appendix D and Fig. D1 for an outline).

C5.1 Basic considerations

Simplifications are necessary for the model to be applicable in a cosmological volume, while the essence of the nuclear burst must be retained. Here we model the distributions of the gas and stars within the SNF nucleus using spherically averaged, smoothed profiles. This takes into account the isotropy expected, at least statistically, in a dynamically hot nucleus. Sub-structures, such as gas and stellar clumps, are missed in such a treatment. Thus, star formation, gas inflow and interaction between feedback and gas in the model are treated as effective processes representing the average over diffuse and clumpy structures.

The basic equation is the continuity equation of gas density $\rho_g(r, t)$:

$$\frac{\partial \rho_g}{\partial t} = -\frac{\partial \rho_*/\partial t}{v_{\text{out}}^2} - \frac{(\dot{e}_{\text{AGN}} + \dot{e}_{\text{SN}} + \dot{e}_{\text{SE}} + \dot{e}_{\text{ext}}) \rho_g}{v_{\text{out}}^2} - \frac{1}{r^2} \frac{\partial}{\partial r} (r^2 \rho_g v_r), \quad (\text{C45})$$

where v_r is the radial velocity of inflow. This equation describes the reduction of the gas density at a given BH-centric distance r due to different sinks, each quantified by one term in the right-hand side. Here, $\rho_*(r, t)$ is the stellar density, and the term $-\partial \rho_*/\partial t$ describes the consumption of the gas by star formation. The terms, $\dot{e}_{\text{AGN}}$, $\dot{e}_{\text{SN}}$, $\dot{e}_{\text{SE}}$, are the specific (i.e. per unit mass) energy injection rates by the feedback from the BH, SN explosions and stellar evolution within the nucleus, respectively, while $\dot{e}_{\text{ext}}$ is from the star formation outside the nucleus. The quantity v_{out} denotes the velocity of the outflow driven by all the feedback effects. The total baryon density will be denoted as $\rho = \rho_g + \rho_*$. The time domain of this equation is $t \geq t_{\text{burst}}$, with t_{burst} being the starting epoch of the dispersion stage. The radial domain of the equation is $R_{\text{in}} \leq r \leq R_{\text{nuc}}$, with the inner boundary at $R_{\text{in}} \ll R_{\text{B}}$ in the beginning so that the time evolution of the Bondi radius can be self-consistently determined by its definition (Eq. C21) and the structure equation at $r \leq R_{\text{B}}$ can be determined by the model of the accretion disk (Appendix C4).

To estimate the relative importance of each term in Eq. (C45), we consider the depletion timescale, $t_{\text{dep},x} = M_{\text{nuc}}/\dot{M}_x$, for a process x that changes the gas mass at a rate of $\dot{M}_x$. The last sink term in Eq. (C45) corresponds to a continuous accretion flow through the nucleus towards the BH. The corresponding mass inflow rate can be estimated using the Bondi accretion rate, $\dot{M}_{\text{B}}$ (Eq. C28), and the depletion timescale is:

$$t_{\text{dep,bhacc}} = \frac{M_{\text{nuc}}}{\dot{M}_{\text{B}}} = \frac{M_{\text{nuc}}}{(2q+1)\Omega_{\text{B}}\dot{M}_{\text{BH}}}. \quad (\text{C46})$$

The first sink term in Eq. (C45) describes the gas consumption by star formation. The corresponding depletion timescale may be approximated by the free-fall time of the gas within the nucleus

(Eq. C15),

$$t_{\text{dep,SF}} = t_{\text{ff,nuc}} = \frac{\pi}{2\sqrt{2}\Omega_{\text{nuc}}}. \quad (\text{C47})$$

The ratio between the two depletion timescales is

$$\frac{t_{\text{dep,bhacc}}}{t_{\text{dep,SF}}} = \frac{2\sqrt{2}\psi_{\text{nuc}}^{2\gamma_{\text{nuc}}-1}}{\pi(2q+1)} \left(\frac{M_{\text{nuc}}}{M_{\text{BH}}} \right)^{\gamma_{\text{nuc}}+1}. \quad (\text{C48})$$

The prefactor is of the order of unity, while the mass-dependence is $(M_{\text{nuc}}/M_{\text{BH}})^{3/4}$ in the default case ($q = 1/2$). The timescale $t_{\text{dep,SF}} \ll t_{\text{dep,bhacc}}$ if $M_{\text{BH}} \ll M_{\text{nuc}}$, and $t_{\text{dep,SF}} \gg t_{\text{dep,bhacc}}$ if $M_{\text{BH}} \gg M_{\text{nuc}}$. This suggests a competitive scenario between BH accretion and star formation in the SNF nucleus, with the dominant process determined by $M_{\text{nuc}}/M_{\text{BH}}$.

The second sink term in Eq. (C45) groups the effects of different feedback channels of gas depletion. The energy injection by the SN feedback, $\dot{e}_{\text{SN}}$, is expected to be negligible before SN explosions from dying massive stars ($t - t_{\text{burst}} \lesssim t_{\text{snf}}$; see the texts below Eq. C2), but to be sufficiently strong afterwards to totally disperse the gas from the nucleus. SNe thus place a hard limit on the duration of the nuclear burst. The depletion timescale by the SN feedback is thus $t_{\text{dep,SN}} = t_{\text{snf}}$, comparable to that by the star formation ($t_{\text{dep,SF}}$). The energy injection by stellar evolution, $\dot{e}_{\text{SE}}$, and by external star formation, $\dot{e}_{\text{ext}}$, are expected to be negligible owing to the high density of the nucleus, as discussed in Appendix C2. Thus, during a burst of an SNF nucleus, AGN feedback is the only possible channel of feedback that is effective to deplete gas.

The exact effects of AGN feedback on the nuclear gas, however, are difficult to quantify from first principles. The difficulty comes from uncertainties in, e.g. the ‘structure’ of the BH (e.g. spin; see Huško et al. 2022), the state and emissivity of the gas in the accretion disk (Yuan & Narayan 2014; Koudmani et al. 2024; Hopkins et al. 2024c) and in the circum-BH medium (Dutta et al. 2024), the potential launch of ultra-fast outflows (Wagner et al. 2013; Gaspari & Sądowski 2017), the coupling of the feedback energy (in radiation and outflow) with the nuclear gas (Su et al. 2023, 2025), and the interplay among them. To simplify the problem, we do not attempt to differentiate channels of the AGN feedback (e.g. jet, wind, radiation and cosmic ray; see Shi et al. 2024a). Instead, we parameterize the total mass loss rate of the SNF nucleus produced by the AGN feedback as

$$\dot{M}_{\text{AGN}} = \chi_{\text{AGN}} L_{\text{Edd}} / v_{\text{out}}^2, \quad (\text{C49})$$

where $L_{\text{Edd}} \equiv M_{\text{BH}} c^2 / t_{\text{Sal}}$ is the Eddington luminosity; χ_{AGN} is the Eddington ratio of feedback energy, taking into account the net effect from both the energy conversion efficiency of the BH accretion and the coupling efficiency of the feedback energy to the nuclear gas. As suggested by e.g. Inayoshi et al. (2016), a thickened accretion disk in the super-Eddington regime is radiatively inefficient due to ‘photon trapping’, and is able to radiate at the Eddington level. Given the high density in the SNF regime, most of the emitted radiation is expected to be coupled with the nuclear gas. This implies that AGN feedback is inefficient in radiating but efficient in coupling, very different from the sub-grid recipes commonly implemented in unresolved hydrodynamic simulations (e.g. Schaye et al. 2015; Weinberger et al. 2017; Davé et al. 2019). We thus adopt $\chi_{\text{AGN}} = \chi_{\text{AGN},\infty} = 1$ as our default value at the high accretion rate (see below for the treatment of low accretion rate). The default outflow velocity is $v_{\text{out}} = 10^3 \text{ km s}^{-1}$, as suggested by observations (e.g. Sturm et al. 2011; Greene et al. 2012; Perna et al. 2017, 2019; D’Eugenio et al. 2024) and feedback models (e.g. Costa et al. 2014; Davé et al. 2019). With the adopted parameterization, the

depletion timescale for the AGN feedback is

$$t_{\text{dep,AGN}} = \frac{M_{\text{nuc}}}{\dot{M}_{\text{AGN}}} = \frac{t_{\text{Sal}}}{\chi_{\text{AGN}}} \frac{M_{\text{nuc}}}{M_{\text{BH}}} \left(\frac{v_{\text{out}}}{c} \right)^2. \quad (\text{C50})$$

The ratio of the depletion timescale between the BH accretion and the AGN feedback is then

$$\frac{t_{\text{dep,bhacc}}}{t_{\text{dep,AGN}}} = \frac{\psi_{\text{nuc}}^{2\gamma_{\text{nuc}}-1} \chi_{\text{AGN}}}{(2q+1)t_{\text{Sal}}\Omega_{\text{nuc}}} \left(\frac{c}{v_{\text{out}}} \right)^2 \left(\frac{M_{\text{nuc}}}{M_{\text{BH}}} \right)^{\gamma_{\text{nuc}}}, \quad (\text{C51})$$

where $(c/v_{\text{out}})^2 \approx 300^2$ and $t_{\text{Sal}}\Omega_{\text{nuc}} \approx 450 \times 1.8 \approx 800$. In the default case ($q = 1/2$, $\gamma_{\text{nuc}} = -1/4$), the factor $(M_{\text{nuc}}/M_{\text{BH}})^{\gamma_{\text{nuc}}} \geq 0.01$ for any $M_{\text{BH}}/M_{\text{nuc}} \geq 10^{-8}$. Thus, $t_{\text{dep,bhacc}} \gg t_{\text{dep,AGN}}$ almost always holds.

The above estimates yield a number of conclusions: (i) the depletion of gas in an SNF nucleus is dominated by the AGN feedback; (ii) if the AGN feedback does not disperse all the gas until $t - t_{\text{burst}} = t_{\text{snf}}$, SN feedback kicks in and clears the remaining gas; (iii) a short time window, $t - t_{\text{burst}} \in [0, t_{\text{snf}}]$, is left for intense BH accretion and star formation, which leads to bursty growths in mass for both the BH and in the stellar component. The temporary absence of SN feedback in the time window thus gives the BH a chance to grow differently from that expected in the continuous mode where SNe constantly regulate the gas content around the BH. Such a bursty mode may be missed in numerical simulations that are incapable of resolving dense structures at $n_{\text{g}} \gg n_{\text{snf}}$. The gas medium around BHs in these simulations may be numerically over-smoothed, so that the SN feedback is always effective to regulate the BH growth (see e.g. Bower et al. 2017; Habouzit et al. 2017, 2021; Li et al. 2025b). In the following, we present a detailed numerical treatment based on these conclusions.

C5.2 A refined treatment

To solve $\rho_{\text{g}}(r, t)$ and $\rho_{\text{*}}(r, t)$ for an SNF nucleus during the dispersion stage, we need to explicitly formulate each sink term in Eq. (C45). The term of star formation is parameterized as

$$\frac{\partial \rho_{\text{*}}}{\partial t} = \frac{\alpha_{\text{SF}}}{f_{\text{Q}}} \frac{\rho_{\text{g}}}{t_{\text{ff}}}, \quad (\text{C52})$$

where $\alpha_{\text{SF}} \sim 1$ is a free parameter, $t_{\text{ff}} = \sqrt{3\pi/(32G\rho)}$ is the free-fall timescale of gas. The suppression factor,

$$f_{\text{Q}} = \exp\left(\frac{Q - Q_{\text{B}}}{\delta_{\text{Q}}}\right), \quad (\text{C53})$$

takes into account the suppression of star formation by the large Toomre Q parameter in the accretion disk, which allows a rapid but continuous decrease of the star formation towards the center of the accretion disk; $\delta_{\text{Q}} \sim 1$ controls the sharpness of the decrease; Q is obtained from Eq. (C41) at $r \leq R_{\text{B}}$, and $Q_{\text{B}} \equiv Q(R_{\text{B}})$; we set $Q = Q_{\text{B}}$ for $r > R_{\text{B}}$, meaning no suppression outside the Bondi radius.

The inflow rate at the Bondi radius is based on the turbulence-modified Bondi accretion (Eq. C20), with a free parameter $\alpha_{\text{BH}} \sim 1$ characterizing the normalization:

$$\dot{M}_{\text{BH}} = \alpha_{\text{BH}} \dot{M}_{\text{B}}. \quad (\text{C54})$$

The inflow term $\partial(r^2 \rho_{\text{g}} v_r)/\partial r/r^2$ in Eq. (C45) depends on conditions at the outer boundary, R_{nuc} , and the dissipation processes within this boundary. Here we model it by a stable solution, $\dot{M}_{\text{g}}(r) \equiv \dot{M}_{\text{BH}}$, in which the inflow rate $\dot{M}_{\text{g}}(r)$ is constant over the entire domain of $r \leq R_{\text{nuc}}$. This solution implies that the inflow term in Eq. (C45) vanishes and the density profile, without being affected by star

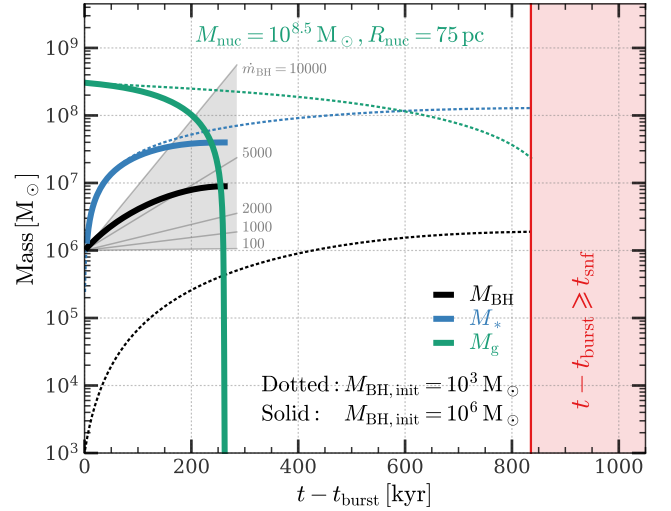


Figure C1. Evolution of masses in nuclear bursts predicted by the refinement engine. Here we show two events of nuclear bursts as examples, both with a BH placed in the center of the SNF nucleus with $M_{\text{nuc}} = 10^{8.5} M_{\odot}$ and $R_{\text{nuc}} = 75$ pc, typical for MW-size galaxies and their progenitors (see e.g. Fig. F1). The initial BH mass is $M_{\text{BH}} = 10^3 M_{\odot}$ (dotted) and $10^6 M_{\odot}$ (solid), respectively, in the two bursts. In each burst, black, blue and green curves show the masses of the BH, stars and gas, respectively, as a function of elapsed time since the beginning of the burst. Red shaded region indicates the truncation of the burst by SN feedback, which sets a time window with a duration of t_{snf} for the BH growth and star formation. Grey lines in grey shaded region represent the evolution tracks expected for BHs with constant Eddington ratios, $\dot{m}_{\text{BH}} \equiv \dot{M}_{\text{BH}}/\dot{M}_{\text{Edd}}$. Larger initial M_{BH} produces stronger feedback that depletes the gas more quickly, and a self-regulated growth pattern of BH is thus expected (see Fig. 11 and §3.2.3). A refinement engine is incorporated into the model to numerically obtain the evolution of masses during each nuclear burst. See Appendix C5 for details.

formation or feedback, does not change with time. This solution relies on the findings in Appendix C5.1 that the feedback term dominates the gas depletion at $R_{\text{B}} < r \leq R_{\text{nuc}}$ so that an arbitrary choice on the inflow rate has negligible effect, and that star formation is suppressed at $r \leq R_{\text{B}}$ so that inflow rate is preserved throughout the zone of influence of the BH (see Appendix C4).

The specific energy injection rate by SNe is modeled by a delayed flare, as

$$\dot{e}_{\text{SN}}(t) = \begin{cases} 0, & \text{if } t - t_{\text{burst}} \leq t_{\text{snf}}; \\ \infty, & \text{otherwise,} \end{cases} \quad (\text{C55})$$

so that a hard limit of t_{snf} is set for a complete ejection of the gas in the SNF nucleus and the end of the dispersion stage. Other forms of stellar feedback, such as those represented by $\dot{e}_{\text{SE}}$ and $\dot{e}_{\text{ext}}$, are expected to be negligible and are ignored.

The specific energy injection rate by AGN is determined implicitly according to the parameterization of the total feedback energy (Eq. C49), as

$$\chi_{\text{AGN}} L_{\text{Edd}} = \int_{R_{\text{in}}}^{R_{\text{nuc}}} \dot{e}_{\text{AGN}} \rho_{\text{g}} 4\pi r^2 dr, \quad (\text{C56})$$

corresponding to a specific mass loss rate of $\dot{e}_{\text{AGN}}/v_{\text{out}}^2$. The Eddington ratio of the feedback energy, χ_{AGN} , is parameterized as a function of the Eddington ratio of the accretion rate:

$$\chi_{\text{AGN}} = \chi_{\text{AGN},\infty} \frac{\alpha_{\chi} \dot{m}_{\text{B}}}{1 + \alpha_{\chi} \dot{m}_{\text{B}}}, \quad (\text{C57})$$

so that χ_{AGN} is saturated at $\chi_{\text{AGN},\infty}$ for $\dot{m}_{\text{B}} \gg 1$, and $\alpha_{\chi} \sim 0.1$ in the sub-Eddington regime where the accretion disk is expected to become a normal α -disk (Shakura & Sunyaev 1973) that is often used as a sub-grid model in cosmological hydrodynamic simulations. The radial dependence of $\dot{e}_{\text{AGN}}$ is determined by the details of how the outflow mass and energy are coupled to the nuclear gas. Since these are largely unknown, we parameterize $\dot{e}_{\text{AGN}}$ as

$$\dot{e}_{\text{AGN}} \propto \rho_{\text{g}}^{-\beta_{\text{out}}}, \quad (\text{C58})$$

which accounts for the expectation that both coupling and cooling (dissipation) should depend on the gas density. We adopt $\beta_{\text{out}} = 0$ as the default, which implies a radially invariant specific mass loss rate caused by the AGN feedback. The calibration using hydrodynamic simulations, and the quantification for the effects of other relevant state variables, such as gas clumpiness and metallicity, are worthy of being explored further. Note that the outflow launched from the SNF nucleus can be very strong due to the Eddington-level luminosity of the BH. For $M_{\text{nuc}} \sim 10^8 M_{\odot}$ and $t_{\text{dep}} \lesssim 1$ Myr, we obtain $\dot{M}_{\text{w}} \gtrsim 100 M_{\odot} \text{yr}^{-1}$. When propagating outwards, the outflow can continuously couple with the gas in the SGC, and potentially launch galaxy-wide outflows.

Fig. C1 shows two examples of nuclear bursts. In each, a BH with a given initial mass is placed at the center of the SNF nucleus that is typical for MW-size galaxies and their progenitors. The evolution of mass components (BH, stars and gas) are shown as functions of elapsed time since the beginning of the burst. In both cases, the reduction of gas mass is substantial, much larger than the growth of BH and stellar masses. The main source of gas depletion is the AGN feedback. In the case of small initial BH mass, $M_{\text{BH},\text{init}} = 10^3 M_{\odot}$, about 1/10 of the gas remains in the nucleus at ≈ 800 kyr before SNe kick in to clear the gas. The central BH grows by more than three orders of magnitude in mass, from $10^3 M_{\odot}$ to more than $10^6 M_{\odot}$. A nuclear star cluster with $M_{\star} > 10^8 M_{\odot}$ also forms in the burst, indicating a strong competition with the BH in consuming the gas. In the case with $M_{\text{BH},\text{init}} = 10^6 M_{\odot}$, the feedback efficiency (energy output per unit accreted mass) is much higher, and the gas depletion terminates the burst at ≈ 200 kyr. Star formation here is less competitive than in the former case, and the growth of both M_{BH} and $M_{\star}$ is about $10^7 M_{\odot}$. The final M_{BH} is higher than in the former case, but the final stellar mass is lower. The competition between the BH growth and star formation thus depends strongly on $M_{\text{nuc}}/M_{\text{BH}}$, consistent with the expectation based on the depletion timescales (Eq. C48). For comparison, grey lines in Fig. C1 show the evolution of the BH mass expected from a constant Eddington ratio ($\dot{m}_{\text{BH}}$) between 10^2 and 10^4 . The average growth rates of the two cases during the entire bursts fall in the range of $\dot{m}_{\text{BH}} = 10^3$ to 10^4 , suggesting that the bursty growth of the BH is super-Eddington over a wide range in the parameter space. Interestingly, observations indeed indicate that a small fraction of AGNs are experiencing super-Eddington accretion, either based on measurements of the BH mass and accretion rate (Liu et al. 2021), or based on the weak/variable X-ray emission expected from the obscuration of thick disk and/or clumpy wind surrounding a BH with super-Eddington accretion (Wang et al. 2022; Zhang et al. 2025b).

The diversities in the mass ratio, $M_{\text{nuc}}/M_{\text{BH}}$, in the efficiency of the AGN feedback and in the BH growth-star formation competition are thus expected to produce diversities in the growth path among different galaxy/BH systems. This can be explored further to explain the observed diversity of high- z galaxies/AGNs (e.g. Mezcuza et al. 2024; Onoue et al. 2025; Li et al. 2025a), and our refined treatment for nuclear bursts provides a basis for such explorations within the current cosmological framework, as we will show in §3.

C5.3 Comparison with other growth channels of black holes

To see that the nuclear burst is indeed an efficient channel of BH growth, here we compare it with other viable channels.

The continuous mode of BH growth, as derived in Paper-I using the loss-cone argument (Shapiro & Teukolsky 1986), predicts an accretion rate of $\dot{M}_{\text{BH}} \propto (M_{\text{BH}}/M_{\text{SGC}})\dot{M}_{\text{g,cool}}$. Thus, the Eddington ratio of BH accretion rate is

$$\begin{aligned} \dot{m}_{\text{BH}} &\equiv \frac{\dot{M}_{\text{BH}}}{\dot{M}_{\text{Edd}}} = \alpha_{\text{cap}} t_{\text{sal}} \frac{F_{\text{cool}} f_{\text{b}}}{f_{\text{gas}}} \frac{\dot{M}_{\text{v}}}{M_{\text{v}}} \\ &= 3.67 \alpha_{\text{cap},2.5} \frac{F_{\text{cool}} f_{\text{b}}}{f_{\text{gas}}} M_{\text{v},9.5}^{0.14} (1+z)_{10}^{2.5}, \end{aligned} \quad (\text{C59})$$

where we have adopted Eq. (A8) for an approximation of the halo assembly rate, $\dot{M}_{\text{v}}$; F_{cool} is fraction of gas associated with the halo assembly that can cool and flow into the SGC; $\alpha_{\text{cap}} \equiv 2.5\alpha_{\text{cap},2.5}$ is a free parameter, adopted to be 2.5 by default. Most halos at $z \approx 10$ have mass $M_{\text{v}} < 10^{12} M_{\odot}$ (see Fig. 2), implying effective cooling ($t_{\text{cool}} \lesssim t_{\text{v}}$; see fig. 2 of Paper-I) that leads to $F_{\text{cool}} \approx 1$ and $f_{\text{gas}} \approx f_{\text{b}}$. The dependence of $\dot{m}_{\text{BH}}$ on halo mass is quite weak, while the factor $(1+z)^{2.5}$ implies a rapid decline of $\dot{m}_{\text{BH}}$ with cosmic time. The expected value of $\dot{m}_{\text{BH}}$ is $\lesssim 5$ at $z \approx 10$, marginally touching the Eddington rate, and falls rapidly to $\lesssim 0.1$ at $z \lesssim 3$. This implies a much less efficient accretion for BHs in the continuous mode than in the bursty mode. The relation between the BH growth and the halo assembly implied by Eq. (C59) is $\dot{M}_{\text{BH}}/M_{\text{BH}} \propto \dot{M}_{\text{v}}/M_{\text{v}}$ and that between the BH growth and the star formation is $\dot{M}_{\text{BH}}/M_{\text{BH}} \propto \dot{M}_{\star}/M_{\star}$ if star formation can catch up with the BH accretion in the early Universe. Given the relation between the BH mass and the galaxy mass, $M_{\text{BH}} \propto M_{\star}^{\gamma}$ with the observed range of $\gamma \gtrsim 1$ in the local Universe (Kormendy & Ho 2013; Greene et al. 2020; Graham & Sahu 2023, e.g.), the continuous mode alone is not sufficient to drive the BH seed onto the observed relation, unless the seed is produced above the relation. In contrast, the bursty mode may be able to drive the BH seed onto or above the relation, as we will discuss in Appendix G.

In the case where turbulence is negligible at the scale of the SGC, rotationally supported disk is expected to form as a result of angular-momentum mixing (the dashed curve in Fig. 4a), while thermally supported gas processed by the feedback and left in the spheroid are too hot and too diffuse to form stars. This corresponds to a classical picture of gaseous components of galaxies in the local Universe. The rate at which the hot gas is accreted into the zone of influence of the BH can be described by the original form of the Bondi accretion (Eq. C20, with $\sigma_{\text{eff}} \approx c_{\text{s}}$):

$$\dot{M}_{\text{B}} = \frac{4\pi G^2 M_{\text{BH}}^2 \rho_{\text{g}}}{c_{\text{s}}^3}. \quad (\text{C60})$$

With the approximation that the sound speed $c_{\text{s}} \approx V_{\text{c}}(R_{\text{B}})$, the factor $4\pi G^2 M_{\text{BH}}^2 / c_{\text{s}}^3$ is similar to that of the bursty mode, but the low density ρ_{g} makes the accretion of hot gas much less efficient than the bursty-mode accretion. On the other hand, the rate at which cold gas in the disk is accreted into the zone of influence of the BH can be formulated with an additional factor accounting for the viscosity, as:

$$\dot{M}_{\text{B}} = \frac{4\pi G^2 M_{\text{BH}}^2 \rho_{\text{g}}(R_{\text{B}})}{V_{\text{c}}^3(R_{\text{B}}) C_{\text{visc}}}, \quad (\text{C61})$$

where $C_{\text{visc}} = 2\pi [\alpha_{\text{visc}}(H/R)^2]^{-1}$; α_{visc} is the viscosity number; $H/R \ll 1$ is the height-to-radius ratio that describes disk geometry. This formulation was derived by Rosas-Guevara et al. (2015) by comparing the original Bondi timescale and the viscosity timescale, and was adopted as a sub-grid model in the EAGLE simulation

(Schaye et al. 2015; Crain et al. 2015). The density ρ_g of the cold gas can be large, but the viscosity suggested by them is quite small. This gives a value of $C_{\text{visc}} \sim 10^6$ that largely suppresses the accretion rate of the BH, supporting the assumption made in Paper-I that BH accretion is negligible in a thin galactic disk expected in the slow phase of the halo assembly.

APPENDIX D: DETAILED IMPLEMENTATION OF THE MODEL

In Fig. 1, we have briefly outlined the steps to implement the model. Here we describe the implementation in detail following the logical order shown in a more detailed flowchart in Fig. D1. For the physical contexts for the seeding and growing procedures, refer to Figs. 3 and 4, respectively. The code for the implementation is publicly available (see). For reference, we also list the components of the model and the default parameters adopted in this paper in Table D1.

The model starts with a set of subhalo merger trees constructed from a N-body simulation (TNG100-1-Dark in this paper; see §2.1). A cosmological simulation covering such a volume often has limited resolution, and is thus insufficient to resolve the assembly histories of halos down to the masses required by the seeding procedure. To tackle this issue, we extend the simulated merger trees by joining them with those generated by a Monte Carlo algorithm based on the extended Press-Schechter (EPS) formalism. The Monte Carlo trees need to be generated with sufficiently high resolution, so that the assembly history of any simulated halo, after the extension, can be reliably traced back to a mass below the H_2 -cooling limit at which the first generation of stars and BH seed may form (Eq. 1; see also the purple dotted lines in Fig. 2a,c,d).

D1 Extension of subhalo merger trees

We follow the algorithm detailed in Chen et al. (2023a, see their §3.2, the ‘Central-Stage Completion’ part) to perform the extension. We list the steps below.

- (i) We decompose each simulated subhalo merger tree into a set of disjoint branches by a recursive method. Starting from the forest F that consists of all the simulated subhalo merger trees, we arbitrarily select a tree T from F . We find the root subhalo, h_{root} , of T , and trace back the main branch, B , of h_{root} (defined as the branch with ‘most massive’ history in TNG; see e.g. Rodriguez-Gomez et al. 2015). We remove B from T , which leaves T with a set of disjoint subtrees, $\{T_i\}$, each rooted in a subhalo h_i whose descendant is h_{root} . We put all T_i back into F , and repeat the above process until F is empty.
- (ii) For each branch B obtained above, we start from the leaf subhalo (i.e. that with the highest redshift), walk along the branch towards lower redshift to find the ‘anchor’ redshift z_{anc} at which the halo mass M_v becomes larger than a threshold $M_{v,\text{anc}}$ for the first time. The subhalo at z_{anc} so obtained is referred to as the ‘anchor subhalo’ of B , and is denoted as h_{anc} .
- (iii) We start from h_{anc} , walk along the branch B towards higher redshift to find the ‘half-mass’ redshift, z_{form} , of h_{anc} , at which the halo mass drops below $M_v(z_{\text{anc}})/2$ for the first time.
- (iv) We follow Parkinson et al. (2008) to generate a set of N_{MC} Monte Carlo merger trees with a sufficiently high resolution, each having a root halo h'_{root} with a mass and redshift equal to that of h_{anc} . We compute the half-mass formation time, z'_{form} , for the root halo of each Monte Carlo tree along the main branch.

- (v) We search for the Monte Carlo tree whose root halo has a z'_{form} that is the closest to z_{form} among all the Monte Carlo trees, and we denote the main branch of this root halo as B' . Physical quantities directly related to M_v (e.g. R_v , V_v and T_v ; see Eq. A3–A5) are assigned to the halos of B' according to their definitions. The profile of any halo in B' is assumed to be an NFW profile, with a concentration parameter (c_v ; see Eq. A6) computed from the mass assembly history by the analytical fitting of Zhao et al. (2009, with slight adaptation to account for the difference in the definition of halo boundary), and the maximal circular velocity (V_{max}) is computed from the profile.
- (vi) We prune all halos at $z > z_{\text{anc}}$ from B , and join B' to B at z_{anc} .

Note that the above steps ensure that the Monte Carlo branch B' not only has exactly the same M_v as B at z_{anc} , but also has similar first-order derivative, since the matching of z_{form} approximately provides such a constraint. The Monte Carlo algorithm proposed by Parkinson et al. (2008) is adopted here. This algorithm is re-calibrated by N-body simulations to closely match the simulation results (see Jiang & van den Bosch 2014 for details), and is highly efficient due to their carefully chosen bound function for the rejection sampling. The number of Monte Carlo trees, N_{MC} , should be sufficiently large so that the sampling of z'_{form} is dense enough to give a close match to z_{form} . By trial and error, we find that $N_{\text{MC}} = 512$ is sufficient to produce a smooth extension. The threshold mass $M_{v,\text{anc}}$ that defines the anchor point should be at least twice the minimal mass that can be resolved by the simulation used, so that z_{form} of h_{anc} can be reliably determined. Based on the resolution of TNG100-1-Dark, we set $M_{v,\text{anc}} = 5 \times 10^8 h^{-1} \text{M}_{\odot}$, which is the mass of about 100 dark-matter particles in the simulation. Since the seeding procedure requires modeling the radiation and matter feedback from any galaxy to its large-scale environment (see the flowchart Fig. D1c2), the locations of halos in each B' have to be assigned. Here, we assign the location of h_{anc} in B to all halos in the matched B' . This means that the peculiar motion of a halo at $z > z_{\text{anc}}$ is ignored and that the accuracy of small-scale radiation and matter feedback is compromised.

With the extension, the assembly history of each simulated subhalo can be smoothly extended down to the H_2 -cooling limit, and the extended merger trees will be used as the input of subsequent modeling for galaxy formation. The steps for obtaining the simulated merger trees and extending them are outlined in the flowchart of Fig. D1a.

In Fig. D2, we show examples of mass assembly histories (MAHs, i.e. M_v at different redshifts) of simulated subhalos before (black) and after (black plus red) the extension, over-plotted with the regime in which Pop-III stars are expected to form (purple). As expected, the MAHs now cross the Pop-III band, thus allowing the formation of the first generation of stars and BH seeds to be directly modeled. Note that the Monte Carlo trees themselves can be used as the input of the model, if LW radiation and IGM enrichment are available through other means. This is the case for the idealized and controlled experiments presented in §3.2.3, where the environment is set to the extreme so that $M_{\text{BH,seed}}$ is minimized or maximized. A similar approach was used by Spinoso et al. (2023, see their §2.4 and fig. 1) to marginally extend the simulated merger trees to the H_2 -cooling limit at $z \gtrsim 15$, thereby allowing some light-weight seeds to be reared. The semi-analytical model of Ventura et al. (2024) for Pop-III stars was built on merger trees from a N-body simulation that can resolve halos down to $\approx 1.5 \times 10^5 h^{-1} \text{M}_{\odot}$, but with a compromised volume of $(10 h^{-1} \text{Mpc})^3$. Our mixed approach of using a large-volume simulation and a Monte Carlo extension thus allows us not only to trace the formation of Pop-III stars, but also to have a large sample that can reduce the effects of cosmic variance (e.g. Somerville et al.

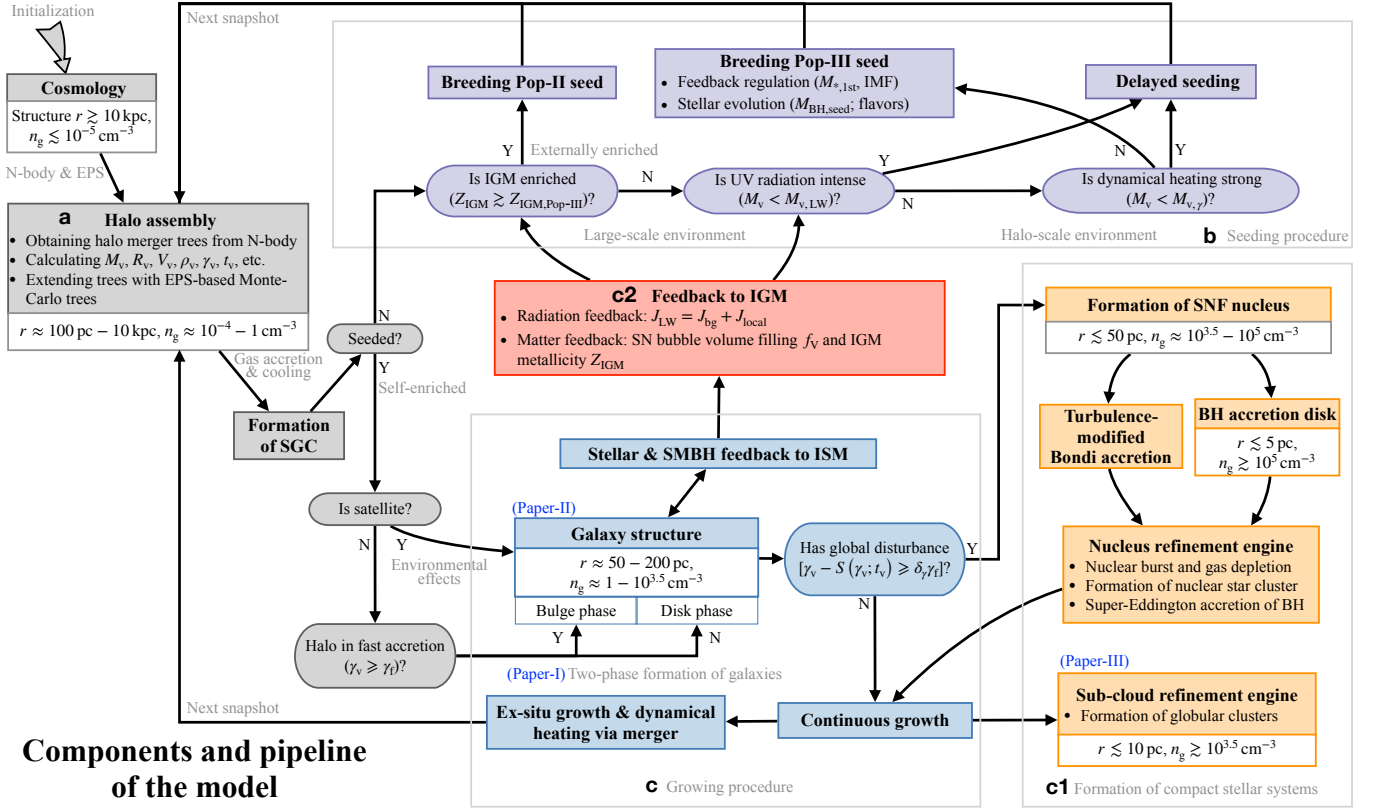


Figure D1. Components and pipeline of the model. The model is built upon the Λ CDM paradigm, takes halos and their assembly histories as input (a), and uses two main procedures, the **seeding procedure** (b; see Fig. 3 for an outline of the relevant physical processes), and the **growing procedure** (c; see Fig. 4 for the relevant physical processes), respectively, to seed and grow SMBHs with their host galaxies and halos. Inset c1, sub-procedure for modeling compact stellar systems, including the refinement engines for nuclear bursts and for the formation of GCs. Inset c2, sub-procedure for modeling radiation and matter feedback of galaxy formation to the large-scale environment, which provides environmental conditions for the seeding procedure and serves as an interface connecting the two main procedures. Gas densities (n_g) of structures at different scales are listed, taking the progenitors of MW-size galaxies at $z = 9$ as an example (see Fig. A1 for the profile). See Fig. 1 for a brief version of this figure, and Appendix D for the implementation details.

2004; Moster et al. 2011; Chen et al. 2019, 2023b; Jespersen et al. 2025).

D2 The global integrator

The seeding procedure in our model is built on two factors in the large-scale environment of subhalos: the LW radiation and the IGM metallicity. Both factors are self-consistently modeled by tracing the radiation and matter feedback from all other subhalos that have already formed stars by the time step in question. Entanglement among subhalo merger trees is thus inevitable and the integration of the formation histories of galaxies must be performed globally, rather than individually on each subhalo merger tree. Here we refer to this integration method as the ‘global integrator’, and we implement it as follows.

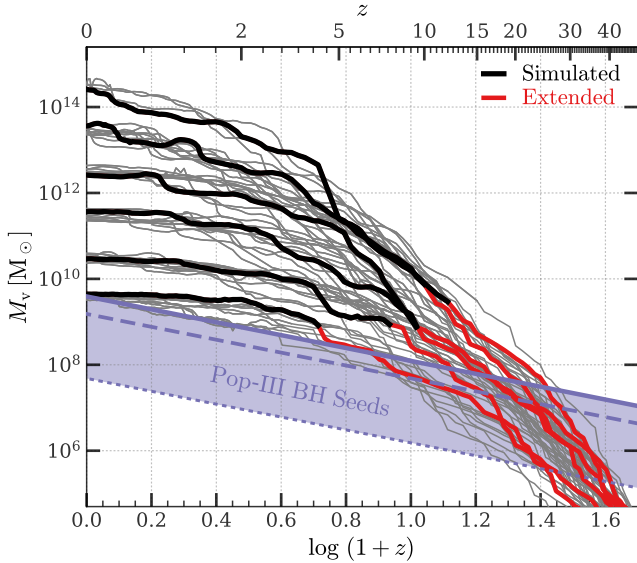
- (i) Starting from the snapshot with the highest z , we iterate over all snapshots towards the lower z until $z = 0$.
- (ii) For each snapshot s (at redshift z_s), we take all the N_s subhalos, $H_s \equiv \{h_i\}_{i=1}^{N_s}$, belonging to s from the extended merger trees constructed in Appendix D1. For each subhalo $h \in H_s$, we follow the algorithms to be detailed in Appendices D5 and D6, respectively, to find the LW intensity (J_{LW}) and the IGM metallicity (Z_{IGM}) at $\mathbf{x}_h$, the location of h . This step is schematically shown in Fig. D1c2 as a feedback loop between the two main procedures (seeding and growing).
- (iii) We check whether h has already formed stars at $z > z_s$. If so, we

assume the galaxy in h has been self-enriched, and branch to the growing procedure to be detailed in Appendix D4 (see also Fig. D1c, including the inset c1, for an outline) to model the post-seeding growth of the galaxy. If not, we follow the seeding procedure to be detailed in Appendix D3 (see also Fig. D1b for an outline) to check whether the condition for the collapse of gas in h is satisfied, and breed a BH seed if the condition is satisfied. Otherwise, we assume that h remains ‘dark’, without any star or BH formed at z_s .

With this global integrator, each subhalo is assigned with a seeding epoch at which the first generation of stars and BH seed is bred, and the post-seeding growth is driven by different channels that gradually build up the BH, stellar and gas contents of the subhalo. As demonstrated by Chen et al. (2023b), both physical and observational uncertainties in the predicted quantities can introduce biases in statistical properties of galaxy populations, especially for rare objects. Accurately accounting for such uncertainties is almost impossible, and here we provide a tentative solution by injecting stochasticity into the predicted quantities. As shown in e.g. Chen et al. (2021a) and Chen et al. (2021b) using hydro simulations and regressors with successively refined predictors, many nuanced factors other than halo properties can produce galaxy-to-galaxy variance in the growth, and such variance may thus be missed by a halo-based model. To account for such missed variance, we introduce stochastic noise, with a log-normal distribution that has a zero mean and a standard deviation

Table D1. List of model components and default parameters for each component. See Fig. D1 for a flowchart outlining the components and pipeline, and Appendix D for the implementation details and the choices of parameters in the *Default* model.

Model Component	Default Parameters	Outline and Implementation
Extension of subhalo merger trees	$N_{\text{MC}} = 512$; $M_{\text{v,anc}} = 5 \times 10^8 h^{-1} \text{M}_\odot$	Fig. D1a and Appendix D1
The global integrator	$\sigma_{\text{M}} = 0.2 \text{ dex}$	Appendix D2
The seeding procedure (Appendix B)	Pop-III threshold: $Z_{\text{IGM,Pop-III}} = 10^{-4} Z_\odot$; Delay of seeding: $J_{\text{crit},21} = 7.5$; $\beta_{\text{J}} = 1.5$; $\gamma_{\text{crit}} = 10$; $\beta_\gamma = 2$; Masses of seeds: $\eta_{\text{f}} = 25$; $\beta_{\text{m}} = 1$; $M_{\text{BH,Pop-II}} = 10 \text{M}_\odot$; $M_{*,\text{Pop-II}} = 100 \text{M}_\odot$; $M_{\text{g,seed}} = 10 \text{M}_\odot$	Fig. D1b and Appendix D3
The growing procedure (Appendix C)	Halo assembly: $\gamma_{\text{f}} = 3/16$; Gas cooling: $M_{\text{cool,f}} = 10^{13} h^{-1} \text{M}_\odot$; $\beta_{\text{cool,f}} = 4$; $M_{\text{cool,s}} = 10^{11.9} h^{-1} \text{M}_\odot$; $\beta_{\text{cool,s,0}} = 3.6$; $\gamma_{\text{cool,s}} = -0.72$; Feedback (continuous mode): $\alpha_{\text{SN,f}} = 0$; $\beta_{\text{SN,f}} = 2.5$; $V_{\text{w}} = 250 \text{ km s}^{-1}$; $R_{\text{s}} = 10^{-0.96}$; $\beta_{\text{SN,s}} = 1.9$; $\alpha_{\text{AGN,f,0}} = 5 \times 10^{-3}$; $\beta_{\text{AGN,f}} = 2$; $\beta_{\text{AGN,s}} = 1.8$; $f_{\text{ej,f}} = 0.75$; $f_{\text{ej,s}} = 1$; Star formation and BH accretion (continuous mode): $\epsilon_{*,\text{f}} = 0.75$; $\epsilon_{*,\text{s}} = 0.32$; $R = 0.4$; $\alpha_{\text{cap,f}} = 2.5$; $\alpha_{\text{cap,s}} = 0$; Satellite evolution: $\tau_{\text{sat}} = 4 \text{ Gyr}/\ln 10$; SNF nucleus: $\delta_\gamma = 1$; $n_{\text{snf}} = 10^{3.5} \text{ cm}^{-3}$; $p = 0$; $q = 0.5$; BH accretion disk: $\zeta_{\text{t}} = \zeta_{\text{A}} = 1/3$; $\psi_{\text{t}}/\psi_{\text{A}} = 1$; Nuclear burst: $\xi_{\text{AGN},\infty} = 1$; $v_{\text{out}} = 10^3 \text{ km s}^{-1}$; $\beta_{\text{out}} = 0$; $\alpha_{\text{SF}} = 1$; $\alpha_{\text{BH}} = 1$; $\delta_{\text{Q}} = 1$; $f_{\text{dry}} = 6.8 \times 10^{-3}$; Merger and dynamical heating: $f_{\text{merge}} = 0.5$; $f_{\text{heat}} = 0.5$; Metallicity: $V_{\text{esc}} = 75 \text{ km s}^{-1}$; $\beta_{\text{esc}} = 2$; $z_{\text{esc}} = 5$; $\gamma_0 = 1$; $\beta_{\text{mix}} = 2$	Fig. D1c and Appendix D4
The model of LW radiation field	$f_{\text{esc}} = 1$; $\eta_{\text{LW}} = 4 \times 10^3$; $Q_{\text{LW}} = 10^{36} \text{ erg} \cdot \text{s}^{-1} \text{M}_\odot^{-1}$; $J_{\text{local,th},21} = 0.01$	Fig. D1c2 and Appendix D5
The model of IGM enrichment	$A_{\text{sh}} = 1$; $\epsilon_{\text{SN}} = 10^{51} \text{ erg} \cdot \text{M}_\odot^{-1}$; $l_{\text{sm}} = 250 h^{-1} \text{ kpc}$	Fig. D1c2 and Appendix D6

**Figure D2.** Extension of subhalo merger trees. Here we show the main-branch mass assembly histories (MAHs) for examples of central subhalos at $z = 0$ with masses separated by 1 dex in the range of $10^{9.5} - 10^{14.5} \text{M}_\odot$. For each curve, **black** part is obtained directly from the simulation, which is truncated at the anchor point due to limited resolution; **red** part shows the extension by our EPS-based Monte Carlo algorithm that smoothly joins the simulated part at the anchor point. Each bundle of **grey** curves shows the MAHs of up to 10 additional halos with masses at $z = 0$ similar to the example. The extension ensures that the assembly histories can be traced back to sufficiently high z and sufficiently low M_{v} so that the formation of Pop-III stars and BH seeds can be captured, as indicated by the **purple** band (the same as that in Fig. 2a). See Appendix D1 for details.

of $\sigma_{\text{M}} = 0.2 \text{ dex}$, to the modeled amount of growth, ΔM_{BH} and ΔM_* , in every time step, at every growth stage (fast or slow phases, or satellite stage; see below) and through every growth channel (seeding, or bursty or continuous modes). In comparison with observations, when appropriate, we additionally add stochastic noise to mimic observational uncertainties, which will be clearly stated when the comparison is made (see Fig. 12 and §3.3).

D3 The seeding procedure

The seeding procedure takes the internal and environmental properties of a subhalo $h \in H_s$ that has not yet formed any star as input, determines whether or not the first generation of stars and the BH seed can form in h at z_s , and breeds stars and BH seed. The procedure follows prescriptions presented in Appendix B, and is implemented through the following steps (see Fig. D1b for an outline).

- (i) We check the IGM metallicity, Z_{IGM} , at $\mathbf{x}_h$. If $Z_{\text{IGM}} \geq Z_{\text{IGM,Pop-III}}$, we populate h with a Pop-II star cluster, breed a BH seed from the cluster, and end the seeding procedure for h . The mass of the BH seed is set to be $M_{\text{BH,Pop-II}}$, and the mass of the remaining stars in total is set to be $M_{*,\text{Pop-II}}$.
- (ii) Otherwise ($Z_{\text{IGM}} < Z_{\text{IGM,Pop-III}}$), the subhalo is in the Pop-III regime. We take the LW intensity (J_{LW}) at $\mathbf{x}_h$ and the specific accretion rate (γ_{v}) of h , and use Eq. (5) to find the threshold of halo mass ($M_{\text{v,th}}$) for the gas in h to collapse. If the mass of h is below the threshold ($M_{\text{v}} < M_{\text{v,th}}$), we assume that h forms no star nor BH at z_s , and we end the seeding procedure for h . Otherwise ($M_{\text{v}} \geq M_{\text{v,th}}$), we assume that a Pop-III star cluster forms in h , and proceed to the next step to breed a BH seed from the cluster.
- (iii) To determine the properties of the Pop-III star cluster, we first evaluate the properties of the SGC in h according to Eqs. (B1)–(B5), assuming that a pristine halo has a baryon fraction equal to the cosmic value, f_{b} (see texts around Eq. A1). We then use the radiation-regulated scenario (Appendix B2) to determine the mass

of the dominant star in the cluster. Specifically, we use Eq. (B17) to compute the zero-age main sequence star formation efficiency (ϵ_{ZAMS}) and luminosity (L_{ZAMS}) of the dominant star. If this luminosity does not exceed the Eddington luminosity ($L_{\text{ZAMS}} < L_{\text{Edd}}$), the mass of the dominant star is set to be $M_{*,1\text{st}} = \epsilon_{\text{ZAMS}} M_{\text{SGC}}$. Otherwise ($L_{\text{ZAMS}} \geq L_{\text{Edd}}$), the star reaches the Eddington-level accretion, and we turn to Eq. (B20) to find the corresponding star formation efficiency (ϵ_{Edd}) and set $M_{*,1\text{st}} = \epsilon_{\text{Edd}} M_{\text{SGC}}$.

- (iv) With the mass of the dominant star, the IMF of the entire Pop-III cluster is obtained by Eq. (B23), which is a power-law function with an index of $-\beta_{\text{m}}$ and an exponential cutoff at $M_{*,1\text{st}}$. The normalization of the IMF is obtained by solving Eq. (B24) that equates the integral of the IMF over $[M_{*,1\text{st}}, +\infty]$ to 1. This implies that the instantiation of stars from the IMF is a process of ‘optimal sampling’, which is supported by observations of star clusters (Yan et al. 2023), and is also consistent with theoretical expectations under our assumption of self-regulation (Kroupa et al. 2013; Schulz et al. 2015).
- (v) The evolution of the dominant star is treated by a simplified piecewise model (see Appendix B2), with two mass thresholds, $M_{*,\text{PISN}}$ and $M_{*,\text{DCBH}}$. If $M_{*,1\text{st}} \geq M_{*,\text{DCBH}}$, the star ends its life as a DCBH, which we use as the BH seed. The mass of the seed is set to be $M_{\text{BH},\text{seed}} = M_{*,1\text{st}}$, assuming no significant mass loss during the stellar evolution and the direct collapse. If $M_{*,\text{PISN}} \leq M_{*,1\text{st}} < M_{*,\text{DCBH}}$, the star ends its life by PISN which totally disrupts the star and leaves no remnant. In this case, we seek for the most massive star (with a mass $M_{*,\text{surv}}$) that can survive PISN by requiring the integral of the IMF over $[M_{*,\text{surv}}, M_{*,\text{PISN}}]$ to be equal to 1 (Eq. B27), again implying optimal sampling. The remnant of this star is treated as the BH seed, and the mass of the seed is set to be $M_{\text{BH},\text{seed}} = f_{\text{rem}} M_{*,\text{surv}}$, accounting for a fraction of $1 - f_{\text{rem}}$ of mass loss during the stellar evolution. If $M_{*,1\text{st}} < M_{*,\text{PISN}}$, the dominant star ends its life by CCSN, and leaves a BH remnant with a mass of $M_{\text{BH},\text{seed}} = f_{\text{rem}} M_{*,1\text{st}}$, which is adopted as the BH seed. Once the BH seed is determined, the evolution and remnant mass of any other star in the cluster is obtained similarly by the piecewise model, weighted by the IMF, and integrated over $[M_{*,\text{min}}, M_{*,1\text{st}}]$ (for DCBH or CCSN) or $[M_{*,\text{min}}, M_{*,\text{surv}}]$ (for PISN) to obtain $M_{*,\text{seed}}$, the stellar mass bred together with the BH seed.

The default values of parameters are chosen as follows. The critical metallicity is set to be $Z_{\text{IGM,Pop-III}} = 10^{-4} Z_{\odot}$ (see Appendix B2). The parameters describing the dependence of the collapse mass on LW radiation are set to be $J_{\text{crit},21} = 7.5$ and $\beta_{\text{J}} = 1.5$ (see Appendix B1.3), and on dynamical heating are set to be $\gamma_{\text{crit}} = 10$ and $\beta_{\gamma} = 2$ (see Appendix B1.4). The coupling efficiency in the radiation-regulated scenario is set to be $\eta_{\text{r}} = 25$ (see Appendix B2). The index of the IMF of Pop-III stars is set to be $\beta_{\text{m}} = 1$ (see Appendix B2). The mass of a Pop-II BH seed is set to be $M_{\text{BH,Pop-II}} = 10 M_{\odot}$, a typical value for the remnant of a massive star in normal star clusters. Unfortunately, the stellar mass associated with a Pop-II BH seed is unknown, and so an assumption has to be made. We notice that the bursty mode dominates the growth of galaxies after the seeding, erases the difference produced by the seeding method, and sustains a ratio of $M_{\text{BH}}/M_{*} \approx 1/10$ (see §3.2.1, Figs. 9 and 12). Thus, we manually set $M_{*,\text{Pop-II}} = 10 M_{\text{BH,Pop-II}}$, so that a Pop-II seed appears in an $M_{\text{BH}}-M_{*}$ relation similar to that shaped by the post-seeding growth without introducing an artificial discontinuity. In both Pop-II and Pop-III cases, the gas density at fragmentation is expected to be high due to the lack of effective metal cooling, and thus the seeding process may be bursty. This implies a strong gas depletion during seeding, and a short period of quiescent phase after seeding, as discussed in the beginning of §2.4. For this

reason, we set the remaining gas mass after seeding to be negligibly small, $M_{\text{g,seed}} = 10 M_{\odot}$.

With the seeding procedure, the halo h either remains ‘dark’, or is populated with initial values of $M_{\text{BH,seed}}$, $M_{*,\text{seed}}$ and $M_{\text{g,seed}}$ so that the post-seeding processes modeled by the growing procedure are provided with proper initial conditions.

D4 The growing procedure

The growing procedure takes a subhalo $h \in H_s$ that has already been seeded as the input, and updates the properties of the galaxy in h by modeling the evolution within the current time step (i.e. the duration since the previous snapshot, $s - 1$, to the current snapshot s). The growing procedure is based on the last version of our model presented in §2 of Paper-III, with a major extension to include the growth in the bursty mode proposed in this paper and a few updates on the other components of the model to accommodate the extension. Unless otherwise specified, the default parameters adopted in this paper are taken from Paper-III.

In terms of the underpinning of BH growth, the assembly of dark matter halos (quantified by γ_{v} ; see Eq. B9) is expected to be the key factor that determines the gas dynamics of the central galaxy hosted by the halo, and thus also determines how the central BH hosted by the galaxy accretes the gas and reacts via AGN feedback (see the discussion in Appendix C1 and summary in Fig. 4). Specifically, the BH in a halo during the fast phase grows efficiently by capturing the sub-clouds formed out of the dynamically hot (turbulent) SGC, while the BH in a halo during the slow phase grows inefficiently due to the barrier of angular momentum (j) associated with the formation of dynamically cold disk. In addition, a global disturbance, quantified in this paper by a temporary excursion of the halo assembly to high γ_{v} , induces violent gas inflow that causes the formation of an SNF nucleus, followed by a nuclear burst that increases the mass of the BH.

The above physical considerations motivate us to treat galaxy formation as a multichannel process (see §3.2.1, Figs. 8 and 9). To aid the implementation of the model, we separate each branch B of subhalo merger trees constructed in Appendix D1 into several stages/phases, so that numerical prescriptions for modeling galaxies hosted by subhalos during each stage can be applied separately. For this, we first identify the ‘infall snapshot’ (s_{infall}) of B , defined as the last snapshot in which the subhalo in B is still a central subhalo. Subhalos at and before s_{infall} are defined to be in the ‘central stage’, and those after s_{infall} are defined to be in the ‘satellite stage’. For numerical convenience, we set the value of M_{v} for any subhalo in the satellite stage to be that at s_{infall} , effectively assuming that there is no growth nor excursion of halo assembly in the satellite stage. For the entire branch, we smooth the MAH (i.e. values of M_{v}) by a running Gaussian kernel with a standard deviation equal to the dynamical timescale of the halo (t_{v} ; see Eq. A7), and denote the smoothed MAH as $\mathcal{S}(M_{\text{v}}; t_{\text{v}})$. We then perform a numerical differentiation to obtain the smoothed version of the specific accretion rate from $\mathcal{S}(M_{\text{v}}; t_{\text{v}})$, and denote it as $\mathcal{S}(\gamma_{\text{v}}; t_{\text{v}})$. If $\mathcal{S}(\gamma_{\text{v}}; t_{\text{v}}) \geq \gamma_{\text{f}}$, where the default value of the threshold is $\gamma_{\text{f}} = 3/16$, the subhalo is defined to be in the fast phase; otherwise, it is defined to be in the slow phase. Note that this separation of fast and slow phases is slightly different from the one with a single transition redshift z_{f} used in previous papers of this series: the excursion of the halo assembly to the slow (fast) phase on a timescale of t_{v} can now take place at higher (lower) z , thus allowing the model to capture some recurrent transition of galaxy dynamics that has been evident in observations (Parlanti et al. 2023; Rowland et al. 2024; Scholtz et al. 2025; Danhaive et al. 2025; see also the

discussion in Appendix C of [Paper-I](#). For convenience, we use B_{fast} , B_{slow} and B_{sat} to denote the sets of subhalos in B during the fast phase, slow phase and satellite stage, respectively.

During the current time step, we determine whether the subhalo h has a temporary excursion to high γ_v by examining the condition $\gamma_v - \mathcal{S}(\gamma_v; t_v) \geq \delta_\gamma \gamma_f$, with the default value of $\delta_\gamma = 1$ (see Eq. C3). If the condition is not satisfied, we assume that the growth in the bursty mode is not triggered during this time step. Otherwise, we assume that a global disturbance has been imparted to the SGC, leading to the formation of an SNF nucleus in the SGC center, followed by a nuclear burst that depletes the gas from the nucleus. Note that such a burst takes place in a much shorter timescale ($\lesssim \text{Myr}$; see Fig. C1) and within a much smaller volume (see Fig. A1) than those modeled using the global properties for the host galaxy. To properly treat this small-scale process, a refinement engine is designed and applied on top of the large-scale modeling (see Appendix C5). Another example of small-scale processes is the formation of individual sub-clouds and star clusters, which is modeled using a similar refined treatment, as detailed in [Paper-III](#). Since these small-scale processes are critical in determining the growth of compact objects, such as GCs, NSCs and SMBHs, which are of particular interest in recent observations at high z , we represent them as a separate inset c1 in Fig. D1.

Other unresolved processes, such as stellar evolution and baryon cycling in individual galaxies, are treated by an instantaneous approximation given that their timescales are short compared with the assembly of halos. Hence, the stellar mass formed in any galaxy during any time step, ΔM_* , should be understood as the remaining mass of stars after stellar evolution. On the other hand, if instantaneous (unevolved) star formation rate (SFR) is required, we obtain it by

$$\text{SFR} = \frac{1}{1-R} \frac{\Delta M_*}{\Delta t}, \quad (\text{D1})$$

where $R = 0.4$ is the returned fraction of mass assuming a Chabrier IMF ([Bruzual & Charlot 2003](#)) for stellar populations other than Pop-III. Note that in Fig. 10, we have used this unevolved SFR to produce the cosmic star formation history.

With the above definitions, the steps of the growing procedure for a subhalo $h \in H_s$ are as follows (see also Fig. D1c for an outline).

- (i) We find the branch B that contains h , and determine which stage/phase h is in ($h \in B_{\text{fast}}$, $h \in B_{\text{slow}}$ or $h \in B_{\text{sat}}$). If $h \in B_{\text{sat}}$, we jump to step (ix); otherwise, we continue with the next step to model the growth of the BH with its host galaxy and subhalo in the continuous mode.
- (ii) We compute the total amount of gas in h that is available for subsequent processes:

$$\Delta M_{g,\text{avail}} = \begin{cases} f_b \Delta M_v, & \text{if } h \in B_{\text{fast}}; \\ \frac{f_b M_v}{\tau(z)} \Delta t, & \text{if } h \in B_{\text{slow}}, \end{cases} \quad (\text{D2})$$

where $\tau(z) = (1+z)^{-3/2}/(10H_0)$ is an approximation to the dynamical timescale of the halo (Eq. A7), Δt is the duration of the current time step, and ΔM_v is the change of halo mass in this duration. Note that the equation in the second line is expressed by M_v , instead of ΔM_v , which takes into account that halo growth ceases in the slow phase, and the gas processed earlier and retained in the halo is recycled back to fuel star formation and BH growth (see e.g. §3.2.2 and Fig. 10 for a discussion).

- (iii) The fraction of available gas in h that can cool down is determined

by a cooling factor, F_{cool} , which is

$$F_{\text{cool}} = \begin{cases} \left[1 + \left(\frac{M_v}{M_{\text{cool},f}} \right)^{\beta_{\text{cool},f}} \right]^{-1}, & \text{if } h \in B_{\text{fast}}; \\ \left(1 + \frac{M_v}{M_{\text{cool},s}} \right)^{-\beta_{\text{cool},s}(z)}, & \text{if } h \in B_{\text{slow}}. \end{cases} \quad (\text{D3})$$

The factor takes into account the fact that cooling is effective only in halos with mass smaller than a threshold of $\sim 10^{12} M_\odot$, and that only a fraction of gas in halos with masses larger than the threshold can cool down and flow into the galaxy (see e.g. the cooling curves in fig. 2 of [Paper-I](#)). The default parameters are calibrated to be $M_{\text{cool},f} = 10^{13} h^{-1} M_\odot$, $\beta_{\text{cool},f} = 4$, $M_{\text{cool},s} = 10^{11.9} h^{-1} M_\odot$, $\beta_{\text{cool},s} = \beta_{\text{cool},s,0}(1+z)^{\gamma_{\text{cool},s}}$, $\beta_{\text{cool},s,0} = 3.6$ and $\gamma_{\text{cool},s} = -0.72$. Note that the redshift dependence of both M_v and $\beta_{\text{cool},s}$ makes F_{cool} a function of redshift. With the cooling factor, the amount of gas that can cool down and flow into the galaxy is

$$\Delta M_{g,\text{cool}} = F_{\text{cool}} \Delta M_{g,\text{avail}}, \quad (\text{D4})$$

and the remaining fraction, $1 - F_{\text{cool}}$ is referred to as the ‘hot’ gas. Note that the reduction of the specific growth rate of gas mass in the massive example at low z in Fig. 9c8 is partly due to the reduction of F_{cool} .

- (iv) Owing to feedback effects from star formation and AGN, only a fraction of the cooled gas can cool further and become star-forming gas. We model the amount of star-forming gas by modulating the amount of cooled gas $\Delta M_{g,\text{cool}}$ with two additional factors describing the two sources of feedback, as:

$$\Delta M_{g,\text{SF}} = F_{\text{SN}} F_{\text{AGN}} \Delta M_{g,\text{cool}}. \quad (\text{D5})$$

Here, F_{SN} describes the effect of supernova feedback and is modeled as

$$F_{\text{SN}} = \begin{cases} \frac{\alpha_{\text{SN},f} + (V_g/V_w)\beta_{\text{SN},f}}{1 + (V_g/V_w)\beta_{\text{SN},f}}, & \text{if } h \in B_{\text{fast}}, \\ \left(\frac{M_v/M_{\text{cool},s}}{R_s + M_v/M_{\text{cool},s}} \right)^{\beta_{\text{SN},s}}, & \text{if } h \in B_{\text{slow}}, \end{cases} \quad (\text{D6})$$

where the default parameters are set to be $\alpha_{\text{SN},f} = 0$, $\beta_{\text{SN},f} = 2.5$, $V_w = 250 \text{ km s}^{-1}$, $R_s = 10^{-0.96}$, $\beta_{\text{SN},s} = 1.9$, and $V_g = V_{\text{max}}$ is the typical velocity dispersion of the galaxy. F_{AGN} describes the effect of AGN feedback, and is given by

$$F_{\text{AGN}} = \begin{cases} \exp \left[-\frac{\alpha_{\text{AGN},f}(z) M_{\text{BH}} c^2}{M_b V_g^2} \right], & \text{if } h \in B_{\text{fast}}; \\ \left(\frac{R_s + M_v/M_{\text{cool},s}}{1 + M_v/M_{\text{cool},s}} \right)^{\beta_{\text{AGN},s}}, & \text{if } h \in B_{\text{slow}}. \end{cases} \quad (\text{D7})$$

Here, M_b is the total baryon mass (gas, stars and BH) of the bulge of the galaxy, and the default parameter $\beta_{\text{AGN},s} = 1.8$. The exponential function for the fast phase is similar to the one adopted in [Paper-I](#) at small M_{BH} , but approaches to zero more smoothly at large M_{BH} . Different from previous papers of this series, the efficiency of AGN feedback in the fast phase, as quantified by $\alpha_{\text{AGN},f}(z)$, is now modeled to be redshift dependent. This is partly motivated by the theoretical consideration that the fraction of feedback energy that can be dissipated depends on the cooling timescale of the gas affected by the feedback, and thus also depends on z , owing to the cosmic evolution of the SGC density (Eq. B2). This is also supported by empirical evidence that star formation efficiency in high-mass halos is redshift dependent (e.g. [Lu et al. 2014](#); [Behroozi et al. 2020](#)). Hence,

we parameterize the factor $\alpha_{\text{AGN},f}(z)$ as

$$\alpha_{\text{AGN},f}(z) = \alpha_{\text{AGN},f,0}(1+z)^{-\beta_{\text{AGN},f}}. \quad (\text{D8})$$

The value of $\beta_{\text{AGN},f}$ is not known, but the bound of it may be obtainable from a timescale argument that takes into account the competition of energy injection and dissipation. If we consider an extreme case that the accretion of BH smoothly tracks the growth of the host SGC and thus the timescales of BH growth and injection of feedback energy follow the free-fall timescale of the SGC ($t_{\text{ff},\text{SGC}}$; Eq. B5), the feedback efficiency may be approximated as $\alpha_{\text{AGN},f} \sim t_{\text{cool},\text{SGC}}/t_{\text{ff},\text{SGC}} \propto n_{\text{SGC}}^{-1/2}$, where the cooling timescale $t_{\text{cool},\text{SGC}} \propto n_{\text{SGC}}^{-1}$ holds for any two-body radiative process (see e.g. Appendix B1.4 of Mo et al. 2010). In the other extreme, if the accretion of BH is a highly localized process with a microscopic timescale irrespective of the SGC density, the feedback efficiency may be approximated as $\alpha_{\text{AGN},f} \propto t_{\text{cool},\text{SGC}} \propto n_{\text{SGC}}^{-1}$. The above timescale argument suggests a bound for the feedback efficiency as $\alpha_{\text{AGN},f} \propto n_{\text{SGC}}^{[-1, -1/2]}$, and a more realistic value may be somewhere in between. In this paper, we adopt the default as $\alpha_{\text{AGN},f} \propto n_{\text{SGC}}^{-2/3}$ (i.e. $\beta_{\text{AGN},f} = 2$; see Eq. B2 for n_{SGC}), which falls between the two extremes. The default value of the normalization factor is set to be $\alpha_{\text{AGN},f,0} = 5 \times 10^{-3}$, so that $\alpha_{\text{AGN},f}$ at $z \approx 2$ (the redshift at which massive halos with $M_v \gtrsim 10^{13} M_\odot$ ends their fast phase; see Fig. 2b) is comparable to the value calibrated in Paper-I.

- (v) Part of the feedback-affected gas is expected to be expelled from the galaxy (referred to as ‘ejected gas’), while the rest remains hot in the galaxy and is prevented from forming stars (referred to as ‘prevented gas’). To separate the two components, we introduce a parameter f_{ej} , which is the fraction of the feedback-affected gas that is ejected from the galaxy. Thus, the amounts of ejected and prevented gas are

$$\Delta M_{\text{g,ej}} = f_{\text{ej}}(1 - F_{\text{SN}} F_{\text{AGN}}) \Delta M_{\text{g,cool}}, \quad (\text{D9})$$

and

$$\Delta M_{\text{g,prev}} = (1 - f_{\text{ej}})(1 - F_{\text{SN}} F_{\text{AGN}}) \Delta M_{\text{g,cool}}, \quad (\text{D10})$$

respectively. Following Paper-III, we set the default value to be $f_{\text{ej}} = f_{\text{ej},f} = 0.75$ for the fast phase, so that a galaxy well regulated by feedback has a gas fraction comparable to the typical spin parameter, 0.04, of dark matter halos, and thus can transit to a stable disk in the slow phase. For the slow phase, we set $f_{\text{ej}} = f_{\text{ej},s} = 1$, meaning that all feedback-affected gas is ejected from the disk, with some recycled back later on a dynamical timescale of the host halo (see Eq. D2).

- (vi) The amount of stars formed in the galaxy is related to the amount of star-forming gas, $\Delta M_{\text{g,SF}}$, by a star formation efficiency, ϵ_* , as

$$\Delta M_* = \epsilon_* \Delta M_{\text{g,SF}}, \quad (\text{D11})$$

where $\epsilon_* = \epsilon_{*,f} = 0.75$ and $\epsilon_* = \epsilon_{*,s} = 0.32$ for the fast and slow phases, respectively, as calibrated in Paper-I. The stellar mass so obtained is added to the bulge component of the galaxy if $h \in B_{\text{fast}}$, or to the disk component if $h \in B_{\text{slow}}$.

- (vii) The growth of the mass of the central BH is obtained by the fraction of low- j sub-clouds that can approach the center of the galaxy and be captured by the BH, which is derived in Paper-I as

$$\Delta M_{\text{BH}} = \alpha_{\text{cap}} \frac{M_{\text{BH}}}{M_{\text{b}}} \Delta M_{\text{g,SF}} \quad (\text{D12})$$

where $\alpha_{\text{cap}} = \alpha_{\text{cap},f} = 2.5$ in the fast phase, and $\alpha_{\text{cap}} = \alpha_{\text{cap},s} = 0$ in the slow phase due to the full mixing of gas. Note that the ‘enhancement factor’ is now removed from the equation, as it is replaced explicitly by the nuclear burst (see steps x and xi below).

- (viii) The change of gas mass in the galaxy can be obtained by mass conservation, taking into account the contributions by cooling, consumption

by star formation and BH accretion, and ejection by feedback, as

$$\Delta M_{\text{g}} = \Delta M_{\text{g,SF}} - \Delta M_* - \Delta M_{\text{BH}} + \Delta M_{\text{g,prev}}. \quad (\text{D13})$$

Similar to the treatment for M_* , the update of gas mass is performed on the bulge and disk components, if $h \in B_{\text{fast}}$ and $h \in B_{\text{slow}}$, respectively. Note that the mass and specific rate of gas in Fig. 9 (rows 7 and 8) are obtained with this equation.

- (ix) If the subhalo is in the satellite stage ($h \in B_{\text{sat}}$), the star formation efficiency is expected to be low due to the environmental effects, such as ram-pressure stripping (Gunn & Gott 1972; Du et al. 2019; Kulier et al. 2023; Carnall et al. 2023) and tidal stripping (Binney & Tremaine 2008; Read et al. 2006; Drakos et al. 2022). Motivated by the observational results of e.g. Peng et al. (2015), we model the SFR in this stage by an exponential decay, as

$$\dot{M}(t) = \dot{M}(t_{\text{infall}}) \exp\left(-\frac{t - t_{\text{infall}}}{\tau_{\text{sat}}}\right), \quad (\text{D14})$$

where $t_{\text{infall}} \equiv t(s_{\text{infall}})$ is the cosmic time at the infall snapshot, and $\tau_{\text{sat}} = 4 \text{ Gyr}/\ln 10$ is the timescale of the decay. The growth of BH is assumed to be stopped in this stage. The amount of gas is updated according to the ‘close-box’ model with instantaneous recycling: the supply of gas is stopped by the environmental effects, and the feedback-affected gas is instantaneously recycled into the ISM. With these assumptions, the only sink term of gas is the consumption of star formation.

- (x) If a nuclear burst is triggered within the current time step, we take M_{g} of bulge (obtained from Eq. D13) as the mass of SGC (M_{SGC}), derive the radius fraction, $f_r \equiv R_{\text{SGC}}/R_v$, by the self-gravitating condition, as $f_r = M_{\text{SGC}}/M_v$ (see Eq. 26 of Paper-I), and use Eqs. (B1)–(B5) to compute the properties of the SGC. Using the properties of the SGC, we move a step inward to the galactic nucleus according to double power-law density profile of the entire SGC (Eq. C5; see also Fig. A1 for examples), and compute the properties of the SNF nucleus using Eqs. (C6)–(C9). The default parameters to define the SNF nucleus include the supernova-free density, $n_{\text{snf}} = 10^{3.5} \text{ cm}^{-3}$, and the structural parameters, $p = 0$ and $q = 0.5$. Moving a step further into the zone of influence of the BH ($r \leq R_{\text{B}}$), we follow Eqs. (C38)–(C41) to derive the structural properties of the accretion disk. The default parameters we adopt for the accretion disk correspond to those describing a turbulent disk with flux-frozen magnetic fields: $\zeta_t = \zeta_A = 1/3$ and $\psi_t/\psi_A = 1$.

- (xi) Using the properties of the SNF nucleus, we numerically solve the continuity equation, Eq. (C45), that describes the evolution of gas profile in the SNF nucleus during the nuclear burst, with specific choices of functional forms detailed in Appendix C5.2. The default parameters controlling the burst include those describing the AGN feedback to the nucleus, $\xi_{\text{AGN},\infty} = 1$, $v_{\text{out}} = 10^3 \text{ km s}^{-1}$ and $\beta_{\text{out}} = 0$; star formation efficiency in the nucleus, $\alpha_{\text{SF}} = 1$ and $\delta_0 = 1$; and BH accretion efficiency, $\alpha_{\text{BH}} = 1$. The numerical solver, referred to as a ‘refinement engine’ in this paper, works with a time step of the order of a year, much finer than the global integrator so that the $\lesssim \text{Myr}$ timescale of the nuclear burst can be resolved. Examples for the evolution of BH mass, stellar mass, and gas mass during individual nuclear bursts are shown in Fig. C1. Here we take the changes in these masses, $\Delta M_{\text{BH,burst}}$, $\Delta M_{*,\text{burst}}$ and $\Delta M_{\text{g,burst}}$ during the entire burst as outputs of the refinement engine, and add them to the BH mass, stellar mass and gas mass of the bulge of the galaxy, respectively. BH growth through gas accretion in this way is referred to as the ‘wet’ channel in the bursty mode.

- (xii) The energy of AGN feedback released in the nuclear burst could have long-lasting effects on the gas on larger scales, which is also modeled here by a formulation similar to the AGN feedback adopted for the

continuous growth of the BH. Specifically, we remove a fraction

$$F_{\text{ej,burst}} = f_{\text{ej}} \left\{ 1 - \exp \left[-\frac{\alpha_{\text{AGN},f}(z) \Delta M_{\text{BH,burst}} c^2}{M_{\text{b}} V_{\text{g}}^2} \right] \right\} \quad (\text{D15})$$

of gas from that remaining in the bulge, with the parameters $\alpha_{\text{AGN},f}(z)$ and f_{ej} exactly the same as those used in Eqs. (D7) and (D9), respectively. This implies that the coupling of AGN feedback with the SGC due to a nuclear burst is assumed to be similar to that due to the continuous growth of the BH, which is certainly a simplified treatment and aims to avoid an over-complication of the model.

- (xiii) After a nuclear burst, stars formed in the galactic center may continuously adjust their orbits via dynamical relaxation, causing some stars to migrate to the center and be captured by the BH. The fraction of stellar mass that can sink and be captured depends on the detailed mass and phase-space distribution of the stars, which has not yet been known. To account for this process in a simple way, we move a fraction f_{dry} of stellar mass formed in each nuclear burst ($\Delta M_{*,\text{burst}}$) to the BH. The default value is set to be $f_{\text{dry}} = 6.8 \times 10^{-3}$, derived with analytical approximation by Dekel et al. (2025b, see their §3.1.1). We refer to this process of BH growth as a ‘dry’ channel in the bursty mode, as opposed to the ‘wet’ channel modeled in the above step (xi). There should be an order-unity uncertainty in f_{dry} . However, as we have shown in Fig. 9, the dry channel is unimportant compared to the wet channel, and the exact value of f_{dry} thus does not significantly affect our results.
- (xiv) If the subhalo h merges into another subhalo h_{mp} to form a descendant subhalo (i.e. h_{mp} is a main-branch progenitor of the descendant, while h is not), we add a fraction, f_{merge} , of M_{BH} to that of h_{mp} . Similarly, we add a fraction, f_{merge} , of M_* and M_{g} of h to those in the bulge of h_{mp} . The remaining stars and gas are assumed to be tidally stripped or scattered into the halo to become intracluster baryons. Since the merger may also dynamically change the orbits of baryons in the disk of h_{mp} and transform its morphology, we move some stellar and gas masses in the disk of h_{mp} to those in the bulge. The moved masses are assumed to be a fraction, f_{heat} , of M_* and M_{g} of the infalling galaxy hosted by h , since a more massive infalling galaxy is expected to have a larger dynamical effect on the disk. The exact values of f_{merge} and f_{heat} are very uncertain. The reported values in the literature differ significantly (see e.g. Lu et al. 2014; Behroozi et al. 2019), and depend on how the boundaries of galaxies are defined (He et al. 2013; Bernardi et al. 2013). In this paper, we set $f_{\text{merge}} = f_{\text{heat}} = 0.5$ as the default only for demonstrating the effects of mergers. However, even with such a large value, the contribution of mergers to the growth of M_{BH} and M_* is found to be only significant in massive halos at low z (see §3.2.1, Figs. 8 and 9) – a conclusion also reached by hydro simulations (see e.g. figs. 2 and 10 of Li et al. 2025b).
- (xv) Gas-phase metallicity of galaxies is modeled according to the ‘gas-regulator’ scenario, with certain modifications accounting for the high- z environment (e.g. cold-mode accretion, high clumpiness) that produces different metal transport from that in the local Universe (see §2.4 of Paper-III). Note that the model of metallicity remains highly uncertain due to large uncertainties in both observations and theories. For example, for the galaxy in the subhalo h , the change of the oxygen mass, ΔM_{Z} , during the current time step is given by the continuity equation that combines the effects of stellar yield, star

formation consumption, and feedback ejection, as

$$\Delta M_{\text{Z}} = y_{\text{eff}} \frac{\Delta M_*}{1 - R} - (\Delta M_* + \Delta M_{\text{BH}}) Z; \quad (\text{D16})$$

$$y_{\text{eff}} = y f_{\text{mix}} (1 - f_{\text{esc}}); \quad (\text{D17})$$

$$f_{\text{mix}} = \frac{1}{1 + x_{\gamma}}; \quad (\text{D18})$$

$$f_{\text{esc}} = \frac{1 + \alpha_{\text{esc}} x_{\text{v}}}{1 + x_{\text{v}}}. \quad (\text{D19})$$

Here ΔM_* and ΔM_{BH} are the change of stellar mass and BH mass, respectively, during the current time step, as modeled in the previous steps; $Z \equiv M_{\text{Z}}/M_{\text{g}}$ is the metallicity of the galaxy; y_{eff} is the effective oxygen yield, which incorporates the intrinsic yield of stars, the escape of decoupled fraction, the ejection by stellar feedback, and the recycling due to cooling, all approximated to be instantaneous; $y = 0.0163$ is the intrinsic oxygen yield of star formation; f_{mix} is the efficiency of metal mixing between the inflow gas and the existing ISM, and $x_{\gamma} = |\gamma/\gamma_0|^{\beta_{\text{mix}}}$ characterizes its dependence on γ ; f_{esc} is the escaped fraction of metal from the galaxy, $x_{\text{v}} = (V_{\text{max}}/V_{\text{esc}})^{\beta_{\text{esc}}}$ quantifies its dependence on the gravitational potential well h , and $\alpha_{\text{esc}} = z/(z_{\text{esc}} + z)$ characterizes its evolution with redshift. With the above parameterization, the metal mass ejected from the galaxy is

$$\Delta M_{\text{Z,ej}} = f_{\text{esc}} y \frac{\Delta M_*}{1 - R}, \quad (\text{D20})$$

Note that the metal mass brought in by inflow gas from the IGM is ignored in Eq. (D16) given that the metallicity in IGM is much lower than that in ISM. The default parameters adopted here are $V_{\text{esc}} = 75 \text{ km s}^{-1}$, $\beta_{\text{esc}} = 2$, $z_{\text{esc}} = 5$, $\gamma_0 = 1$, and $\beta_{\text{mix}} = 2$.

With the above steps, we can update the global properties – such as the masses of the BH, stars and gas – of the galaxy in h during the current time step. The growth of these mass components contributed by different channels can be separated, as shown in Figs. 8 and 9 for examples of halos. The evolution of galaxies also provides the basis for modeling their radiation and matter feedback to the large-scale environment, as to be detailed in the following.

D5 The model of Lyman-Werner radiation field

As shown in §3.1.3 and Fig. 7, the radiation feedback in the LW band from galaxies to their large-scale environment is a critical factor that determines when and how the first generations of stars and BHs in halos illuminated by the radiation can form. An accurate modeling of the radiation field requires full spectral information from galaxies and a precise algorithm for radiative transfer, both of which are computationally prohibitive in a cosmological volume. Here we describe a simplified prescription suitable for our purpose.

We first note that the modeling of radiation feedback can take advantage of the characteristics of the LW photons: these photons have a large mean free path, up to $\approx 100 h^{-1} \text{ Mpc}$ at $z = 20$ (see Eq. D24 below and Ahn et al. 2009), comparable to the spatial scale of the box of the N-body simulation used in this paper. The intensity of LW radiation can exhibit strong variation (both spatially and temporally), due to the clustering (i.e. biased distribution) of galaxies in the cosmic volume, the strong temporary variation of star formation rate of high- z galaxies (especially when our bursty mode is included; see e.g. Fig. 9), and the sharp (r^{-2} ; see below) radial decay from individual sources.

To account for both the large mean free path and the strong variation in spacetime of LW radiation, we follow Agarwal et al. (2012) to decompose $J_{\text{LW}}(x, t)$, the total LW intensity illuminating

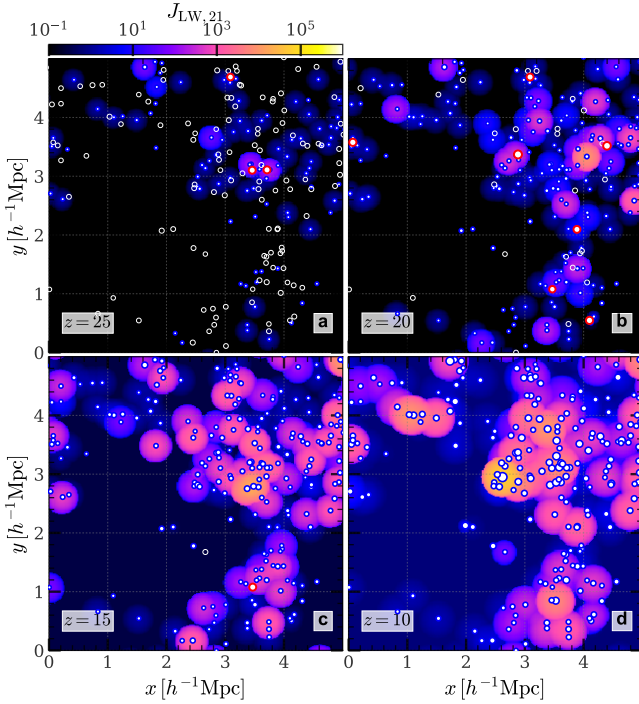


Figure D3. Lyman-Werner radiation field predicted by the model. a–d, LW intensity, $J_{\text{LW}} = J_{\text{bg}} + J_{\text{local}}$, color coded according to the color bar on the top left, in a $5 \times 5 \times 2 \text{ (} h^{-1} \text{cMpc)}^3$ slice of the simulation box at four different redshifts. The slice is a part of that shown in Fig. E1g–i (lower-left corner in x - y , and middle slice along z). Each dot represents a subhalo, sized proportionally to R_v , and styled according to the status of the galaxy it hosts: seeded (blue edge), unseeded and unilluminated ($J_{\text{LW},21} < 1$, white edge), unseeded and illuminated ($J_{\text{LW},21} \geq 1$, red edge). LW photons represent a type of radiation feedback from galaxy formation to the large-scale environment, and play a critical role in delaying the formation of the first generations of stars and BH seeds in subhalos illuminated by the radiation. See Appendix D5 for the detailed steps to model the LW radiation.

at an arbitrary location $\mathbf{x}$ and a cosmic time t , into a global term of radiation background, J_{bg} , and a local term, J_{local} , mainly contributed by nearby star-forming sources, as:

$$J_{\text{LW}}(\mathbf{x}, t) = J_{\text{bg}}(t) + J_{\text{local}}(\mathbf{x}, t). \quad (\text{D21})$$

The background term includes the contribution from all radiation sources in the Universe, and can be obtained by using the total stellar mass density $\rho_*(t)$ of all surviving stars in the simulated volume, as

$$J_{\text{bg}}(t) = f_{\text{esc}} \frac{hpc}{4\pi m_p} \eta_{\text{LW}} \rho_*(t) (1+z)^3. \quad (\text{D22})$$

Here, $f_{\text{esc}} \leq 1$ is the fraction of LW photons that escape from galaxies; h_p is the Planck constant; c is the speed of light; m_p is the proton mass; η_{LW} is the number of LW photons emitted per proton mass; $z = z(t)$ is the redshift at time t ; the factor $(1+z)^3$ accounts for the expansion of the universe. The background is uniform and isotropic, as implied by the cosmological principle. This formulation implies that ρ_* obtained in the simulated volume is a fair representation of the entire Universe, which is satisfied as long as we are using a simulation with a large volume and a fairly sampled initial condition (§2.1). The default parameters we use are $f_{\text{esc}} = 1$ (Dijkstra et al. 2014; Spinoso et al. 2023), assuming that the LW emitters have been depleted of H_2 by recent star formation, and $\eta_{\text{LW}} = 4 \times 10^3$ (Agarwal et al. 2012) for Pop-II stars. We have not incorporated the contribution from Pop-III stars, since their star formation efficiency (SFE) is low

(see Appendix B2), and the post-seeding growth in the bursty mode dominates the growth of galaxies soon after the seeding (see §3.2.1 and Fig. 9). We have also not incorporated the contribution from AGNs, but we note that the recently observed high number density of faint AGNs at high z suggests that they should be seriously taken into account (see e.g. Fig. 3.2.2 for the cosmic BHAR density).

The local term is modeled by summing up the contribution from individual source galaxies with a spherically symmetric approximation for the emission, as

$$J_{\text{local}}(\mathbf{x}, t) = \sum_i \frac{f_{\text{esc}}}{4\pi \Delta v_{\text{LW}}} \frac{Q_{\text{LW}} M_{*,\text{alive},i}}{4\pi d_{\text{L},i}^2} f_{\text{mod},i}. \quad (\text{D23})$$

Here, Q_{LW} is the LW luminosity per unit alive stellar mass; Δv_{LW} is the frequency range of LW radiation (11.2–13.6 eV); $M_{*,\text{alive},i}$ is the mass of alive stars in the i -th source galaxy; $d_{\text{L},i}$ is the luminosity distance between the source galaxy and the target location $\mathbf{x}$; $f_{\text{mod},i}$ is a modulation factor that accounts for the attenuation of LW radiation by IGM. To avoid over-complication of radiative transfer, we use the fitting formula obtained by Ahn et al. (2009) for $f_{\text{mod},i}$. Adapting it to our cosmology, we obtain

$$f_{\text{mod},i} = 1.7 \exp \left[- \left(\frac{d_{\text{c},i}}{110 \alpha_{\text{mod},i} h^{-1} \text{Mpc}} \right)^{0.68} \right] - 0.7 \quad (\text{D24})$$

for $d_{\text{c},i} \leq 92.4 \alpha_{\text{mod},i} h^{-1} \text{Mpc}$, the LW horizon, and $f_{\text{mod},i} = 0$ otherwise. Here, $d_{\text{c},i}$ is the comoving distance between the source galaxy and $\mathbf{x}$, the scaling factor $\alpha_{\text{mod},i}$ is

$$\alpha_{\text{mod},i} = \left(\frac{\Omega_{\text{m},0}}{0.3089} \right)^{-1/2} \left(\frac{1+z_i}{10} \right)^{-1/2}, \quad (\text{D25})$$

and z_i is the redshift of the source galaxy. The default value we adopt is $Q_{\text{LW}} = 10^{36} \text{ erg} \cdot \text{s}^{-1} \text{M}_{\odot}^{-1}$, obtained by averaging the emission of massive O/B stars over their lifetime of $\approx 10 \text{ Myr}$ (Dijkstra et al. 2014, see their Eq. 6). The corresponding mass, $M_{*,\text{alive},i}$, is computed by including only these young massive stars. The summation should be performed over all galaxies in the past light cone of $(\mathbf{x}, t)$, leading to a computational complexity of $O(N^2)$ that is prohibitive for a large number (N) of galaxies. To proceed, we first note that the light speed is large enough so that the emission and reception of LW radiation between two subhalos in the simulated volume can be simplified as synchronous events. Thus, the summation can be performed by using sources within the current snapshot. This also leads to an approximation $d_{\text{L},i} \approx d_{\text{c},i}$. To reduce the number of summations, we limit the zone of influence of each source galaxy to a sphere with a radius at which the LW intensity emitted by this galaxy drops to a negligible threshold, $J_{\text{local,th}}$. The default value we adopt is $J_{\text{local,th},21} = 0.01$, which is about a factor of 10^{-2} – 10^{-3} smaller than $J_{\text{crit},21}$, the value to produce a significant delay in gas collapse (see Appendix B1.3; see also Appendix A of Spinoso et al. 2023 for a similar approach). With these simplifications, the computational time for the LW intensity in a single time step is reduced to a few seconds for our sample of subhalos (§2.1).

Fig. D3 shows the LW intensity, J_{LW} , in a slice of the simulation box at four different redshifts, $z = 25, 20, 15$ and 10 . Subhalos with seeds having been bred are represented by dots with a blue edge, which are the sources of LW radiation. Subhalos without seeds are represented by dots with a white ($J_{\text{LW},21} < 1$) or a red edge ($J_{\text{LW},21} \geq 1$). At $z \approx 25$, the universe is in the seeding era (see §3.2.2), and most subhalos are still on their way to form the first generation of stars and BH seeds. The large number of unseeded subhalos in panel a reflects this infancy era of galaxy formation. The seeding era ends

at $z \approx 20$ when most subhalos have been seeded, which can be seen from the reduction in the number of unseeded subhalos from panel a to panel b. The rapid post-seeding star formation through the bursty mode produces intense but localized LW radiation from the seeded galaxies, manifested as individual lamp-like illuminated zones in panel b. A small fraction of unseeded subhalos located in these zones are illuminated by the LW radiation, and are delayed in forming their first generation of stars and BH seeds. At $z \lesssim 15$, the rising of cosmic star formation rate density (see also Fig. 3.2.2) lifts the global background. However, by this time, most subhalos have already been seeded (see also Fig. 3.1.2), providing harbors of star formation and BH growth that are well shielded from external radiation.

D6 The model of intergalactic-medium enrichment

The enrichment of the IGM is a type of matter feedback of galaxy formation to the large-scale environment. Compared with the radiation feedback described above, matter feedback also plays a role in determining the formation of seeds in subhalos surrounded by the enriched IGM; a major difference is that the diffusion of metal is much slower than that of radiation, so that the expansion of the zone of enrichment produced by the matter feedback from any galaxy has to be tracked over a long timescale.

To simplify the model of IGM enrichment so that it can be efficiently implemented into a large volume, we make the following assumptions: the intergalactic diffusion of metal is assumed to be a spherically symmetric process around each source galaxy, and the enriched volume is described by a spherical bubble; the energy powering the expansion of the bubble is assumed to come from the blast wave generated by SNe, while the contribution from Pop-III stars and AGNs is ignored. All these assumptions are consistent with and explained in our treatment of the radiation feedback (Appendix D5).

To estimate the sizes of SN bubbles produced by galaxies, we adopt a self-similar solution of blast wave (see e.g. §8.6 of Mo et al. 2010):

$$R_{\text{sh}} = A_{\text{sh}} \left(\frac{E_{\text{SN}}}{\rho_{\text{g}}} \right)^{1/5} (t - t_0)^{2/5}, \quad (\text{D26})$$

where R_{sh} is the physical radius of the shock wave, $A_{\text{sh}} \approx 1$ is a normalization constant whose value is determined by energy conservation, E_{SN} is the energy released by SNe during a burst of star formation, ρ_{g} is the density of the IGM swept by the shock wave, and $t - t_0$ is the elapsed time since bubble generation at time t_0 . We express the energy released by SNe as $E_{\text{SN}} = \epsilon_{\text{SN}} R \Delta M_{*,\text{new}}$, where the energy efficiency $\epsilon_{\text{SN}} = 10^{51} \text{ erg} \cdot \text{M}_{\odot}^{-1}$ (Dijkstra et al. 2014; Spinoso et al. 2023), R is the fraction of returned mass (Eq. D1), and $\Delta M_{*,\text{new}}$ counts all stars newly formed in the past duration of t_{v} (Eq. A7) in which the replenishment of the gas may be accomplished to trigger the next episode of star formation. If there are multiple bubbles generated in the main branch of a galaxy, we only consider the largest (Ventura et al. 2024). However, since SFRs of high- z galaxies often increase rapidly with time (see examples in Fig. 9), the largest bubble is most likely produced in the recent star-forming episode.

Before we implement the IGM enrichment model in our pipeline, we provide an analytical estimate for the bubble sizes. We take $\rho_{\text{g}} = \epsilon_{\rho} \rho_{\text{b}}(z)$, where ρ_{b} is the cosmic baryon density (Eq. A1), and $\epsilon_{\rho} = 60$ (Dijkstra et al. 2014), roughly the density contrast at R_{v} of any halo at z . The stellar mass formed in the recent t_{v} is:

$$\begin{aligned} \Delta M_{*,\text{new}} &= \epsilon_{\text{SF}} f_{\text{b}} \dot{M}_{\text{v}} t_{\text{v}} \\ &= \left[1.73 \times 10^7 \text{ M}_{\odot} \right] \epsilon_{\text{SF},0.1} M_{\text{v},9.5}^{1.14} (1+z)_{10}, \end{aligned} \quad (\text{D27})$$

where $\epsilon_{\text{SF}} = 0.1 \epsilon_{\text{SF},0.1}$ is the star formation efficiency, with its typical

value of 0.1 taken from observations (e.g. Wang et al. 2025) and empirical models for high- z galaxies (e.g. Nusser 2024), and $\dot{M}_{\text{v}}$ is the halo accretion rate, approximated using Eq. (A8). Substituting the above equation into Eq. (D26) and taking $t - t_0 = t_{\text{v}}$, we can obtain the bubble radius in comoving units as

$$R_{\text{sh},c} = (1+z) R_{\text{sh}} = 231 \text{ kpc} \left[\chi_{\text{sh}} M_{\text{v},9.5}^{1.14} \right]^{1/5}, \quad (\text{D28})$$

where we have defined $\chi_{\text{sh}} = \epsilon_{\text{SN},51} \epsilon_{\text{SF},0.1} R_{0.4} / \epsilon_{\rho,60}$, $\epsilon_{\text{SN},51} = \epsilon_{\text{SN}} / (10^{51} \text{ erg} \cdot \text{M}_{\odot}^{-1})$, $R_{0.4} = R/0.4$ and $\epsilon_{\rho,60} = \epsilon_{\rho}/60$. Thus, the comoving size of an SN bubble is only a weak function of halo mass, and is redshift insensitive, if χ_{sh} does not vary significantly. Another useful quantity is the ratio between R_{sh} and R_{v} (Eq. A3) of the host halo,

$$R_{\text{sh}}/R_{\text{v}} = 5.05 \chi_{\text{sh}}^{1/5} M_{\text{v},9.5}^{-0.105}, \quad (\text{D29})$$

which again depends weakly on the halo mass and is largely insensitive to redshift, suggesting that the radius of an SN bubble is nearly proportional to the size of its host halo.

We can also estimate the global volume filling factor, f_{v} , of SN bubbles (i.e. the fraction of cosmic volume filled by the bubbles). For this, we multiply the volume of bubble (in comoving unit) with $\Phi(M_{\text{v}})$, the halo mass function (defined similarly to the mass function of BH seed in Eq. 7), and integrate it over the full range of halo mass:

$$f_{\text{v}} = \int \frac{4}{3} \pi R_{\text{sh},c}^3 \Phi(M_{\text{v}}) d \log M_{\text{v}}. \quad (\text{D30})$$

Since the power index on M_{v} in the halo mass function is smaller than -0.8 (see §3.1 of Tinker et al. 2008), the integrand scales with halo mass as $R_{\text{sh},c}^3 \Phi \sim M_{\text{v}}^{\beta}$, with $\beta < 1.14 \times 3/5 - 0.8 \approx -0.12 < 0$. Thus, the volume filling factor is dominated by low-mass halos. This signifies the need for modeling the full population of galaxies in a cosmological volume, particularly in the early Universe where halo masses are typically small, so that the environment that affects the seeding can be fully captured.

In our numerical implementation, we follow Ventura et al. (2024) to model the spatial distribution of metal on coarse grids, and ignore the sub-grid distribution. The reason for adopting such an approach is to take into account the limitations arising from the simplifications we have made: (i) the assumptions of spherical symmetry and blast-wave expansion are not able to capture the sub-structure of bubbles; (ii) interactions among bubbles generated by different galaxies or by one galaxy at different episodes of star formation cannot be resolved; (iii) the locations of subhalos extended by our Monte Carlo algorithm (Appendix D1) are not exactly known. Thus, a grid-based approach with spatial resolution comparable to these physical limitations is sufficient for our purpose. The steps of our numerical implementation are as follows:

- (i) We partition the simulation box into cubic cells with a fixed side length (in comoving unit) that is sufficiently fine compared with the spatial scale to be used to smooth the spatial distributions (see step iv below). Any field in the simulation box is then defined by a set of discrete values in the cells. The first field to compute is ρ_{m} , the mass density field of the current snapshot, whose value in each cell C is obtained by summing up the masses of all the simulated dark matter particles located in C , and dividing it by ΔV_{c} , the volume of the cell.
- (ii) For each subhalo, we track the evolution of R_{sh} of the SN bubble generated by the galaxy hosted by the subhalo, and update it over time steps. In the current snapshot s , we compute $\Delta M_{*,\text{new}}$, the mass of newly formed stars over the past duration of t_{v} in the subhalo $h \in H_s$. We compare $\Delta M_{*,\text{new}}$ with those evaluated in the previous time steps along the main branch of h , and take the largest value among them to

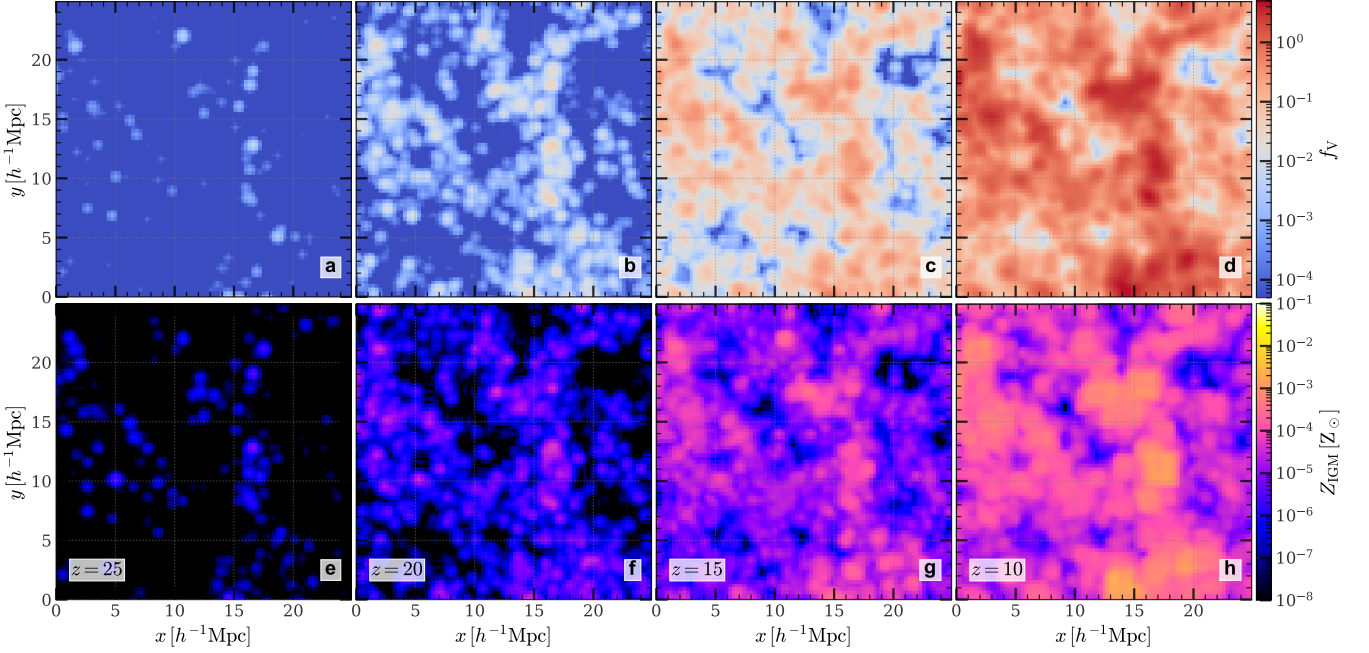


Figure D4. Enrichment of intergalactic medium predicted by the model. **a–d**, distribution of volume filling factor (f_v) of SN bubbles in a $25 \times 25 \times 2$ ($h^{-1} \text{cMpc}$)³ slice of the simulation box at four different redshifts. **e–h**, similar but for Z_{IGM} , the metallicity of IGM. Note that Z_{IGM} shown here is the average value within each entire cell (i.e. $Z_{\text{IGM}} = \Delta M_Z / (f_b \rho_m \Delta V_C)$), without an f_v in the denominator, not just within the bubble-filled fraction that is used to sample the IGM metallicity for target galaxies (steps iv and v in the enrichment modeling). For both f_v and Z_{IGM} , mean values along z -axis are taken, and are color coded according to the color bar on the right. The slice shown here is exactly the middle $2 h^{-1} \text{Mpc}$ along z -axis of that shown in Fig. E1g–i. At $z \approx 20$, a noticeable fraction of the cosmic volume has been enriched by metals ejected from early-formed galaxies, leading to the significant formation activity of Pop-II seeds; by the time of $z \approx 10$, most cosmic volume has already been enriched to some level, marking the prevalence of Pop-II seeds (see e.g. Fig. 6a for the cosmic seeding history and Fig. 7 for the seeding atlas). See Appendix D6 for the detailed steps to model the IGM enrichment.

power the expansion of the SN bubble. We find the derivative of R_{sh} with respect to t using Eq. (D26), as

$$\dot{R}_{\text{sh}} = \frac{2}{5} A_{\text{sh}} \left(\frac{E_{\text{SN}}}{\rho_g} \right)^{1/5} (t - t_0)^{-3/5}. \quad (\text{D31})$$

Here E_{SN} is computed using the largest value of $\Delta M_{*,\text{new}}$ and t_0 is the time when this value is achieved. The gas density ρ_g is evaluated at halo-centric distance R_{sh} by assuming an isothermal profile, and a total mass of $f_b M_v - M_g - M_* - M_{\text{BH}}$ (i.e. the available baryon mass subtracted by the gas, stellar and BH masses in the galaxy hosted by h) for the gas within h (Appendix A and Fig. A1), if R_{sh} is smaller than R_v of h ; otherwise, we use ρ_m at the location of h on a scale of R_{sh} to find the gas density as $\rho_g = f_b \rho_m$. We then use $\dot{R}_{\text{sh}}$ to update R_{sh} for one time step.

- (iii) We define the volume filling factor f_v of any cell as the fraction of the cell volume that has been filled by SN bubbles, and we define the metal mass ΔM_Z in any cell as the total mass of metal ejected by galaxies into the cell. To compute the contributions from a source subhalo $h \in H_s$ to these fields, we identify the intersection of its SN bubble with any cell, and accumulate the contribution from h to f_v of the cell by using the volume of the intersection; similarly, we compute the total mass of metal ejected from h and all of its progenitors, assuming this mass is uniformly distributed within the SN bubble of h , and accumulate the contribution from h to ΔM_Z in any cell intersected with the bubble.
- (iv) We smooth the fields, ρ_m , f_v and ΔM_Z by a Gaussian kernel with a comoving size of $l_{\text{sm}} = 250 h^{-1} \text{kpc}$, a scale comparable to the typical size of SN bubbles estimated in Eq. (D28). We compute the metallicity of the IGM in the bubble-filled volume of any cell as

$Z_{\text{IGM}} = \Delta M_Z / (f_b f_v \rho_m \Delta V_C)$, and as $Z_{\text{IGM}} = 0$ outside the bubble-filled volume.

- (v) For any target subhalo $h \in H_s$, the metallicity of the IGM around the galaxy hosted by h is determined by a stochastic algorithm. Specifically, we draw $u_Z \sim U[0, 1]$, a random number with a uniform distribution over $[0, 1]$, and compare it with the value of f_v in the cell containing h . If $u_Z \leq f_v$, the galaxy is considered to be located in the bubble-filled volume of the cell and the value of Z_{IGM} is assigned accordingly; otherwise, it is located outside the bubble-filled volume and $Z_{\text{IGM}} = 0$ is assigned.

Fig. D4 shows the volume filling factor and IGM metallicity of the cells in a slice with a thickness of $2 h^{-1} \text{Mpc}$ (in comoving units) in the simulation box at four different redshifts, $z = 25, 20, 15$ and 10 . At $z = 25$ (panels a and e), a few SN bubbles can be seen around galaxies whose post-seeding star formation has just started owing to the onset of nuclear bursts (see Fig. 9 for example galaxies). The sizes of these bubbles are small, as expected from the low halo masses (see e.g. Fig. 7) and the proportional relation between R_{sh} and R_v (Eq. D29). The metallicity of the enriched volume is as low as $\sim 10^{-6} Z_{\odot}$, which is a consequence of the low star formation efficiency in low-mass halos (see e.g. Fig. 10b for a comparison between $\dot{\rho}_*$ and virialized $\dot{\rho}_b$). From $z = 25$ to $z = 20$ (panels b and f), the bubble-filled volume expands significantly from the sites densely populated with galaxies that have been newly seeded and experiencing intense star formation through the bursty mode. The metallicity in some bubble-filled zones increases beyond $10^{-4} Z_{\odot}$, and the unseeded subhalos swept by the bubbles have increased chances to be fueled with enriched gas and to form Pop-II stars and BH seeds. The rapid increase of cosmic seeding activity contributed by Pop-II seeds (Fig. 6a, green shade) at

$z \gtrsim 20$ can thus be explained. At $z \lesssim 20$, most subhalos had already been seeded, while their SFRs through the bursty mode continue to increase due to the fast halo assembly (Fig. 2a). The SN bubbles continue to expand and overlap with each other to form extended structures, leaving only a small fraction of the cosmic volume pristine (panels c and g). By the time of $z = 10$ (panels d and h), most cosmic volume has been filled by SN bubbles, with some hot spots having $Z_{\text{IGM}} \gtrsim 10^{-3} Z_{\odot}$. In this era, Pop-II seeds are bred at a rate comparable to late-formed Pop-III seeds, as can be seen by a comparison among cosmic BH seeding histories contributed by different flavors in Fig. 6a.

APPENDIX E: SPATIAL DISTRIBUTION OF SEED POPULATIONS

As introduced in §3.1.3 with Fig. 7, the regimes in the seeding atlas are shaped by the environmental processes affecting the seeding. Since the three environmental processes take effect via mediators with different domains of influence, mini-halos affected by each of the processes, and seeds bred in these mini-halos, are expected to have a distinct clustering pattern. Owing to the preservation of the matter distribution on large scales, the clustering pattern is expected to be persistent, and may be retrieved later to provide an archive of seeding histories for halos that have grown massive enough to host observable galaxies.

To quantify the spatial distribution of a given target sample of galaxies at a given redshift z , we measure the two-point cross-correlation function (2PCCF) of the sample with respect to a reference sample. The 2PCCF is defined as the expected excess number density, relative to the mean number density in a representative volume, of the galaxies in the reference sample at a given distance r from a galaxy in the target sample. Numerically, the 2PCCF can be estimated as

$$\xi(r) = \left\langle \frac{n_{\text{ref}}(r)}{\bar{n}_{\text{ref}}} \right\rangle_{\text{target}} - 1, \quad (\text{E1})$$

where $n_{\text{ref}}(r)$ is the number density of galaxies in the reference sample within a spherical shell of finite thickness at distance r from a galaxy in the target sample, $\bar{n}_{\text{ref}}$ is the mean number density of galaxies in the reference sample, and the average is taken over all galaxies in the target sample. Note that we adopt the 2PCCF, instead of the two-point auto-correlation function (2PACF), because the former is more stable for small target samples, such as those hosting Pop-II seeds, provided that the reference sample is sufficiently large. With our choices, the 2PCCF of the reference sample, which we denote as ξ_{all} , is the 2PACF of the reference sample itself. The ratio between any ξ and ξ_{all} , either evaluated at a given r or averaged over a range of r , is referred to as the relative bias at the corresponding spatial scale (e.g. Zhang et al. 2025a).

Here we measure the 2PCCFs of various target samples of galaxies at a given z selected according to the masses and flavors of the seeds that have been bred in the main branches of the subhalos hosting the galaxies. A common reference sample consisting of all galaxies at z is adopted to maximize the statistical stability. In Fig. E1a–c, we show $\xi(r)$ of the target samples at $z = 15, 5$ and 0 , respectively. The corresponding results for the relative bias are shown in panels d–f. For a demonstration, panels g–i show the locations of individual galaxies using scatter points with styles marking the mass and flavor of the seed.

For galaxies with Pop-III seeds, the 2PCCF is higher at all scales if the seeds are more massive (purple curves in Fig. E1a–c). At $z = 15$, the 2PCCF of galaxies with $M_{\text{BH,seed}} \gtrsim 10^4 M_{\odot}$ (mostly DCBHs with Eddington-accreting progenitor stars; see Fig. 5a) is a factor of

2–4 higher than that of galaxies with $M_{\text{BH,seed}} \lesssim 10^3 M_{\odot}$ (mostly remnants after CCSNe and PISNe). The difference in the 2PCCF between the target samples with different $M_{\text{BH,seed}}$ becomes less and less pronounced over time. As shown in Fig. 6a and d, most seeds have already been bred by $z \approx 15$ – 20 , and the cosmic seeding rate density is negligible at $z \lesssim 5$. Thus, the decline of the $M_{\text{BH,seed}}$ -dependence of the 2PCCF at $z \lesssim 15$ is due to the evolution of the cosmic large-scale structure rather than the cosmic seeding activity. Panels g–i clearly show that galaxies are less clustered at $z = 15$, but become more clustered later as structures form.

As shown by the orange and green curves in Fig. E1a–f, halos affected by each of the three environmental processes have a distinct clustering pattern at a given redshift and such a pattern shows a distinct trend of evolution. Fast halos (with $\gamma_v \gtrsim 3$ at the seeding epoch) at $z = 15$ have a relative bias of about 2–3, quite independent of the scale at $r \gtrsim 0.3 h^{-1} \text{Mpc}$. At $z = 5$, the relative bias is comparable to that at $z = 15$ at $r \gtrsim 0.3 h^{-1} \text{Mpc}$. Such a high relative bias reflects the requirement of an excessively high γ_v , and thus an excessively over-dense environment, in comparison to galaxies in normal halos, for the seeding to be significantly delayed. Interestingly, the relative bias of fast halos at small scale ($r \approx 0.1 h^{-1} \text{Mpc}$) shows a clear increase with time and reaches about 2–3 at $z \lesssim 5$. The build-up of such a ‘one-halo’ term (e.g. Jing et al. 1998; Peacock & Smith 2000; Zheng et al. 2005) in the clustering pattern reflects the large number of satellite subhalos assembled into the host halo during the fast accretion. At $z = 0$, the relative bias of fast halos at large scale declines due to the cessation of cosmic seeding activity and the evolution of the large-scale structure (seen from the decline of the number of fast halos in panels g–i), similar to the evolution of clustering of galaxies bred with massive DCBHs. The small-scale upturn of the relative bias is persistent and extends to scales up to a few Mpc, reflecting the continuation of halo growth and the assembly of satellite subhalos.

The clustering strength of LW halos (with $J_{\text{LW},21} \gtrsim 7.5$ at the seeding epoch) at $z = 15$ is strong at the scales displayed in Fig. E1a. The relative bias is about 2 at $r \gtrsim 1 h^{-1} \text{Mpc}$, and reaches about 7 at $r \approx 0.1 h^{-1} \text{Mpc}$. At such high redshift, the global LW radiation field has not been established yet, and strong LW radiation is only available from nearby galaxies with intense star formation (see Appendix D5). The strong clustering at large scales is thus expected since the production of LW photons requires the presence of galaxies with intense star formation, which is more likely to happen in over-dense regions harboring early halo formation. The excessively strong clustering at small scales reflects the properties of the domain of influence of the LW radiation. Since the local term of LW radiation declines rapidly with distance as r^{-2} (see Eq. D23), halos affected by LW radiation (LW halos) tend to have close neighbors and thus enhanced small-scale clustering. Note that we have only seen a small number of LW halos in the slice of panel g, suggesting that the chance to get such a close pair of galaxies is small at $z = 15$ (see also Fig. 6d for n_{seed}). Towards lower redshift, the global LW background gradually builds up, and the requirement of close neighbors for LW halos is relaxed. This can be seen from the decline of the small-scale upturn of the relative bias of LW halos at $z = 5$ and 0 , and the less pronounced large-scale clustering of LW halos than that of fast halos at these redshifts. The large-scale clustering of LW halos declines with time, again due to halo mergers, as seen from the decline in the number of LW halos from panels h to i.

Similar to fast halos and LW halos, the large-scale clustering of galaxies with Pop-II seeds (green in Fig. E1) is higher than ξ_{all} at $z \gtrsim 5$, implying that Pop-II seeds are embedded in over-dense regions. This is consistent with the aforementioned finding that the

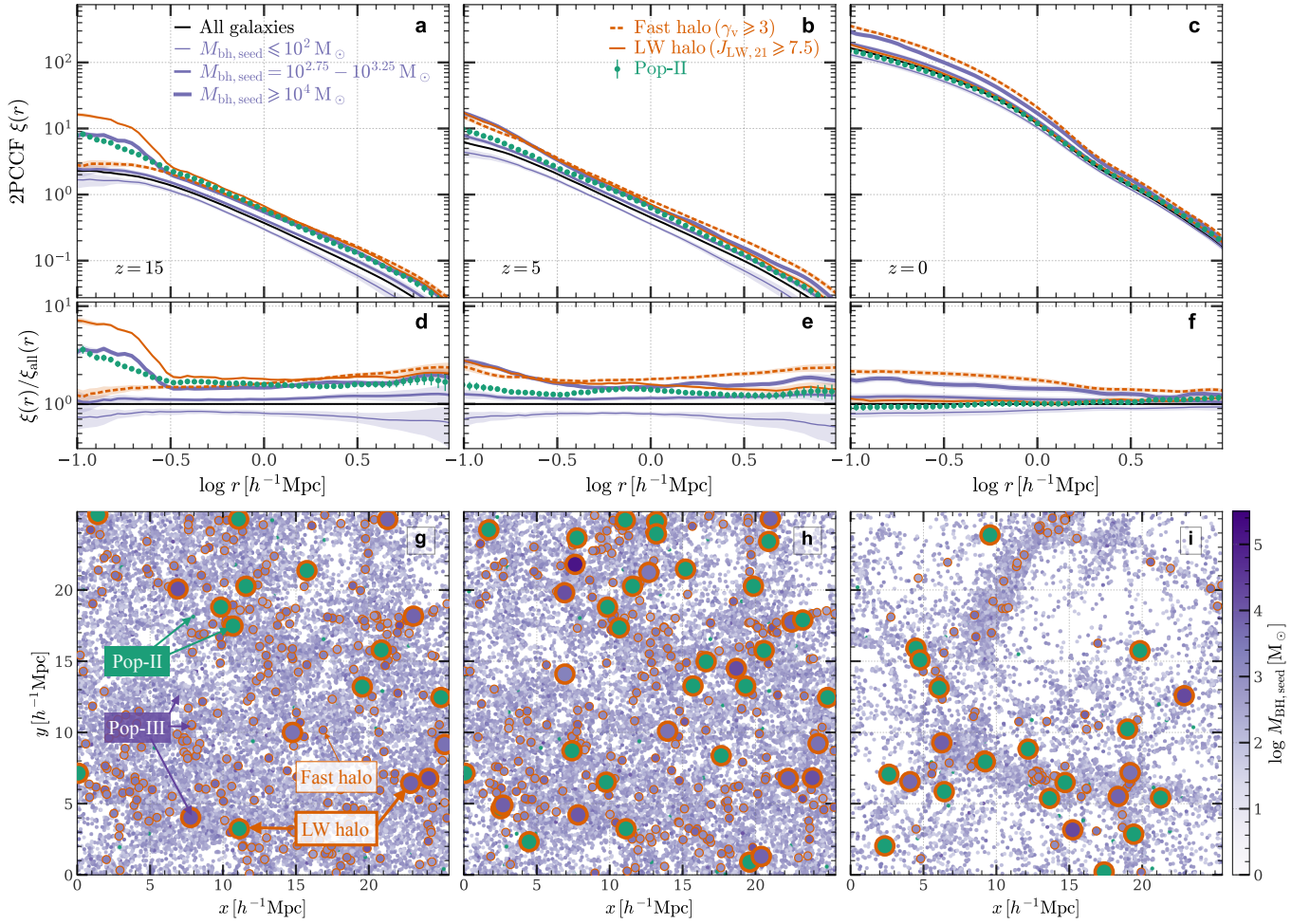


Figure E1. Spatial distribution of galaxies that bred different black-hole seeds. Three columns show the results at $z = 15$, 5 and 0 , respectively. **a–c**, 2PCCFs (ξ) as functions of comoving separation r for different target samples of galaxies. At each given z , the 2PCCFs of the target samples are measured with respect to a common reference sample that includes all galaxies at z . The target samples displayed are: all galaxies (**black**; equivalent to the 2PCF of the reference sample; denoted as ξ_{all}); galaxies with Pop-III seeds that have different seed masses (**purple**; $M_{\text{BH,seed}}$, the mass of BH at the breeding epoch); galaxies hosted by fast halos (**orange dashed**); galaxies hosted by LW halos (**orange solid**); galaxies with Pop-II seeds (**green**). **d–f**, relative bias of each target sample, defined as the ratio between the 2PCCF of the sample and ξ_{all} . Curves with error bars (or shaded regions) show the means and standard deviations from bootstrap resampling. **g–i**, dots represent galaxies in the same slice with a thickness of $10 h^{-1} \text{Mpc}$ at the three redshifts, respectively. Style of a dot is encoded according to the seed bred in the main branch: **purple** if the seed is a Pop-III remnant (more massive seed is darker); **green** if the seed is a Pop-II remnant; with **orange edge** if the seeding is in a fast (**small size**) or a LW halo (**big size**). This figure shows that BH seeds with different flavors exhibit distinct clustering patterns. Some of the patterns are persistent over cosmic time, and may be retrieved at low z . See §3.1.3 for details.

two components of Pop-II seeds shown in Fig. 7a overlap the regimes of dynamical heating and LW radiation, respectively, and that both factors preferentially select over-dense regions for the seeding. At $z = 15$, the small-scale clustering of galaxies with Pop-II seeds is exceptionally strong, as expected from the typical bubble sizes of star-forming galaxies that enrich the IGM with metals (see Eq. D28 and Fig. D4). Towards lower redshift, the relative bias of galaxies with Pop-II seeds declines at small scales, and even becomes slightly lower than that of the reference sample at $z = 0$. Such a distorted pattern of clustering is caused by the competition between two factors, one is that the generation of Pop-II seeds requires a crowd of galaxies to enrich the IGM with metals, and the other is that Pop-II seeds tend to be bred in low-mass halos (Fig. 7) that have a small number of satellite subhalos. At $z \lesssim 5$, the evolution of the clustering of galaxies with Pop-II seeds again follows that of the large-scale structure.

APPENDIX F: EVOLUTION OF THE STRUCTURE HIERARCHY

In Appendix C1, we described the structure hierarchy (Fig. 4) that leads to BH growth and shown the ladder-like gas profile for a halo of MW size and its progenitors at different redshifts (Fig. A1). Here we present the properties of individual levels on the ladder, and the evolution of these properties, in the histories of halos. In applications, one can start from any level on the ladder, use it as a large-scale boundary condition for the processes that lift gas to higher levels, and resolve the formation and growth of structures on smaller scales.

Fig. F1 shows the evolution of the properties of SGC and SNF nucleus in the assembly histories of halos with different masses at $z = 0$ ($M_{\text{v},z=0}$). To produce these properties, we first take the median assembly history of halos with a given $M_{\text{v},z=0}$ in the same way as that used to obtain Fig. 2. We then assume that a global disturbance has been induced in the SGC hosted by a halo at a redshift

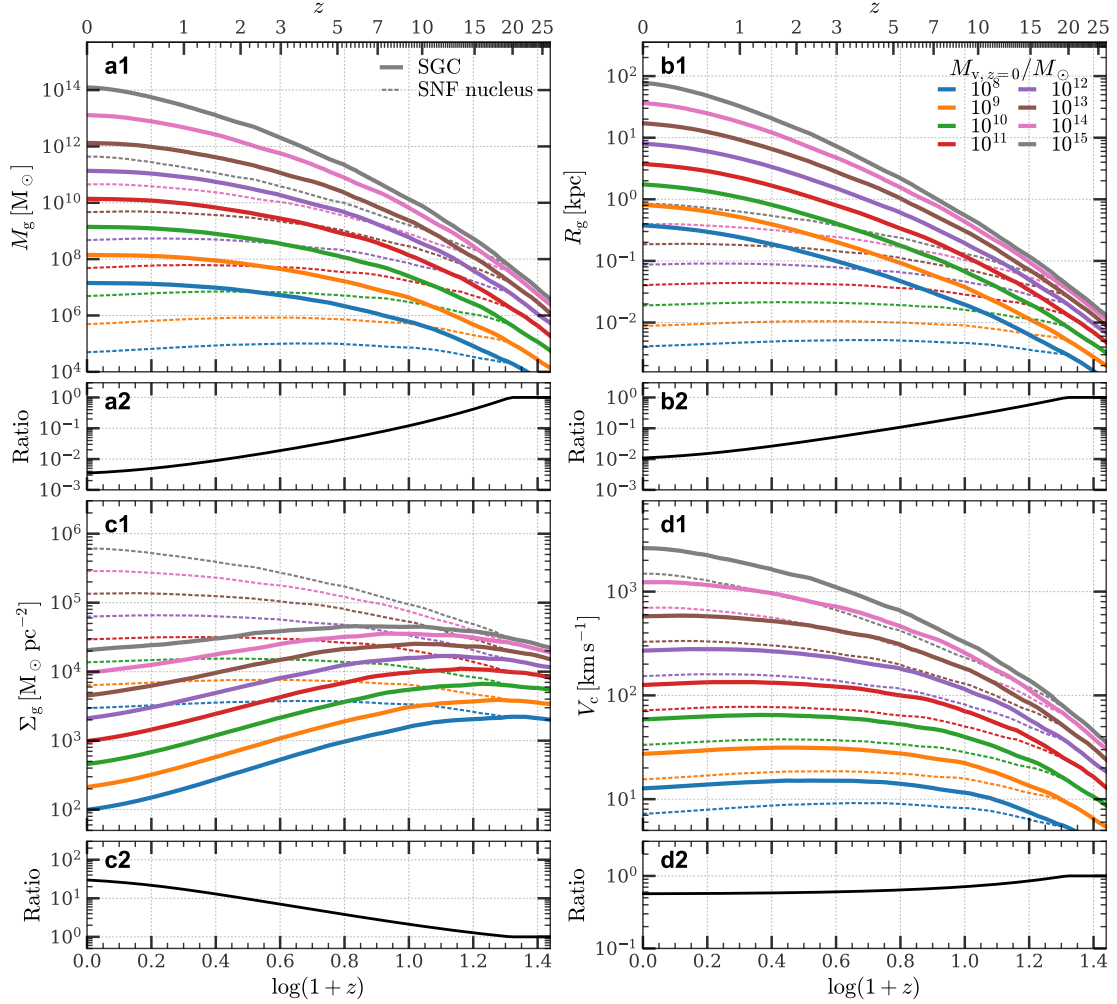


Figure F1. Properties of self-gravitating gas cloud and supernova-free nucleus in the histories of halos with different masses. Four properties for SGC (solid) and SNF nucleus (dashed) are shown: gas mass (M_g ; **a1**), gas radius (R_g ; **b1**), gas surface density (Σ_g ; **c1**) and circular velocity (V_c ; **d1**). **a2–d2** show the ratios of the corresponding properties between SNF nucleus and SGC. Each property is shown as a function of z evaluated based on median assembly history of halos with a given mass M_v at $z = 0$ ($M_{v,z=0}$). The assembly histories of halos used in this figure are exactly the same as those shown in Fig. 2. Analytical approximations for the properties of SGC can be found in Eqs. (B1)–(B5), and of SNF nucleus can be found in Eqs. (C12)–(C17). See Appendix F for details.

z , and that the disturbance generates an SNF nucleus in the center of the SGC. We compute the properties of the SGC according to Eqs. (B1)–(B5) (the first equality in each of them) and the properties of the SNF nucleus according to Eqs. (C1)–(C10). We perform the calculation for halos of any $M_{v,z=0}$ and at any z , and show the gas mass (M_g), radius (R_g), surface density (Σ_g) and circular velocity (V_c) in panels a1–d1, respectively, for both SGC (solid) and SNF nucleus (dashed). For reference, we also show the ratio between the properties of SNF nucleus and SGC in panels a2–d2. The gas fraction assumed here is $f_{g,0.157} = 1$, which describes a gas-rich SGC that has not been depleted by feedback; the radius fraction is $f_{r,0.04} = 1$, roughly the critical value at which rotation is still unable to provide enough support; the feedback-free threshold is $n_{\text{snf},3.5} = 1$, which corresponds to a free-fall timescale of about 1 Myr. Halos with values of $f_{g,0.157}$, $f_{r,0.04}$ or $n_{\text{snf},3.5}$ other than 1 can also be analyzed by the equations listed above. Note that at $z \gg 1$ (when the universe is approximately an Einstein-de Sitter universe), the dependence of SGC properties on M_v and z can be analytically approximated using Eqs. (B1)–(B5); if, in addition, $z \lesssim 10$ (i.e. the concentration $c_g \equiv R_{\text{SGC}}/R_{\text{nuc}} \gg 1$), the dependence of the properties of SNF

nucleus on M_v and z can also be analytically approximated using Eqs. (C12)–(C17).

The most obvious feature of the SNF nucleus compared to the host SGC is the redshift evolution. At $z \gtrsim 20$ (when the age of the universe is less than 200 Myr), an SGC is entirely supernova-free, as seen from the ratio of M_g (or R_g) between the SNF nucleus and the SGC. At lower redshifts, the SNF nucleus occupies a smaller fraction of the SGC: at $z \approx 5$, the SNF nucleus takes about 10% of the SGC mass, and at $z = 0$, the fraction drops to about 1% (panel a2); a similar trend is seen for the radius (panel b2). The surface density of the SNF nucleus, however, remains higher than $10^3 \text{ M}_\odot \text{ pc}^{-2}$ at any redshift for halos with any mass shown in Fig. F1, and the ratio of Σ_g between SNF nucleus and SGC continuously increases with time. The evolution of the properties of SNF nucleus and SGC reflects that the importance of SNF nucleus in supplying gas to the central BH of a galaxy declines with time. This, together with the less frequent global disturbance at lower redshift, explains the decrease of BH growth through the bursty mode with time (see §3.2.1 and Fig. 9).

Climbing one level higher on the ladder, one can deduce the Bondi properties of the central BH hosted by an SNF nucleus. The properties

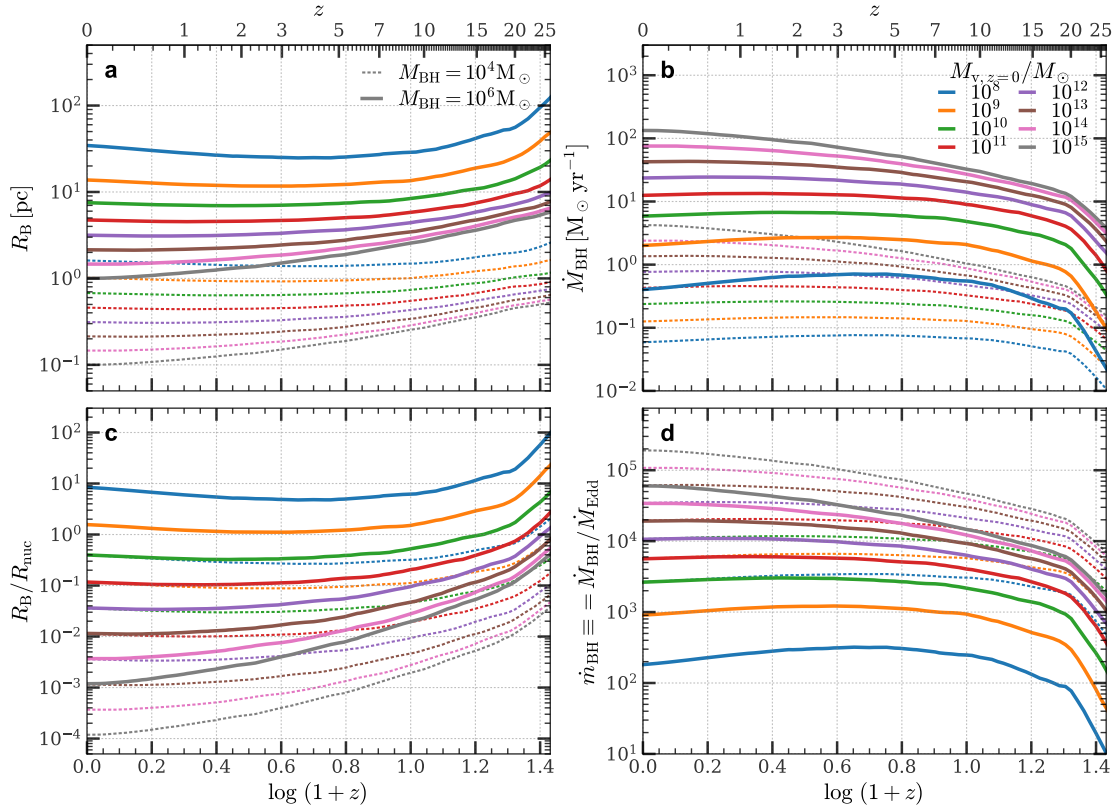


Figure F2. Bondi properties of central black holes in the histories of halos with different masses. Here we show four properties: Bondi radius (R_B ; **a**), Bondi accretion rate ($\dot{M}_B$; **b**), the ratio between R_B and the radius of the SNF nucleus hosting the BH (**c**), and the Eddington ratio of Bondi accretion rate ($\dot{m}_{BH} \equiv \dot{M}_B/\dot{M}_{Edd}$; **d**). Each property is shown as a function of redshift evaluated based on properties of the host SGC and SNF nucleus (Fig. F1) in halos with a given mass M_v at $z=0$ ($M_{v,z=0}$), assuming a BH with a mass of $M_{BH} = 10^4 M_\odot$ (dashed) or $10^6 M_\odot$ (solid) is placed at galactic center. See Appendix F for details.

most relevant to the modeling of BH growth are the Bondi radius (R_B ; Eq. C21) and the Bondi accretion rate ($\dot{M}_B$, or $\dot{m}_B \equiv \dot{M}_B/\dot{M}_{Edd}$; Eq. C20), which form the boundary condition to determine the structure of the accretion disk around the BH that experiences the bursty mode accretion. In Fig. F2, we show the evolution of R_B and $\dot{M}_B$ in panels a and b, respectively, for a BH with a mass of $M_{BH} = 10^4 M_\odot$ (dashed) or $10^6 M_\odot$ (solid) placed in the center of an SNF nucleus in the history of a halo with a given $M_{v,z=0}$. At $1 \ll z \lesssim 10$, R_B and $\dot{m}_{BH}$ can also be approximated analytically by linking them to the properties of either the SNF nucleus, or the SGC, or the halo, using the chained equalities in Eqs. (C29) and (C30). As expected from these approximations, R_B increases with M_{BH} , reflecting the larger zone of influence of more massive BHs, and decreases with M_v , due to the deeper gravitational potential well in the centers of more massive halos. On the other hand, $\dot{M}_B$ increases with both M_{BH} and M_v . Note that in some cases where a too massive BH is placed in a too small SNF nucleus, R_B is obtained by extrapolating the SGC density profile (Eq. C5) outward beyond R_{nuc} , which can lead to $R_B/R_{nuc} > 1$. In realistic applications where BHs grow from well-defined initial and boundary conditions, such cases should not occur.

APPENDIX G: VARIANTS OF THE MODEL

The bursty mode requires a global disturbance to generate the hotness of the gas. This is more likely to happen if the merger rate of the host halo is high, and is thus expected to be more common in the

early Universe and in a halo with higher mass. The accretion rate of a BH in the bursty mode is stronger if M_{BH} is smaller, so that the Eddington ratio is initially higher and the feedback regulation is less effective, or if M_v of the host halo is larger, so that the SNF nucleus is more massive to supply more gas to the BH (see texts below Eq. C30 for an estimate). Thus, the growth of a BH through the bursty mode is expected to be more significant in the early Universe (before the BH has grown massive) or in a halo with higher mass. A similar conclusion applies to star formation in the bursty mode, since a less massive BH is less competitive compared to star formation (see Appendix C5.1), and a more massive halo can supply a more massive nucleus to form stars. However, the continuous mode depends on the fast accretion of the host halo and the global properties of the SGC. Such a mode is thus expected to be efficient during the entire fast phase of the halo, even if the SNF nucleus is not formed or is negligibly small. Finally, the contribution from merger dominates if other channels are not active. This is more likely in later times and/or in a halo with higher mass where galaxy quenching suppresses the growth in the other modes.

As described in §3.2.1, the growth of stellar and BH masses in galaxies are modeled as a multichannel process, with each channel governing a certain stage of the growth. To cover the broad scales of time, space and mass involved in the growth, we have incorporated certain assumptions for each channel, and used the parameters listed in Table D1 to characterize the degree of freedom (DoF) in the model. Some parameters have already been assigned values or ranges based on either analytical estimates, numerical simulations or

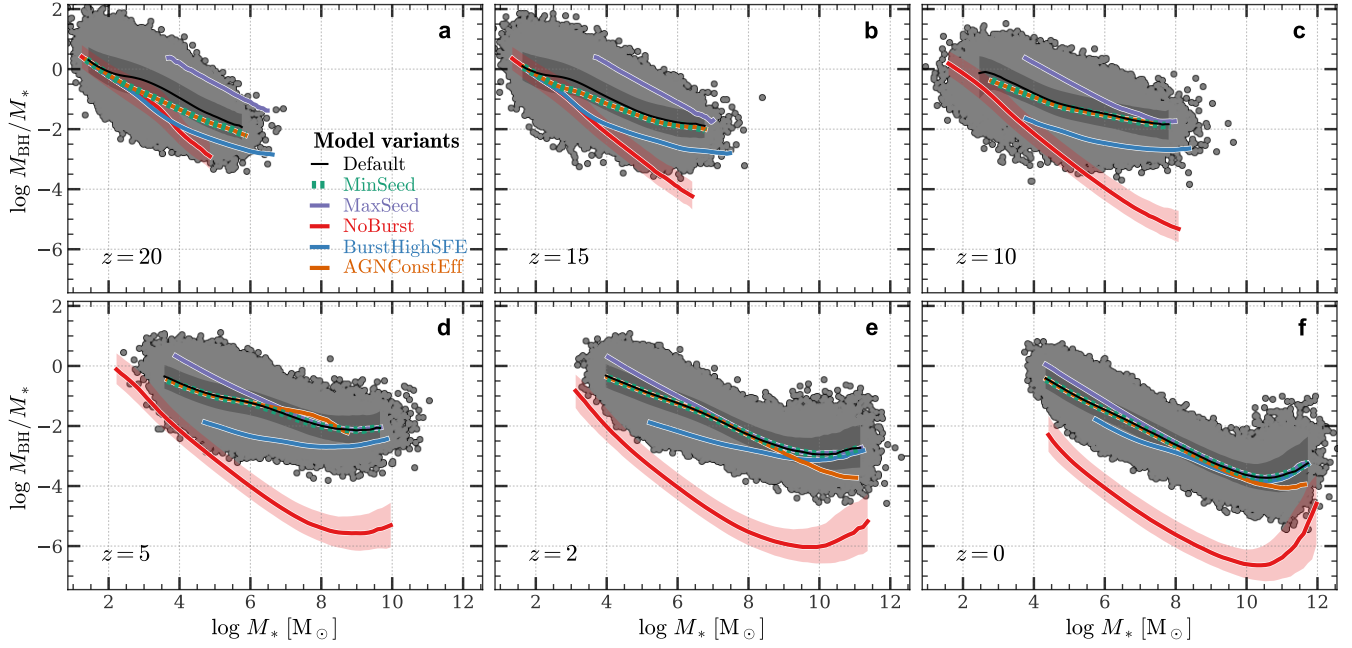


Figure G1. M_{BH}/M_* - M_* relations predicted by the variants of the model. **a-f**, M_{BH}/M_* versus M_* predicted at different z . In each panel, curves show the median values of M_{BH}/M_* as a function of M_* predicted by six variants. **Scatter** points show the predictions for individual galaxies by the *Default* variant. For *Default* and *NoBurst*, we also show the $1-\sigma$ range (16th–84th percentiles) by shaded regions. The variants demonstrate the effects of adapting the physical recipes for different growth channels in the build-up of BH and stellar contents of galaxies. See Appendix G for details.

observational/empirical constraints, while others remain uncertain. Here, we perform a set of controlled experiments by introducing a number of variants of the model. The purpose is neither to perform an exhaustive exploration in the full parameter space nor to find a best-fit calibration, but to demonstrate the effects of different physical recipes on model predictions for the stellar and BH contents of galaxies and to explore implications for future observations.

To maximize the coverage of the parameter space, the variants to be considered include adaptations for each of the growth channels. To minimize the entanglement of the effects of different processes, we limit ourselves to changing the parameters of only one process at a time to produce a new variant. Note that the modifications made on one process may affect the consequences of other processes. For example, a change in the star formation efficiency in any channel also modifies the LW radiation field and IGM enrichment, thus affecting the seeding in other halos affected by the radiation or matter feedback. To avoid such complication, new variants introduced here are based on the *MinSeed* variant (see §3.2.3), which pins down the seeding process by always producing minimal masses of BH seeds in mini-halos. All variants are applied to the same set of merger trees (obtained from TNG100-1-Dark; see §2.1) as used for the main texts.

The variants considered here, including those defined in the main texts, are listed below.

- (i) *Default*: this is the default case used to produce the results in the main texts (except those shown in Fig. 11), with the values of parameters listed in Table D1.
- (ii) *MinSeed*: this variant is the same as *Default*, except that Pop-III seeds with minimal masses are forced to form in all mini-halos. To achieve this, we set the parameter $\gamma_{\text{crit}} \rightarrow +\infty$ and force $J_{\text{LW}} = 0$ and $Z_{\text{IGM}} = 0$ for all halos. This ensures that the factors f_{J} (Eq. B8) and f_{γ} (Eq. B12) are both zero, so that the delay of gas collapse in mini-halos is completely disabled.
- (iii) *MaxSeed*: this variant is the same as *Default*, except that Pop-III

seeds with maximal masses are forced to form in all mini-halos. This is achieved by setting the parameter $\gamma_{\text{crit}} \rightarrow 0$ and forcing $J_{\text{LW}} = +\infty$ and $Z_{\text{IGM}} = 0$ for all halos, so that the delay of seeding and the masses of seeds are maximized.

- (iv) *NoBurst*: this variant is the same as *MinSeed*, except that nuclear burst is completely turned off. This is achieved by setting $\delta_{\gamma} = +\infty$ (see Eq. C3), so that the formation of SNF nucleus never occurs.
- (v) *BurstHighSFE*: this variant is the same as *MinSeed*, except that the star formation efficiency in nuclear burst is boosted, so that the competition between star formation and BH accretion is tilted towards the former and thus BH cannot grow by the wet channel in the bursty mode. This is achieved by setting $\alpha_{\text{BH}} = 0$ and $\delta_{\text{Q}} \rightarrow +\infty$ (see Appendix C5.2). Note that the dry channel in the bursty mode is still open in this variant.
- (vi) *AGNConstEff*: this variant is the same as *MinSeed*, except that the efficiency of AGN feedback is set to be redshift-independent. This is achieved by setting $\beta_{\text{AGN},f} = 0$ (see Eq. D7).

Among the above variants, *MinSeed* and *MaxSeed* are designed to bound the effects of seeding, and are used to demonstrate the degeneracy between seeding and growing (see §3.2.3). IGM enrichment is disabled (by forcing $Z_{\text{IGM}} = 0$) in both variants to avoid the entanglement introduced by Pop-II seeds (see the seeding atlas, Fig. 7).

Fig. G1 shows M_{BH} and M_* predicted by the above model variants at different redshifts. To limit the dynamic range of displayed values, we show the ratio, M_{BH}/M_* , instead of M_{BH} itself, as a function of M_* . Obvious differences can be seen among the variants with adaptations for the seeding methods (*Default*, *MinSeed* and *MaxSeed*). At $z = 20$ (panel a), about the ending point of the seeding era (see §3.2.2 and Fig. 10), the *MaxSeed* (*MinSeed*) variant predicts the highest (lowest) M_{BH}/M_* at any M_* , and the *Default* variant predicts a value in between the two extremes. The largest value of M_* produced by the *MaxSeed* is also the largest among the three variants, while that produced by

the *MinSeed* variant is similar to that produced by the *Default* variant. The largest values of M_* have already exceeded $10^6 M_\odot$ at $z = 20$, implying that significant growth through the bursty mode has occurred at $z \gtrsim 20$ (see Fig. 9 for some examples). Such a growth is expected to be most significant in the *MinSeed* variant due to the least effective AGN feedback, thus explaining the similar ranges of M_* produced by the *MinSeed* and *Default* variants. The differences among the three variants become less significant when galaxies evolve to higher masses, and vanish at $M_* \gtrsim 10^7 M_\odot$. This is a direct consequence of the self-regulation growth of BHs, as demonstrated by the converged evolution of M_{BH} and M_* in individual galaxies using a series of idealized and controlled experiments done in §3.2.3. The differences in M_{BH} and M_* produced by the seeding methods are thus observable only for extremely low-mass galaxies ($M_* \lesssim 10^7 M_\odot$) or galaxies at very high redshift ($z \gtrsim 10$), for which current observations are still very limited (see §3.3 for a data compilation).

In contrast to the seeding methods, adaptations for the bursty mode can produce larger and longer-lasting effects on the BH and stellar contents of galaxies. This can be seen by comparing the predictions from the three variants with different treatments of nuclear bursts (*MinSeed*, *NoBurst* and *BurstHighSFE*). As discussed in §3.3, the growth through the bursty mode plays an essential role in boosting the masses of low-mass BHs to the regime of SMBHs ($M_{\text{BH}} \gtrsim 10^6 M_\odot$). The complete absence of the bursty mode in the *NoBurst* variant thus produces the lowest M_{BH} at any M_* and any z , and leads to an $M_{\text{BH}}-M_*$ relation that is significantly below the observation results (see §3.3). However, disabling only the wet channel in the bursty mode (*BurstHighSFE*) yields smaller deficiency in M_{BH} , reinforcing the conclusion reached phenomenologically in Paper-I (see their §5.2 and fig. 7) that moderate boosts of BH masses are sufficient to bring the BHs into the regime where the continuous mode takes over the responsibility of growing BHs to higher masses. Thus, the dry mode alone seems sufficient to grow BHs to the $M_{\text{BH}}-M_*$ relation observed at $z \approx 0$ (panel f). However, at $z \approx 5-10$, the BH masses predicted by the *BurstHighSFE* variant is more than 0.5 dex lower than those predicted by the *MinSeed* (or *Default*) variant at $M_* \lesssim 10^9 M_\odot$, as expected from the inefficiency of the dry channel in growing BHs compared to the wet channel (see §3.2.1). Another interesting feature that can be expected for the *BurstHighSFE* variant is the peculiar properties of nuclear star clusters (NSCs). This expectation is built on the consideration that AGN feedback dominates the depletion of gas in SNF nuclei in the *Default* variant, and that little BH accretion during a nuclear burst also implies little competition from the AGN feedback (see Appendix C5.1). Intense nuclear starbursts are thus inevitable in this variant, which may produce NSCs that are much more massive than those produced in other variants. The presence of such massive NSCs may significantly bias the total light of galaxies towards their centers and produce dot-like morphologies. Fortunately, this regime overlaps with the faint AGN population that has recently been probed by JWST (see §3.3), thus providing a potential way to distinguish the contributions to BH growth through the wet and dry channels in nuclear bursts, and to constrain the coevolution of galaxies, NSCs and BHs. We will come back to this topic in an upcoming paper.

The final variant considered here is *AGNConstEff*, in which we impose a constant efficiency of AGN feedback at all redshifts (see Eq. D7). Compared with the *MinSeed* variant, *AGNConstEff* produces significantly stronger AGN feedback in the continuous mode at $z \gg 0$, and thus prevents the growth of BH and stellar masses in massive galaxies. Noticeable differences can be found at $z \lesssim 5$ when galaxy formation has transited to the continuous era (see §3.2.2). At $z \approx 5$, the upper limit of M_* produced by the *AGNConstEff* variant (panel d, orange curve) is about 1 dex lower than that produced by the

MinSeed variant (green curve). The self-regulation of BH growth also leads to a reduction in M_{BH} larger than that in M_* , so that the ratio M_{BH}/M_* at given M_* is also lower in the *AGNConstEff* variant in the high-mass end at $z \lesssim 2$ (panels e and f). The deficit in the number of over-massive galaxies and BHs at $z \approx 5$, and the low values of M_{BH}/M_* at $z \lesssim 2$ suggest that *AGNConstEff* may not be able to match observations of massive galaxies and BHs, such as high- z quasars and local massive ellipticals. Instead, these observations seem to prefer a redshift-dependent efficiency of the AGN feedback, such as the one adopted in the *Default* variant.

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